MEANINGFUL, SUSTAINABLE, HUMANITY CENTERED

DESIGN FOR A BETTER WORLD

DON NORMAN

Author of *The Design of Everyday Things*

Design for a Better World

Also by Don Norman

Textbooks

Memory and Attention: An Introduction to Human Information Processing (first edition, 1969; second edition, 1976)

Human Information Processing (with Peter Lindsay: first edition, 1972; second edition, 1977)

Scientific Monographs

Models of Human Memory (edited, 1970)

Explorations in Cognition (with David E. Rumelhart and the LNR Research Group, 1975)

Perspectives on Cognitive Science (edited, 1981)

User-Centered System Design: New Perspectives on Human–Computer Interaction (edited with Steve Draper, 1986)

General Interest

Learning and Memory, 1982

The Psychology of Everyday Things, 1988

The Design of Everyday Things, 1990 and 2002 (paperbacks of *The Psychology of Everyday Things* with new prefaces)

The Design of Everyday Things, revised and expanded edition, 2013

Turn Signals Are the Facial Expressions of Automobiles, 1992

Things That Make Us Smart, 1993

The Invisible Computer: Why Good Products Can Fail, the Personal Computer Is So Complex, and Information Appliances Are the Solution, 1998

Emotional Design: Why We Love (or Hate) Everyday Things, 2004

The Design of Future Things, 2007

A Comprehensive Strategy for Better Reading: Cognition and Emotion, 2010 (with Masanori Okimoto; my essays, with commentary in Japanese, used for teaching English as a second language to Japanese speakers)

Living with Complexity, 2011

CD-ROM

First Person: Donald A. Norman. Defending Human Attributes in the Age of the Machine, 1994

Design for a Better World

Meaningful, Sustainable, Humanity Centered

Don Norman

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To my students:
Past, Present, and Future

Contents

I
Artificial
Almost Everything I See Is Artificial

1
Almost Everything Artificial Has Been Designed

As I sit down to write, I peer to my left out the window looking south to the scene outside. I live on a steep hill in San Diego, California (the local name is Mount Soledad), so my view extends for miles, across the trees and vegetation outside my window, populated by bees, lizards, and hummingbirds, juncos and crows, Red-breasted Hawks, and other creatures I cannot yet name. Even the occasional rabbit. In the distance, I can see the waters of Mission Bay with its many small boats, the Pacific Ocean to the right, with its larger boats and ships, and on a clear day even the hills of the Coronado Islands and the mountain range in the adjoining country, Mexico.

Figure 1.1
The view from my window: everything is there by design. Photo by the author.

Almost everything I see is artificial, designed. The house is designed and made by people, but even the yard was carefully crafted out of the hills and ravines to be level and smooth by excavating some parts and building up other parts with dirt, some of it from the building of Mission Bay, which was created from what once was a wetland and marsh. The result of this transformation is the largest human-made aquatic park in the United States, where, as is common with this kind of project, ecological concerns were overridden by the requirements for recreation.

The plants and animals are natural but carefully tended by people, who kill or remove those we disapprove of. The houses and roads are clearly designed; the plants are carefully planted and maintained, from the grass to the towering palm trees, some more than 100 feet (30 meters) tall. The weeds were not planned; they are side effects, sometimes called the “unexpected consequences,” except that weeds are always expected.

What about the animals? The wildlife is natural, but its habitat and survival depend entirely on the protection offered by this quiet community and its plants and buildings, which offer shelter and places for burrows and nests. All are aided by the availability of food: leaves and flowers for some of the birds; seeds, insects, and worms for some; and small animals and birds for raptors such as hawks to feed upon. So you could say that the presence of all this living life is another side effect: I suspect some of the animals were both expected and welcomed, but some of them, like the weeds, are considered nuisances and pests, including the gophers and rattlesnakes.

Note that even the terms *weed* and *pest* are artificial, for they are not descriptions that nature would use. A weed is a perfectly reasonable plant, just as an animal that is a pest is simply living the way it has evolved to live. It is people who label these natural lives depending on their opinions of plants such as palm trees and dandelions or of animals such as hummingbirds and gophers. People prefer grass, carefully trimmed and maintained: weeds are ungainly. People like animals that do not disturb the environment (or at least the animals should do so out of sight). Gophers make holes in the ground, destroying the “naturalness” of the artificially planted, cropped, and maintained lawns. This is not a bad metaphor for many of the ailments of the world. We like diversity and naturalness, but only as long as they do not interfere with our lives.

We live in a world designed by people. Artifacts abound, from our homes and clothes to our tools, our books. The concept of countries and forms of government are artificial, designed by people. Even things that we think of as natural, a part of nature—such as the earth, the environment, animals, and plants—have been shaped and impacted by people's creations and activities. And design hasn't affected just objects. People invented—designed—organizational structures and ways of governing themselves. Hunting and farming. Ways of preparing and cooking food for eating. What else? Name it, and you realize it is artificial, designed. Money, laws and lawyers, clothing, the notion of a country, how people are named. And just as these designs have shaped, molded, and constrained these things, they have in turn shaped us, so we are no longer natural.

The things we design—the artificial—change the way we behave and act so that just as we have changed things through our designs, those things then change us, affecting how we behave and live: we design the world, and it, in turn, designs us. Each of us is but one of the many parts of a complex, interacting, dynamic system that encompasses all of humanity, all of the earth, and, for that matter, all of the solar system, for our lives, even our genetics, and the existence of the things around us have developed to fit within the cycles of weather, ocean tides, daylight, and climate—all strongly impacted by the location of Earth within the solar system. We can neither survive nor even act alone, but only within the constellation in which we exist.

Why Does Our Artificial World Seem so Natural?

While Britain ruled India, it imposed severe restrictions and rules of behavior upon the Indians, treating them as second-class citizens. In 1947, Jawaharlal Nehru became the first prime minister of India after the Indians forced the British to leave the country. But although children of the Indian upper classes were allowed to be educated at the top institutions of learning in England (Eton, Harrow, Winchester, Cambridge, and Oxford), "their race [still] relegated them to the second rank." Afterward, Nehru wrote that it was surprising "that we, or most of us, accepted it [this hierarchical system] as the natural and inevitable ordering of our lives and destiny." The same holds true for all of us. Our way of existing, our way of being, seems natural to us. After all, it is what we have experienced since birth. It is very difficult to sit back and think through the alternatives, difficult to realize that it doesn't have to be this way. All too often when we have trouble with technology, we blame ourselves. "No," I frequently say, "it's not your fault; it's due to bad design." Well, today the world is a mess, and when people complain, I have the same answer: "Bad design.” I don't mean design as practiced by today's professional designers but rather design practiced by those throughout the history of humankind who invented, constructed, and developed the world as we know it today.

We have long assumed that the earth was so huge that its resources were for all practical purposes infinite. In a similar way, the ruling classes simply assumed that not all people were equal—some solely by being defined to be in upper or lower classes, some because of their skin color, religion, place of origin, or belief system. Human beings seem to have an ability to form small, cooperating, supporting groups but also to distinguish themselves from other human groups in numerous ways in a process that in the 1930s the anthropologist Gregory Bateson noticed in the communities he was studying. He called this tendency “schismogenesis.” Rather than borrow and copy so that neighboring societies become more alike, many seek to amplify the differences. The result can be extreme distortions, altering societies that actually are quite similar into ones that are quite different. The resulting differences can be extreme, artificial distinctions where before there were none.

People are able to take almost any noticeable difference and through schismogenesis or similar mechanisms force divisions and prejudices where before they did not exist. This tendency seems to be the basis for discrimination and prejudices based on skin color, gender, religion, country of origin, language, or accent. It, too, is artificial, shaping almost anything that distinguishes one group from another into a defining characteristic of why the two groups are different. The amplification of small, sometimes insignificant differences then can lead to major areas of disagreement.

We must change the very fundamentals of our lives and being. We must change our way of being on this earth, recognizing that we all are in this together—that we, nature, and the environment form a complex system where changing any part can impact the whole. To understand this system, we must also consider its history, for the current state of our system is path dependent, highly determined by its history. Then we must change the very fundamentals of our lives and our way of living on this earth.

To change our way of being means changing almost everything: how we live, what we make, what we believe, and how we behave. We must question and reexamine all the things that we take for granted, no matter how ordinary, sensible, and logical they may seem. For many of these beliefs and activities, there may very well be completely different frameworks. Only after we recognize that many of the ideas and things we take for granted are artificial constructs fashioned by the people who preceded us can we make a start at a new beginning.

If Design Got Us into Today's Mess, Can Design Get Us Out?

The world is in a mess. Climate change, of course, is at the forefront of everyone's mind, but it is a symptom not the cause of our difficulties. Climate change is the result of many other fundamental ailments that constitute the root causes of today's difficulties. If we want to address climate change, yes, we must address the symptoms, but unless we also address the underlying root causes, the problems will continue. The main difficulty is our way of being, our way of residing on the earth, our belief that we, people, have free rein to use the earth's resources for our own purposes, which means to enhance our comfort and our livelihood. And that difficulty is compounded by the fact that this “we” is not everyone: it is the ruling classes, those with power and wealth that enable some to dominate over others, to enhance their lives, to conquer and rule over territories. The ruling countries have dominated the weaker ones, colonizing them physically at first and then economically, robbing them of their resources.

We inhabit a complex sociotechnical system that encompasses all aspects of the globe: societies, economies, beliefs and behaviors, commerce, education, health, the spread of disease, and ecological disasters. Some relations are direct and obvious, but most are indirect, slow, and not very visible. The latter interactions can be the most dangerous, for they creep up on us, silent and invisible, and by the time we become aware of them and try to act, it may be too late. Examples abound: despite earlier predictions of future pandemics, when the COVID-19 pandemic struck in 2019, the responses were slow and confused, and then the recovery itself has been slow and confused, cycling between periods of optimism and periods of depression, with mixed guidance to people from different countries and different political organizations—even within a given political domain. Meanwhile, the coronavirus responsible for the disease continually changes, introducing more and more variants that evade the medical treatments and cause the scientists to change their approaches and advice. When the virus changes, it is only natural that the scientific advice needs to change as well, but the resulting continual change in advice is blamed on the scientists, not on the virus. The result is decreased trust in scientific advice.

The rise of prejudice, extreme nationalism, and social injustice is also an indirect result of this complex system in which we live. The wide discrepancy in incomes and standards of living between the small percentage of those people who are living comfortably and the vast majority who are not often causes changes in governments, led by people who take advantage of the discontent among the majority, promise improvements, but then use the power they receive for their own power and profit.

The practice of modern capitalism is one of the culprits. The concept of capitalism is not bad in itself. Rather, it is the way capitalism has become deformed and practiced by large corporations, some of them richer than many nations, and by the large financial markets and banks of the world. Here, a search for profit dominates these institutions' behavior, regardless of the cost to people and the environment. The profits don't necessarily have to be based on anything substantial, simply the shifts in the valuation of things from moment to moment (sometimes measured in tiny fractions of a second). Models generated by economists govern this activity, measuring success as an increase in monetary value, independent of the impact on human lives or the environment, and solely in terms of monetary values, not societal ones. Moreover, the formal models impact human behavior even though the models themselves have incorrect assumptions about that behavior.

A new beginning is not something that can be done by a single person or a single group. It will take a mobilization of many to change our way of being so that we can change the world, a change in which we measure success not in money or other indexes such as gross domestic product (GDP) but in the wellness and happiness of people—all people.

Given all these complex problems, how can design be part of the solution? To most people, including me at first, this doesn't make sense. However, as I did the research required to write the book, and as I discussed these issues with numerous people from across the world and rethought my own basic assumptions, I realized that design was at the center of the issues that got us here, so a redefinition and restructuring of design could get us out of them. The beliefs and behaviors that structure our lives are designs created by the early developers of government and religions. The resulting social classes and prejudices are artificial. The design profession also has its defining history. It developed as a tool of modern capitalism to help sell the products of the Industrial Revolution. As a result, design and design education today are driven by the need for the companies and clients that hire designers to be profitable.

The modern field of design has therefore unwittingly added to the problems, designing goods that are harmful to the ecosystem through the mining operations required for the materials; harmful through the manufacturing process, through the use of energy during the use of these products, and through the way people became seduced into usage; harmful because they were designed to be difficult to repair or upgrade; and harmful because of the destruction to the environment when they are disposed of—all discussed in part III of this book, “Sustainability.” The harm was and is not deliberate but rather a direct result of the lack of system thinking, the lack of understanding the impact on existing societal attitudes and behaviors, especially in cultures different from those of the Western world.

All these behaviors are artificial in the sense that they are a result of human behavior, which means they all can be changed, but only by rethinking and redefining the role, education, and activities of the design profession, by rethinking the roles that design plays in life, governance, and industry. These are the themes that I address in this book. The task is made easier by many designers' growing recognition of the need to reframe the field. Mind you, early warnings of the problems were sounded in the late 1800s and in the early and mid-1900s. Today, more and more designers are joining those voices of concern. I add my voice to the others. My framing of the issues is unique, with my major emphasis on human behavior, but the fact that so many people, designers and nondesigners, are voicing their concerns and starting to take action provides a reason for us to be optimistic that we can change things.

2
Our Artificial Way of Life Is Unsustainable

I started this book with the statement "*Almost everything I see is artificial, and almost everything artificial has been designed*." Why? Because the very act of designing is the creation of artificiality. The science of design is one of the "sciences of the artificial," a phrase developed by the Nobel laureate Herb Simon and used as the title of his book. Moreover, even though the field of design exists as both an academic discipline and a practice, long before the discipline was given a name, throughout the history of humankind, people have designed the tools and structures that have altered their lives and ways of living.

Everyone designs. By this, I mean that any deliberate decision to modify the way some activity is done to make it easier or more effective is a design decision. The result is some sort of artifact: a physical device (e.g., a tool), a way of controlling natural phenomena (e.g., to harness fire for warmth, protection, light, and cooking), or a way of governing groups of people, recording events, counting objects, and manipulating ideas.

Professional designers also do these things, but they are trained to apply these skills to larger, more fundamental issues or perhaps to make the idea available to large numbers of recipients in ways that are understandable, effective, and economical. Designs, whether by professional designers or mechanics and tinkerers, scientists and engineers or artists, royalty or servants, have created today's world. A world that is unsustainable. A world that pits the rich and powerful against the poor and the weak.

The root causes are political and economic, deeply embedded in our systems of government, economics, laws, and procedures, all of which are artificial structures and modes of practice designed over many centuries. Many of these structures and modes resulted from the colonization of the world by the European powers. This colonization replaced Indigenous cultures with European ones—most importantly, the economic structure of capitalism as practiced in Europe (and the United States), the emphasis on measurement in science, and the need to supply the materials and labor to feed the factories of the Industrial Revolution. Because cultures are human made, they can be changed. Many of the problems are the unanticipated outcomes of the early design work, often done by people outside of the design profession, so that even though such designs often had well-thought-through physical structures, there was little or no consideration of their impact on the environment or people and their lives, no consideration of the waste products that might result. This time around, we need to get professional designers involved, people trained to consider the impact on people and society.

Fortunately, designers, schools, and foundations are starting to address these issues. More and more designers are working on major societal issues. It is time to change how the world operates, to design sustainable systems, to take account of a design's impacts on the people, to ensure that the design is done with everyone in mind, and to hold sustainability and equity as the goals.

We Are All Designers

Because everyone designs, we are all designers, so it is up to all of us to change the world. However, those of us who are professional designers have an even greater responsibility, for professional designers have the training and the knowledge to have a major impact on the lives of people and therefore on the earth. Many of the ailments facing us today—the problems of waste, the mountains of trash full of devices that cannot be repaired and cannot be disassembled to reuse the parts—are partially the responsibility of the design profession. We professional designers have a greater responsibility to correct the problems that our ways of living and behaving have caused.

If Design Is to Save the World, It Must Be a Different Kind of Design

If design got us into today's mess, perhaps it is design that can get us out, though not the way design is conceived of and practiced today. We need a new form of design, one that can work with the huge variety of issues, people, politicians, and businesses across the world. We need design as a way of thinking, of approaching large sociotechnical systems, of recognizing each person as a component in the complex system of the world that comprises all living things, the earth, land, and sea, where each component impacts the others. For humankind to exist, this complex system must be sustainable, resilient, repeatable. Today, it is none of these.

Design must change from being unintentionally destructive to being intentionally constructive: repairing what's gone wrong, collaborating with marginalized voices, and sustaining the earth's limited resources. The massive changes we must make require us to rethink the fundamentals of our way of life, which means our industries, schools, governments, and the economic systems that measure and define what is considered “good.” Because our way of life is artificial, designed by people over the course of history, it can be changed—changed by people over the course of the future.

3
Why History Matters

Those who cannot remember the past are condemned to repeat it.
—George Santayana, *The Life of Reason,* vol. I

Wrong. History does not repeat itself, whether the past is remembered or not. This lack of repetition does not diminish the importance of history: how people have acted in the past, in history, does have a significant impact on the future. Our beliefs, our sense of right and wrong and of our place in the world, and what actions we believe are appropriate derive from our own experiences and personal history, which in turn are affected by the experiences of our families and those around us, all of which were influenced by history. We do not repeat history, but we are shaped by it.

Path Dependence: The Past Does Matter

History doesn't repeat itself, but the past matters. In the physical sciences, the simplest models obey a principle called “path independence.” The current behavior is independent of its history and is entirely determined by the current state of things. There may have been multiple paths to the current state, but how that state was reached does not matter: it is path independent.

Path independence, like many scientific assumptions, makes the computations easier and is correct for classical physics. If I throw a ball, the path it takes is not affected by the ball's historical experience but only by its current state and the forces applied to the ball as it leaves my hand. The physical world is in many ways easier to understand than the world of human beings. We should not make the mistake of believing that the methods used in the sciences of the physical world are appropriate to our understanding of the biological world, and these methods are especially inappropriate for our understanding of the world of human beings. We are not balls being thrown in the air. People have memories and histories that impact beliefs and emotions. The behavior of people and societies is path dependent, not independent. How people arrived at their current beliefs and thoughts can often be traced back for centuries.

Look at some of the long-lasting feuds between countries or even within countries. Insults and catastrophes or evil acts from hundreds or even thousands of years ago are still remembered in ways that shape current actions and beliefs. The history of racism and prejudice, of the colonization and inexcusable treatment of so many colonized people, the destruction of Indigenous cultures, and the rape of the earth are all part of our history.

This is why we need to understand the history of colonialism, capitalism, and the unbridled actions of the powerful upon the weak.

The history of the world's societies, empires, and nations is not path independent. Whether we remember the past or not, the current state in which we now exist has been affected by the past. As a result, the only way to overcome the injustices of the past is not just to remember them but also to understand them, to understand how they have left an impact on the world today, and therefore to begin to understand what must be done to alter that path. Otherwise, yes, we are condemned to repeat the sins of the past.

Except for scholars who study and trace the path of history, the rest of us are often unaware of its impact. We are born into the world, and our early experiences and belief systems seem so natural and obvious that it is difficult to imagine any other possibility. People take for granted the basics of their everyday life: living in a family, going to school, learning the topics taught in a certain way there, getting a job, and so on. In many countries, jobs take one away from family for most of the daytime hours and oftentimes into the night. All of this is taken for granted.

Why must the demands of work separate families, though? Why must some cultures demand long hours of work, often from dawn to dusk, six days a week from its citizens, leaving partners and children to struggle throughout the day on their own? The need for a large number of workers at one site started early in history—for example, in the massing of people by the Egyptian pharaohs to build the pyramids. In part, it was a natural result of the rise of cities and nations, where people were concentrated. Providing food for large numbers of people required workers to congregate, whether in Baghdad's cooking competitions in the ninth century or in Venice's building of ships in the early twelfth century. The standardization of work in factories took place at the start of the Industrial Revolution in the early 1700s for the weaving and spinning of silk and cotton as well as for the manufacturing of household goods in Great Britain.

The path taken by Western countries of the world and exported globally controlled the lives of the workers, treating them as if they were machines to be used until they wore out and then replaced. Time dominated the lives of the workers, with bells and whistles telling them when to start and end the day, when breaks were permitted, and when they could eat lunch. Time dominated the work, and work dominated wellness, satisfaction, and family. Did it have to be this way? No.

Why is history important? Because it demonstrates the artificiality of many of these beliefs. Artificial does not mean wrong; it means the development of the work structure came from practices spanning the entirety of recorded history, roughly 5,000 years, and from some arising in the past century or past few decades. The critical part of history is that which occurred before each of us was born. When we are born, we grow up in a world that has been shaped by historical events, but we are unaware of all that transpired to create the world we are born into. We take the way we live as natural. Some of us are born into a comfortable, accommodating life, others into a life full of struggles and prejudices. The more comfortable our lives, the more difficult it is to recognize the artificiality. Even when we see the excesses, the problems, and the impact on others, we tend to think of them as symptoms that can be directly addressed.

Many of the major problems of today's world appear to have simple answers. Climate change? Let's reduce the pollution of the atmosphere. Droughts? Use water more efficiently and find new sources. Rising water levels, heavy rains, and flood? Build higher dikes. Prejudices based on race, skin color, religious beliefs, country of origin, and class distinctions? Harder to deal with but still solvable if only we put our minds to it. Alas, these simple solutions are, well, too simple. Some of them treat the symptoms but not the underlying causes, and some are far more complex than the simple statements that appear to be the solution.

To make a meaningful change, we must reframe the underlying causes, which are deep and difficult to attack. Some causes involve learned beliefs and prejudices so strong that those who hold them do not see them as prejudices or biases but rather as simple truths that people need to respect. These beliefs involve our methods of governance, our lifestyles, our artifacts, the way we manufacture and distribute, the way we reward, promote, and punish. Change does not come about by attacking the symptoms: we must attack the causal factors—the system.

Ways of life can change, but such change is difficult. Learning history helps us understand how artificial so many beliefs are, but one must be careful. History, it is said, is written by the victors, by those who were successful, and so the victors' ancestors reject changes that to them seem artificial (which they are) and harmful to their beliefs and well-being (which they may be). Those who benefited from the historical path see little reason to change, not necessarily because of self-interest but because they cannot imagine any other way of being.

How do we change the way we act today? We need to consider the history of the world from all viewpoints—from those who lost as well as from those who won so we can understand different value systems, ideas, ways of governing, and ways of living. Just because something is old does not mean it is beneficial. Yet alternative histories can be just as biased and misleading as the histories written by the victors. Fairness requires considering all points of view, even though they are apt to contradict one another, often with very little evidence available today to sort out the stories.

Today's view of history is dominated by European thought or, if you will, ways of thinking that come from the Western, technological countries of the world, sometimes called the "Global North." Note that these names, "European," "Western," and "Global North," are not geographical terms; they are political. The United States and Canada are part of the "European" tradition. Australia is part of the Global North, Western, and European. In general, Western thought has dominated the world of ideas, commerce, and educational practices. European countries have dominated the world since the beginning of the Industrial Revolution in the mid-1700s, setting up commercial ventures, taking over territories, and colonizing them. The colonization movement destroyed the existing Indigenous cultures and replaced them with European/Western ways of thinking and behaving. This was accomplished by sheer force in most cases.

Take, for example, the world of cotton. In *Empire of Cotton*, the Harvard professor Sven Beckert traces the history of British merchants' worldwide domination of cotton in the late 1700s, starting huge trading companies, taking the cotton from the East (Indonesia and India), and shipping it to Britain, where it was manufactured into clothing, which was then sent out to be sold across the world. The demand for cotton led to the colonization of India and other countries by Great Britain and European nations. One type of cotton plant is native to the New World and was cultivated by settlers in what is today Florida in the mid-1500s. American cotton plantations started supplying the English textile industry in the 1700s. Growing and preparing cotton for shipment by removing the seeds is extremely labor intensive. The necessary labor came from captives of the tribal wars in western Africa, enslaved people who were shipped to the American South. Southern law considered these enslaved individuals property, and so they were legally bought and sold on the marketplace.

The story of cotton is just one part of the general history of the European exploitation of non-European territories. Britain dominated primarily because it had the largest, most powerful navy in the world. The English East India Company, "formed for the exploitation of trade with East and Southeast Asia and India," played a major role in both the trade of goods (cotton, silk, and spices) and the slave trade. Enslaved individuals from Africa were an important part of the story of the importance of cotton in US trade, which involved shipping the cotton to England, getting back cloth and clothing, as well as acquiring newly enslaved people from Africa. Although slaves have long existed in the world—they are discussed in the Old Testament of the Bible—because the enslaved people in the United States came mostly from Africa, they could be distinguished by skin color, which is one of the causes of prejudice.

The history of cotton is long and complex but covers many of the origins of today's problems: military power, colonization, slavery, and racial prejudice. And so the history of the world does impact the behavior of people and nations today.

The Role of Technology and Modernity

A major theme of this book is the dominant role that technology and modernist philosophy play in our lives. People have become second-class citizens, having to fit almost everything we do into the dictates of our technologies, whether it is when we decide (or, rather, are told) to wake up in the morning, what we are induced to buy, what kinds of jobs we do, how much we like them or not. However, technology is not the real culprit; rather, it is simply part of the dominant way of life in the industrialized countries of the world.

The philosophy that ties all of these components together is *modernity*. The themes of the Chicago World's Fair in 1933 spell out modernity's fundamental principles: “Science discovers, genius invents, industry applies, and man adapts. . . . Individuals, groups, entire races of men fall into step with the slow or swift movement of the march of science and industry.”

Yes, this was (and in many ways, still is) the West's prevalent view. Modernity stands for science and technology, for rational thought, and, above all. for progress, with progress defined in terms of technology and commerce. The *Encyclopaedia Britannica* provides a more complete description of modernity's impact: “Modernity was associated with the rapid-pace movement of global capital, satellite transmission of images, and immediate transglobal communication. Despite those developments, however, modernity also came to define savage inequities on a global scale, from industries run for the profit of invisible shareholders to cultural biases in the provision of public education." The philosophy of modernity covers almost all facets of Western life, including politics, governance, business, economics, equity, treatment of workers, social classes, and the interactions between the Global North and the Global South, the latter often serving as a source of basic materials and low-cost labor for the former.

To understand the dominance of technology over people requires us to understand the history of our civilization, for from the time history was recorded, the technology of warfare, especially machines and weapons, has enabled tribes, societies, regions, dynasties, and countries to dominate others, to establish agriculture, trading, cities, and structures for governance. This way of life empowered the wealthy, establishing the notion of classes, a caste system where some people dominate over others from birth to death. It also gave rise to the continual seeking of "progress," measured by the size of the group, the amount of wealth it had accumulated, and the power of its rulers. The rich lived one set of lives, and the poor lived "modernism." To many people of the past, especially in the mid- to late 1900s, modernism was the driver of progress. Today, however, many consider it the driver of evil, for the progress of one group is invariably accomplished through the disenfranchisement of others, whether they be enslaved individuals or simply low-paid workers in the farms, fields, or factories.

The Curse of Modernity

> There are professions more harmful than design, but only a very few of them.

This quotation is the very first sentence of the book *Design for the Real World* by Victor Papanek, a brilliant (and controversial) twentieth-century designer. He advocated designing for real people in real situations. Not the gaudy, expensive, resource-wasteful stuff in the countries of the West but stuff for places that have few resources.

Why is Papanek saying this? Because, he says (in the same paragraph), “by creating whole species of permanent garbage to clutter up the landscape, and by choosing materials and processes that pollute the air we breathe, designers have become a dangerous breed.”

Disclosure: I changed the first quoted sentence by deleting the word *industrial* in the phrase “industrial design”: in the original, he blamed “industrial design,” but I broadened the criticism to apply to all of design. Why did I take this liberty? Because I am certain that Papanek would have approved. When he wrote the book, industrial design was indeed the culprit, but today, with so many designers plying their craft in systems, services, software, and digital products (“nontangibles,” they are sometimes called), the blame can be attached to the entire field. It would not be fair to let these other important subdisciplines of design avoid blame.

Although Papanek's condemnatory statement is widely circulated today, he was not the first person to have tried to convince the world of this message. Siegfried Giedion made a similar argument in his massive and influential book *Mechanization Takes Command* more than 20 years earlier in 1948, and, even so, he was not the first.

Make It New

In the mid-1930s, the poet Ezra Pound wrote a series of essays under the title *Make It New*. New was good: old was, well, inferior. Pound was speaking to artists and poets, arguing that they “must break with the formal and contextual standards of their contemporaries.” “Make it new” was a fitting motto of modernism, but the phrase and the beliefs it summarizes are almost a thousand years old, tracing back to the neo-Confucian scholar Chu Hsi (1130–1200 CE), whose works Pound had been reading.

The term *modernism* refers to developments in aesthetics, the arts, architecture, and a broad cultural movement aimed at throwing out old ideas, breaking with tradition, and developing new forms of art, dance, theater, and literature. Ezra Pound was fighting for modernism, at least for art and poetry.

The term *modernity* is related to *modernism* (and often used interchangeably with it), but it refers to the intellectual and philosophical emphasis upon reason, the Enlightenment, and, of course, the great changes taking place in science and technology. Modernism also impacted economics and the role of the merchant (later called the businessperson) in using the scientific methods of accounting to define “progress” as the making of money. Ethical considerations completely disappeared. Victor Papanek fought against modernism, especially the harm done to people and the environment as the design profession aided business to churn out a continual stream of new things for people to purchase. He was a strong opponent of the advertising profession, which convinced people that they needed these new and different objects even if they didn't or even if they had older versions that still worked perfectly well.

Papanek's argument is a worthy one, for it still applies today. But he was wrong to blame the design profession, for designers, too, were the victims of the Industrial Revolution, not the causes of the harms it wrought. Designers occupy mostly midlevel positions in the world, whether in the rankings of university departments, the level of influence they have in companies, or the influence they might have over their clients’ wishes in private consultancies. Papanek points to the symptoms, not to the core issues. The core issues arise from the philosophy of modernity, which creates a heavy dependency on science and technology, on rational thought, coupled with a lesser emphasis on people, humanity, and nature. The new was considered good: the old was bad and inferior. Modernity is also the philosophy of consumerism. It is the notion that neither science nor engineering nor their applications—technology—can do wrong; that the continual building of stuff, whether digital or physical, companies or organizations, is “progress”; and that progress by definition is good.

This kind of “progress” might be good for people with wealth and power, but it is certainly not good for everyone. The large—and increasing—number of unhoused people in cities across the United States and the world is emblematic of the failure of “progress,” as is the increasing income disparity between the very few with power and capital and those without. The financial markets amplify the discontinuities. Companies are urged to aim for short-term profits at the expense of long-term good.

This is not the capitalism that the eighteenth-century economist and philosopher Adam Smith described and aspired to in *The Wealth of Nations*. Indeed, he warned against the excesses we now see. “People of the same trade seldom meet together,” wrote Smith, “even for merriment and diversion, but the conversation ends in conspiracy against the public or in some contrivance to raise prices.” Adam Smith's theory of capitalism should not be confused with the current practice of capitalism: human greed has distorted the theory.

For the points I raise in this book, three of the most important historical influences on today's world are modernity, the Industrial Revolution, and economic theory (Adam Smith is often named the father of economic theory). These three influences are tightly intertwined. All started at roughly the same time, in the mid-1700s, and all in combination created the governments and economic systems we see today, heavily influencing the conduct of business and the employment of workers in industry. The heavy hand of path dependence dominates many decisions and activities, even though there have been many advances in all three of these developments since the 1700s. Modernity is now rapidly growing out of favor, even though the mindset lives on. The Industrial Revolution is now in its fourth iteration but some people are already specifying what the fifth generation might look like.

The first Industrial Revolution. Stage one was the introduction of steam and water-wheel-powered machines, starting with the making of textiles but soon including the making of many home goods and tools. Prices dropped rapidly as the volume of goods that could be produced increased dramatically.

The second Industrial Revolution. Stage two took place from the end of the 1800s to the early 1900s. New methods of communication—especially the telegraph and telephone—and of travel—especially the automobile, truck, and the continent-spanning railway systems—changed many aspects of business and family life. Electricity, gas, oil, and coal became widely used sources of energy. Henry Ford developed the assembly line to produce automobiles much less expensively than had been possible before, thus driving a rapid transformation in family mobility and leading to the dispersion of people from cities to suburbs.

The third Industrial Revolution. Stage three started in the mid-1940s following the end of World War II. It marked the emergence of electronics and the early computers, moving from dominance by mechanical machines to dominance by information-based ones, whose power and flexibility increased rapidly.

The fourth Industrial Revolution. Stage four, which started in the early twenty-first century and is ongoing, marks the emergence of intelligent machines, of worldwide immediate access to information via the internet and related information networks. Autonomous machines and vehicles are starting to appear, and people are being replaced in many jobs that were once thought impossible to be done by machines: jobs done by low-level lawyers, accountants, and even physicians (for example, reading MRI and X-rays and even prescribing treatments).

The fifth Industrial Revolution. Stage five, which at this point is more of a dream than an existing development, unleashes the power of quantum computing, fully automated and highly intelligent systems, better security through encryption and blockchain, and cybercurrency. New advances are occurring in biology with the design of DNA tools of all sorts—sequencers, synthesizers, and snipping. Sensors are dramatically altering the power of machines, of science, and of medicine. Tourists now take space trips (at exorbitant costs). New pandemics are sweeping the world. And despite the rising concern about the dangers of climate change, more money goes into the development of armaments and armies than into societal needs and ways to thwart climate change.

These five stages have been driven primarily by technological advances. As a result, they have neglected many important components of human behavior. First, they ignore the Age of Waste (discussed in part III of this book). They ignore the huge impact on people's lives, as some who lose their jobs because of automation may not be able to develop the skills necessary for other work. They perpetuate the use of mathematical models and the precise measurement of meaningless variables, ignoring the important variables that we cannot measure (discussed in part II). Climate change? A simple glitch of technology, say fifth-generation industrial revolutionists: we can solve it with technology. No, we can't, because the underlying cause of the problem is human behavior.

Modernity, technology, and the current forms of economic theory are impairments to a better life, as the rest of this book demonstrates.

4
Precise—but Artificial—Measurements

We experience time, weather, and all aspects of life in ways that impact our activities, our emotions, our sleep cycles, and our health. These experiences are *subjective*, a term that is often used despairingly by the physical scientists but that in fact simply means "personal"—affected, for example, by feelings, tastes, beliefs, and emotions. Physical scientists are dismayed by the variations in these subjective experiences and therefore have replaced them with highly artificial systems of definitions and measurement. I do not object to the need for precise, repeatable measurements. What is objectionable is that these definitions necessary for the sciences have been used to define and describe our own, intimate, personal experiences and thereby to take over our day-to-day lives, impacting how we live in ways that have nothing to do with the scientific need for precision.

Let me start by examining two highly artificial systems that dominate our lives despite their arbitrary, nonnatural characteristics: the scientific definition of the year's seasons and the scientific measurement of time. Here are two questions to ponder:

1. Why do we look at our calendar to see when summer begins instead of looking at the weather? Isn't the actual day-to-day weather the important variable for people and all living things? Seasons as defined by science are artificial.
2. Why do we look at our clocks to determine when we should eat instead of asking our stomachs? Aren't our body's needs and its signals what we care about, not the time of day? Time as defined by science is artificial.

Seasons as Defined by Science Are Artificial

Why are there four seasons in the year? This question probably never occurred to most of you. If you were born and lived most of your life in the world defined by the Global North, the four seasons seem natural and appropriate, with the starting and ending dates of each one precisely defined by the positions of Earth in its path around the sun. We define the seasons of the year not by using the actual temperature or weather, not by noting the importance of the seasons to agriculture and human activity, but instead by depending on arbitrary astronomical cycles and calendar dates.

The seasons are a natural result of Earth's tilt as it revolves around the sun, which causes different amounts of sunlight to reach Earth and different patterns of weather during the year. It is easy to understand why people in the Northern and Southern Hemispheres might have recognized these differences as representing two seasons: one that was warm and had days longer than the nights, the other that was cold and had nights longer than the days.

For countries close to the equator, the amount of daylight and the average temperatures do not vary much throughout the year. Nevertheless, many of these countries copied the dictates of the Global North to distinguish among the four seasons even though the distinctions are meaningless for them. Many could simply have two seasons, rainy and dry, and some do use such designations.

So why do we have four seasons? Is it because there are natural divisions of the year? We call the warm and light period “summer” and the cold and dark period “winter” and then add two transition periods, “spring” marking the transition to summer and “autumn” marking the transition to winter.

Astronomers have arbitrarily divided the year into four seasons based on Earth's position during its travels around the sun. Note that it is not the distance from the sun that matters but rather the angle between the Earth's tilted axis relative to its elliptical path around the sun, which creates different night and daylight times at different positions on the path (except for people at the equator). Science has ignored our actual experiences and the natural biological behavior of all living things to stipulate that the four transitions are marked by the two solstices and the two equinoxes, which is why we have come to accept the division of the year into four seasons, with artificial starting and ending dates.

A technical note: the four days that mark transitions among the four seasons were chosen because they are significant ratios of the amount of sunlight to darkness—maximum, minimum, and equal. These solstices and equinoxes are correlated with climate, but correlations are statistical averages and don't indicate what is actually happening in any given year. The weather in any particular month or on any particular day is even less correlated with calendar dates, and for any given date the actual weather conditions can have large variation from location to location.

Not everybody in the world uses these “astronomical seasons” to divide up their year. Some countries use what is called “meteorological seasons,” where the seasons begin on the first day of the months in which the equinoxes and solstices appear. In either case, though, the timing/designation of the seasons is artificial, not at all what people whose lives and livelihood depend on the weather need.

Why four seasons? Why don't the seasons match our observations of temperature, plant life, and animal migration? Birds and animals time their migration and plants their seasonal variations by weather patterns, not by a calendar. Similarly, natural human behavior is controlled by weather patterns, not by Earth's precise location on its journey around the sun (which is arbitrary to most people, even though not to astronomers). For example, the Australian Aborigines divided the year differently in different parts of Australia. Some communities designated only two seasons each year, others six. The discrepancies between different tribal groups' labels did not lead to discord. What leads to discord is the discrepancy between human-fixed boundaries and the natural behavior of the weather, plants, and animals.

If defining seasons by their impact on the lives of people, animals, and plants is so natural, why don't people do it that way? They used to. Indigenous people across the earth planned their lives according to the perception of the climate patterns, dividing the yearly weather patterns into groups that fit their particular needs and selecting the dates according to the weather's impact on plants and animals. As noted earlier, many countries near the equator have only two seasons: dry and wet. There is nothing accurate or natural about four.

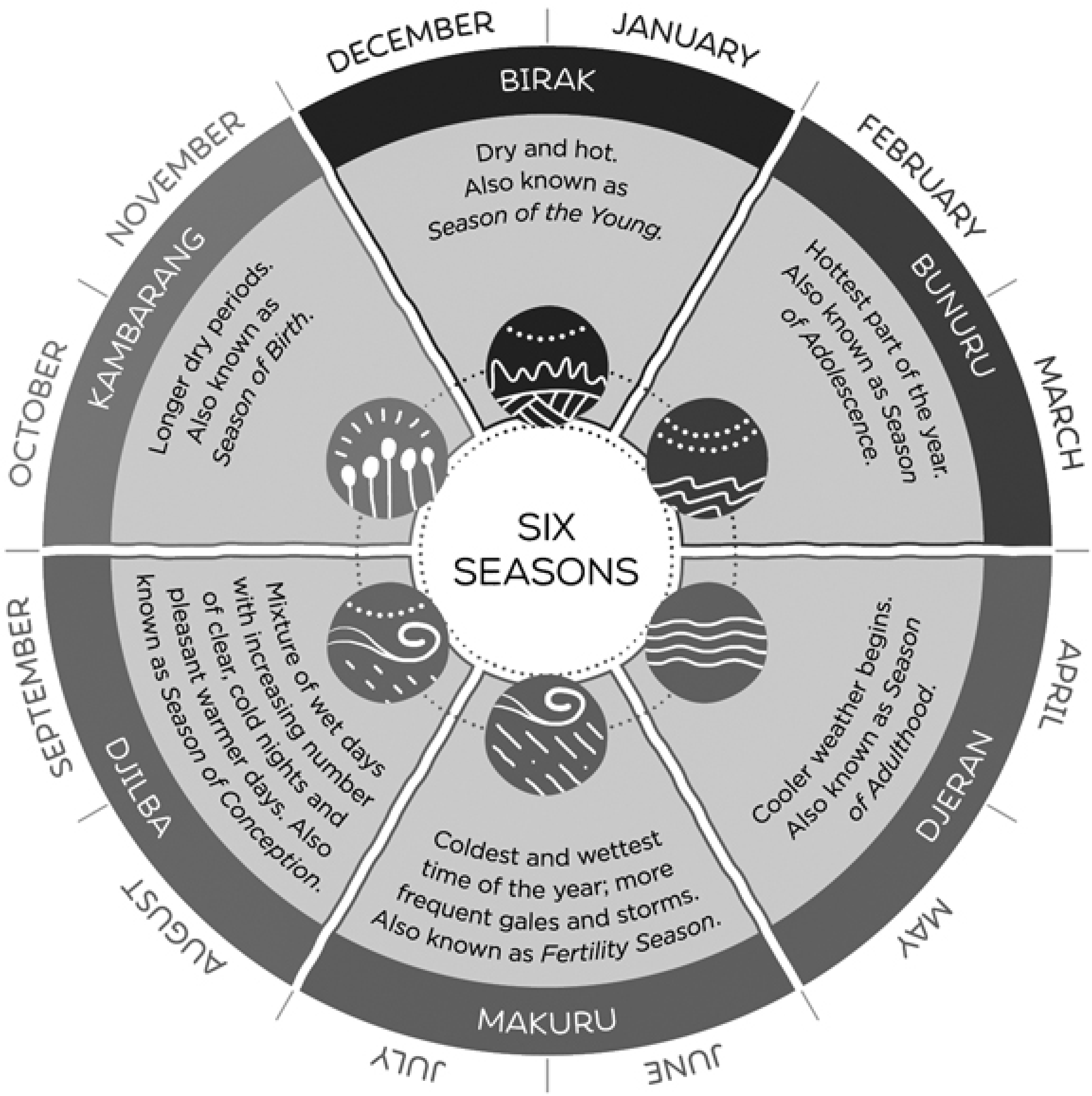


Figure 4.1
"Six Seasons of the Noongar." The Noongar are the Aboriginal people who live in the southwestern corner of Australia. Their calendar shows these six seasons and indicates the typical flowers, other plants, animals, and weather patterns of each season. The seasons help determine when activities such as hunting, gathering of foodstuff, and camping take place. From "Six Seasons of the South West: The World's Oldest Living Culture," Australia's South West, n.d., https://www.australiassouthwest.com/south-west-inspo/six-seasons-south-west. Reprinted by permission from Australia South West.

Of course, if the division of seasons were to follow weather patterns, the start and end of a season would be determined each year by observation and would therefore differ every year, not only in different parts of the world but even in different parts of the same country. Would this be horrible? No. It would accord with local conditions. Would the world become confused? No, such as system would be much more useful for those whose livelihood is determined by the weather.

But wouldn't this system make the scheduling of some recurring events—such as the start and end of various sports, of schooling—difficult? No, because these events could be determined by fixed dates on the calendar. The variability in the actual boundaries of seasons do not have to impact these schedules.

Why was the determination of seasons changed? Because scientists like precision and certainty. They truly dislike the uncertainty of a system governed by the idiosyncrasies of local weather or the behavior of animals and plants. So they have forced their artificial categories upon us, first starting in Europe and then moving on to the colonies as the European countries imposed the same definitions on them.

Time as Defined by Science Is Artificial

In today's world of relativity physics, time and distance are further refined into a single unifying concept of spacetime (one concept that combines space and time, so it is written as one word—spacetime or space–time). Fortunately, the equating of time and space in physics is unlikely to be imposed on our everyday time-measurement system.

Time plays a key role in physics, but it isn't the same "time" that people perceive. For example, people perceive a period with multiple events as passing rapidly but the same period when there are no events as going on for a long duration. Interestingly, people's memory for these periods is just the opposite of experiences during the period. Later on, when asked about the events, people remember the period with multiple events as lasting for a long time, whereas they remember the empty duration as being brief. Why? The debate is still going in the scientific journals of psychology, but my opinion (supported by many psychologists) is that, to people, time perception is highly dominated by how busy they were during the interval: the more activities they took part in (or the more complex the activity), the longer the interval seems to be in their memory. But if the interval was empty of events, there is also nothing to remember.

Because time plays such an important role in science, earlier scientists needed a measure of time that was independent of how it was filled, a measure that would yield the same result no matter who did the measurement or when it was done. This measurement had to be independent of people's experiences. The result of that need was that scientists began to measure time by counting repetitive physical phenomena, starting with the number of pendulum swings between the start and end of a period and culminating today in atomic measurements by using the atomic resonance of the stable isotope of cesium (cesium-133), where 9,192,631,770 pulses define a second (roughly 9.2 gigahertz).

This great precision in defining and measuring time is very important for the progress of many different branches of science and technology. For example, it helps the global positioning satellites (GPS) give our navigation systems precise location information. Yes, the scientific measurement of time has important and useful consequences, but it has lost all connection to human perception and behavior. If physicists had used a different word to define what they were measuring, the result may have been better. The word *time* as used by physicists has little to do the word *time* as used by people.

Today, we have learned to live with the artificial and arbitrary division of a day into 24 hours, further divided into 60 minutes per hour, and 60 seconds per minute. Why does it matter to us if scientists insist on time being defined as a regular, physically unchanging count of a vibrating atom? How does that impact our lives?

Was this precise definition necessary? Not at all. Scientists can define the world any way they like, but they should use those definitions only for science. By imposing this categorization upon all of us, they have dictated when schooling starts and ends, when vacations are taken, when we eat lunch, how long we work, and other regularities that do not require the arbitrary accuracy and specifications of scientific measurement.

Our Lives Are Governed by Arbitrary and Artificial Measurements

The scientific definitions of the seasons of the year and even the measurement of time by precise mechanical and atomic clocks are artificial, delimited to suit the needs of science, not human behavior. So why aren't these definitions used just for science? Why are they imposed on our lives? The concept of a day, month, and year has a natural basis in the rhythms of the physics of the earth, the moon, and the sun. The repeated rhythm of brightness followed by darkness makes the definition of “day” relatively easy, although the time that a day starts and ends is arbitrary. Some cultures start at sunrise, others at sunset, and scientists define the start of each day at midnight.

But what about the week? Where does it come from? And why does it consist of seven days? And which day starts the week? Many cultures did just fine using the cycles of the sun and moon to define years and months, but there is no natural phenomenon that defines the week. Some cultures did not have any concept of the week. Some had a week that lasted five days or other amounts. During the French Revolution, when the metric system tried to make everything be based on a base of 10 (the scientific mind at work), the week was defined to be 10 days long, with 10 hours in the day and 100 minutes in an hour. Obviously, this system did not catch on.

Even today there are numerous attempts to reform the calendar, with a major effort attempting to make each day of the seven-day week always fall on the same numerical day each month. This would simplify many different issues, but it would require adding leap weeks to the calendar every five or six years or interspersing extra days, each of which change would add extra complications.

Because calendars are artificial inventions, a wide variety of calendars have been invented. There are three *natural* markers of time: the day (the period of one light–dark cycle), the month (the number of days between successive full moons), and the year (one revolution of Earth around the sun, determined by plotting the locations on the horizon at which the sun appears to rise or set, coupled with weather patterns because every position except for the two extremes repeats twice a year). Nature, however, does not consciously do arithmetic, so the number of moon months per year is not an integer, so all yearly calendars have had to fiddle with the duration of its months to make things work out, and because the length of the year is not an integral number of days, calendars must create extra days now and then to keep them synchronized with the yearly patterns. Today, most of the world follows a reformed calendar designed by a series of Catholic popes who tried to reconcile the lunar and solar calendars into one solar-based calendar, with rules for adding an extra day every four years (skipping the leap year every 100 years, except when the year is divisible by 400—because the year is not precisely 365¼ days long but rather 365.2422 days). The dates of many religious holidays are still determined by lunar cycles, which is why their fixed time in the lunar calendar translates into a variable date on the standardized solar calendar.

Perhaps the greatest impact of the artificial definition of time came about during the Industrial Revolution with the invention of the factory.

The Demands of Industry Caused Lives to Be Ruled by Clocks

Once upon a time, the life of people was controlled by the intersection of their needs and the cycles of nature. Prior to the Industrial Revolution, time was a psychological variable, experienced differently by people depending on many factors, including the activity they were performing as well as their psychological moods. Families often lived according to the rising and setting of the sun, not to clocks. There was no artificial separation between work and family. “Work” didn't exist as a concept. There were chores to be done, activities to be performed. On the farm, plants and animals had to be cared for. Those who were craftspeople or merchants had jobs to do, but the timing was left up to the individuals. There was no artificial separation of the day into periods when work was performed away from the family and periods when the family was together. Instead, from the family's point of view, these activities were not indistinguishable, for the day's activities and the interactions with family members were intermingled.

Activities were performed when they needed to be done. Different groups of people and families had different cycles of activity. I do not pretend that family life was always comfortable or even pleasant, but their mode of work allowed for the interweaving of family life and work. The timing was natural.

With the Industrial Revolution, however, factories were designed so that people had to be together at the same time. Life was soon regulated by the dictates of factory whistles and church bells, marking the arbitrary dictates of when it was time to wake, be at work, take a meal break, and end the day, not by natural rhythms.

Families were separated, with those employed rushing off to their jobs, often not to return home until late at night. I have seen the workers in South Korea and Japan work long hours every day and attend the quasi-obligatory drinking sessions after work. These sessions end so late that the workers do not have time to make the long commuting trip back home, so they stay overnight at the many hotels that catered to this need. Workers often do not return to their families for days. Their work habits are slaves to both the clock and the perception of doing work. In fact, the long hours mean that the workers are sleep deprived, often falling asleep at conferences and meetings and on the commuter trains. Studies have shown that long hours produce less work than the shorter, more focused hours used in some countries.

Artificiality Creates Societal Antiaffordances

The artificial nature of society controls human behavior in multiple ways. Quite often, the artificial rules and habits are so subconsciously engrained that we are not even aware of how they constrain our behavior, sometimes encouraging certain kinds of activities but more often constraining other activities.

These subtle controls over behavior reflect a concept that has been called “affordance.” The word *affordance* was originally coined by the perceptual psychologist J. J. Gibson in 1966 to refer to the actionable properties between the world and an actor (a person or animal or, today, many artificial devices). In other words, an affordance is a relationship. For example, a chair has the affordance of support, which makes it serve as a chair, but only for the proper-size object. A small chair might not afford support for a giant (or an elephant). Some chairs can be thrown, but this affordance is limited to people strong enough to lift and throw it. Chairs offer support to nonanimate objects—for example, books and papers—but only if of appropriate size and weight.

I introduced Gibson's term to the field of design in 1988 in my book *The Psychology of Everyday Things*. Designers usually care whether a person perceives that some action is possible (or in the case of perceived antiaffordances, not possible). How are affordances perceived? Actors are not necessarily aware of all the affordances. Sometimes they discover new ones by accident, by their observations of other people, or through instruction. In 2008, I proposed that we consider the cues that allow the perception of affordance to be conceptually separated from the relationship itself: I called the cues “signifiers” and the relationship “the affordance.” Appropriate signifiers allow us to behave appropriately with thousands of new things that we encounter in our lives even though we may never have seen them before or never been instructed on how to use them.

Note that although signifiers are usually used to help actors discover the relationship offered by the affordance, at times it is desirable to hide or even eliminate all signifiers to allow for private usage of something, a usage known or discoverable only by those who are told the secret. Similarly, false affordances can be used to mislead in many creative ways, some beneficial, others harmful. In a similar fashion, antiaffordances can be used to prohibit some activities (and false signifiers may make it seem that an activity is prohibited even though it still can be done).

How is the concept of affordances relevant to the artificiality of human artifacts, governments, and ways of behaving? The relevance follows directly from a simple generalization of the concept from the physical world to technological structures such as the use of media, graphics, and the design of technological systems that guide behavior—encouraging some uses, discouraging some, or preventing others. The discussions of the arbitrary dictates of time, the division of the year into fixed seasons, and the rules of government and even of etiquette are artificially constructed societal affordances, permitting some activities, preventing others. The customs and conventions of society (its mores) that determine how each societal group lives, how it organizes its activities, what kinds of social interactions it allows, and, of course, its governance and legal systems are societal affordances.

Want an example? Some cultures eat their food with their fingers but restrict eating to the right hand. Some sip soup from the bowl; others insist on the use of spoons. These differences indicate the artificiality of much of our behavior, but try explaining that to someone whose norms have just been violated.

The difference between the original concept of affordances and generalizations such as societal ones is that in the real world affordances and antiaffordances refer to actions that are physically possible or impossible, respectively. The words *permit* and *prohibit* point out that societal affor-dances are strong suggestions but, unlike the affordances and antiaffordances in the physical world, can be violated.

Societal affordances play an important role in the existence—and persistence—of the artificial. And the contradictions between the affordances of different cultures can be the source of jokes but also of insults, both unintended and deliberate. Affordances, antiaffordances, and signifiers, both real and false, have exerted powerful influences on human behavior, quite often without people's conscious awareness of them, and all are derived from and in many ways creating and reinforcing the artificial nature of human society.

The power of the control that the artificiality of our ways of life has exerted upon the societies of earth cannot be overstated. Not only has this artificiality led to major prejudices, to harmful behavior, and wars, but now it is in danger of destroying the ecological systems that are essential to human life. We are in a new geological age that has resulted from human behavior, the Anthropocene. That's the bad news. The good news is that if human behavior, artificial beliefs, and customs got us into this situation, then we should be able to change beliefs and behavior and get ourselves and the other inhabitants of this planet out of it. That is the theme of chapter 5.

5
If Technology Got Us into Today's Situation, Maybe Technology Can Get Us Out

The demands of the Industrial Revolution and its technologies forced us into rigid compliance with its temporal organization. But many of today's new technologies can seemingly free us from the rigidity of the old technologies. One of the great, unexpected results of the development of cellphones is their ability to let us relax about time. Did we agree to meet for dinner at 7:00 p.m.? With mobile phones we can be flexible, continually negotiating and changing time, place, and even who should join us, explaining that we are late—or early. The mobile phone allows us to live more flexibly, breaking out of the rigid framework of fixed time, coordinating our actions to everyone's satisfaction.

The rise of the global pandemic from the spread of COVID-19 forced people to quarantine, to keep away from their places of work. Fortunately, this occurred just as the technology of internet-enabled video conferencing was possible, and soon millions of people did their work from home and conferred with others via video feeds. Like all rapid introductions of technology, this one had numerous unexpected outcomes, some good, some not so good. Video conferencing enabled the scheduling of meetings all around the world without the need for travel. The reduction in travel had numerous beneficial side effects. For the individual traveler, it reduced the huge amount of time wasted traveling to distant locations (and accommodating to jet lag). It allowed meetings to take place that couldn't have happened otherwise. And the reduction in travel, invariably in vehicles powered by fossil fuels, led to a noticeable reduction of harmful emissions that impact climate change.

In my own life, the substitution of video for travel allowed me to give numerous keynote addresses and teach classes across the United States, Europe, Africa, China, and India while sitting at my home in San Diego, California. Yes, sometimes I had to wake up at very early hours of the morning to accommodate the differences in time zones, but it was still easier and far less expensive and damaging to the environment. Of course, there were negative impacts from video conferencing as well. Education suffered, especially for those in the precollege years. Personal relationships were strained. At the end of a video day, with call following upon call, people reported being exhausted, mentally and physically.

The impact of the COVID distancing and quarantine restrictions applied quite differently to different kinds of jobs. Many jobs require the continual presence of workers: manufacturing, food processing, shipping and warehousing, cooking and serving food at or for a restaurant, and many physical jobs such as cleaning and delivering packages. Although some medical services can be done remotely by video conferencing, most require a full complement of physicians, nurses, and medical technicians. The people in such jobs were in great demand, and so their work hours away from home were often increased, even while at work they faced new stresses and more exposure to the infectious virus. Firefighters, police, social workers, and health-care professionals still needed to be continually available. Many jobs went away because businesses had to close. The workers were then without jobs, which often meant without any income.

For so-called white-collar workers, whose jobs involved writing, reading, and attending meetings, their work could be done just as well at home or through video and computer conferencing and collaboration tools. The arbitrary separation of parents from their children was rather dramatically relieved. During these video sessions, it was common for children and animals to come into view and sometimes interrupt the parent during the meeting. To me, these accidental intrusions revealed the attendees' personal, human side, which helped to relieve the seriousness, tediousness, and stress of a continuous stream of video calls.

Even with the abatement of the pandemic, many of these benefits to business and education will persist. The development of flexible technology can allow us to modify the heavy dependence we have upon the artificiality of many rigid, fixed schedules. We have demonstrated that adding flexibility to our daily lives can enhance the quality of life without disturbing the quality of work. But this flexibility works differently for different kinds of jobs. In some cases, extreme flexibility is already possible. In others, there needs to be some fixed times when everyone can gather together physically. Some jobs today require everyone to be present at the same location during the entire work period. Some of these jobs can be changed; others cannot be. Nonetheless, even if many jobs do not appear to allow for flexibility, some clever new designs might allow changes in the work patterns that would enhance the quality of life for the workers without detriment to the business. Enhancing the quality of life for workers often results in their superior performance in doing their work. We must build on the adaptations we had to make during the pandemic to change other aspects of life.

Combining Technical Standards with Human Needs

Despite their arbitrary nature, scientific time standards are very useful for many people. The major virtues of rigid, formal time standards apply to their use in coordination and planning. Both the strict, scientific definitions of things and the more common, people-centered definition have advantages and disadvantages. Why not be flexible and combine the two approaches?

Consider the coordination required for modern transportation systems or for international collaboration where a meeting requires the attendance of people from around the globe. Use of a formal, strict definition of time is essential. Everyday time used to be defined by the position of the sun in the sky, with noon set to be when the sun was at its maximum height as it appeared to travel across the sky from east to west. But in the 1800s, the development of train travel ran into difficulties—and numerous collisions of trains—because of the lack of a time standard that could allow synchronization across different geographic locations. Hence, the invention of a standard time for the world in the 1800s, originally using the Greenwich Observatory in southeastern London, England, as the world standard location for the time zones. Because, by convention, the day has 24 hours, the original goal was to divide the earth into 24 zones, each marking a 15-degree rotation of the earth (which from the human observer's point of view means how far the sun moves across the sky). Having a formal definition of time made scheduling possible, and because people still wished that noon would occur while the sun was close to its peak height, time zones provided a nice compromise between strict scientific precision and people's preferences.

If the time zones had been rigidly enforced as longitude lines at 15-degree intervals, however, some divisions would have divided cities into two different zones or even passed through people's homes or places of work. As a result, the scientifically determined rigid time zone demarcations were relaxed to allow considerable freedom to place the time zone division where it would minimize disruptions to communities. As a result, the lines of each zone are jagged, puzzling to a foreigner but sensible to those who live in that zone. Some places in the world elected to have time zone offsets of 30 or 45 minutes instead of 60 minutes. China is so large that if it were to have a time zone every 15 degrees, it would need five time zones, so it decided to use a single time for the entire country. (For comparison, the continental United States is divided into four time zones.) Having a single, uniform time zone for China is convenient for business, but in some regions people, especially farmers and stores, use local times for local activities.

What Is the Answer? Both Science and Human Experience Are Correct, so Use Both

The history of time zones exemplifies how the demands of both scientific precision and human normal activities can be accommodated. When the need for precision meets the need for tolerance, it is possible for compromises to provide a comfortable accommodation of both.

There are many occasions when it is necessary for people to synchronize their activities. Similarly, some events need to happen at specified times. In all these cases, we definitely should use the scientific specification of time so that everyone agrees on the precise schedule. In other instances, schedules can be flexible. People can follow their own time preferences, and where informal things need to be scheduled, the presence of always available communication technologies—for example, the mobile phone—makes it easy to be spontaneous in arranging meeting and events and moving the agreed-upon day, time, or place.

When there are multiple approaches to an issue, quite often each approach has its virtues and deficits. We should not insist on the same solution for every person and every occasion. Let people lead a meaningful life by giving them flexibility wherever it is possible. Flexibility of course does not excuse people from doing the required work, but it might get a job done sooner, make the job more pleasurable, and minimize burnout. Compromise can provide the best of all worlds.

Where Capitalism Has Gone Wrong

Some companies manage to avoid having to pay the true costs of manufacturing. Once they discharge fumes into the air or poisonous materials into the sewerage system or local waterways and out of the factory, they call them “externalities” and do not enter them on the balance sheet. So who pays the cost of cleaning up? The public, either in taxes or in many cases through illness and poor living conditions.

Those who purchase the goods of a company are called customers. Individuals who purchase goods for personal use are called consumers: beings who consume. Therefore, companies have invented multiple ways to ensure that their customers consume the produced items in larger and larger quantities and more and more frequently. Those who sell food have an easy time, for food is literally consumed, so there is always a need to purchase new food. But with more permanent things, companies must invent reasons for their customers to continue to consume them. One approach is to make the stuff that people already have obsolete by convincing them that it is no longer fashionable. The entire fashion industry is built to convince people that fashion matters, so they must purchase new clothing, even though the old is still perfectly functionable. Fashion today extends to far more things than clothes: automobiles, mobile phones, computers, household appliances—the list is extended indefinitely, limited only by the limits of the creative minds of the marketing divisions of companies.

A second approach is planned obsolescence, defined as “the deliberate shortening of a lifespan of a product to force people to purchase functional replacements.” Obsolescence can occur if the items continually break down to the point where the repair costs are greater than the purchase price of a new item. Another approach is to keep adding features and then to convince people that these features are absolutely essential to their success, well-being, or job productivity. Other approaches involve changing standards and platforms periodically, so the old stuff no longer works even though it worked fine before the platform changed.

All these strategies to cause artificial desires to purchase were aided by the rise of the influence of the behavioral sciences, whose findings were at first used by marketing divisions of companies, usually to the surprise of the academic researchers whose research findings were used. As soon as researchers became aware of the job opportunities for themselves or their students as well as of new sources of research funding for their work, the practice was intensified, and the result has been a series of developments that make items more attractive for purchase and that make products, stores, websites, and other aspects of a company more addictive and "stickier." The term *sticky* describes the amount of time someone spends on a product: if it is a store, how much time is spent inside, browsing, questioning, and, of course, purchasing; if a website, how much time someone is spending within the confines of the website. And for all these things, how addictive is the behavior? How many times per week does the person return? In extreme cases such as games and social media, stickiness is determined by how many times per day, per hour, or even per minute a person accesses them. Some situations are designed so well, according to the metrics used by these companies, that the person's relationships to others, performance on jobs or at school, and sleep suffer. The results, both the positive features for sales and profits and the negative impact on people, have now been extensively studied, leading to books with such titles as *Evil by Design* by Chris Nodder, *Addiction by Design* by Natasha Dow Schüll, and *The Age of Surveillance Capitalism* by Shoshana Zuboff.

The deliberate efforts to enhance profits do extreme harm to people and to the planet (the planetary harm is the subject of part III, sustainability). Designers design the products that manufacturers churn out as a never-ending stream of new, improved, better-for-you devices, reaping harm on the environment in mining for materials and spewing out waste products from manufacturing. Driven by the science of human behavior, companies manipulate people's thoughts and behavior, all to the cause of greater profits, leaving behind as casualties such attributes as quality of life and benefit to society, the unintended side effects—externalities—in the war to capture people's attention, mindshare, time, and, of course, money.

Yes, all this—and more—is wrong and possibly evil, but the real causes are the way that reward structures and standards of behavior have accrued over generations. Many are no longer appropriate (some never were). But to change these practices to ones that are more beneficial to all, that enhance the quality of life for all living things on the planet, and that add to our health, happiness, and education, we must completely rethink and redo how we decide to live, what we reward, and what activities we stop. And this rethinking moves far beyond the design of stuff to the politics and economics and legal philosophies that encourage these harmful activities. We must change everything.

6
This Book: Meaningful, Sustainable, and Humanity Centered

Our lives are controlled by the measurement tools of science, by the needs of industry. Industry, in turn, is controlled by the economic theories of the world and by the need for currency to pay for materials, equipment, transportation, and, of course, labor. All of this is controlled by laws, regulations, and customs, all of which are artificial procedures devised by people. Those in charge often define these rules and procedures for their own benefit and impose them upon others, who must obey these artificial and often quite arbitrary dictates.

Business schools and the financial community have defined the goals of industry as growth and the promise of increasing profits. Where did these goals come from? They were invented by businesspeople, merchants, and bankers, then codified into mathematical theories by economists that have influenced the laws and regulations that govern our lives.

The push for growth and profits has also meant using the apparently infinite resources of the earth as a source of materials to make products and the apparently infinite waterways, seas, and atmosphere as a place to dump the waste products of manufacturing and usage.

But the earth is not infinite, and today the finite capability of its natural resources has been exceeded. Temperatures are rising, the atmosphere is becoming poisonous, food is less and less healthy as the number of chemicals necessary to grow it in overplanted farms has increased. How can this be changed? Only by a sufficient number of people working together with businesses, educators, and politicians. Change is essential to allow the earth to come back into balance, to allow the lives of people to become more balanced. It is a huge job that requires many people and many changes in the way that we live.

In this book, I address the same issues that many people, organizations, and books have addressed, but from a different perspective. Most of the other approaches focus on either the technological approaches or the policy changes that will help us change our ways. Those approaches are essential, too, but my focus is on the human side, ensuring that whatever activities take place prioritize the impact on people, their ways of living, and the quality of their lives. These changes require the support of the people of the world, for without that support it will not be possible to change policies or implement technological solutions.

What Is Missing?

In this book, I cover many fundamental aspects of our lives that are fueling the current ecological and social crises, behaviors that must be changed to minimize the harm to the ecology and the globe and to minimize the extreme weather patterns that have already started but must be contained. My intention is to focus on the human behavior that has had major negative ecological impacts on the planet and that today impedes the attempts to address the issues. If these behaviors are not changed, everyone's lives will be irreversibly harmed. The planet is not in danger: it has survived many mass extinctions and transformations. It will survive this one, although it will be a changed planet. Even if human beings survive, our lives will be much altered.

This focus means that I don't have the space to cover the major societal issues of prejudices of every sort or the inequities in how people are treated, in where they live and go to school, in where they work, and in how they are treated and paid. I do discuss the financial systems of the world and how they dominate so much of the lives and behavior of governments, companies, and people throughout the world, but I don't discuss how to counter any of these issues.

The governance of the world is artificial, biased; in many places it is arrogant, dictatorial, or corrupt. The governmental structure of most democracies attempts to balance the conflicting interests and points of view of the country's citizens. The result, however, can also lead to stalemates: an inability to get things done. Even when the parties agree to take action, the legislative process requires many accommodations to account for the conflicting opinions and get sufficient votes for passage. These requirements can lead to so much weakening of the critical aspects of legislation that it ends up being powerless and meaningless. I don't talk about that.

I also do not discuss the way technology is being used to control our behavior, to spy on us in part to increase the dominance by companies to seduce us into spending more and more time in their spheres of influence as well as the dominance by governments, police agencies, and political parties to get us to behave in certain ways.

I don't talk about many other important things, even though they are things I personally care about. I believe strongly that we must change how people treat one another. We need to judge people as individuals, not as members of rigid categories. Not by how tall or short they are, thin or fat; not by the color of their skin, or whether they have a physical deformity; not by religion or creed or what nation they come from; not by gender; not by how much money or access to power they have. Our systems today are horribly biased, in many cases unwittingly in part because the factors that lead to these biases and unfair treatment have been in place for so long that we think of them as natural. We seldom question the way things are. And many of the biases are against people who are different, whether by accent or language, gender, race, physical appearance, or abilities. Many of these prejudices and biases are deliberate, though, and completely inexcusable.

Nature is biological, and biological is not digital—or, to be more precise, binary. Summer does not suddenly become fall or winter: there are many subtle in-between states. The differences between plant and animal species are not sharp boundaries: they are graded. So, too, the same can be said of skin color, race, and religious beliefs. The distinction between species is graded, often continuous, the boundaries unclear. The philosopher Ludwig Wittgenstein long ago wrote about the difficulties of forming rigid categories. Language itself is deliberately ambiguous. If I sit on an upside-down wastebasket, a basketball, or a soccer ball, do they suddenly transform themselves into chairs? And what about the distinction that has caused more political difficulty than all—gender? Nature does not have strict boundaries. It is analog, biological, graded. Male and female, yes, but many shades in between, and because male and female are determined by many variables, there are many existing mixtures of the variables. We need to respect people as individuals and not force them into tightly, artificially defined categories.

Although these issues are not within the scope of the book, please do not interpret their absence as a lack of concern. I have benefited from interactions with many people who work to eliminate these injustices, interactions that have taken place in person, via video discussions, and through their writings and books. In the notes to this chapter, I list some of the important books that have changed my understanding of the world. I am thankful to all those people who spent time with me, educating me.

What Is Included: What This Book Is About

I cover some of the fundamental issues of our society:

Artificiality. Because most of the things that sustain our civilization are created by people, they are artificial. Behaviors, beliefs, and cultural values are also artificial. They are difficult to change because people have been guided by their culture's beliefs and values for so many generations, they assume they are natural, proper, and unchangeable. But because these are artificial, they *can* be changed.

Technology by definition is artificial, and its power has constrained and defined the structures of our society. It has often done so unwittingly, but the very invention of many important technologies has made certain societal structures and behaviors seem natural, even while they dominate and change everyday life. The power of technology and of the mathematics invented to serve technology, coupled with the language used to describe and explain the technology, is often unintelligible to non-technologists. Worse, the mathematics and science underlying the technologies are often applied even in situations where doing so is inappropriate. I single out economics, where everything from income to happiness to the worth of people and countries is measured on the same scale: money. The result is that people and societies follow the dictates of science and technology without understanding that many of the guiding principles are arbitrary, often simply to make it easier for the scientists and technologists, with little attention to the negative impact on society.

The emphasis on STEM—science, technology, engineering, and mathematics—leaves out humanity. We have become the servants of technology. Wasn't it supposed to be the other way around, with technology serving people? Unfortunately, the STEM fields do not consider any of the many fields of knowledge that do understand people, society, the arts, literature, and other things that make life worthwhile.

The emphasis on technology as the solution to all that ails us but without any consideration of people, lives, and the ecology of the planet has led us to the Age of Waste, where the waste products of our technological, industrial society, conveniently labeled as “externalities” so that nobody is responsible for them, are leading to the destruction of life on the planet.

We need to put the ecological and humanistic issues and values first and to downgrade everything else.

That's all. Just a few simple fundamental issues of our society. Which is why I didn't have room for all that I left out. If there is one theme to this book, it is that we need to return to human values, to focus on humanity. It is human behavior, guided by STEM and economics, that got us to today's state, so it will have to be human behavior that gets us out. We are faced with numerous technical issues, but the most difficult issue of all is human behavior, the theme of this book.

If the major theme of this book is people and human behavior, why do I talk so much about the role of design? Because more and more the role of designers is to act as the interface among technologies, policies, and people. Design is the major discipline that understands people and focuses its attention on their needs and understanding. Today, that focus must expand to encompass humanity, which means all living things, the world's ecosystems, and the complex system in which we all live. That is the focus of this book.

If this book can make a difference, I will be pleased, but I will not be content until all injustices are dealt with.

The Role of Design

In this book I talk of a new form of design for the twenty-first century that emphasizes the needs of all of humanity coupled with sensitivity to the lives and cultures of the people for whom the designs are intended. I address three fundamental themes: ensuring 1) that the world be described in terms that are meaningful to everyday people; 2) that products are not destructive to the planet's ecological systems; and 3) that community projects are driven by the people themselves with designers acting as facilitators and mentors.

The book is organized into six parts. Part I has introduced the book and, most importantly, the notion that almost everything that concerns human life is artificial. Parts II through IV cover the current issues, and parts V and VI focus on actions that must be taken and the reasons they are so difficult to achieve.

Parts II, III, and IV examine the main themes: we need to transform our lives so that they are meaningful (part II), sustainable (part III), and humanity centered (part IV). Here is a brief overview of these three critical causal foundations of many of the planet's ecological ailments.

Meaningful

People seek meaning in all aspects of life, in their jobs or schooling, in their family and social interactions, and in the artifacts they own or create. Even an old, out-of-focus personal photograph can have deep, significant meaning. We must make more events meaningful.

Many of the metrics that define modern life, especially for governments and corporations, are unintelligible to the average person, and some of these metrics are unintelligible even to the professionals who must use them. Unintelligible things cannot be meaningful, or if they have any meaning at all, it is negative, filled with the frustration and anxiety that they bring.

We need to measure the things people care about and understand—quality of life, not economics. We must change the metrics used to judge countries and companies to those that enhance the quality of life, not profit and growth. We should use meaningful psychological measures that focus on societal criteria such as wellness, health, education, security, and happiness and stop using economic measures such as GDP, monetary profit, and stock market indexes.

Sustainable

The world is a complex network of interactions. As a result, the major societal issues do not have simple, easy solutions. Two of the major issues, societal resilience and sustainability, require action at multiple levels—local, global, technological, organizational, and political. Natural resources are under attack, in some cases being depleted, in others poisoned. The manufacturing processes of the world produce waste and dispose of it in an irresponsible, harmful way.

There are better models. The one I propose in part III invokes a circular economy, where things can be repaired, renewed, and reused. Even better is when all of the materials used are biological, thereby creating a self-sustaining system, much as exists in nature—a circular economy, where everything is resilient, renewable, reusable, and, as much as possible, regenerative.

Humanity Centered

Projects intended to bring improvement to the communities of the world must be done in collaboration with and by the people for whom the designs are intended, providing them with resources and facilitation from a multidisciplinary pool of subject-matter experts. The design community must stop designing for people without their input or to get people to do or want something and instead act as facilitators and resources to the communities. We need a holistic design with a wide range of expertise in government and law, economics and engineering, health, and education. The complexity of large systems is best dealt with through small and incremental projects, where responses to early work inform and direct later work.

Small projects are more acceptable to communities than large ones because they are flexible and responsible. We can address large issues through a multitude of small, flexible projects. Our designs must serve humanity, which also requires that the designs serve the ecosystem.

The goal is to produce a meaningful, sustainable, and humanity-centered world. This requires everyone to rethink and redo the basic premises and ways of being that govern our lives. I have covered the artificiality of lives in this part of the book: part I. Parts II, III, and IV discuss, in turn, the need to make the world meaningful, sustainable, and humanity centered. Now I turn to part II: to what is meaningful.

II
Meaningful
Communicate in Understandable Ways

7
The Need for Meaning

People have naturally evolved to seek meanings and explanation, yet we are forced to live in a world where meaning is buried beneath documents and pronouncements that use abstract and complex, specialized terminology—legalistic, technical, and governmental. Sometimes the lack of clarity is unintentional, when people unused to communicating outside their specialized domains attempt to write for the public, not realizing that the specialized language of any field is often completely meaningless—and, worse, intimidating—to the nonspecialist. Sometimes the complexity is intentional, as in those long, legalistic forms we have to agree to every time we purchase a new application for our computers or log on to a website, pages and pages of legal language, visible only through a tiny window, where the words are displayed in all upper-case letters (which are less readable than lower case), yet we cannot proceed without agreeing to the document. Here, the writers of these documents want you to agree without reading (and even if you attempt to read them, you soon discover they are deliberately and intentionally unintelligible). No wonder people have trouble understanding medical, financial, and safety advice or in trusting or responding to advice about such things as climate change or economic forecasts. Yes, we live in a world that is mostly artificial, created by people. So why has it been shaped to be so unintelligible, so nonmeaningful?

How Scientists Explain the Crisis of Climate Change

It has been very difficult to get people—the general public, business executives, and politicians—to take climate change seriously. The problem is to prevent the global average temperature from rising more than two degrees above preindustrial levels. The words are very easy to understand, but the message is completely lost. I had to search the internet to figure out when “preindustrial” was (answer, roughly 1750). Three hundred years ago? Huh? A two-degree increase? The temperature where I live (San Diego) varies more than that every day: What's the big deal? The same problem with the rise in sea level. The predicted rise for the year 2100 is between 0.9 and 1.4 meters (3 to 4½ feet). Eighty years from now? Where I live, the ocean level goes up even more than that—1.8 meters (6 feet)—every time there is a full moon due to the tides.

As long as scientists insist on talking scientifically (scientese?), why should the general public be alarmed? None of these predictions has any significant meaning to the average person. And why should people who live far from the oceans worry about sea level? Threats have to be described in ways that are meaningful to the audience. Don't talk about average rise in temperature or sea level. Talk about disasters. The world has already faced many of the disasters caused by climate change, so talk about the widespread massive fires; storms at record intensity—hurricanes, cyclones, monsoons, tornados, typhoons, heavy rainfall—leading to massive flooding and disasters far from the oceans; massive droughts leading to severe hardships in the affected communities; massive changes in agricultural crops across the world, where traditional locations for many critical food supplies are no longer able to produce the required quantities.

Warning: do not focus only on the disasters, though. Messages like that are called “gloom and doom.” They cause people to worry, and the more often the dire message is presented, the less attention it gets. Who wants to hear continual statements of doom? Sure, state the prediction but also say what can be done about it. Unless people feel there is something they can do, they will not respond. What are those positive messages? That is what part VI of this book is about. (Turn to it now, if you like: chapters 36–38.)

The world is complex, a huge system where almost everything is connected in one form or another. The complexity of the systems of the world defies complete analysis, complete understanding. Some of the complexity derives from the combination of the multiple physical factors present during the formation of the planet, involving physics, chemistry, and geology. Some of it comes from biology, the nature of life and the evolutionary processes, and from the way that all species of life—animals, plants, and other living things (for example, fungi)—form an interconnected web that supports all life. The question is how the average, everyday citizen can come to understand the world sufficiently well to get along in life as well as to make intelligent and sensible choices and decisions.

Making Complex Topics Understandable

Design can play an essential role in making the complexity of modern technology understandable and usable. That has been my own crusade in design. In my book *Living with Complexity*, I point out that many things in our lives are complex, but if they are understandable, we call them "simple." The key to enabling understanding is to use everyday language and examples.

Consider the way a family organizes cooking utensils and foods in a kitchen. To the family members who determined what went where, the organization is simple and understandable. Yet for someone not in the family, attempting to cook something in this unfamiliar kitchen can be bewilderingly complex. Why is it simple for the family? Because they know why each item is located where it is. It may be due to history; it might be due to the wish to put items used together in similar places; it might be due to the need to make some items difficult for children to reach, others easy for everyone to reach. Each item has a story, and as long as the stories are known, the organization is logical and understandable. But when that family goes to someone else's kitchen, without understanding that kitchen's history and stories, they will find its organization bizarre and unintelligible.

The world is, of course, rather more complex than the kitchen, but the principles are the same: if people understand the basic reasons for things, they can appreciate the world's complexity and many of the major causal reasons underlying climate change. Once people have reached that understanding, they can readily appreciate what must be done to stop the harm. The same principle applies to all the other things our governments expect its citizens to understand. You know what I mean. All those forms and regulations. All the economic variables. All the statistics that governments and financial services love to present with no attempt to make them understandable and meaningful to the ordinary person.

Most technical professions are used to explaining their work to other professionals in the same or closely related fields and fail to recognize that talk directed to colleagues is unintelligible to nonprofessionals. This is not true of all fields. For example, journalists, authors of popular books, documentary film producers, and many others have learned how to communicate in ways appropriate to the audience. These people can help the technical professions explain their ideas.

Designers use diagrams, pictures, and simple illustrations to explain their work, whether it is how to turn the power on or off in a machine or how to use a DNA synthesizer or even a washing machine. Designers understand the need to test their efforts on the people for whom they are intended. Invariably, the tests reveal problems, places where people do not understand, are confused, or, worse, could make serious errors in their use of the device under consideration: perhaps losing all of their work and, in the case of machinery or medical devices, perhaps leading to accidents or serious injuries. The reason for this early testing is to find design flaws before the product is released, allowing the design to be corrected. In design, especially the set of procedures that goes under the general label *human-centered design*, multiple iterations of the design are constructed and tested. Each new iteration—that is, each modification of the design—is guided by the results of tests.

The testing should not start with completed, working objects because then it is too late to make changes. It can start with rough sketches (even on napkins). Then it can move on to diagrams, simple mockups using simple display programs for software and foam or cardboard constructions for physical products. Each iteration does get closer to the final version. These rules are true for complex physical devices, for software, for a procedure or checklist—for everything.

All fields that are concerned with communicating to the general public do similar things. Plays are given tryouts and rehearsals before selected audiences. Films are often tried out on test audiences before being released, and television shows are sometimes first released as “pilots” to assess the interest to their intended audience. Writers test their writing on readers. Iteration invariably improves things, but the critical factor is that the trials have to use the intended audience. If the ideas are tested by presenting them just to friends or relatives, the latter are not apt to give meaningful feedback. I test my ideas by giving talks to my research group, which also means that they think I am a lousy speaker and writer because they see only those first, fumbling attempts. They seldom get to see the final product that was made better by their critiques.

Scientists of the world, take heed: it is very important that the world understand your work, your findings, and your warnings. To do this, you have to take the task of communicating just as carefully as you take your research. Communicating effectively is not easy. It will require multiple revisions: write or build, test, think deeply about the results and then rethink your ideas, rewrite the text, rebuild devices. Repeat again and again.

What? You don't have the time? Sorry, this is time well spent. Unless you do this, you cannot complain that people don't understand your work. They will not be able to understand until you present it in ways they find meaningful.

Measurements

Scientists love to measure things, and as I discuss in chapters 4 and 38, when they can't measure what they want to study, they measure something else that is related but then soon think that “something else” is what they care about. Measuring the increase in global atmospheric temperature is an example: it is not the exact temperature that matters—it is the resulting devastation that is important.

When scientists tell us about atmospheric temperature, that sounds familiar: just look at a thermometer. But as I thought about it, I realized that this temperature is supposed to represent the atmospheric temperature of the entire planet. How is that measured? I decided to investigate how that value is determined. The answer? It's very complicated.

Where do we measure the temperature? Answer: all over the place. From ships and buoys in the ocean to multiple weather stations on land around the globe to balloons and satellites: more than 20,000 places in all. The temperatures are taken daily, averaged, and sent to databases in the United Kingdom, Japan, and the United States. The collected data are fed into climate models, monstrous computer programs devised by different groups of scientists and technologists, and each of those programs is slightly different, so each gives different results. The United Nations uses roughly 30 different climate models and from them extracts the single temperature number that is released to the public. This story is very much abbreviated because the different collections have differing coverage and accuracy. Different databases deal with missing data and other discrepancies in different ways. Satellites measure temperature using infrared radiation and microwaves. Ground and ocean-based locations use various forms of thermometers. The single number eventually arrived at hides a lot of complexity. What does the average atmospheric temperature mean to the average person? Very little. To the climate scientist, it is an important measurement. But it is not the correct measurement to present to the public, who will not understand the severity of what appears to be a tiny change in temperature, measured in a complex, nonintuitive way.

Measurement is very important in science. Some of the most important advances in science have come about by new techniques that enable the measurement of things that before were not measurable or improved accuracy and precision in the measurement of things previously measured. Why is measurement so important? Let me turn to the role of measurement in science.

8
Measurement in the Physical Sciences

If I ask you to name the most powerful tool of science, you might guess mathematics or computers. Although these tools are of great importance, they would not be nearly so powerful without the ability to measure things. Measurement is one of the fundamental principles of science and engineering. Here is what the British scientist William Thomson (more commonly known as Lord Kelvin) said about the importance of measurement in 1883:

> In physical science a first essential step in the direction of learning any subject is to find principles of numerical reckoning and methods for practicably measuring some quality connected with it. I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind: it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced to the stage of science, whatever the matter may be.

I learned Kelvin's dictum about measurement when I was an undergraduate engineering student at MIT. This statement has been influential in most sci-ence and engineering disciplines. It has also been accepted by many of the social and behavioral sciences and extended into the humanities and the arts despite the evidence that once we leave the physical sciences, measurement is neither easy nor always appropriate. Almost everyone ignores the first three words of Kelvin's utterance: "*In physical science*."

How can scientists measure decision-making in a tribal council or a town hall meeting, places where there are multiple agreements and disagreements, interruptions, fights, secret agreements, favors, inducements, and threats? Productivity is a favorite measure of a company's efficiency. It works well when measured in a factory or warehouse, where each worker is assigned a specific task to do repeatedly. Productivity of the worker is measured by the number of tasks performed per hour or day. But what about tasks that require thinking and decision-making? Interestingly, although companies reward senior executives for their management skills, they invariably “measure” the executives by the increase (or decrease) in company profits and the price of the company stock, which not only have little to do with their management skills but often result from the efforts of their predecessors years earlier. The main reason why these measures rather than others are used is that they are easy to get. Faulty measurements quite often reward the wrong people.

How does one measure the productivity of a manager, a scientist, an artist, or a journalist? Computer programmers often have their productivity measured by the number of lines of code they produce each day, just as journalists can be measured by the number of words of text they produce (or, today, by the number of downloads or clicks their articles receive and of reactions in social media).

Notice the absence of the word *quality* in any discussion of productivity. The measures ignore both the quality of the output as well as the quality of life for the worker forced to endure these measurements, continually being asked to be more “efficient,” to increase output, increase quantity. The weird thing is that the more lines programmers write per day, the more errors they make, which then leads to a separate measurement of the productivity of the “debugging team” charged with finding and fixing errors. The productivity of the error-correction team is measured by how many errors they fix every hour (or day). Both teams can receive high productivity marks even though one is producing faulty computer code and increasing the total time required to complete the work. Why measure this way? Because it is easy.

When it comes to human behavior, much is lost in the transformation of complex activities into the abstractions of measurements. Here is where measurements can cause one to lose track of the goal, where the context and the complexities, the personalities of the individuals and their hidden motives and incentives are lost, often deliberately ignored.

We of the Global North who grew up with the theories of modernity, championing science, rationality, and the power of mathematical deduction, love to measure things. Most of these measurements are incomplete, inaccurate, and inappropriate, but that doesn't stop us. I know because this is how I was trained and how I acted in the first years of my professional life.

The problem is that not everything that matters can be measured, or as the famous witty aphorism puts it, "*Not everything that can be counted counts, and not everything that counts can be counted*." But what can be done with important things that do matter but that either cannot be measured or are too difficult or expensive to measure? When the scientific community can't measure the property it is interested in, it tends to do one of two things:

1. Call the attribute unimportant and forget about it.
2. Find something else to measure, something that has a relationship to the variable of interest but that can be measured. Then call the resulting measurement by the name of what is really cared about, forgetting (or ignoring) the fact that the variable of interest is not what is being measured.

There are better answers. First, the measurement schemes used by physical scientists have been greatly extended by social scientists, which has greatly enlarged the range of attributes that are measurable (the topic of chapter 9). Second, qualitative descriptions are just as powerful and important as numerical quantities, and many complex items are far better described this way than with numerical assessments that lack validity.

The Scientist's Creed That Measurement Is Necessary, Even of the Unmeasurable

Lord Kelvin's maxim has had enormous influence far beyond the natural sciences for which it was intended. Scientists measure in part because appropriate measurements enable statistical and mathematical analyses that add insight and deepen our knowledge and understanding. But many of them measure because that is what is expected of them. I could cite many critics of the excessive use of assigning numerical values to things that are not measurable by normal standards, but an interview of the political scientist Ruth Carlitz in the journal *American Scientist* makes the point well. Carlitz was studying policy decisions in authoritarian regimes during the COVID-19 crisis. The interviewer (Scott Gabriel Knowles) asked, What measures are used "to classify a country as authoritarian?" Here is Carlitz's answer:

> Once we put numbers and rankings on things, we take them to be objective, but behind these numbers are a lot of assumptions. Many of these rankings are based on the subjective opinions of experts. But who gets to be an expert? And when you're comparing a country rank of 97 versus 96, what does that mean?
>
> There are a lot of boxes that don't get opened once something has a number on it. The ranking becomes a cold, hard, objective fact. I do use these rankings in my work, so I'm being a little hypocritical, but they do sometimes make me uncomfortable.

The measurements we make have two flaws. The first flaw I have already discussed: the measurements are made of things that are easy to measure rather than of the things or phenomena or activities that are of most importance. Second, once human activities have been transformed into numbers, they lose context and thereby meaning. They become abstractions that are used to make decisions about human behavior without considering the impact of their use on the lives of individual people. We need to measure the things that are important to people in meaningful and useful ways.

9
Measuring What Is Important to People

Lord Kelvin's proclamation that if you can't measure, you lack understanding still rules in most of science, even though, as I said earlier, his first three words, “in physical science,” are ignored. The psychological, social, and behavioral sciences are immensely more complex than the physical sciences. Living organisms are complex, dynamic systems, impacted by many factors that may be remote in either distance or time, dependent upon their past history (that is, they are path dependent), and continually modifying themselves, always changing.

The Greek philosopher Heraclitus is credited with saying that “it is impossible to step into the same river twice,” for, after all, if the river is defined by the water within it, the water is continually changing. In the era of Lord Kelvin, Heraclitus's statement did not apply. It is possible to measure the very same thing many times in the physical sciences, and each time the object and measurement will be the same, ignoring perturbations due to the method of measurement. This is most definitely not the case with living things. People are continually changing their beliefs, performing, and saying contradictory things at different times. People interact through facial and body movements, through their choices of words and tone of voice, and are aided in their communications, of course, by technologies that allow interaction across both time and space. Experience, education, or manipulation can change belief systems, and as beliefs change, so too do people's responses to situations governed by those beliefs. All living things are in constant flux, interacting with one another and the environment, growing over time, changing physically, chemically, and, for those with brains and minds, mentally.

Science has made much of its progress through observation, measurement, mathematics, and experimentation, coupled with the public sharing of ideas and methods. An important part of the process is reproducibility, others' ability to repeat and build on previous results. When the results cannot be reproduced, this is often the start of a scientific debate waged in the journals and conference rooms until either experimental differences are clarified or new interpretations are produced or, sometimes, the earlier results are repudiated. The ability to replicate results has led to almost all advances in understanding in the physical sciences.

Replication of results has worked out quite well for the physical sciences, but when that method is applied to the sciences of the living, difficulties arise. The only way to do careful, well-controlled experiments is in what some critics have called "the white room," a carefully controlled space in a laboratory, usually at a university, that is completely isolated from the outside world and everything not directly concerned with the variables under study. Results from white rooms are often deceptively simple and clear statements of human behavior, statements that can be replicated in other white rooms. But once we move out of the carefully controlled and artificial settings of the laboratory into the real world, the science breaks down. Real behavior is context sensitive and highly determined by each person's individual history. Worse, university scientists often study the population of people that are most accessible to them, young university students. Critics point out that, as a result, much research of human behavior conducted in the universities of the West primarily study WEIRD people: Western (or White), Educated, Industrialized, Rich, and Democratic. In other words, the most carefully controlled studies are of people who are not very representative of the 8 billion people of the world.

Economists are especially guilty of trying to capture the complex behavior of people and society in a few simple numbers. The field of economics has unfortunately gained great influence on government policies, in part because its use of the tools of logic, simple measurements, and mathematics makes it appear to be rigorous and scientific. But economists seldom test their predictions. Repeatability is not one of the major tools of the field. To be fair to economists, it is difficult, perhaps impossible, to perform controlled experiments on real-world behavior. Every situation is different, and once the results of early studies are publicized, this knowledge can impact any attempts to repeat the experiment.

Government officials call upon economists to provide wisdom and advice for complex decision-making, but not only are the wisdom and advice questionable, the economists' impressive-looking mathematical reasoning hides the basic assumptions in the technical language of mathematics. Nonspecialists can easily be swayed by mathematical arguments. Although mathematics can't lie, the assumptions within the mathematics can be just as wrong as incorrect numerical values entered into the computations. The standard saying about the power of mathematics is that the quality of the output depends on the quality of the data and assumptions provided as input: "garbage in, garbage out."

One of the fundamental assumptions in most economic models is that people behave rationally and logically. Despite decades of research by psychologists showing that these assumptions are wrong, economists ignored the evidence until finally, it was so strong that the Nobel Committees started awarding prizes in economics to people who weren't even economists but who were demonstrating the fallacy of some of these basic assumptions. But even this has not fully changed the practice of economics.

The methods of the physical sciences are often used in economic and social science models in an inappropriate manner that is harmful to real people and real societies. The scientific method has many virtues, especially the requirements for evidence, observations, experimentation, and independent replication of ideas and results. Different areas of scientific disciplines require different methods for performing these requirements, but when the methods are followed, the resulting knowledge can be quite effective in determining national policies.

The sciences of human and societal behavior can detail the important issues and variables, with quantitative and qualitative assessment where appropriate. How does one make decisions without the simple answers often provided by the mathematics of today? That is what decision makers are supposed to do: make important decisions on the basis of incomplete, ambiguous, and conflicting evidence. They do make the decisions, but the conflicting and ambiguous values are often hidden from sight behind the opacity of mathematical or computer models. Worse, important qualitative assessments that address societies' and people's true needs are often left out because they do not fit into the existing frameworks.

The field of economics has special difficulties when dealing with large-scale phenomena such as the complexity of modern organizations and countries, and with the international flow of trade, people, and ideas. The job market, the balance of trade, the hiccups in the smooth trades among nations due to tariffs, patent disputes, weather, and more can disrupt international trade. This became especially noticeable when the combination of disease (the COVID pandemic), natural disasters (fires and floods), wars, and trade barriers recently disrupted the management of critical supply chains all across the world.

The Measures Used in Economics

The field of economics has become a field of applied mathematics. Economists' work, models, and opinions are central to many government fiscal policies and international treaties. The most powerful of the economic subfields is macroeconomics, which studies—and tries to predict—the behavior of large systems (hence the prefix *macro*). The study of macroeconomics includes the behavior of countries, governments, and large international corporations—many of the latter having larger revenues than most countries. Economists who study macroeconomics rely heavily on measurements and theories of human behavior and decision-making. These assumptions are treated like the axioms of geometry that many of you may have learned in high school—for example, "two parallel lines never intersect"—statements so obvious that there is no need to test them. As I show in the next section of this chapter, many of these obvious, commonsense axioms about human behavior, axioms you might even agree with, are in fact false. The assumptions represent only the economists' logical opinions about how people should behave. If the economists were to observe people's behavior in the world, they would discover how inappropriate their assumptions really are. But until recently economists did not do this: this is what psychologists, anthropologists, and sociologists did, but, again until recently, their findings have been ignored.

Macroeconomics studies large-scale impacts; microeconomics studies the behavior of individuals making relatively small decisions (hence the prefix *micro*). This field has also been perverted by an emphasis on rationality and perfect decision-making, neither of which applies to actual people. For decades, psychologists have studied the problems of microeconomics, but their studies have long been ignored by economists. I did some of those studies when I was a graduate student in psychology. I remember that as a student I tried to convince a number of my friends who were studying economics that their measurements and theories did not match real human behavior, but they laughed and argued that this didn't matter. The power of crowds overcame the small, local fallacies, they said, often quoting Adam Smith's book *The Wealth of Nations* and his argument about the power of the "invisible hand."

Both macroeconomics and microeconomics are essential components of a deep understanding of economic behavior, and there is no question about their importance. But we need a more meaningful view of economics, one that uses measures readily understood by the general public, one whose models are based on actual human behavior, not on oversimplified assumptions. And it is important to remember that monetary numbers, such as the gross domestic product (the GDP), although at times useful, can be very misleading. Consider the evidence Charles Kenny presents in his book *Getting Better: Why Global Development Is Succeeding—and How We Can Improve the World Even More* that even though GDP in many developing countries of the world has hardly changed over the decades, there have been widespread improvements in health, education, even happiness. Measuring the expenditures of a country turns out not to be the way to value a country's progress or the success of aid programs. Here is a case where macroeconomics' use of a solitary number hides progress in things important to the quality of life.

Examples of Flawed Assumptions

One basic assumption of economic theory is that people maximize utility—that is, the satisfaction they get from a good, a service, or an activity. Most often, utility is specified in monetary terms and therefore interpreted to mean that people seek the least expensive alternative. Suppose you saw two identical products for sale, but one was more expensive than the other. Which would you buy? Traditional economic theory would dictate that, obviously, you would purchase the least expensive one.

Why do people buy expensive wines, whiskeys, and watches, then? Do they taste better or keep time more accurately? Nope. Inexpensive digital watches often keep better time than the extremely expensive hand-crafted, mechanical watches. One way to increase the sale of some products is to raise the price, which violates economists' sensible, logical axioms. This doesn't work for every product or for the most knowledgeable purchasers, but there are numerous times when people will think that a more expensive product is better. Just look at high-priced wines and liquors, luxury automobiles, and expensive meals at highly rated restaurants. In fact, if you were comparing household items that appear to be identical on the surface, but one is half the price of all the others, would you buy it? Or would you be suspicious that there must be something wrong with it—perhaps cheap, inferior components or shoddy workmanship?

Herbert Simon, one of the first people in the field now called behavioral economics, was awarded the Nobel Prize in Economics in 1978 for his work on decision-making. Simon was a true transdisciplinary researcher, making important contributions in management, psychology, computer science, and artificial intelligence. He argued that people's memories and mental computational abilities are limited—which he called “a bounded, limited rationality.” As a result, he said, people cannot optimize decisions, so they “satisfice.” He invented this word, combining “satisfy” and “suffice” to mean that rather than waste time trying to decide among alternatives without sufficient information, they will choose one that is good enough. Simon taught me to follow this rule: I satisfice, for I know that perfection is impossible. I save myself time, energy, and patience: I satisfice.

When I worked for a large electronics company that manufactured laser and ink-jet printers for industry and households, I soon discovered why there are often three versions of many consumer goods, especially large items such as computer printers and so-called white goods: kitchen and laundry appliances that were traditionally manufactured with white enamel finishes. If the manufacturer makes only one version of its product, people who bought it might have been willing to spend more money, so the company is losing some income. If the company offers two versions, one with more features and more expensive than the other, people will compare the two models and still buy the less expensive one. But if the company introduces a third model with even more features and more expensive than the other two, sales of the second model go up. Why? The company doesn't expect many people to buy the third, more expensive model, but its very presence makes them choose among the alternatives. They like the features of the most expensive model, but not the price. The middle item has more features than the least expensive one, and it is less expensive than the fanciest model. They buy the middle item (bragging about how much money they have saved), unaware that they have been manipulated by the presence of the higher-priced item.

Features are also very important in the selling process. The point is to convince buyers that they cannot live without these features, even though they have been doing so for their entire life up to now. After they buy the item, most people will never use the new features. Designers know that adding features that people really don't need makes the product more difficult to use as well as more expensive. However, the marketing people are correct: features sell.

The ability of marketing to use its superior understanding of human behavior to manipulate people should be considered immoral and unethical, but in today's world of commerce it is applauded. The marketing people are promoted.

Cost–Benefit Analysis and Risk Management Base Their Recommendations Using Money as Their Standard of Comparison

Economic variables show up in many aspects of modern life. Here are two more examples: cost–benefit analysis and risk management, schemes for comparing the costs and benefits of a planned course of action. Many divisions of the federal and local governments require cost–benefit analysis and deem an action doable only if the benefits outweigh the cost, or in many cases they rank projected actions by their cost–benefit analyses. Risk assessment in safety-critical industries uses similar methods. It can help determine if the company (or government agency) has prepared itself in case of severe disruption of activities due to storms, strikes, illness, or equipment failure. Many risk assessments are performed to examine the likelihood of accidental injury or death.

Cost–benefit analysis makes good logical sense. Large societal projects can be very expensive. How do we determine if the benefits are sufficiently high to justify the cost? Similarly, suppose a company is manufacturing something that has dangerous potential: an automobile, for example. How much money should companies spend to reduce the number of deaths? How much should governments spend to create safer roads? Automobiles kill roughly one million people every year in the world, so this is an important question.

Obviously, we should do all that is possible to enact laws that require driver training, build roads that have safety features, and design automobiles to reduce the risk of an accident and to minimize the chance of serious injury if an accident occurs. However, some of the proposed remedies are very expensive. At what point should the company decide that adding more safety features will simply be so expensive that people can no longer afford the car? The same question applies to the construction and maintenance of roads. In a city, when should a traffic light be installed at intersections to avoid accidents?

Cost–benefit analysis answers these questions by comparing all the costs with all the benefits. But one cost is the loss of human life. How should that be factored in? Economists say that they need to have *all* costs and benefits on a common numerical scale, and the one scale that is almost always selected is money. If a person is killed, what is that cost? Does it matter how old the person was, how famous? How much income the person earned? Does it matter if the person lives alone or has a family with young children? What is the monetary value of a safe street or the cost of having an ugly signal light in front of a home? All these variables under consideration are complex, multifaceted, and very difficult to quantify on a simple numerical scale. Reducing anything that has multiple dimensions to one single number is impossible, let alone to a number such as dollars, whose entire meaning is derived from other contexts.

Cost–benefit analysis is closely related to the field of risk management, which identifies, analyzes, and then decides what actions to take for the many different kinds of risk that exist in the world. Ideally, one could avoid situations that are risky, except that is impossible to do. Crossing the street is risky, driving an automobile is riskier, but traveling on a commercial airline is far less risky than either, although perceived by many people to be the most dangerous of all. People's perceptions of danger do not necessarily correspond to the statistics.

How much time, effort, and money should be spent trying to mitigate against the negative events that constitute risk? This is where cost–benefit analysis and risk analysis come together. How does one decide? The basic philosophy underlying these methods is important and sensible. Yes, it makes sense to weigh the benefits against the harms and consider the probability of the risk. But why must this be done by assigning numerical cost values to the variables, many of which are not really quantifiable? Why should everything always be transformed into monetary terms? What is the value of life? What is the value of losing an arm, leg, eyesight? Or the value of pollution, of long commuting, of inferior health care or schooling? Many of these variables involve subjective experiences and judgments. How are the latter measured? I discuss these questions in the remaining chapters of this part of the book (especially in chapter 11).

Instead of forcing everything to be defined in monetary terms, we need to measure items that deal with subjective experiences by appropriate psychological variables. Psychologists have developed a wide range of measurement techniques that are just as rigorous as the numerical ones used in the physical sciences but use different kinds of measurement scales. In the end, such measurement does not lead to automatic decision-making through the comparison of two numbers. It instead yields guidelines for the decision makers. The decisions will be left to human judgment. But don't fool yourself that this means substituting an imprecise method for a precise one. It is a well-known secret that when decision makers use highly quantitative, numerical tools to guide their decisions, if the numerical measures of risk analysis or cost–benefit analyses are not to their liking, they will change the numbers (or the weights—importance—assigned to the values). The decisions that result from these methods are not as scientific, precise, and relevant as advertised.

Artificial Measures of Values Clash with the Values of Humanity

The tools of risk management and cost–benefit analysis sound reasonable and logical. They appeal to people who seek quantitative, mathematical decision-making for every possible situation, but they are bad for society.

The topic of designing for humanity is so important that it is part of the title of this book. It is covered more fully in part IV, where risk management and cost–benefit analysis again play a role. But, meanwhile, think about this: Why do we consider some human lives more valuable than others, some homes and businesses more important to society than others? It is as if we said that lions, the kings of the jungle, are more important and worthy of protection than household animals such as people's pet dogs and cats.

Because both risk management and cost–benefit analysis require monetary estimates of the costs and benefits, they automatically tend to favor expensive buildings, businesses, and homes over inexpensive ones, for that is where the cost of damage in a risk analysis is highest and higher benefits are assessed. Both methods assume that money is better spent protecting the homes and beachfronts of the wealthy, for, after all, the cost of protection is the same for these properties as protection for the homes where the poor live. When we measure the benefit by the value of the properties that are protected, the wealthy always win.

Risk and cost–benefit analyses are unwittingly prejudiced against citizens with low income. The result is to protect the wealth of those who need protection the least simply because they are wealthy. The difficulty is that the measurements are restricted to objective standards, such as house and land valuations. The fact that the wealthy might be able to afford to do the repairs without assistance or perhaps were warned against building in a risky zone is not taken into account. Similarly, risk assessment and cost–benefit analysis also do not consider the fact that when lower-income individuals lose their homes and possessions, they may also lose their jobs. These items, whether for high- or low-income individuals, are not considered because there is no easy way to determine the costs.

As a result, those who need help the most, those living below poverty thresholds, are passed over, while those who need help the least get it. Is this a humanity-centered design? All people are equal under the eyes of the law, at least so we are taught to believe. So why shouldn't all people be equal under the eyes of risk management and cost–benefit analysis? Ponder these ideas as you read about sustainability and the Age of Waste in part III and then again in part IV, where sustainability and meaning join together to consider humanity.

10
The Gross Domestic Product

Economists measure the importance, quality, and economic health of a nation by its gross domestic product (GDP). GDP is a computation of the total monetary value of all goods and services produced by the country. In the United States, the Bureau of Economic Analysis, which is part of the Department of Commerce, is responsible for computing the GDP estimates for the country and for each state four times a year and once a year for counties, metropolitan areas, and industries. But, as critics continually point out, the measure does not distinguish between beneficial and harmful spending. It doesn't consider the well-being of the people in the group being measured. It basically assumes that all spending is good, despite considerable evidence to the contrary. Spending that pollutes the air, water, and land is not good. Spending that degrades the quality of life is not good. Nonetheless, these expenditures are added to the GDP in the same way that spending money on hospitals or schools is added. Even though GDP is meant to demonstrate a citizen's ability to purchase goods and services, it fails to do this.

GDP is the most widely used measure of a country's performance. It probably has more critics than supporters, but nonetheless it is still used. Why? Because it is relatively easy to generate, requiring very little subjective judgment. All the alternative measures, including the ones I propose, require subjective measures, something disliked by economists. They are "subjective" because they involve personal experiences and feelings, attributes that social scientists have developed reliable, repeatable ways to measure. Nonetheless, they are not considered "objective," a characteristic defined to be better even though the things we care about most are judged precisely by those subjective feelings. For something to be meaningful to people, it must reflect their experiences.

What is the problem with GDP? Let me give a simple example to start. On January 28, 2022, the *New York Times* had two articles about the economy of the United States. One headline proclaimed that the economy "surged by 5.7% in 2021, best in decades," as measured by GDP. The other article had a different headline: "If GDP Is UP, Why Do Voters Feel so Down?" The article, of course, quoted numerous economists who tried to explain the apparent discrepancy.

I have a much simpler explanation. The GDP has nothing to do with people's lives. To the average person, it is completely meaningless. To the professional in the field of finance, it says something about investment opportunities, but it is such a complex mixture of components, some positive, some negative, that it is meaningless to the average person. In January 2022, people were uncomfortable, feeling frustrated, and confused about the state of the world, the United States, and their own livelihood. The GDP numbers were completely meaningless in addressing those issues.

GDP ranks countries by measuring everything they do in monetary terms, usually US dollars. There are numerous criticisms of this measurement, but I'll reduce them to three:

- It emphasizes material output without considering overall well-being.
- It counts costs and waste as economic benefits.
- It summarizes the economic status of a nation by a single number, hiding the vast complexity underlying the computation. It is "opaque": invisible, mysterious, secretive, and confusing.

Yes, GDP ignores the well-being of the nation. It is simply a count of expenditures. Moreover, if a company spends money on its products but yields a lot of waste and pollution to the world, the money spent to produce these harmful substances is counted as a good thing, adding to GDP. Then, if government has to spend money to undo the harm, that money also adds to the GDP. These waste products are counted as "externalities," which means they do not count against the company that produced them. Cleaning up these destructive waste products requires the expenditure of money, so, as a result, this negative act counts twice as good for the GDP: first, the spending that went into making the products in such a destructive manner and then the costs of cleaning up the resulting mess. The power of externalities: the costs are born by society, not by the groups that give rise to them, and those costs are then considered beneficial to GDP. Externalities play a major role in the destruction of the earth, the waterways, and atmosphere. They are a major contributing factor to global climate change. Yet they boost a nation's GDP and therefore international prestige.

Charles Kenny's book *Getting Better*, subtitled *Why Global Development Is Succeeding*, takes a detailed critical look at the factors that went into his title's positive statement about global development. In many of the low-income countries of the world, despite years of effort by both the countries themselves and foreign aid, GDP often remained relatively flat: no meaningful improvement. This flat line has led many to say that foreign aid is wasted. But when Kenny looked more closely at these countries, he noted that although GDP may have stayed the same, measures of health, education, political freedom, access to infrastructure, and technology went up, as did happiness. Quality of life and human well-being: Aren't those attributes of great importance? GDP doesn't measure them. It instead restricts itself to measures that are easy to get. Which should economists prefer: the correct way or the easy way?

Numerous attempts have been made to redefine the measure of a country's quality of life. “Money isn't everything,” said Justin Fox in an article in the *Harvard Business Review*, “but for measuring national success, it has long been pretty much the only thing.” David Pilling wrote that “if GDP were a person, it would be indifferent, blind even, to morality. It measures production of whatever kind, good or bad. GDP likes pollution, particularly if you have to spend money cleaning it up. It likes crime because it is fond of large police forces and repairing broken windows.” British students in economics have also criticized their chosen profession, voicing their complaints eloquently in a club, a website, and a book: *The Econocracy: The Perils of Leaving Economics to the Experts*. They even got Andrew Haldane, chief economist at the Bank of England, to write a long, supportive foreword to the book.

Economists have managed to become extremely influential in national and world governance in part because the financial and economic status of nations is of critical importance. But they are basing their judgments on the wrong kinds of models, a point that many senior and well-respected economists continue to make.

Why are our lives dominated by the economists? Why do the newspaper headlines continually shout the stock market figures at us, showing how far the stock market has risen or fallen on any day? Why is this a measure of the average person's contentment? Why do economists think that two of the most common measures of a country are (1) gross domestic product and (2) the various stock market indexes?

What role do stock market indexes play? The price of stocks reflects the goals of the professional traders who are anxious either to make more money or to minimize losses. In other words, traders are driven by avarice and fear. Yes, the value of stocks impacts many people because retirement funds and the cost of loans are driven by these numbers. But paying attention to the variability every day is counterproductive for all but the professionals who mine the differences for profit and thus make money without providing any substantive value to the world.

The Quest to Replace GDP

GDP represents the value of goods and services produced by a country. Why does this matter? Economists say that it is important to determine whether an economy is growing or not. True, but this determination is arrived at using a faulty measure of “economy.” As I have already said, GDP ignores health, happiness, and quality of life. Why is the complexity of a country's economy reduced to a single number, a faulty one at that? Many different measurements go into the monetary measure of GDP. Why is it summarized by a single number when in truth it is a complex multidimensional mélange of many different numbers and determinations? How can one number summarize all that complexity?

There are three major flaws of the GDP: one, the higher the value, the better, yet the costs of many harmful things go into the number; two, it doesn't even try to measure the things that are of most importance to people; three, it takes a very complex set of measures and tries to reduce it to a single value.

I've already covered point one. For point two, suppose we measured the quality of life not in terms of money but in terms of human values. I don't have to invent any new measurement tools. There are many to choose from, and some are already in use, but because they are not widely accepted by economists or by officials who depend on economists, they do not play a meaningful role in economic decisions of countries and businesses. The World Economic Forum pointed to five measures of growth that are better than GDP:

- Good jobs
- Well-being
- Environment (including sustainability)
- Fairness (and equity)
- Health

Here is my list of other measures that might be added.

- Inclusiveness
- Resilience
- Education
- Societal values
- Quality of governance
- Happiness
- Freedom from hunger

The two lists are illustrative. With so many alternatives to choose from, how should a country be measured? My recommendation is simple: don't even try. Do not combine disparate measures in the vain hope of ending with a meaningful number. Yes, the end result will be a number, but will it be meaningful? Why combine the values for each of the attributes to get one summarizing number? Instead, show the values for *each*, ideally with a scale that indicates how well each one stands in relation to its desired value. One way of doing this is shown in the doughnut model of economics (figure 11.3) in chapter 11. Very much like the diagram of threats to societal needs in figure 11.2b, the doughnut model of the economy displays multiple attributes along with an indicator of how distant the value of that attribute is from the ideal. This gives an excellent picture of where the country stands with respect to the critical variables. By keeping the attributes separate, it is easy to determine precisely where each country should focus its attention.

GDP numbers and the rankings of countries don't tell us about the actual lives of the people who live there, and that is what matters. Other indexes measure things of great concern to many people. For instance, the Gini index measures income inequality, a concern among many low-paid workers. Named after the Italian statistician Corrado Gini, who developed it in 1921, the Gini index rates countries on a scale from 0 to 100, where the lower the number, the better the equity: a rating of 0 indicates perfectly equal distribution of income across the population; a rating of 100 means the country has reached the highest possible level of inequity. All these alternative measures still use economic indicators of satisfaction, not because everyone believes that money really says anything about comfort or satisfaction but because the data are easy to get.

GPI and HDI

Two of the more popular alternative metrics for the worth of a country are the Genuine Progress Indicator (GPI) and the Human Development Index (HDI). The GPI includes measures of environmental and social factors as well as societal contributions in an attempt to determine whether the economy is providing meaningful benefits to people. The HDI is a similar attempt to examine societal values. According to the United Nations Development Programme,

> The HDI was created to emphasize that people and their capabilities should be the ultimate criteria for assessing the development of a country, not economic growth alone. The HDI can also be used to question national policy choice. . . .
>
> The Human Development Index (HDI) is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable and have a decent standard of living. The HDI is the geometric mean of normalized indexes for each of the three dimensions. . . .
>
> The HDI simplifies and captures only part of what human development entails. It does not reflect on inequalities, poverty, human security, empowerment, etc.

How do the three indicators—GDP, HDI, and GPI—compare? In "Alternatives to GDP in Measuring Countries," the newsletter *Economics Online* states:

- Essentially, GDP focuses too much on total wealth while ignoring other important humanly and environment aspects.
- The HDI is a prime alternative to the GDP system, factoring in life expectancy, education length and quality, and standards of living.
- Another alternative is the GPI system, which factors in ecology to measure a country's total value. A country that emits less pollution but has moderate economy will have a better GPI index overall.

Although both the HDI and the GPI attempt to assess critical variables of human concern, they still emphasize the financial measures. Then they combine all the measures, which requires that they all be along the same kind of measurement scale. And what is the scale? Money, of course. The HDI measures "life expectancy," and the GPI measures "pollution," but they translate those measures into monetary terms. This means that the multiple components are reduced to a single number, which provides very little insight into which attributes have contributed (or failed to contribute) to the value.

11
What Measures Are Truly Important to People?

Economic measures such as GDP, the Consumer Price Index, employment and unemployment rates, and others are clearly important, but they fail to measure the critical values that are important to citizens. Measures such as the GPI and HDI begin to address these issues, but they still are constrained by the methods of traditional economic measurement and reporting.

The Seventeen Sustainable Development Goals of the United Nations

What are the things that matter to people? The United Nations provides an excellent start in its list of seventeen Sustainable Development Goals for the world: “The Sustainable Development Goals are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including those related to poverty, inequality, climate, environmental degradation, prosperity, and peace and justice. The Goals interconnect and in order to leave no one behind, it is important that we achieve each Goal and target by 2030.”

The Sustainable Development website describes each of the goals in considerable detail, including the progress that has been made and the challenges that remain. However, all of these goals are complex. How will we know if we are making progress on any of them? We must measure. Some of the goals can be measured by traditional economic means: goals 1 (poverty) and 2 (hunger), for example, are relatively easy. Goal 3 (well-being) is a psychological attribute, not readily measured by traditional methods. You can examine the other goals and determine which you think are well measured by traditional techniques. Many of the goals have mixed characteristics, containing some attributes that are easily quantified and others that, once again, are psychological variables that in general require measurement of subjective experience, and to many people in the hard sciences *subjective* is an obscene word.

Well, I received my PhD in a field called “mathematical psychology,” where among other things I was trained in measurement theory and how it is quite possible to do rigorous, repeatable measurement of many subjective variables. Psychologists have refined the various scales that can be used in measurement and the way in which they are able to measure meaningful subjective experiences such as well-being and happiness.

Measuring Subjective Experiences

It is possible to measure many subjective attributes. Psychologists have devised numerous tools for this purpose, from asking people to rate items on category scales to having them weigh paired comparisons of two variables, simply asking simple judgments such as “Which is more pleasant?,” “Which is heavier?,” “Which makes you feel sadder?, Happier?,” and so on. Such simple, subjective measures, when combined appropriately, can produce surprisingly robust, multidimensional numerical ratings. There is a science of measurement that covers both the physical *and* the behavioral sciences.

Figure 11.1
The United Nations' seventeen Sustainable Development Goals. The UN points out that these seventeen goals are not isolated issues but part of the large sociotechnical system of the world. Goals 1 through 16 point to the desirable attributes of a healthy and happy citizenry. Goal 17 points to the need for societies, countries, and organizations around the world to unite in a global partnership to work on the goals, so this goal is a metagoal: it is the organizational structure and focus required to accomplish the other 16 goals. From United Nations Development Programme, "United Nations: Sustainable Development Goals."

For example, the traditional number scale can be used when making precise numerical measurements of some things such as a person's height and weight. Measurements such as height and weight specify things on a ratio scale, a "ratio" because someone whose numerical weight is twice that of another is indeed twice as heavy. Sometimes we can measure things, but we don't know where the zero point of the scale is located—for example, the everyday measurement of temperature. The zero point is arbitrary—it means the freezing point of water in the Celsius scale and a rather arbitrary point on the Fahrenheit scale. We can indeed take averages of temperatures, but we cannot say that an object with a temperature of 100 degrees is twice as hot as one that is 50. Measurement scales such as Celsius and Fahrenheit for temperature are called “interval scales” because all the intervals are of equal value, but the lack of a zero point means ratios are meaningless. It wasn't until 1848 that William Thomson (Lord Kelvin—whom we first met in chapter 8) figured out where the zero point should be: –273.2°C (–459.7°F). Today, physicists often use the resulting Kelvin scale of temperature (where the zero point is at absolute zero) because it is a ratio scale.

Another kind of measurement scale comes from simply lining items up in a row according to some attribute—for example, arranging names by alphabetical order or arranging apples by their “redness”: reddest on the right, palest on the left. The result is an ordinal scale because only the order is known, not the intervals between items or the ratios among the items. Computing the average makes no sense for an ordinal scale: instead, we can use the median—where half the items are on the left, half on the right. In an ordinal scale, we know that value is increasing when we go from the lowest ranked to the highest, although not necessarily by equal steps. The Mohs Hardness Scale used in the physical sciences—geology, for example, provides a prime example: a mineral's hardness relative to another mineral's hardness is determined by seeing which can scratch the other, then ordering the items in terms of which one could scratch the others.

Measuring Happiness and Well-Being

Consider some of the critical variables that I suggested are missing from measurement of gross domestic product, such as happiness and well-being. Happiness and well-being sound like weird things to measure: How could it be done? Martin Seligman, a psychologist at the University of Pennsylvania, established the Positive Psychology Center with the goal of expanding the meaningful measurements of people and their well-being. The center's website defines its work like this: “Positive Psychology is the scientific study of the strengths that enable individuals and communities to thrive. The field is founded on the belief that people want to lead meaningful and fulfilling lives, to cultivate what is best within themselves, and to enhance their experiences of love, work, and play." The Positive Psychology Center recommends that its method for assessing the "well-being of nations" be used for public policy.

Well-being, of course, is only one of many other important measures of people's livelihoods. The Better Life Index of the Organization for Economic Cooperation and Development (OECD) uses eleven different categories for its measure of a "better life." Although the index collapses the ratings of the eleven categories into a single number, the users of the scale can manipulate the relative importance of each of the categories, which allows them to evaluate how the rankings are affected by the relative importance of the individual ratings.

I have already mentioned the method of "paired comparisons," where people judge a pair of items at a time, deciding which has "more" of whatever attribute is of interest. Not all possible pairs have to be compared: the number of pairs to be judged can be much reduced through appropriate statistical techniques. Although each pair is producing judgments on an ordinal scale, the different assessments can be combined to give quite accurate, meaningful interval or even ratio scale values. This combination is done because if the items being ranked come from different dimensions, each judgment produces a constraint on the possible placement of the items in a multidimensional space. With enough constraints, the location of the items in the space can be reasonably precise. These techniques are part of the field of inquiry known as "multidimensional scaling."

The ability to measure subjective experiences has revealed interesting results—for example, that happiness does not increase with the amount of money one has. Oh yes, if a family has barely enough money to survive, as their income rises, they do indeed become happier, but once they have sufficient income that they do not worry each month how they will get by, many other factors come into play, including fairness (which can also be measured). If workers doing equivalent jobs have similar incomes, they rate that everyone is being treated fairly. However, if some people have much higher incomes for no appreciable reason, the fairness rating goes down and unhappiness increases.

What about the happiness of wealthy people? Studies show that the very rich are not necessarily happier than those with modest income and savings. The real determinants of happiness lie in things other than money.

One of the best methods to understand the underlying mechanism of behavior is to understand what happens when something goes wrong. This follows a well-known engineering and scientific principle as well: it is difficult to understand something when it is working perfectly; imperfections give clues to the underlying mechanisms. I learned and used this principle when I was an electrical engineer, in both my courses and my first few jobs. Once I transitioned to psychology, the same principle applied. It works well in understanding complex machinery or computer programs or human behavior, including my studies of human error.

Clinical psychologists study neuroses and other failures in behavior. Before the days of advanced imaging techniques such as MRIs, neuroscientists studied patients who had brain damage, which gave powerful clues to the organization of the brain. Seligman also focused his early work on the study of abnormal behavior: he became famous for his work on “learned helplessness.” But one day he wondered why he was always studying the negative aspects of life: Why shouldn't psychologists study the positive things? Hence, the Positive Psychology Center was born.

I chose to describe this work as an example of how subjective attrib-utes not only can be measured but also must be. For just as life should never be about amassing more money, the success of a nation should never be about its economic measures. It should be about the subjective well-being of its citizens. We should not evaluate and rank countries by how much money they spend. Instead, we should look at the values that people care about, such as the first sixteen UN Sustainable Development Goals. Then we could see how well each nation is performing on each value. And yes, we could rank the nations, but not with a single composite number for the entire country. Instead, we should be evaluating them through separate rankings for issues such as poverty, health, education, and inequity—and, yes, for happiness.

Today, a time when the world's ecosystem is damaged, leading to climate change, which in turn is leading to droughts and floods, famine, excessive high and low temperatures, and fierce storms, why not ask how well each country is doing to protect the ecosystem of the globe and thus to ensure a positive future for everyone? How can complex information be presented in a way that is easy to understand, allowing senior decision makers to get a quick sense of the state of the systems that concern them, letting them quickly assess which components need attention and which are OK (for now, anyway). The same diagrams are useful for the everyday person who wants to keep informed but not by having to delve deeply into the detailed technical jargon and details. This is where the tool known as a *dashboard* comes into play.

Information Dashboards

In an automobile, the driver needs to know a few critical pieces of information. Over time, the displays in front of the driver have evolved to present critical, important, and sometimes simply useful information: the display is called a "dashboard." The point of an automobile dashboard is to make information readily available at a glance, without distracting the driver. In the field of information technology, dashboards summarize in a simple and clear form the key variables that are essential for decision-making. For example, decision makers need quick and authoritative assessments of conditions, allowing them to know where their attention should be focused. Numerical values might be necessary behind the scenes to determine what to display, but in the dashboard itself numerical values are not critical. To decide where to focus attention requires only a qualitative indicator of where things are going smoothly, where conditions are already in trouble, and the intermediate, marginal states where the situation has to be monitored to prevent worsening. This quick assessment allows executives and other decision makers to know where to drill down more deeply into the details of the situation, getting more precise information and driving detailed strategies for action. The dashboard makes it easy to monitor the situation and to decide whether things are under control or more work needs to be done.

What kinds of dashboards might we create for the state of an economy or a society or the ecosystem? No single representation of a complex system can possibly serve all needs or provide important info at a glance. When it comes to the assessment of the entire planet, obviously the total picture is so complex that it is probably impossible to compile all the required information in one location. Nonetheless, dashboards would be useful for policy makers and citizens alike to quickly understand the state of the ecological or societal systems and what factors need more attention. Good dashboards display all the critical information needed to assess the state of the system in one quick glance, and they can be combined with easy ways to delve more deeply into the data relevant to the attributes that need attention, such as the data given in reports by governments and intergovernmental agencies.

A Meaningful Approach to Climate Change

The first presentation of a graphic display that could serve as a dashboard for monitoring climate change came in 2009 when the climate scientist Johan Rockström and 28 colleagues published a critical paper on climate change in the journal *Nature*. (The British journal *Nature* and the American journal *Science* are considered two of the most important scientific journals in the world.) It was a short paper—just four pages—that summarized the issues (and thus the data behind their findings) succinctly. It used an image of the globe, with wedge-shaped segments radiating out from the center of the globe, each wedge representing a different variable and its length showing the severity of the current state. The goal was that each of the variables represented by the wedge would have a scientifically determined boundary threshold: exceeding the boundary "could have disastrous consequences for humanity." (Some variables did not yet have well-defined thresholds.) Rockström and his colleagues had created a dashboard, although they did not call it that: a simple graphical illustration of the major ecological issues facing the earth, showing which attributes had already exceeded the safe limits that science had established. The authors updated their work in 2015 with a slightly different graphical representation. Both examples used color to indicate the severity of the problems, but the black-and-white versions required by the publishing restrictions for my book were not as compelling, so I have not included them here.

But the graphic designer David Conover and I used the *Nature* authors' depiction as inspiration to develop a simple, meaningful depiction of the issues surrounding climate change that avoids the need for color. In figure 11.2a, the longer the arrow, the more danger its category presents. When the arrow crosses the circle labeled "Ecological Boundary," it is in the danger zone. No numbers are needed: it is easy to see where more attention and action are required. This is a dashboard. When the dashboard indicates areas of concern, then it is time to turn to the scientific data. Do not pay much attention to the actual lengths of the arrows because the data we used are old. Moreover, in the ecological diagram, the limits for all the values were not yet determined. Our diagram simply illustrates the power of the dashboard as an easy-to-digest format.

What about societal categories? Again, David Conover and I applied the same framework used in figure 11.2a to illustrate societal categories in figure 11.2b. In both figures, arrows that remain inside the boundary circles are in safe territory, showing minimum threats to ecological behavior (figure 11.2a) or societal needs (figure 11.2b). Arrows that pass through the outer boundary signal danger: excess ecological harm in 11.2a and insufficient societal need in 11.2b In other words, in both diagrams the longer the arrows, the worse the state, and once the arrows pierce the boundaries, harm is being done. Long arrows are bad.

In a dashboard, all relevant and essential data are presented, usually in graphs and diagrams. Thus, figures 11.2a and 11.2b would be presented side by side so that both ecological and societal variables are visible readily. Why have separate diagrams, though? Why not combine them? The Doughnut Economics Action Lab (DEAL) at Oxford University has done just that, combining the planetary boundaries diagram (from the version developed by Will Steffen and his colleagues in *Science* in 2015) with the critical societal values.

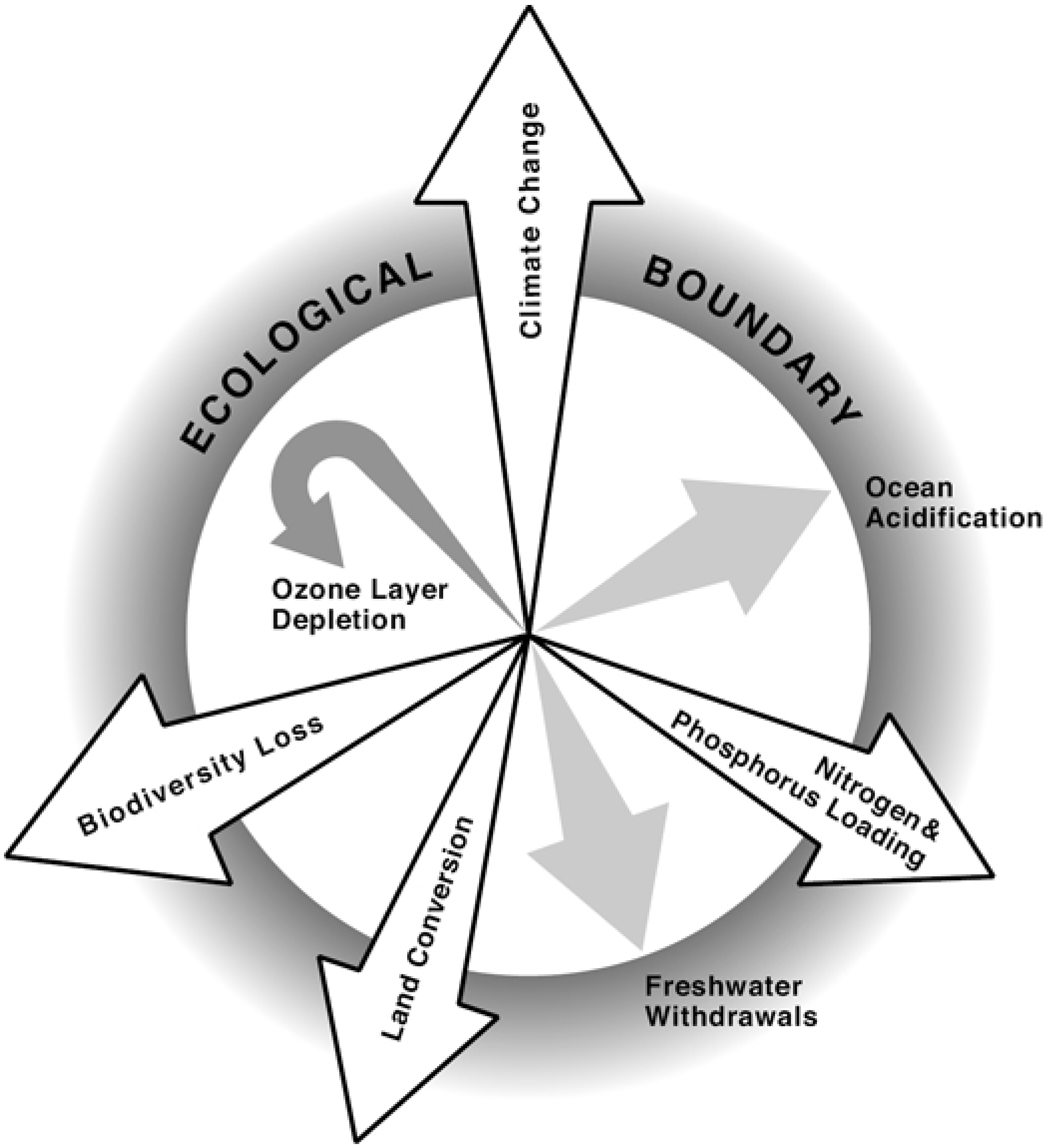


Figure 11.2a
Threats to the earth's ecology. The length of each arrow radiating outward from the center of the circle indicates the amount of potential danger for each category and thus the size of the threat to the earth's ecology. The arrow for ozone-layer depletion shows that this category is receding and the danger starting to decrease.

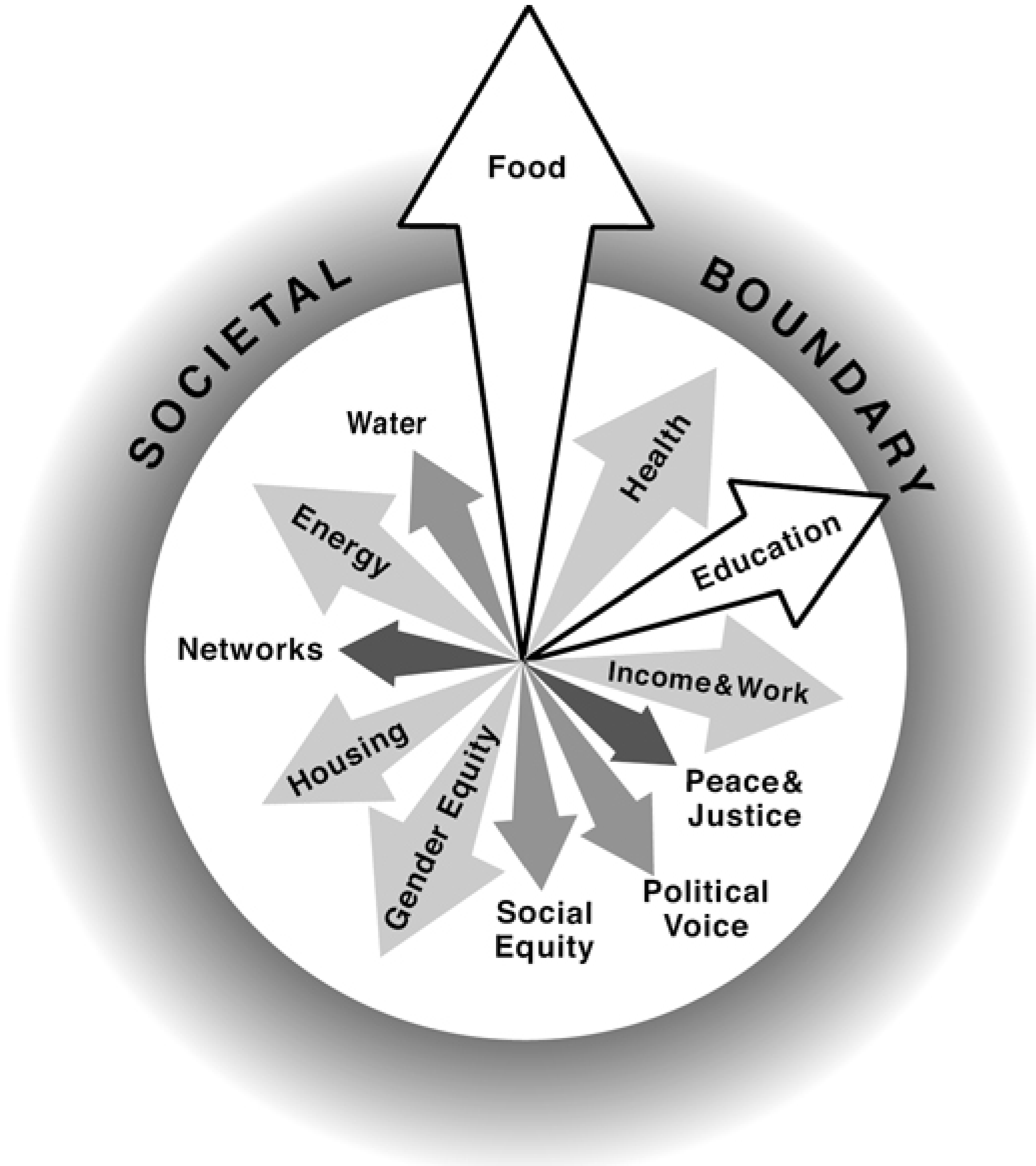


Figure 11.2b
Threats to societal needs. The length of each arrow radiating outward from the center of the circle indicates the amount of potential danger for each category: the shortfall in providing for the societal category—hence, the longer the arrow, the more society is threatened. Arrows that cross the outer “Boundary” circle are in the danger zone and are white outlined in black to emphasize the danger they pose.

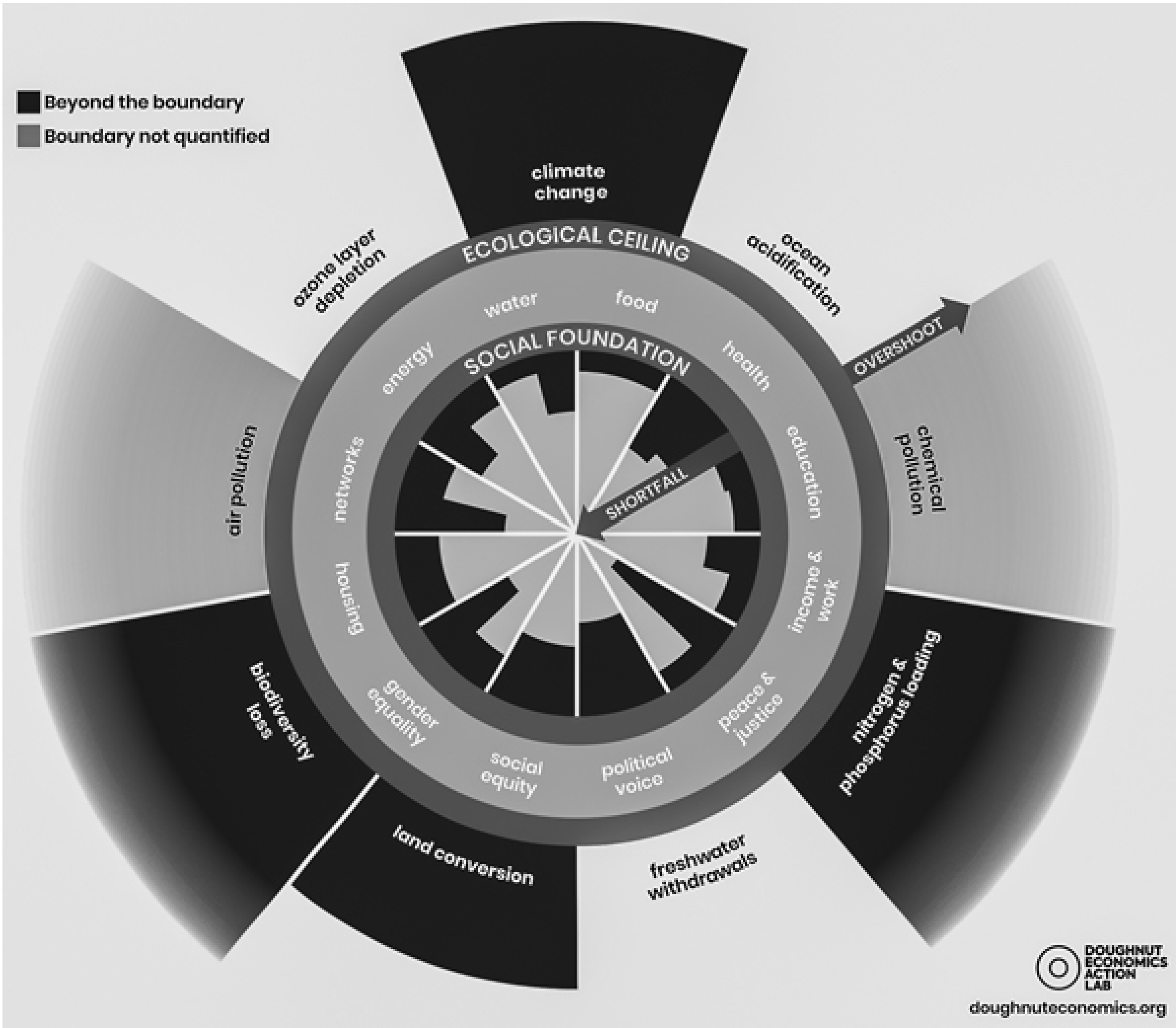


Figure 11.3
The Doughnut Model of Social and Planetary Boundaries: a quick “dashboard” of the state of the planet. Any variable that projects outward from the Ecological Ceiling ring has become destructive. Any variable projecting inward from the Social Foundations ring is detrimental to society. The goal is to avoid (or at least minimize) both forms of projections. In other words, keep the values within the ring that represents the doughnut, not in the hole and not outside the outer boundaries. Reproduced from the Doughnut Economics website. Licensed under Creative Commons CC BY-SA 4.0 License. The figure has been changed from color to black and white.

The Doughnut Model: An Effective Dashboard

The Doughnut Model by the Oxford economist Kate Raworth and DEAL shown in figure 11.3 is an example of how to present a meaningful, important display of information in a way that completely rejects the need for a single, aggregate numerical value. More importantly, it provides a meaningful structure for a complex interplay of data. It was the inspiration for adding figure 11.2b to the ecological representation of figure 11.2a.

The two concentric rings that form the doughnut make it possible to show both the ecological and social foundation values on a single diagram. The outer ring is the “Ecological Ceiling,” with the major factors that impact the earth's ecology labeled along the ring's circumference just outside the ring. The similarity of the variables for the ecology in figure 11.2a with the ecological variables in figure 11.3 is deliberate. Values that are too high for earth's safety, thereby harming the ecological balance, are shown projecting outward from that outer ring.

The innermost ring represents the social foundation necessary for a well and just society, with the major factors contributing to the social values labeled in the middle ring just outside the Social Foundation ring. Attributes that fail to meet the minimum requirements are shown within the central hole of the doughnut, projecting inward. The Doughnut Model is most effective when seen in color, a feature not available to us in this book. (In the online version of the diagram, some of the wedges in the Social Foundation part have multiple wedges for a variable to allow separate plotting of subcomponents for the variable, and the subcomponents are identified in pop-up windows when the cursor is dragged over the image.)

Figures 11.2a, 11.2b, and 11.3 are qualitative presentations. The lack of numbers in these figures is their strength. Numbers are not necessary to determine where to pay attention. When needed, numbers can be accessed in more detailed tables or charts or even as numerical labels on the doughnut. (In the case of the doughnut economics circles, when the items are displayed on a website, the viewer can roll over any of the sectors with mouse, pen, or finger, and a small pop-up window will display the numerical values.) For a quick overview, however, the relative lengths of the radiating bars and arrows provide a fast and efficient way of determining where more effort needs to be exerted. The purpose of a dashboard is to show what needs to be attended to and what does not: items that require further attention can then be examined with alternative displays.

Why care if country A is beating country B on any of the single-value metrics? For example, a diagram allows us to determine that a country might be very strong on education and biodiversity but weak on factors that lead to ocean acidification and ozone-layer depletion. For another country, the diagram might show that it is strong on education but weak on biodiversity. Then, by comparing the current year's diagram with diagrams produced in the past, we might notice that the country has made vast improvements in decreasing the release of pollutants into the waterways and air. That's the information we need. Ranking may be good for some egos but serves no useful purpose.

These examples of dashboards have several powerful features:

1. The graphical depiction makes it very easy to see which of the variables are within the safe limits of planetary boundaries, which have reached the point of doing harm, and which need more work.
2. Dashboards do not make the error of trying to combine all the measures into one single number: each attribute can be individually and quickly assessed.
3. By combining ecological and societal issues in one location, a doughnut model makes it easy to get a quick assessment of the state of the system under study (country, state, city, company). The doughnut cleverly puts these values into concentric circles; the other type of dashboard requires two side-by-side circles.

Which do you prefer? The doughnut of 11.3 or the paired circles of 11.2a and 11.2b? In our testing, we found that some people had difficulty with the doughnut model because the arrows in the outer and inner rings moved in opposite directions (outward and inward, respectively) to illustrate harm. However, the doughnut diagram has the virtue of projecting all the critical variables in one compact image rather than the two that our scheme requires. The purpose of including the different types of diagrams here is not to judge which is superior but to make the point that spending time and effort to develop meaningful ways of presenting information to nonspecialists is worthwhile. Otherwise, the importance of the message can get lost in complexity. The best way to present information depends on the audience: different audiences will have different preferences.

What is missing? Subjective, psychological measures such as quality of life, happiness, and a feeling of being in control of one's own destiny. This chapter has discussed how these other attributes might be measured: now someone has to add them to the dashboard, probably in separate diagrams. Automobile dashboards (and the equivalent displays in aircraft and process-control plants) summarize the variables in a variety of ways. Ideally, all the critical variables fit on one piece of paper or computer screen, but this is not necessary. The important point is that all the critical information can be scanned quickly, so that in a few seconds or minutes the observer has a good sense of the state of the system.

The dashboard is an excellent tool for information that can be presented graphically, either with numerical precision or, as in the examples presented here, with qualitative indicators. Full understanding, however, requires additional important information—the underlying story or narrative.

12
Human Behavior and Economics

When I was a graduate student in psychology, I did experiments in utility theory, a part of microeconomics. The experiments were artificial, conducted within the psychology laboratories and asking the people we were observing to make choices between several bets, with varying probabilities. Ward Edwards, a psychologist at the University of Michigan, expanded these studies to Las Vegas, where the casinos allowed him to do similar experiments by observing the behavior of bettors playing with slot machines. The casinos had already designed wonderful conditions for the study because they had a variety of games available, where the amount of money inserted into the machine for each bet differed, the probability of a payoff differed, and the sizes of the payoff differed. These variables allowed Edwards to look for the differences in behavior. Still, slot machines and their addicted players constitute a rather unique environment, not a good substitute for natural behavior among a more diverse set of people.

I became disillusioned with utility theory and with the state of economics theory in those years because it made overly simplistic assumptions about human behavior (this was a long time ago—in the early 1960s). I didn't persist with utility theory studies and turned instead to research in auditory psychophysics, a better complement to my engineering background and interests in that period of my life. In 1978, the Nobel Prize committee awarded the Nobel Prize in Economics to a noneconomist, Herb Simon, for his concept of bounded rationality, demonstrating that people did not operate in simple, logical fashion. I discussed his work in chapter 9 in the section "Examples of Flawed Assumptions." "Satisficing," for example, is not in the normal vocabulary of economists. When I learned that Simon was awarded the Nobel Prize, I rushed to share the news with my colleagues on the faculty of the economics department, only to be told, “What a waste of the Nobel.” Yes, to economists of 1978, giving the award to a noneconomist was heresy.

Herb Simon was an amazing person, producing important work in many different fields: management, psychology, problem solving, artificial intelligence, and economics. He was also one of the founders of the field of artificial intelligence. How did he do all this? When he accompanied me and others to conferences, he would never show up at breakfast or even at the opening talks of a conference. We asked him why not. He explained that he used his mornings to write papers. He also added that he never wasted his time by reading newspapers or listening to news. Why? He told us, “If anything important is happening, my friends will tell me.” His work that has had the most impact on designers is the book *The Sciences of the Artificial*.

In 2002, 24 years after Simon received his award, Daniel Kahneman, a psychologist, received a Nobel Prize in Economics for the work he did with his lifetime collaborator, Amos Tversky (who died in 1996; the Nobel is never given posthumously). Their joint work exposed the foibles of even simple decision-making problems by people, including by senior economists. Kahneman has written a best-selling book about his work: *Thinking, Fast and Slow*.

Now let me credit yet a third person who is an actual economist, Richard Thaler. He received the Nobel Prize in 2017, 15 years after Kahneman and 39 after Simon—39 years for the field to pay attention, although the headline about Thaler's award in the British newspaper the *Guardian* was “Richard Thaler Is a Controversial Nobel Prize Winner—but a Deserving One.” Thaler is best known to the general public for his work using simple, practical psychology to change behavior: *Nudge: Improving Decisions about Health, Wealth, and Happiness*. I first met Thaler when he visited me in my office at Northwestern University, explaining that he had just written a book about design and was getting it published. He asked me to write a back-cover blurb, and I was delighted to comply. *Nudge* is a sensible, psychology-based approach to human behavior. Thaler wrote the book with Cass Sunstein, a legal scholar who did not share the prize—which was not given for the book but for a much larger body of work by Thaler. Sunstein went on to advise the Obama administration on how to implement the principles of *Nudge*. He is now head of the behavioral economics group at Harvard Law School.

Many people have helped establish the field of behavioral economics. The psychologist Ward Edwards, for example, whom I mentioned at the start of this chapter for his work studying gamblers, was called the “father of behavioral decision making.” Edwards had been working in this field since the early 1950s, training a large number of students (including Amos Tversky) and collaborators. But not until the Nobel committee awarded its prestigious prize in economic sciences to people in what is today called behavioral economics did the work finally become an established field, part of the mainstream. However, it still has not made much difference to the economic models that control governmental policy. It can take a long time for ideas to make their way from academic journals into practical application, especially when it involves changing long-term policies and practices of governments.

The signs are positive, though. More and more economists are now studying real behavior, not just the statistics gathered from economic data, surveys, and questionnaires. Some call themselves “behavioral economists,” others call themselves “neuroeconomists.” The coupling of neuroscience with cognitive scientists and psychologists has added considerable depth to our understanding. Neuroscience has enabled us to understand which parts of the brain handle the different components of decision-making. This is an excellent example of the power of working across disciplinary boundaries.

The Power of Stories

People are storytellers. Stories explore and explain the world and its sometimes-idiosyncratic behavior. People tell stories to and about themselves, to and about others. Human experiences form coherent narratives with goals, constraints, impediments to be overcome, and causal chains of events and behavior.

Stories are as important as scientific investigations to our understanding of the world. Science provides verifiable evidence and explanations, but the latter are usually presented as abstract principles, divorced from any individual occurrence. This abstraction is necessary, though, for science aims to provide general principles that apply to many different individual occurrences. Stories, in contrast, provide context. They show how behavior can be impacted by the principles under discussion. The combination of scientific knowledge and the way that it might affect everyday life provides a deeper understanding of phenomena.

Science can talk only about what it can study, yet many of the phenomena important to people are beyond the capabilities of today's science. Moreover, scientific replication depends heavily on measurement of the topics under investigation, and as I have already described, scientists measure what they can measure, which is seldom the thing that they actually care about.

The storyteller can portray the elements of the situation, especially the aspects that cannot be measured. Stories can cover important aspects of a situation that are left out of a scientific report, and quite often these missing parts become critical to understanding. Stories also have limitations. They are good at providing context and concrete examples of the phenomena of interest. However, they are very limited in scope. The power of stories to provide concrete examples also restricts the generality that is provided: each story can address a small number of issues and individuals or groups. Stories can also be biased by the teller's point of view, the choice of examples, and other factors, many of which are subconscious, not recognized or intended by the storyteller.

Formal measurements and evaluations are important, but by their very nature they also have several weaknesses. First, they are restricted to things that can be readily measured, which may not truly reflect the aspects of a situation that are of greatest importance. Second, their abstraction reduces the discussion of large, complex problems to numbers, graphs, and illustrations. The economist Morton Schapiro teamed up with the historian Gary Morson to discuss the importance of stories, arguing that economists could learn much by paying attention to great novelists. Robert Shiller, another Nobel laureate in economics (although the Nobel award he received in 2013 was for empirical analyses of asset prices) also worked in behavioral economics. In his book *Narrative Economics: How Stories Go Viral and Drive Major Economic Events*, Shiller points out that the ability of economists to create one-year forecasts has “been on the whole worthless.” It is rare, he says, for economists to take into account what people think about the economy, but his book argues for taking people's beliefs seriously, especially in the stories that become popular. The point is so important that Shiller elevated it to be the subtitle of his book: “stories go viral and drive major economic events.” I return in more detail to the work of Shiller and the team of Morson and Schapiro in part V of this book, but, for now, I focus on how stories provide significant information about real human behavior.

In the social sciences, stories can provide useful supplements to formal measures. Here is where the stories and observations of anthropologists, journalists, and writers give us the context and background information necessary for understanding how people live and behave. Designers often create short descriptions and stories about the lives of the people they are designing for: “personas.” Each persona consists of a short one- or two-page statement about a person's life and activities. Although the personas are fictional, when created properly they can be extremely realistic, stemming from the evidence of numerous field studies and observations of the people being depicted, although modified to preserve privacy. Those personas are often made into posters, placed in the work areas so that the designers can see them every day, the better to remind them of the people for whom the design is intended.

The real insights into human behavior occur when measurement is combined with stories. Measurement provides precision but is an abstraction of the situation. The lack of context can lead decision makers to make perfectly rational, logical choices that are fundamentally wrong once implemented in the real world. Stories provide context and understanding of the impact of decisions or actions. Stories are selective and limited in coverage, which is their weakness, so the combination of stories and measurements works because each makes up for the other's weaknesses.

Let me share two examples that show some of the values of narratives. They come from the world of hospital health care and are based on my observations, observations that would never be captured in traditional measurement-driven science.

Story 1: How Measurements Transform Patients from Human Beings into Numbers

I was once involved in a study of medical record systems being conducted by the American National Academies of Science, Engineering, and Medicine. Our group visited hospitals across the United States and observed a wide range of activities within the hospitals, from admission to bed assignment (which is far more complex than any of us had realized) and, of course, physicians and medical personnel at work. One early morning we tagged along as the attending physician made the rounds of patients in the surgical ward. The physician was accompanied by a flock of nurses, resident physicians, and interns. One of the nurses who accompanied us pushed a computer workstation on wheels. As we got to each room, the attending physician knocked on the door and greeted the patient: “Good morning, Mrs. Johnson.” Then the physician turned to the rest of us—everyone still in the hall—while the computer displayed the test results for the patient. The residents relayed to the attending physician the patient's vital statistics and the current state of medication. The physician would listen carefully, ask questions, give suggestions to the nurses and residents, and then knock on the door once again and say cheerfully, “Goodbye, Mrs. Johnson. Have a good day.” The physician never entered the patient's room: the entire assessment was done through numbers.

Medical professionals know this phenomenon well: thinking of patients not as people but rather as their ailments. When physicians think only of their own specialties, they treat the test results and instrument readings and neglect to treat the patient, diagnosing and prescribing remotely simply on the measured values, without ever interacting with the patient.

What can be done to overcome this tendency? One step could be to make sure that patients’ records include a brief story of their lives, family, children, hobbies, jobs, and interests. The stories need to be short and poignant because busy medical personnel do not have time for long stories. Stories have the power to transform patients from numbers and symptoms into people. They also can put many observations and treatments into a cohesive story. In theory, the patient's medical history has this information, but in practice this story is difficult to read: it is a compilation of past ailments and treatments coupled with the current event, medical tests, diagnoses, and medical condition. The problem is that physicians do not have time to read the entire medical record. Moreover, it is usually organized in temporal order by when the patients were examined, laboratory results were received, and diagnoses and treatments made. The information is not put together in an easy-to-digest form, emphasizing the critical points. We need a way of making these histories easier to read, more cohesive, interesting, and relevant to the patient's medical condition. Would this be difficult?

Some people are looking for technology to fix this problem. In the field known variously as automated, algorithmic, or robot journalism, computers are writing stories automatically based on sports results and market performance. I informally tested one such system by asking it to review usage of my website, and, to my surprise, it provided me with insights I wasn't able to get from looking over the same statistics the artificial intelligence system used. Could it do the same for making sense of the clinical histories of patients? Probably.

Story 2: How Medical Recording Equipment Can Literally Hide the Patient

During the same study that yielded the observation reported in story 1 but at a different hospital, I saw a room packed with instruments in the pediatric acute-care ward for premature births. There were multiple infusion pumps, computer readouts, and monitors. The entire room was filled with the red glowing lights of display readouts and the dim white of graphs on computer screens.

"Fascinating," I said. "You've brought all of the monitors into one place so you can see how all the patients are doing."

"No," said one of the physicians. "What do you mean?"

"So where are the patients?" I asked, expecting to be told that they were in rooms adjacent to the instruments.

“Right there,” said the physician, obviously puzzled by my question. “Right there in the room, right in front of you.”

I looked closely and still couldn't see a patient. One of the nurses walked over and pointed. “Oh,” I said.

There were so many medical devices, so many readouts and displays, that I couldn't even see the patient until someone showed me. This was an infant ward, so this particular patient was tiny, but, even so, the scenario here is a good illustration of modern medicine. Patients have been transformed from human beings into numbers derived from medical tests and the readouts of monitoring equipment. The patients themselves have literally disappeared from sight.

Stories 1 and 2 are real. They illustrate the complexity of modern health care. The measurement techniques have gotten to an advanced state, as have the sensors and computer-equipped medical devices that interpret the sensors and provide readouts and, where necessary, the alarms and alerts. All these measurements are useful but insufficient. (Actually, the alarms and alerts are yet another issue, for when there is a major problem, the cacophony of the multiple alarms so distracts medical attendants that they sometimes waste valuable time turning them off, so they can concentrate upon the problem. But I have already covered this type of problem in my other books.)

Measurements Give Facts, Stories Provide Meaning

My two stories show the abstraction that has taken place in medicine, where specialists see their specialties, not the patient, and even the generalists who are responsible for the patient see primarily the numbers. The two stories illustrate concrete examples. The stories are powerful, but their weakness is that each is unique, representing one example and one point of view from the many that are necessary to understand health care.

Historians face similar issues. They need to report on past events, but just presenting the statistics from the written record does not capture what really transpired. The official records don't tell the stories of the people, the way they felt and behaved, and, perhaps most importantly, what they perceived and believed or what their goals and needs were. Thus, historians often present their results as narratives, combining stories of lives and actions with the records from the times being depicted.

Stories and measurements are two complementary forms of representing our understanding of a phenomenon. We should celebrate their differences and combine them to offer the best of the two worlds. This is the power of diversity—each element covers a deficit in the other. It is the combination that wins. In the science of climate change, narrative articles, even in scientific journals, are more influential than expository writing—even to the scientists. Moreover, it has been found that the journals with the highest impact rating feature more narrative articles than other journals. Why? As Ann Hillier, Ryan Kelly, and Terri Klinger point out in their article “Narrative Style Influences Citation Frequency in Climate Change Science,” “Narrative writing tells a story through related events, whereas expository writing relates facts without much social context. Presenting the same information in a more narrative way has the potential to increase its uptake—an especially attractive prospect in the context of climate science and scientific writing generally—and consequently, narratives are widely recognized as powerful tools of communication.”

Numbers, statistics, dates, and historical records are important, but they are abstract, devoid of meaning. Stories are the key to adding context, the environmental setting, and meaning to the otherwise dry, impersonal abstraction of data. People have difficulty when presented with incoherent observations or too much data. How many times have you read a report or article with a table of numbers? Fascinating? Maybe to the small number of people who relish details and who work in areas related to the report. They have the correct background and experiences to provide the meaning necessary to understand the numbers. But the rest of us need meaningful structure.

Here is a simple example, taken from my life years ago when I was an experimental psychologist. Can you remember a string of 48 letters, just the letters but not in any particular order? Here you are—learn these letters:

ngflinuaemstneanceeisevryyaseforoneyanotmerermeb

People skilled in the art of memory and the various mnemonic methods or the relatively few people who qualify as having a "photographic memory" might be able to memorize the string quickly, but most of us take one look and don't even try.

So here is a second test: try to remember the next string of letters. Once again, all you should care about is remembering as many of the letters as possible. They don't have to be in the same order as in the string.

A meaningful sentence is very easy for anyone to remember.

I suspect many of my readers will think this is an unfair trick. Of course, a meaningful sentence is easier to remember than a string of 48 meaningless letters. That's the point. The fact that this second string is dramatically easier to remember than the first (and, as a bonus, to recall them in their exact sequence as presented) is no surprise; after all, it is a meaningful sentence. But the sentence has the very same letters as the first string. Many scientific articles read like meaningless streams of letters, numbers, and complex, difficult-to-understand sentences—even to experts in the field. Put the information into a meaningful narrative, and it becomes easy to understand and remember.

People are meaning-seeking creatures. Meaningful experiences lead to understanding and ease of remembering. How do designers transform complex technology into systems that people find so easy to use that they say it is simple? The secret is to make it meaningful. It doesn't matter how complex something is. If people find it meaningful and understandable, they will judge it to be "simple." "Complexity is a fact of the world, whereas simplicity is in the mind."

III
Sustainable
Reverse and Repair the Harm Done to the Ecosystems of the World

13
We Live in the Age of Waste

The twentieth century was the age of waste: pervasive, ubiquitous, and harmful waste—a plague that is still with us in the twenty-first century. The *Waste Age* exhibit at the London Design Museum made the point quite dramatically. When the exhibit opened in October 2021, I spent three days discussing the issues with people from the Design Museum, the UK Design Council, and the Ellen MacArthur Foundation. We all agreed that design was a major source of the problem, as already discussed in the opening pages of this book.

The exhibit's subtitle echoes one of the major themes of this book: *What Can Design Do?* What *can* design do to reverse this state of affairs? What roles should designers play? Shouldn't designers alter the behavior of commerce, starting with the places where they work and with the design of things that do not contribute to the body of waste? Alas, today few designers are in a position of sufficient authority to have this kind of power. To be able to make a difference, designers themselves need to change, to become knowledgeable on topics normally thought to be outside the domain of design but essential to understanding the world: economics, business models, history, humanities, technology, ethics, management, and politics. This is massive change—change in how designers are educated, in the amount of decision-making authority they have, in the business models of companies, and in the structure of society.

No single discipline can eliminate the problems we are experiencing today. They require everyone to contribute, especially economists and managers, politicians and officials in government organizations, organizers of foundations and nongovernment agencies, and key leaders of the world's nations. The problems of today will affect everyone on earth, so the leaders and representatives of all people must help in the recovery.

Our wasteful ways are destroying the entire ecosystem—poisoning people, animals, plants, irrevocably changing the upper atmosphere, and leading to massive global warming of the entire planet, something that has gone on for so long that harmful impacts have already been felt around the world. Multinational meetings are in progress to discuss the issues. That sounds promising, except that the talks started two decades ago, are simply continuing today, and, I suspect, will still be going on for a long time. Lots of discussions and promises, but not nearly enough action. Even if the leaders of the world were to completely agree on what needs to be done, when they return home, they are often unable to convince their legislatures of the need to act. And even if the legislatures do act, it will still be difficult to convince the major industries and the citizens. The leaders are surprisingly powerless to stop the waste (especially in democracies).

The world relies on industry for clothing, food, and entertainment. Travel in the twenty-first century allows people to visit most of the world easily and more comfortably, more quickly, and less expensively than ever before possible. This makes it possible for many of us to travel around the world to meetings to discuss the polluting impact of travel. The economy of the Global North is so dependent on these facilities and capabilities that the activities are difficult to stop and the means of production and travel difficult to replace, even though they are destroying the planet. Today's infrastructure supports people's lives across the globe and cannot be changed rapidly without severe harm to many people, especially the poor and underprivileged. Many goods are no longer made at one location. Instead, their production requires supply chains that cover the world.

The burning of fossil fuels (coal, oil, natural gas) is destructive to the atmosphere and is a major factor in the global rise in temperature. But so much of the world's infrastructure depends on these energy sources for travel, electric power, heat for homes and cooking, as well as for the making of products such as automobile tires for vehicles, pharmaceuticals, fertilizers, and, of course, plastics, that stopping the use of fossil fuels will be extremely difficult and take many decades. Yes, today's power plants can be replaced with other means of producing energy that is nondestructive—for example, hydroelectric, wind, and solar power as well as perhaps hydrogen power, if we can isolate hydrogen in an economical, nondestructive way. Nuclear power is an ecologically sound electricity producer, but nobody has yet been able to solve the problem of safely disposing of the radioactive waste products. Engines that are powered by fossil fuels can be replaced with motors powered by electricity, although this replacement helps only if the electric power and the motors themselves are created from ecologically sound sources of energy.

To replace the existing sources of energy with new, renewable sources will require considerable time: the most optimistic predictions say it will take several decades, not finishing until 2050; other predictions say it will take even longer. The world gets so much of its energy from fossil fuels—estimates say more than 84 percent —that the magnitude of the effort required to change the sources of energy is overwhelming. “Overwhelming” does not mean impossible, but the magnitude of the task requires a substantial workforce, a considerable amount of funding, and a large number of construction sites, including new long-distance power transmission lines. Completing such a task will take a long time and, moreover, face considerable political opposition because it will impair the lives of all the people who depend on the traditional energy industries and live in the construction zones. The world will thus rely on destructive technologies for a long time.

The tension between the long-term damage that is being done to the ecosystem of the world and the short-term harm to the lives of many people if we change too many things too rapidly is a major challenge. The short-term harm delays the quest for long-term stability. Nonetheless, it is possible to start immediately to change the reliance on today's methods of production as well as to change many of our activities, ways of living, and ways of doing business in order to reduce our dependence on harmful practices and thus speed the transition to a more sustainable future.

This trade-off between short-term harm and long-term benefit will show up frequently in the remainder of this book. Here is where human needs and behavior conflict with the very neat and powerful solutions that numerous technologists have proposed. The technological solutions are well thought out and, although difficult and expensive, are useful, important, and probably crucial to overcome today's issues. But these technologists invariably ignore the horrible toll their adoption will create for the lives of many societies and people; although some lives will benefit, many others will be harmed. Political decisions are typically biased toward the short-term, especially in the democracies of the world, where leaders are elected based on existing situations, not on massive, expensive actions that will not have any impact for many years. As a result, the political will to take drastic action is greatly restrained.

The way we got to today's state and what can be done about it are the themes of this book. I discussed the artificial nature of civilization in part I, suggesting that because so much of what we do and how we do it is artificial, it *can* be changed. To do this, we must present this case to the people of the world in meaningful, understandable ways, the theme of part II. One of the major things that must be changed is the way we are depleting the world's most valuable resources while polluting the environment in ways that are unhealthy to all living things, that destroy many species of plants and animals, and that are artificially modifying the higher levels of the atmosphere in a manner that is changing the climate. These topics are discussed in this part. Part IV discusses how these needed changes can be accomplished in a humanity-centered way. And parts V and VI review the obstacles and opportunities, with the emphasis on the human behavior that is both the impediment to change and the source of creative, practical solutions.

14
How Did the World Get into Today's Quandary?

The twentieth century was marked by advances in the STEM fields: science, technology, engineering, and mathematics. It was a century of progress, if you can ignore two world wars; the Great Influenza Pandemic in 1918, which infected one-third of the world's population; the vastly increasing inequity between the extremely wealthy and the rest of the population as well as be-tween technologically advanced nations and the rest of the world; the ever-continuing local wars all over the globe; and the considerable political and societal unrest. The development of mass manufacturing and plastics are considered two of the greatest technological accomplishments of the twentieth century, yet they are responsible for the Age of Waste that now plagues the twenty-first century. The ease of manufacturing plus the ubiquity of plastics created an onslaught of products to be sold inexpensively to people. Marketing professionals called these people —us—“consumers,” for that was people's (our) obligation to the world of commerce: to consume again and again and again. The twenty-first century has continued the practices that spread over the eighteenth, nineteenth, and twentieth centuries, while amplifying the huge economic disparities between the wealthy few percent and the rest.

The Throwaway Economy

A major shift in lifestyles took place during the rise of the Industrial Revolution, starting in the 1700s in England. For the first time, factories in England—and then the world—could mass-produce items at relatively little cost. Industrial nations could produce clothing, tools, machines, reading matter, and foodstuffs for everyone with ingredients coming from all over the world; these items could be sold all over the world and then discarded when their usefulness ended. Two hundred years later, at the end of World War II in the mid-1900s, people who had suffered and were deprived of many necessities of life, to say nothing of the luxuries, were anx-ious to make up for the lack of goods during the war. New jobs provided income. New manufacturing methods and the rise of the plastics industry led to the production of huge quantities of inexpensive goods. The economic recovery of war-ravaged countries was based on increased, continual spending by consumers. Many items, especially those made of plastic or paper, were so inexpensive that they could be discarded after a single use. Since the start of the Industrial Revolution, advances in technology, the development of new materials, especially the wide range of plastics, and the development of complex, massive, automated factories have led much of the world to adopt a throwaway economy.

Today, many of the items in daily use are designed for a short life. One extreme example is eating utensils—plates, cups, and tableware made of paper and plastic—all designed to be used once and then discarded. Advertisements extoll their virtue in making life easier and more convenient. Even today (literally as I wrote this sentence), I could find multiple products advertised with the statement "No need to wash dishes after a party ends. More time to have fun with family and friends." This is not a new phenomenon. In 1960—more than a half century ago—Vance Packard wrote about this trend in his book *The Waste Makers*, which was advertised as "an exposé of 'the systematic attempt of business to make us wasteful, debt-ridden, permanently discontented individuals.'"

Other items, such as home appliances and automobiles, were designed with obsolescence in mind, so that they too would be discarded and replaced by newer versions of the same thing. All these throw-away objects gave rise to mountains of trash, and unlike nature, which makes productive use of its waste, we didn't know what to do with ours. The problem is not in the single usage: it is in the materials used to make these items, materials that once used can no longer be reused and in most instances not even recycled or that take anywhere from 20 to 500 years to biodegrade. The Design Museum approached this issue by exhibiting numerous items, from tools to dresses, that can be made out of materials produced from algae, not petroleum (as plastics are), and thus are designed for temporary usage but without being harmful to the environment when they are discarded.

Until we reach the point where everything is designed so that its discards do not contribute to the Waste Age, we need to stop the march toward both single-use and, even worse, planned obsolescence. Just as many things are deliberately designed for single usage, others are deliberately designed to become obsolete: planned obsolescence is a desired and valuable technique for any business that requires repeated sales of its commodities to stay in business.

For some businesses, this is not a problem. Food is literally consumed by its purchasers, and because food is necessary for life, food stores will never run out of customers. The same is true for newspapers: reading the news transforms the new news into old news, of interest only to scholars. Newspapers and other printed matter are readily disposed of when they are printed on paper, which, in theory, can be recycled (although, as with all recycling, there are many ifs and buts to recycling paper).

Reading a book or newspaper on a computer screen doesn't appear to create waste: simply keep the item forever in the apparently infinite storage space called “the cloud” or delete the electronic storage of the item. No spoiling of the environment. Except how and where and of what were the devices used to display the information manufactured? What happens to them when they break or are discarded? Ask the same questions about the computers that are the physical infrastructure behind cloud storage. Each machine is called a “server,” and the constellation of multiple servers, sometimes tens of thousands of them at a single location, is called a “server farm.” The world's server farms use an estimated 200 terawatt hours of energy each year (a tera is a trillion), which is more energy than used annually by many nations.

Today, we still use traditional manufacturing and sales methods. Most manufacturers deliberately design for a fixed, relatively short lifetime for their products. And even if the products can (and do) last a long time, industry has developed other ways of convincing their customers to discard their existing item and purchase a new one by making the existing device seem old and obsolete. How are things made obsolete? Here are the top three techniques used to design for obsolescence: breakdown, progress, and fashion.

Obsolescence through Breakdown

We make things that not only break but are designed to be unrepairable because the components are so well integrated within the product that they cannot be extracted economically. It has become less expensive to discard the old and buy the new than to repair. Even if the parts are replaceable, they are inaccessible or inordinately expensive or so tightly integrated or fused with other components that attempting to remove the broken part to replace it can damage nearby components.

Obsolescence through Progress

Refrigerators used to last for decades. No longer. Today, they are controlled by computer chips with electronic displays, so that today's wonderful new systems seem outdated and obsolete after a few years. The manufacturers of automobiles and many consumer goods, especially electronic and computer-based devices, deliberately bring out new models with new features every year so that the old devices quickly become relics favored by collectors but not able to compete with the new technologies. Today's computers and mobile phones are designed to last only a few years as the relentless march of technology invents new operating systems, new chips, new standards, new communication protocols, new this and new that, making the old seem worthless and often unable to function because they are not compatible with new standards, cannot be maintained, and cannot be upgraded.

Obsolescence through Fashion

Fashion, by definition, means a deliberate change in the appearance of things so that the old looks old and is deemed to be undesirable (unless it is *very* old and rare). Fashion is perhaps strongest in the selling of women's clothing, but the automobile industry extended the model perfected in the clothing industry to vehicles by releasing a new model every year, with new, enticing features within the same body form (changing the body is expensive). However, every few years, there is a major model change, where now the differences in body styles are obvious. People who own the older body style now believe their car to be out of date (even though it works just fine). These major body changes have been effective in increasing sales.

Many industries, especially consumer electronics, have amplified the use of fashion in an attempt to lure their customers into the world of consumerism. Do you have the latest smartphone? Tablet? Video game player or game? Are you signed up to the latest social media apps? The latest cooking device? Up to date on the latest musical group? The actual changes in products are often insignificant, but this lack is hidden by the addition of new features so wonderful that consumers are led to believe they cannot live without them.

Is It Really Necessary for Businesses to Use Planned Obsolescence?

The deliberate manufacture of items for immediate discard after just one or a few uses or for a short lifetime (artificial obsolescence) is bad for the planet but good for business and for people's jobs. From a company's point of view, obsolescence is essential: if companies could make a device that never breaks or gets out of date, then people would purchase only one in their entire lifetime. So how does a company stay in business once everyone who needs the product has gotten one? Once a company stops making that product, the jobs and incomes of all its employees stop. One solution is for companies to recognize that for most customers, their goal is not the product itself but what the product delivers in wonderful experiences. The product is an essential component of a service, but instead of focusing on just the product, companies could start providing services that enhance its value. Services transform the business model from an emphasis on products to an emphasis on the reasons that people want the product—a business model that does not require new products every few years. It replaces the purchase with a continuing subscription providing new opportunities for both customers and companies. Moreover, emphasizing services takes away the drive for planned obsolescence, thereby reducing waste and providing many benefits to the environment and the quality of life for everyone.

Today, many companies are beginning this transformation from physical goods to services. Automobile companies are a perfect example. Where once their business was primarily the sale of automobiles, many now call themselves transportation companies. Instead of selling automobiles to individuals, they plan to treat the availability of automobiles as a service—TaaS, it is called, "Transportation as a Service."

When I was a consultant for BMW in Munich, Germany, I tried its test of this new business model. BMW automobiles were available at parking spots scattered through the city. As my friends and I walked around the city in the evenings after work, if we decided we needed an automobile, anyone in the group who had subscribed to the service could use their mobile phone to find where the nearest one was located, walk there, and then use the phone to unlock the car. Off we could go. The car could be returned at any location in the city used by BMW for returns. Some service plans allowed the customer to choose the kind of car: perhaps a small sedan on weekdays; perhaps a larger sedan for groups, like the group of five I was in; perhaps a van for a family outing on the weekend; or a small truck for carrying large loads. Today, this same strategy is used around the world for scooters and bicycles both small and large, some with electric motors, others manual. They are especially valuable for students.

The range of extensions seem limitless. Add the possibility of supplying drivers (the "ride-share" model) or offering a delivery service for goods that are too heavy to carry comfortably. These subscription services guarantee the companies a continual return on their investment. Subscribers will no longer have to provide storage space for their car, search for parking spots, or worry about maintenance. It is also more beneficial to the planet. The yearly cost could even be lower than the cost of purchasing and maintaining a private vehicle. Today, it is estimated that cars are parked most of the time, being driven only a little every day (I have seen estimates from some European countries that the average car is used only 1.4 hours a day). The amount of space a city must provide for parking is huge. If TaaS expands, fewer cars and parking spaces would be needed. This would also reduce the waste resulting from manufacture and then disposal of all those seldom-used cars. Would fewer cars being required be a problem for the manufacturing companies? Not necessarily. Each car would be used by multiple subscribers throughout the day, dramatically more than the average car is in use today.

15
Sustainability Has Multiple Components and Implications

Sustainability means sustaining—maintaining—the current status indefinitely. When the term is used with reference to the planet, it means to maintain life in a stable state. Sustainability of the planet can occur only if everyone stops the activities that are leading to more harm, ideally while also reversing the harm that has already been done. To be sustainable, we cannot continue to sustain the unsustainable.

This is the Age of Waste, where harm is an unexpected by-product. Now in the twenty-first century, the sins of the past have caught up with humanity: racial and cultural tensions have finally been recognized, and the planet has rebelled with vast ecological disasters—floods, fires, and storms in some parts of the world and drought, heat, and famine in others. A plague came upon the earth when the coronavirus pandemic (COVID-19) killed more than 6 million people in its first two years. In addition to human lives, the rate of loss of species is accelerating, and more than one million plant and animal species are now threatened with extinction.

Sustainability is essential but not enough. It is too late to stop the ecological damage, but not too late to slow it down and reverse its progress. It is not too late to prepare for a different life, a different way of living. In fact, we must, for the planet itself is in grave danger.

Science, Technology, Engineering, and Mathematics Are Not the Answer

STEM is highly touted as the source of great innovations in civilization. And it is true that since the eighteenth century and the start of the Industrial Revolution, on the whole, people's lives have gotten better. Today, people all over the world have better housing, clothing, food, education, and health care than people 100 to 200 years ago. But we cannot look at one set of components in isolation. The benefits are distributed unequally, with many areas still suffering from hunger and starvation, poor health care, lack of education, lack of access to clean water and food, and lack of good government. What they have in excess are diseases and homelessness.

Those who work in STEM may be experts in their respective fields, but they are often ignorant of the impact of their work on society. Worse, many suffer from ignorance of their ignorance. Some people have argued that an A for “art” must be added to the abbreviation STEM, but even STEAM is not enough. Many who are in the arts are also specialists in their own field but ignorant of others. What is needed is a balanced approach that combines the best of our understandings of the workings of the world. Yes, STEM, yes, STEAM, but talking about whether we should have four or five letters completely misses the point. *All* disciplines of knowledge are essential, not just four or five: STEAM plus the humanities, social sciences, economics, politics, and law. Disciplines that are not taught in today's universities. Deeper understanding of how societies function and the wide varieties and ways of government, of the appropriate reasons for and use of law, both local and global, of police, and of the military.

A major source of problems is the manner in which advances in STEM have been used. The S, E, and M fields (science, engineering, and mathematics) are primarily research fields populated by specialists who are appropriately focused on the details of the issues before them. The T field (technology) is different. It involves not research but applications, a field of implementation and commercialization. Here, the primary driving forces are economics and business models, focusing on innovations that have the potential to be a commercial success and earn a profit for the investors. Researchers often think of the long term, the far future. Technologists tend to focus on the near term: often just the next few years. Not only is it difficult to convince technologists to be concerned about longer-term implications of an innovation (or, for that matter, the morality of it), but even if they agree that they should be concerned, these kinds of judgments cannot be made with any assurance. Many of the least beneficial products of today were believed to be a great, beneficial solution to numerous social needs. And, indeed, they often were—at first.

Predicting the harm that may result 100 years after the introduction of a technology is extremely difficult, probably impossible. The first commercial automobiles were thought to be a great boon to humanity, not only easing the requirements of travel but reducing the pollution of city streets. Before the advent of the automobile, the streets of many cities around the world were contaminated by horse manure. "In New York in 1900, the population of 100,000 horses produced 2.5 million pounds of horse manure per day." This claim about cars proved to be true; today's streets are not littered with horse manure. Nobody, however, predicted that there would eventually be more than one billion cars, trucks, and buses in the world producing their own form of manure: exhaust gases that result in air pollution.

The need for raw material created many issues, some social and some ecological, as land was destroyed to mine for materials or forests were destroyed to plant lucrative crops, and waste was produced at all stages of the process, including the disposal of unwanted material from each stage of the operation up to the disposal of the manufactured goods at the end of their useful life.

The pairing of great benefits with great harm accompanies almost every technology. STEM has brought many benefits in worldwide connectivity, telecommunications, new sensors, and powerful computers and their advanced algorithms. Most people live better today than people lived 100 to 200 years ago. Today's harm has also increased, though. I have already talked about the waste and the environmental destruction, the loss of species, and climate change and its impacts. Many of the old sins of technology and the ways of business have gone away, only to be replaced by new sins. Now we also have a society of surveillance, where normal human values are subsumed in the quest for profits and power. Robust communication media make it possible for small groups of extremists to amass great power by attracting people from across the world to their causes. Perhaps such people have always existed, but without the power of today's social networks (coupled with encryption) they could not interact so easily with others who held similar views.

I speak as one with advanced degrees in STEM fields, years of working in technology companies, and membership in multiple STEM societies, including the American National Academies of Science, Engineering, and Medicine, which is, not surprisingly, a strong proponent of STEM education. Is STEM important? Yes, but not to the exclusion of everything else. It must be understood and applied within the proper perspective: the perspective of wide-ranging knowledge and understanding of humanity.

Universities are part of the problem, producing narrow specialists, superb in their knowledge but lacking the broad overview of humanity that comes from exploring a wide range of topics and areas. The university's reward system for faculty members tends to penalize generalists, those with a wide range of knowledge: promotions are based on depth of detailed knowledge of a particular topic. Generalists may have just as much total knowledge, but because it is spread out over many disciplines, it often fails to pass through the promotion process.

Part of the problem with the educational system is its fixation on the early years of life to the exclusion of adulthood. For children and young adults, the 10 to 20-plus years of education they must have seem to be endless, and somewhere in the mid-higher-education years many students wonder why they are still in school. For them, they have spent more than three-fourths of their (still young) life attending school: Is this all there is to life, they ask? Most people get educated from childhood until they either graduate or give up. Education should not cease at the end of formal schooling. Why not provide education over a lifetime, breaking up the years of a concentrated block of education in people's early years into smaller blocks over a lifetime? This would enable people to learn in ways that excite, motivate, and educate them throughout their entire lives. They could take classes when they need them or when they get interested in a topic. In today's formal education, students are forced to take courses even if they have no interest in the topic: no wonder so many do poorly. Later in life, many people discover an interest in the topics they hated while in school. This is when they need the education—when they are willing.

Education should draw upon all sources of wisdom, not just on the tech-nological fields and not just from the monoculture that pervades the world's major universities. It is also time to allow regional and cultural differences to emerge, to take advantage of the ways of learning and teaching of many of the Indigenous peoples of the world, whose views and practices were trampled upon during the colonization of the territories of the earth.

Human life is in grave danger. Many causes of this danger are societal, leading to inequities in the treatment of people. Others are ecological, leading to a world that is becoming increasingly unsustainable for civilization as we know it. The planet, of course, will survive. After all, it has witnessed the rise and death of countless species. And because *Homo sapiens*, our species, is the one that has been most destructive of the earth, the seas, and the atmosphere, perhaps the planet might give a sigh of relief if humankind were to disappear.

Not all points of view or belief systems are beneficial, whether from the Global North or from the Global South. We need wisdom and courage to sift through different lifestyles, to move from a monoculture of beliefs created and propagated throughout the world by the colonizing nations of the West to the acknowledgment that we are living in a world with considerable diversity, that we inhabit a pluriverse. We must practice respect and tolerance toward those whose beliefs differ from ours yet be mindful of the philosopher Karl Popper's admonition that to maintain a tolerant society we must be intolerant of those who are intolerant.

16
Design, Products, Sustainability, and the Circular Economy

The Take-Make-Waste Philosophy of Manufacturing and Usage

Take → Make → Waste

Waste accompanies the entire linear chain of current product design: manufacture, sales, usage, repair, and, finally, end of life. Start with the extraction of raw materials from the earth by drilling or mining, thereby devastating the ecology of the area. Even if the material is from plants, the extraction in commercial quantities can harm the environment. The raw materials then have to be transformed into the basic components of the eventual product, which can involve large quantities of energy, often from fossil fuels. During this stage of processing, the useable parts of the raw materials are separated from the unusable, which are then discarded as waste. Products are usually manufactured in factories that need considerable amounts of energy, which is produced either by burning fuel or from electricity. Either form of energy produces more waste and pollution. The pollution from electricity is hidden from sight because it takes place at the remote location where the electricity was generated.

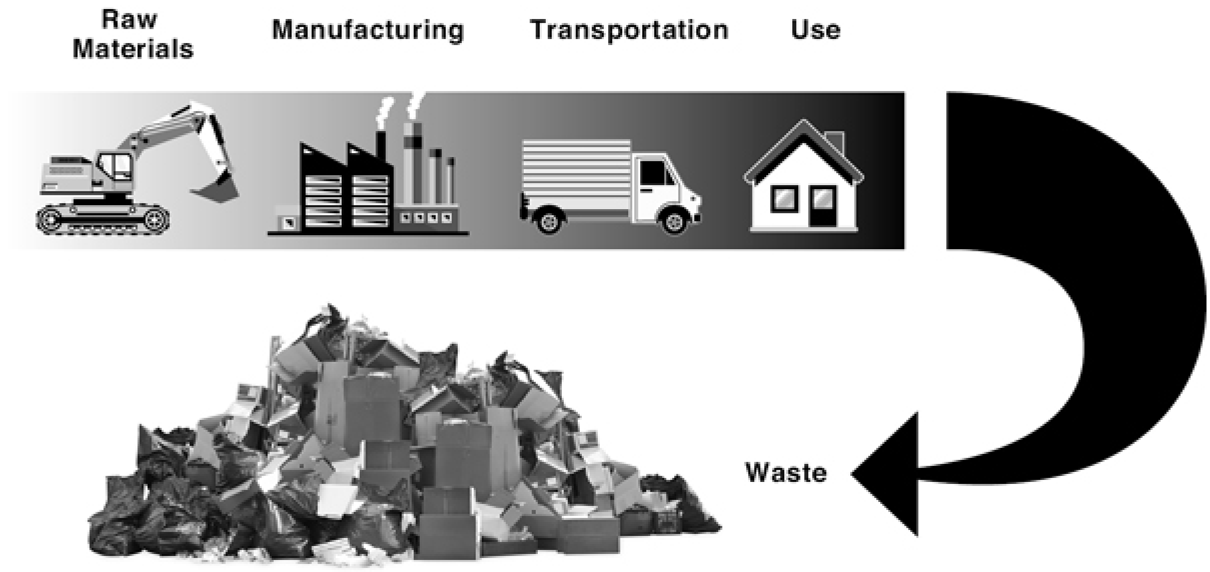


Figure 16.1
The “Take-Make-Waste” linear style of production. This is the basic pattern today: take the materials from the earth, make them into something useful through manufacturing, transport the manufactured product as needed, and when the product's useful life is over, discard it as waste. Image by David Conover.

The designs often make it difficult to repair or replace components—for example, batteries in laptop computers or mobile phones—which leads to premature obsolescence. There is waste at every stage of the life of a product, even after it has been discarded. Beginning during an era when almost everyone was insensitive to these side effects of our product-driven society, any waste seemed like a natural result of the manufacturing and product process. The ecological harm kept building, unnoticeable to most people until today, when it is no longer possible to ignore.

Because these methods of mining, design, manufacture, sales, and disposal seemed so natural, they were built into the fundamentals of modern society. Existing practices are very difficult to change without completely disrupting the system. Even the companies most dedicated to change have difficulties in making the transition. Product lines are built around the harmful methods, and changing these methods into more ecologically sensitive ones is expensive and time consuming. It often requires complete redesign of the products, the materials being used, and the manufacturing process. Customers who are used to the old products may rebel against the new ones.

The resistance to change, especially if such change involves the way people do things, is usually resisted. It doesn't matter if the change is to a superior method or system. Even if people actually do prefer the new system after they have used it for a while, the difficulty is getting them to try it in the first place and then sticking to it long enough to appreciate it. If they do, often they prefer the new over the old. The transition period may cause considerable hardship for a company as public opinion against any changes rises and sales drop. In the best case, customers start to appreciate the new designs, and as the word spreads, the company is able to stay in business. In the worst case, customers flock to competing products, even if those companies are causing ecological harm. By the time some customers start to appreciate the changes, it is too late. The company has suffered unrecoverable losses and soon goes out of business. Nobody ever claimed that change is easy.

Many companies resist bringing about dramatic changes in a product because of this problem. In some cases, they try to introduce the new item slowly, in small, incremental steps. In other cases, they simply give up trying.

The premise of modernity is that society is always advancing, using the outcomes of scientists, engineers, and inventors to create a steady stream of new and enhanced items. Growth was thought to be both good and necessary. Modern business is predicated upon growth to increase profits. Investors in a company's ownership (through their holding of shares) insist on continued growth so that investments generate large returns. The investors benefit, the major leaders of the companies benefit—the employees, the customers, and the world suffer.

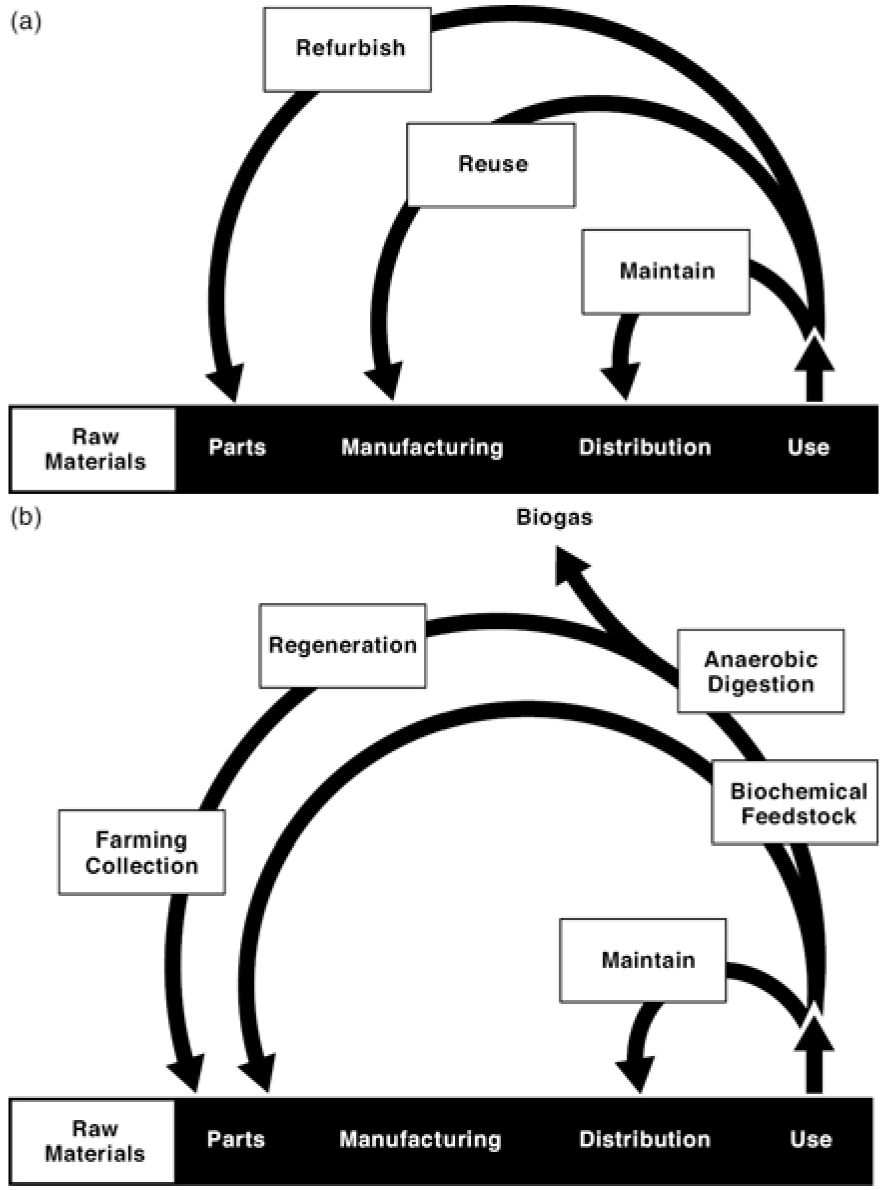
Note that the case against growth is growing, even among distinguished economists. Thus, Herman Daly, former senior economist of the World Bank, founded the journal *Ecological Economics* and has written about *steady state economics*. He now serves as chief economist (emeritus) of the Center for the Advancement of Steady State Economics, a center that states that there is a conflict between “economic growth and various goals for society," and that the notion of "continuous economic growth on a finite planet is wishful thinking." The center's argument for basing economics on principles of ecology and the need for a sustainable, steady state economy is a nice lead-in to the next topic: the circular economy.

The Circular Economy of Repair, Reuse, Regenerate

One powerful method to transform some of our most common problems into sustainable systems, especially in manufacturing (but also in agriculture for both plants and animals) is the *circular economy* championed by the Ellen MacArthur Foundation of Great Britain. What is the circular economy? It takes its message from biology, which has very little waste. Animals and plants gather their nutrients from the soil, water, and the sun, and when they die, their decomposition returns all the substances to the earth, where they can be used as food for other living things, fertilizer for plants, and even building materials for animals to create their nests. When this approach is applied to human-made artifacts, the goal is to design out waste and pollution by making objects repairable, upgradeable, and reusable when their life is over in the sense of having components that can be reused in other devices (see figures 16.2a and 16.2b). The circularity of this approach is a direct repudiation of the traditional "take-make-waste" linear style of manufacturing.

The Ellen MacArthur Foundation uses nature's circular regeneration of biological material as a model for how we might create a circular structure for the regeneration of artificial items.

The renewables diagram in figure 16.2a shows products made from biodegradable materials, so that some components can be reused, and others can be decomposed for valuable, usable ingredients or for rebuilding the agricultural infrastructure to sustain new growth. The goal is to restore nutrients into the biosphere while rebuilding natural capital. The finite-materials diagram in figure 16.2b shows how the circular economy works for nonbiological materials, wherein the product is kept in the economy as long as possible by maintenance, repair, and reuse in other products and services, minimizing the need for new mining and materials manufacturing.



[Figure 16.2](#)
(a) The circular economy for renewable material. In the circular-economy process, the initial building of a product is similar to that of the linear economy of [figure 16.1](#), but as shown here, when the product is made of renewable material (usually biological) it can be continually renewed, repaired, and reused. After the product's life is over, components that cannot be reused can be extracted biochemically to produce other useful materials. There is little or no waste.
(b) The circular economy for nonrenewable materials (e.g., minerals, plastics, metals). It starts by being designed for easy repair and reuse or to be recycled by the manufacturer to reuse the materials. The amount of waste is minimized. Both figures are redrawn from the Ellen MacArthur Foundation's “butterfly diagram” for renewable materials, “Butterfly Diagram Animation,” n.d., https://ellenmacarthurfoundation.org/circular-economy-diagram and https://youtu.be/Lc-FQvPO89Y, based generally on McDonough and Braungart, *Cradle to Cradle*.

Biologically based material lends itself naturally to the circular economy. Natural, biological material can be grown and harvested, and when its use is over, it is easy for natural processes to attack the discards, transforming them into items for growing other organisms, for fertilizing plants, and for enriching biology. Some material can be transformed by microorganisms into biogases that can be burned to provide a renewable source of energy.

The process is more complex for items constructed from material that must be mined and undergo processing or from materials synthesized in large chemical-engineering facilities. Both kinds of materials are required for certain products. Metal and materials such as plexiglass, carbon fiber, and the metals and elements used for such things as our electronic devices—from toasters and clocks to radios, telephones, and all our communication devices—cannot be constructed out of biological material. Circularity sounds simple, but figures 16.2a and b show that it is a complex system.

The Circular Economy Requires Circular Design

The philosophy of the circular economy is simple, but putting it into practice is difficult. The design and manufacturing process starts with material that is as ecologically sound as possible, and the product and its component parts are then repaired and reused as much as possible, thus prolonging its useful life. When the product has reached the end of its life, the reusable parts are sent to manufacturing plants. The design process is critical because it is here that the materials are selected and the ability to repair or upgrade is built into the construction and product architecture. If the design process includes carefully selected materials that have minimal impact on the environment, most or all of the materials not used in the product can be disposed in a way that enhances the ecosystem through decomposition or as a source of useful energy. We call this process *circular design*.

Circular design is based on three principles:

1. Minimize waste and pollution.
2. Keep products in use as long as possible.
3. Regenerate natural systems.

Here are the design implications of the three principles:

1. **Design so as to minimize waste and pollution of greenhouse gas emissions.** Waste and pollution should be thought of as design flaws rather than inevitable by-products of the things we make. By changing our mindset and harnessing new materials and technology, the designer can greatly reduce waste and pollution.
2. **Keep products and materials in use as long as possible.** In the design process, products need to be designed so that they (or their parts) can be reused, repaired, or remanufactured. Packaging should be designed for reuse either as packages or for other uses: the important point is to keep them in circulation, so they don't end up in the landfill.

    a. Design so as to preserve the reuse of the material, which means stopping the clever bonding of materials that enhances their appearance and mechanical properties but also makes it expensive, difficult, or impossible to separate them for reuse. For example, coffee cups and cartons used for juices and milk are designed for single use. They are often made of paper or cardboard with a plastic, waterproof lining that is not easily separated from the paper, requiring these single-use products to be discarded as waste even though the individual components could be reused.

    b. It is possible to make recyclable cups and cartons by changing the materials or, instead of coating recyclable material with plastic, by providing reusable pouches that can go inside containers, allowing the outside to be recycled and the pouch to be washed and reused. Bottles made in this way, where the outer material is paper or cardboard, are recyclable and much lighter than glass, reducing shipping costs for the manufacturer. Plastic does not have to come from fossil fuels. Some companies are experimenting with plastics made from seaweed and, as a result, easily compostable. Will any of these ideas work? It is too soon to know, but it shows that with clever design we can reduce the tremendous waste and add to the circular economy.

3. **Regenerate natural systems.** Nature minimizes waste. Almost all natural products, whether the excretions from living bodies or the remains of the dead, are reusable. Scavenger animals eat some of the remains of dead animals. Small organisms, detritus feeders, feed on decomposing organic material. The products excreted by one organism are put to use by others. Nature recycles materials in so many different ways that not all of them have yet been discovered.

17
The Practical Difficulties of Implementing Circular Design

The circular business model is exciting and plausible, but it requires very different ways of manufacturing, different business models, and many other changes. As a recent *Harvard Business Review* article by Atalay Atasu, Céline Dumas, and Luk Van Wassenhove points out, the circular business model is difficult to achieve, and errors in implementation can be expensive. These concerns make companies hesitant to try.

The London Design Museum asks, “What can designers do?” The challenge is clear: produce less waste; ensure that the waste created is less harmful to the environment; develop new methods of transforming any remaining waste into useful material and energy. By designing and creating the stuff in the first place so that the sourcing of its components is less harmful to the environment, so that it has a longer life that is repairable, upgradable, and capable of easily contributing its component parts for reuse, designers can do much to make the waste that degrades the environment disappear.

What does this require? New training of designers and suppliers in the way that materials are selected and used. It requires new methods of fabrication, ensuring that the parts are easy to replace and do not degrade the environment and thereby creating a clear and easy path to reuse.

The current business models of corporations will suffer. They are based on continual sale of products, and if products last twice as long (the ideal goal would be that they should last 10 times as long), they might have to settle for half the sales (or even one-tenth the number of sales). Companies will resist. The solution is to be creative, to find new opportunities for the use of the products, new opportunities for the repaired or replaced parts, new opportunities to service the world's economy without destroying it in the process. The discussion at the end of chapter 14 of how companies could transform themselves by emphasizing services based on their products rather than just the physical product itself describes one path forward.

Will this be easy? No, of course not. But it is becoming more and more necessary.

Moving from a Throw-Away Society to a Circular Economy of Repair, Reuse, Regenerate

Recycling can be an important part of the circular economy, but it is subject to many challenges. I once wrote a two-article series in the business magazine *Fast Company* explaining why I was a failure at recycling: I could not understand what could and could not be recycled. This is due to major problems in both the plastics industry and the recycling companies. The recycling industry suffers from massive technical difficulties as well as a lack of standards. I have never talked to anyone who fully understood what can be recycled. Why? Because it all depends on the equipment and the processes that each recycling company uses, so the rules for recycling differ from place to place. The stuff you throw away after eating some food from a street vendor might be recycled by different rules than those followed to recycle the very same items you use at your place of work or home. But even if the recycling companies had consistent rules, easy to explain and understand, recycling would not be the answer.

Recycling

Although recycling can never be the complete solution, it will always be a part of the total picture. How can we make it work better?

Standards would help to ensure that materials are easier to recycle. The plastics industry has launched a massive campaign to argue for the use of plastics: plastics, it says, are good for the planet because they can be recycled. The industry is not being very honest in that statement. What it means by “can be” is “it is theoretically possible to recycle if the particular company that does your recycling has the correct equipment and if the material is not contaminated.” The universal triangle with the difficult-to-read numeral inside it that many materials display is meaningless. The numeral indicates what type of plastic the item is made of, but even an expert cannot say whether the local plant can recycle that kind of material. All recycling companies must be required to recycle some minimum list of materials, and then those materials should have a unique mark that indicates that they are indeed recyclable everywhere. The instructions for recycling must be understandable so that people can actually follow them. For example, many recyclers reject containers contaminated by food, but what constitutes contamination? Consistency of the procedures are essential: if every recycling plant were to have the same rules, people would be able to learn them.

Recycling will always be useful and needed, but it cannot be the full solution because the molecular structure of the material degrades with each reincarnation. Recycled paper and plastics are seldom as good as the original, the material that is called “virgin paper” or “virgin plastic.” After a second or third cycle, they must be discarded. High-grade paper eventually deteriorates so that it can be used only to make low-grade products. Some materials—for example, glass and metals—can be recycled indefinitely because they can be readily cleaned, melted, and remade into new objects. Recycling is a business, but quite often the recyclers find that the costs of picking up recycled goods, sorting them to weed out the material that cannot be recycled, and then processing the remaining material exceed the income that they can get from resale. Until there are standards, and until there are stable markets that make it possible for recycling companies to stay in business, recycling is simply the wrong answer. The better answer is to implement the circular-design principles, designing and manufacturing in ways that completely eliminate the need to recycle, except perhaps for items made of easily recyclable materials (e.g., glass and metals).

A caveat: new advances are continuing to improve the recycling process. Chemical methods promise no loss of quality in plastics. Some researchers are investigating the growth of new microorganisms that can decompose plastics into valuable agricultural products such as fertilizer or topsoil. I have seen plastics made from biological material in the research laboratories: “bioplastics” they are called. They hold the promise of creating plastic material that has all the advantages of today's plastics yet is biodegradable. However, they are not yet commercially viable, and their manufacture still produces too much carbon emissions. Some bioplastics degrade naturally in water, but, unfortunately, the biodegradation can be poisonous to aquatic organisms. But as more and more laboratories work on bioplastics, the likelihood is reasonable that the negative features will be overcome. So there is hope.

The best solution is to eliminate most recycling and instead focus on direct reuse of the materials. Recycling generally means transforming the material so that it can be repurposed as something else. A better solution is to make it possible to use the recycled materials just as they are. An even better solution is to do away with the waste that needs recycling.

The Legislative Route to Minimize Waste

One approach to minimize waste is to make those who create the waste pay for the costs of cleanup. This is a direct way of redefining externalities to be internalities—that is, part of the cost of manufacturing. So if a company's method of production and choice of materials increase the cost of disposal or recycling (or perhaps make it uneconomical to recycle), why not ask the company to pay that increase? Why are these extra costs sent to the public or to the government?

In many countries, companies are charged for these costs. In the United States, as of mid-2022 only two of the fifty states—Maine and Oregon—had enacted this policy. These are relatively small states, but their examples have compelled fifteen other states to start the process of enacting similar legislation. This policy needs to be nationwide (and, ideally, worldwide) or at least enacted by the governments of locations whose companies produce the most waste. The point is not to raise money but rather to use the charge as incentive for manufacturers to restructure their design and manufacturing processes so as to reduce that cost. The cost should not be paid for by a deteriorating environment and by taxpayers.

On July 5, 2022, as I was making the final edits to this book, the state of California passed what the *New York Times* reported to be "the most sweeping restrictions on plastics in the nation," thus adding to the number of states in the United States that have passed such restrictions. California is important because it is a large, influential state. However, it is also considered to be an outlier, not necessarily in line with the views of more conservative states. Across the world, there are signs of progress. In March 2022, the assembly of the United Nations Environment Programme voted to craft a global treaty by 2024 with the aim of ending plastic pollution; representatives from 175 countries agreed. (The history of actually forging a resolution and then getting countries to abide by it is more dismal, but the agreement to create the treaty is a necessary first step.)

Why Businesses Might Balk at Implementing the Circular Economy

Although the circular economy and its variants seem sensible and essential, they are very likely to receive a difficult reception in most companies. Two kinds of objections are raised. One is technical, doubting how much can really be done. Among other things, half of the waste is produced before the product is even manufactured: in the mining, refining, and transportation of the basic materials and components that will become part of the product. Can we use circular thinking for this part of the process? Today, we do not know the answer.

The second kind of objection comes from the businesses themselves, contemplating the massive changes they will have to make and wondering if they will ever recover the extra initial costs, especially as it is possible that their sales will suffer because of the products' longer life. A conversation with such companies might go as follows:

Objection: There is nothing new here. We already recycle our goods.
Answer: False. Recycling degrades the material, and so recycled material cannot always be reused. Many companies advertise that they recycle, but only a fraction of items is returned for recycling, and of those, only a small fraction is actually recycled into a reusable form. Many things simply cannot be recycled because the design and manufacturing process intermix different materials in ways that prevent their separation into usable or recyclable parts (even if each component on its own is recyclable, once several components have been bonded together, they cannot be).

Objection: Conforming to the circular-economy way of manufacturing would destroy our company.
Answer: Only if you do not meet the challenge, which offers an opportunity to develop new product lines and business models.

Objection: It is too expensive.
Answer: Clever design and manufacturing could reduce the cost. Many companies find that they can save money because reuse lowers the cost of materials.

Objection: If I do these things and my competitors do not, I will lose sales and lose my business.
Answer: Some of this will be true, especially in the early days. Eventually, it will be those who do not follow these principles that will lose sales. Meanwhile, it may be necessary to have government step in with legislation and regulations to enforce the same requirements on everyone. When automobile safety became a big issue in the mid- and late 1900s, auto companies resisted seat belts, crumple zones, and airbags. Finally, legislation forced them to conform. The companies that implemented safety features early on bragged that "we are the safety company." Companies that stepped out ahead of the crowd ended up with an advantage because when regulations forced all companies to comply, they were already there.

Objection: We are too deeply invested in our current, linear methods to change.
Answer: This is never true. Companies change all the time in response to market forces. If the circular economy were implemented over a 10-year process, the change would not create unmanageable expense. Consider how the automobile industry managed the change from internal combustion engines to hybrid automobiles to fully electric propulsion. The change was spread out for more than a decade. The lag allowed new, competitive companies to start and dominate the early days of electric vehicles. Now the world's leading automobile makers are no longer the leaders. Do you want your company to be defeated by a Tesla in your space?

Can companies actually recycle? Yes, here is one example: Canon recycles their color laser cartridges, first by reusing the box that the new cartridge came in as the carton for returning the old cartridge (they pay the postage). Second, they state that their "cartridges are taken apart, sorted, and then the parts reused, recycled, or put through an energy recovery process that produces plastics, metals, and reconditioned parts that can be used in the manufacturing of new cartridges and other products. Absolutely no landfill waste is generated."

18
Sustainable, Robust, and Resilient Systems

In the circular economy, products must have four different properties: sustainability, robustness, resilience, and the ability to regenerate themselves.

Sustainability alone is not sufficient because it seeks only to maintain the current state, not to repair the parts that are wrong or broken. Similarly, robustness alone is not enough because robust structures often resist until they break (think of a wooden beam that resists forces that try to bend it, but when the forces are too great, the beam breaks and cannot recover). Resilience allows for recovery, but only if the product has sufficient strength to be able to bounce back to its previous state. In addition to combined sustainability, robustness, and resilience, we also need one more property: the ability to regenerate what is needed. With all of these properties together, systems can be recoverable and survivable.

This, in simplified form, is how nature has always operated. The system of living things maintains a state of homeostasis—that is, a stable equilibrium of the system's elements. Many nations have shown great resilience in the face of severe shocks such as economic crises, floods, or famine and then go through regenerative processes during the period of recovery. Often the experience makes them even stronger and more resilient than ever, strengthened by the lessons learned during the survival and as they regenerated the systems of governance and essential infrastructures for living. Nations that lack these required qualities can suffer for multiple generations, and some will never recover.

Resilience

Resilience is a system's ability to maintain its operation through adverse conditions. If the situation is so bad that the system fails, a resilient system can often recover. Resilience is of critical importance in almost every system that serves human needs: for example, family life, communication (e.g., keeping in touch with family members, friends, and essential services), medical services, education, and commerce. Systems have to be designed with resilience in mind: it is usually too late to try to modify an existing system for resilience.

Resilience through Redundancy

One of the simplest ways to develop resiliency is through redundancy. Many commercial aircraft are designed with three different systems for each critical aspect of control. Why three? Because it is always useful to compare the output of one system against the output of a second doing the same analysis. If the two agree, everything is fine. But what if they disagree? Then, the third system comes into play: the faulty system is the one that disagrees with the other two.

The problem with this method of providing resilience is the high extra cost of providing the redundancy. In aircraft where failure can mean deaths, perhaps measuring in the hundreds, the extra cost is worthwhile. The trade-off between cost and efficiency on the one hand and reliability, resilience, and safety on the other is one of the most difficult issues within system design.

Loosely and Tightly Coupled Systems

One way of viewing systems is to ask if the elements are loosely coupled or tightly coupled. A tightly coupled system has a clearly developed line of control, usually a hierarchical structure so that goals can be issued from the highest level and transformed into requested actions as the goals are passed down the organizational structure. Tightly coupled systems are fast, efficient, and easy to control, so that if a new situation arises, the system can rapidly change its behavior. However, some authorities caution that resilience is difficult for tight coupling, so at the most only three or four values should be tightly coupled.

Tightly Coupled Systems Are More Controllable and Efficient Than Loosely Coupled Ones—Until They Aren't

Toyota Motors is famous for the way it cut costs and increased efficiency through “lean manufacturing,” a part of the Toyota Production System, which was envied and copied throughout the automobile industry. One critical component was “Just in Time” (JIT) management of parts for its assembly lines. Instead of having a large inventory of parts, Toyota kept only a small supply at each stage of the assembly line. Each time a worker installed a part, it reduced the small inventory, signaling the appropriate subcontractor who produced the parts to replenish the supply. This system obviously required close collaboration with and close proximity to subcontractors. JIT techniques are now widely used in many different domains: for example, in many forms of manufacturing and even the stocking of grocery stores.

Just-in-time systems are very tightly coupled. This coupling is extremely efficient but requires a robust supply chain, which Toyota successfully managed for many years. But when the COVID-19 pandemic struck, supply chains all over the world collapsed. Critical suppliers had to stop their operations because workers became ill and either were hospitalized or had to remain isolated in quarantine. There also were some natural disasters—fires and storms. On top of all these difficulties, Russia's invasion of Ukraine and the resulting response by Western powers further constrained the supply chain. Transportation among units slowed and sometimes halted because of a lack of drivers. Supplies piled up at ports due to a lack of people to load and unload ships and then a lack of drivers to transport the material. The slowdown led to a shortage of goods and a resulting inflation of prices over much of the world.

Just-in-time methods have little or no redundancy, which means they have little resilience, little robustness. The collapse of supply chains crippled automobile production lines all over the world, a problem that is taking many years to correct. The problem was compounded by another set of miscalculations in the semiconductor industry, so the critical computer chips required in almost all modern automobiles were not available. Along with supply chain problems, in 2021 a major fire in Japan in the plant of one of the biggest suppliers of chips to automobile companies added to the difficulties, followed by a major earthquake in 2022 in Japan that caused additional problems for the already severe difficulties in the global supply chain. Again, a lack of redundancy and the use of tight coupling had enormous ramifications. The supply chain problems for computer chips used in automobiles started the summer of 2020 and in 2022 is expected to continue until 2023 (unless more problems emerge, prolonging the recovery).

Tightly coupled systems are brittle and fragile: fragility is the opposite of resilience. In an extreme case, if one thing breaks, everything must stop. Many modern technology companies are tightly coupled, making it possible for them to shift course rapidly, but, alas, it is also possible for them to fail rapidly and efficiently.

Loosely Coupled Systems Are Difficult to Control but Resilient and Self-Repairing

An example of a loosely coupled system is a modern research university where senior faculty lead a quite independent life, somewhat isolated from the routines of the university. They control their own activities, select research problems, find their own funding, and teach courses in their own areas of knowledge. Senior faculty can select most of the courses they teach, with young professors assigned the introductory, required courses that few senior faculty like to teach. The whole structure is resilient because it is composed of so many independent parts, and if some parts fail to perform, the others are readily available to replace them. But this very same independence and self-determination of activities means that it can be very difficult to manage faculty.

The internet was designed to be extremely resilient by having no central control at all. Proponents of the original design often talked about having the intelligence at the edges instead of the usual command-and-control systems at the center or, in a typical organizational structure, at the top. On the internet, each local node of the network of interconnected sites determines which route it will use to send a packet of information to the destination. Any message is divided into numerous packets, and each packet might very well take a different path. This means that they might arrive at the destination out of order, but they are tagged to make it easy to reconstruct the proper ordering. If a packet is missing, a request can be sent back to the originating location to resend it. Even if components fail, the internet automatically can route around the broken areas. It is highly resilient. It is loosely coupled.

I said "proponents of the original design" because today many players have tried to exert control over the internet, even though it was designed not to have such central control. The result is a system that gets more and more unwieldy every year. This is a familiar malady of many complex systems: it eventually becomes time to start over again, discarding the old and rebuilding based on lessons learned. But to discard the internet, despite its many flaws, will create new problems in the worldwide communication network. Once massive systems are in place, they are difficult to change.

Today the internet is threatened by another movement: countries that wish to completely isolate themselves from the international community, allowing access only to carefully controlled information. This control is exerted primarily to limit citizens' access of information from outside the country or to information that the government considers detrimental. The many false, inciteful, hateful posts on the internet and calls for violence make the wish to exert this kind of control attractive to many locations, even countries that allow free speech.

Large organizations and governments offer another example of loose coupling. Every year new regulations are added to the old, and in the case of governments new laws are added as well. Seldom are the old regulations or laws discarded, though. The result is a huge, cumbersome morass that defeats all sensibilities but allows skilled operators to always find some unusual route to accomplish their deeds. In some sense, the more the organization tries to become tightly coupled with edicts continually coming from the top leadership, the more individuals invent work-arounds and strategies to maintain loose coupling at the bottom.

The standard metaphors often used to describe these tight systems have to do with the herding of animals. Sheep are tightly coupled—as are schools of fish and flocks of birds. As a result, they act in harmony. All an animal herder has to do is get the lead animals moving in the proper direction, and most of the others will follow. Then the herder's job is to find the stragglers and get them into the herd. If the animals were cats, it would be almost impossible to control them. Each cat is independent, doing whatever it wishes. A herd of cats would be loosely coupled—in fact, so loosely that the notion of a herd of them defies credulity.

Managing the faculty in any of the top research universities of the world is like herding cats. Or, as I often explain, if you ask faculty members who their boss is, they don't understand the question. So it is not a surprise that academics prefer the autonomy and flexibility of loose coupling. That's the essence of the extreme of loosely coupled systems.

Loosely coupled systems are composed of many different parts that are autonomous but that nonetheless do obey certain behavioral rules or protocols. Ants are often used as favorite example of success. A single ant is not very powerful, but a group is extremely intelligent, robust, resilient, and resourceful. Groups of them, often called colonies or armies, can cross rivers and ravines by creating bridges from items found on the ground as well as by using their own bodies. An ant army can build complex structures and survive changes in the weather and food supply. Ants sacrifice many lives along the way because it is the army that is smart, not the individual. Ant colonies are extremely robust, able to operate in extreme conditions. However, the individual ants themselves are not robust They are considered disposable.

Both loose coupling and tight coupling have their advantages and disadvantages. Tight coupling is extremely efficient and easy to control. However, often a single failure can bring the entire system down. Loose coupling is extremely robust and resilient. It allows for great creativity, in part because creative things often require going outside normal boundaries, breaking normal rules. This behavior is encouraged in loosely coupled systems. The negative side is the inability to control the system and its inefficiency in that it requires more people than tight systems.

Many creative people prefer the freedom that loosely coupled systems permit. Administrators and management usually prefer rules, guidelines, and, most important of all, rigid schedules and budgets. In my experience, even the most creative and free-wheeling groups benefit by some rigidity. Enforcing a strict deadline quite often creates the sort of tension on people that provides the necessary impetus for them to complete creative work. (I even do this to myself, setting a deadline and announcing it publicly because once it is public, I have to live up to my commitment.)

The earth is extremely resilient. Even though the continents shift, landmasses fall into the sea, and others emerge from the middle of the ocean (the Hawaiian Islands, for example); even though some species of animals die, and others emerge; even though severe storms, fires, earthquakes, and volcanoes occur, these changes do not challenge the existence of the planet. Climate change? A meteorite strike that wipes out an entire species? It has happened before. The earth will survive. Human life is not as resilient, however.

Human life is tightly coupled to today's artificial infrastructure. People construct artificial shelters for protection from the weather; artificial farms, ranches, and fisheries to provide food; artificial sources of energy. Early humans had none of this infrastructure, but in the millennia that humans have existed, many people, especially those in urban communities, have become so dependent on the infrastructure that they cannot survive without access to electricity, water, sewerage systems, a place to get food, and adequate shelter. This is why changing human activities to produce a sustainable, resilient, regenerative infrastructure and way of life is so critical. We, the people of earth, must build and enhance the resilience of human society so that as the environment and climate change, humans will continue to exist.

19
People's Understanding of Systems

People instinctively think linearly, believing that causality is obvious: an action followed by a reaction. As a result of this natural instinct and thought pattern, they view the world in a highly, oversimplified manner.

Causality

A lifetime of experience combined with evolutionary brain mechanisms has led people to understand a simple but strong relationship between cause and effect. Touch a hot stove burner, and you get burned: a single instance may be sufficient for you to learn not to do it again. Here, a simple cause–effect relationship holds. The same logic applies to food: eat a novel food and later fall sick. Even if the sickness does not develop for hours, the obvious cause is the novel food. As a result, people avoid that food in the future and develop a strong aversion so that the slightest taste—and possibly just the odor—can trigger nausea and loathing.

These causal implications are not always correct. Yes, the burn was caused by touching the hot stove, but the food aversion is often wrong. Blaming the most recent novel food eaten seems to be a genetic property of people and a wide variety of animals: it's called “conditioned taste aversion.” It's probably a useful heuristic for surviving in the wild, where many plants are poisonous. The existence of a false positive, or a false attribution of danger, is much safer than a false negative, failing to recognize danger.

This kind of causal understanding forms a simple, linear chain. An event occurs followed by some effect: cause → effect. Flip a light switch, and the light comes on. Turn a car's steering wheel clockwise, and the car turns to the right; step on the brake, and the car slows. It's simple, except when it isn't: when the expected result does not occur or when the simple cause-and-effect chain is not at all obvious.

Simple models of causality seldom apply in real life. In most situations, everything has multiple causes. In a system with multiple feedback and feedforward loops, causality may be impossible to ascertain. The Japanese method of Five Whys, asking "why" multiple times until the ultimate cause is discovered, exemplifies this dilemma. Invented by Sakichi Toyoda, the founder of Toyota Industries, the Five Whys process is a fundamental part of Toyota's much admired and imitated Toyota Production System, which helped establish the company's reputation for efficient, high-quality manufacturing. Basically, when something goes wrong, ask what went wrong, and when told, ask why it happened. But don't settle for that answer: ask why again. Keep asking until the underlying root cause has been found. The number five is not to be taken literally but rather to remind the investigator not to stop at the early, easy answers.

The Five Whys method is clearly based on a linear model of causality. It works fine on the tightly coupled, linear assembly line, but in an investigation of natural disasters or even unnatural ones—such as the crash of a modern passenger airplane, an accident at a large chemical factory, or a failure of the electrical power grid—linear causality is oversimplistic. In other words, it is wrong. Worse, using the Five Whys method will lead to the erroneous determination of a single cause, often blaming the wrong people or fixing the wrong process. In complex systems where there are many contributing factors and multiple feedback loops, linear causality simply does not work.

Instead of Toyoda's method, I prefer the excellent analyses of accidents conducted by the National Transportation Safety Bureau (NTSB), an independent federal agency in the United States that examines every accident in civil aviation as well as significant accidents in transportation, broadly defined as highway, railroad, marine, and even pipelines (which transport such materials as gas and oil). The bureau's reports seldom blame a single cause. Instead, after a very thorough analysis of an accident (which can take a year or more), the report discusses a wide variety of issues, demonstrating that if any one of numerous items had not occurred, the accident would not have happened. What is the cause? That's the wrong question: all the factors are causes. The important purpose of the accident analysis is to identify and fix all the deficiencies that are identified. This is the only practical way of understanding and dealing with complex systems.

Many factors contribute to major accidents. The NTSB final reports of major accidents almost always find problems with poor design, maintenance, procedures, and training. The one common theme of many of their findings is poor management throughout the system. What does poor management mean? Management, especially of anything as complex as a transportation system, is very difficult. There are often many different organizations involved, each of which has multiple divisions, with multiple levels of authority, and often many lengthy, written procedure manuals. To make matters worse, the manuals are seldom kept up to date and, in any event, cannot possibly consider every combination of factors that might occur. During the incident, the people responsible for maintaining control (the pilots, in most commercial aviation incidents) waste valuable time poring through the different manuals, trying to find the relevant case. Modern computer systems attempt to help by automatically diagnosing the situation and either responding autonomously or offering operators the instructions to be followed, but the diagnosis or recommended course of actions is not always appropriate (because in each complex system, most accidents involve different unique factors). Different organizations might be involved: police and firefighters (often from different municipalities), company safety representatives, multiple teams from different divisions of a company, and different companies or government and regulatory agencies who must coordinate their decisions and actions. It is rare that the result is smooth, flawless management.

Major accidents involve multiple interacting systems. Simple linear analyses fail. And despite the firm belief of many journalists, executives, and politicians, “common sense” is frequently wrong and misleading. People—and law courts—want to find a single cause so they can blame someone. Trying to explain the complexity of the situation does not help. People want a simple answer to a complex problem.

To confound the issue of causality, when an error or accident occurs, quite often people do a quick analysis after the fact, find one critical issue that indicates the triggering of the accident, and then blame the people who were involved in that one issue for their failure, even though a thorough analysis often demonstrates that the people responding at the very end of the incident were victims of the situation. After an event it is easy to find courses of action that would have prevented or minimized the damage done: this is analysis by hindsight. During the event, there are usually so many things going on, so many different signals, that it is not easy—often not possible—for the people involved in controlling the system to determine what is happening and then decide what to do.

History has multiple examples of this issue. When a war starts or one country does harm to another, it is easy to look at the event and list all the obvious signs predicting the event and then blame whatever politician was closest to the decision-making. However, if these same people who found the signals so obvious were asked before the event to predict what was about to happen, they would fail. How do I know? Because psychologists have demonstrated this again and again. Hindsight is a lot easier than foresight.

Causality is a tricky business.

The Myth That the Earth Is Infinite

Until recently, most people thought the earth was too large for their simple, everyday acts to have any impact on it. Even today, many people still hold onto that belief. In other words, they believe that for all practical purposes the earth is infinite. It's not an unreasonable belief: if there were only a few people on earth, it would be a fair approximation. However, there are now more than 8 billion people. Even small actions done every day for a few years add up to make a difference, even on our large planet.

Starting with the Industrial Revolution in the eighteenth century, the newly formed mining, pumping, and weaving companies had a similar attitude, and today the cumulative effect of that continued attitude and the resulting abuse to the ecosystem for almost 300 years across many of the countries of the world has created major problems. The earth is not infinite: human and corporate behavior has already succeeded in extracting resources to the point of depletion and producing more waste than the world can hold without suffering severe ecological damage. The impacts from all the years of destructive processes have affected the climate, sources of food, and lives.

People's Understanding of Systems

It is easy to think of ourselves as individuals moving through the local environment, where our individual actions have minor impact upon that world. But people are not outside observers to the world: they are integral participants in the vast, intertwined system of connections and dependencies. Everything people do impacts the world, while simultaneously the world impacts us.

The concept of complex, interconnected networks is not difficult to understand, but the details and implications can be surprising. Full understanding requires one to delve into the mathematics of complexity theory, feedback systems, second-order cybernetics, and agent-based computer simulations. But there is another, much more fundamental source of difficulty: the limitations of human comprehension. Consider the everyday, common air conditioner.

How an Air Conditioner Heats the Entire Planet When It Cools Our Rooms

A home air conditioner provides a good example of the surprising impact of the interconnected world. On the one hand, the home air conditioner is a fairly simple object, designed to give comfort to those who live in hot climates and can afford to have an air conditioner (or, for that matter, can afford to live in a home).

Air conditioners serve as an excellent example of nonobvious causes and effects. They are partially responsible for the rise in sea levels and the increase in severe storms across the entire world. Why? Consider the entire global system of 8 billion people and 2 billion air conditioners. Even the small air conditioners hanging outside the windows of office and apartments all over the world do not work in isolation. Each air conditioner adds more heat to the local atmosphere than it removes from the room. Why? The compressors and fans in the air conditioner generate heat in their operation, heat that is then transferred to the atmosphere along with the heat extracted from the room's air. The result is that more heat leaves the house than is removed. This lack of perfect efficiency is a fundamental property of physics: the second law of thermodynamics, which limits the efficiency of machines (the reason why perpetual-motion machines are not possible).

Consider the entire system. How is the electricity generated and transmitted to the air conditioner? After all, the generation source might be some distance away. The energy required to generate the electricity can come from the burning of oil or gas, which adds to the carbon dioxide (CO_2) levels in the atmosphere. Even if the electric power comes from renewable sources, such as wind turbines or solar cells, there is still considerable waste during the construction of air conditioners, the power plant, the solar panels, and the wind turbines as well as in transporting everything to their appropriate locations. All these activities add to the waste that contaminates the air and environment, impacting both the CO_2 level and the ozone layer, which can lead to warming the earth, which in turn changes the weather patterns, making storms more frequent and severe.

Finally, air conditioners "use refrigerants that can impact climate change in two ways: through their influence on energy-related emissions and from refrigerant leakage." Refrigerants are usually hydrofluorocarbons that are extremely powerful greenhouse gases. Air conditioners throughout the world use more than 2,000 terawatt hours per year, "nearly 20% of the total energy used in buildings around the world." Generation of this electricity leads to sizeable emissions. These chains of causality are neither simple nor linear. They have numerous complex loops and interconnections.

Turn on an air conditioner to cool a room, and you heat the planet. How can one insignificant air conditioner heat the world? Yes, the amount of excess heat pumped into the atmosphere by a single air conditioning unit is small, but there are at least 2 billion air-conditioning units in the world. Moreover, the impact accumulates over time. In 10 years, there are 3,653 days. Not all of these days are hot, but even if only one-third are, that is more than 1,000 days. Two billion air conditioners used for 1,000 days is trillions of hours of use. Does the amount of excess heat produced by air conditioners still seem small?

A Note on Terminology

A greenhouse gas (usually abbreviated as GHG) is any of the gases that contribute to the earth's warming by trapping heat in the atmosphere instead of letting it radiate into space. The most important GHGs are CO_2, water vapor, methane, nitrous oxide, and ozone. Hydrofluorocarbons (abbreviated as HFCs) are used in most refrigerators and air conditioners. When they leak into the air they are classified as super GHGs because they are thousands of times more harmful than CO_2.

If the world's air conditioners use 2,000 terawatt hours of energy per year, what does that mean? Two thousand terawatt hours is 2,000 trillion, or 2 quadrillion watt hours. An LED ceiling light for a typical dining room uses about 30 watts of power and if on for 6 hours per day in winter and 3 in summer, it uses 50,000 watt hours per year. Divide the watt hours per year used by air conditioners by the usage of a ceiling light and we find it takes 40 billion ceiling lights to equal the energy usage of air conditioners. Air conditioners are energy hungry.

Climate scientists predict grave difficulties from every 0.5°C (0.9°F) increase in the average global temperature. At the United Nations Climate Change Conference in 2021, the prime minister of Barbados said, “Two degrees is a death sentence for the people of Antigua and Barbuda, Maldives, Dominica, Fiji, Kenya, Mozambique, Samoa and Barbados.” Today's level is already 1.1°C above that baseline. The prime minister was arguing that the climate change conference should lower the target because even if the world could keep the temperature rise to 2°C and then just maintain it, it would still mean death to those nations. How much does the use of air conditioners raise the world's temperature? A study in the city of Phoenix, Arizona, showed that air conditioners could increase night-time temperatures by 1°C (almost 2°F). If extended to the world, that 1°C rise in Phoenix would mean 2.2°C above preindustrial levels: a death sentence to (at least) nine nations. The minister understood that simply discussing the relatively meaningless small temperature rise would not have much impact, so she translated the focus on temperature into meaningful terms: great harm across multiple nations.

Showing the Planet-Wide Impact of a Single Air Conditioner

Most illustrations of cooling systems show only the home-temperature control loop, completely ignoring the impact on the planet, in part because this impact has seldom been recognized. After all, the feedback loop in which heat is added to the atmosphere by the billions of air conditioners in the world is not visible to the average person, and many years may pass before its impact is felt. But even this is only a small part of the story. Air conditioners require electricity, and the generation of electricity increases the CO_2 in the atmosphere, which traps excess heat, making the world still hotter. Most air conditioners use a liquid/gas refrigerant (chlorofluorocarbons and, more recently, hydrofluorocarbons), and if the refrigerant is released into the atmosphere, it is far more harmful than CO_2, trapping even more heat. The refrigerant isn't supposed to leak, but the amount that does leak into the atmosphere is substantial. The world gets hotter.

Finally, there are long lags between setting the temperature on an air conditioner and any noticeable change in room temperature, which makes both controlling the feedback loops and understanding the impacts more difficult. (A similar long lag also impacts the control of heating systems and ovens). As the air conditioning units operate, they increase the atmospheric temperature not just from the one unit that you turned on but from the billions in the world (indicated in figure 19.1 by the heat rising from all the houses). This lag is measured in the years or even decades it takes to impact the air temperature around any particular air conditioner. So this feedback loop is very difficult to detect by individuals or even communities.

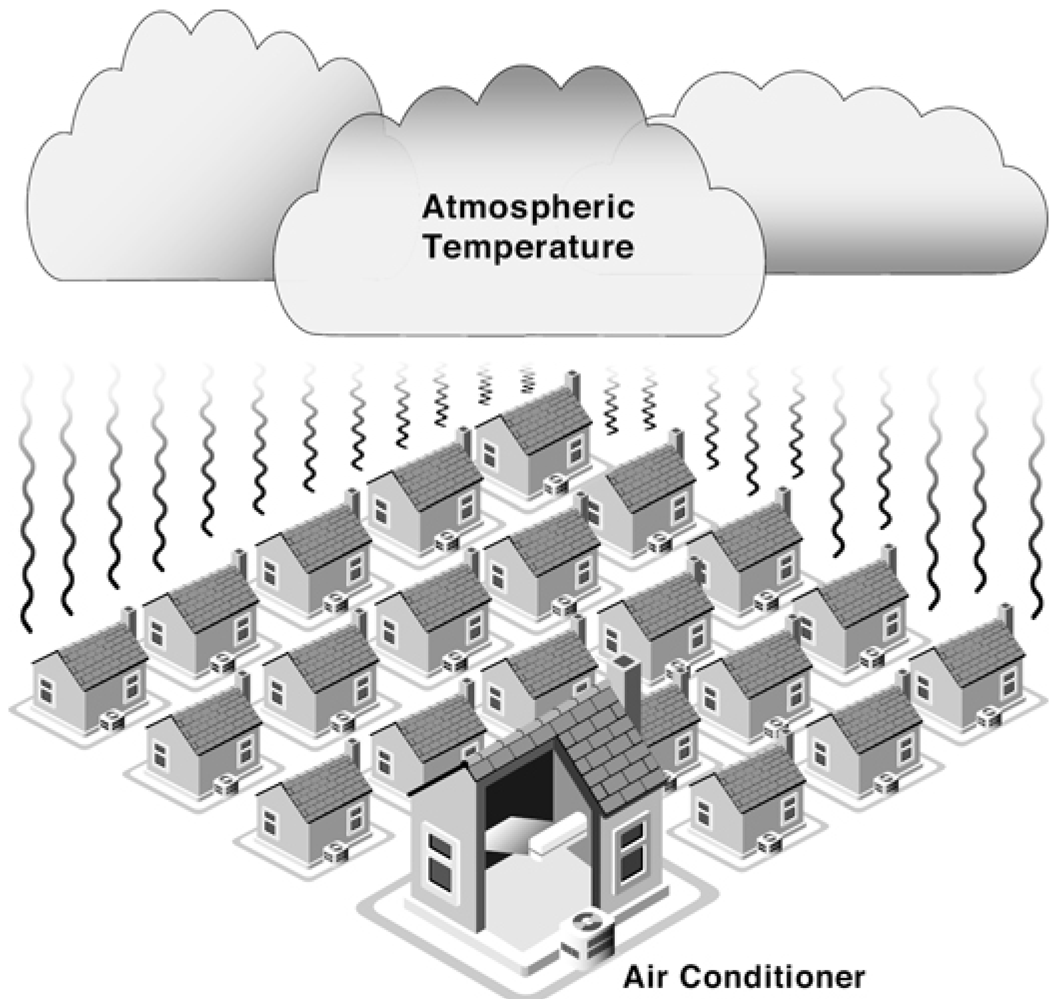


Figure 19.1
How an air conditioner heats the planet. Most people are aware of a simple feedback loop that controls the temperature of a room. A thermostat measures the room temperature, and if it is too low, it turns on the heating system; if too hot, it turns on the air conditioner. But they are often completely unaware of a second invisible feedback loop. The hot air from the home (plus heat generated through the air conditioner's inefficiency) goes into the atmosphere. This heat emanating from billions of people's air-conditioned homes warms the earth's atmosphere, which increases the need for air conditioning. This figure illustrates that second invisible loop. But there are even more invisible loops—for example, the greater requirements for electricity cause electricity producers to put not only more heat into the atmosphere but, if that electricity production is powered by fossil fuels, noxious gases as well. Image by David Conover.

The scenario is actually far worse than what I have described so far because I have neglected to mention three other things.

1. First, the electricity required to operate the air conditioner and the energy waste that occurs in power generation and transmission are significant.
2. Second, air conditioners are extremely inefficient. To keep manufacturing costs low, they are not designed for efficiency. They use perhaps two times (or more) as much energy as an efficient machine would. Manufacturers fear that if they make their units more efficient, they would have to increase the price they charge, thus decreasing sales. To force appropriate behavior requires a coordinated effort by governments and manufacturers.
3. Air conditioners completely lack any elements of circular design. These appliances are designed with minimum efficiency and great difficulty in repair, reuse, or even recycling. But the very exciting good news is that although I failed to find air conditioners that followed circular-design principles in mid-2022, CLASP, a nongovernment organization dedicated to developing "efficient appliances for people and the planet," is working with many appliance manufacturers across the world to employ circular-design principles for all their appliances. CLASP works on appliance design in 12 of the world's top-20 carbon-emitting economies, and it has launched an initiative in Thailand to develop circular-design principles for air conditioners. We need more cross-company, global initiatives like this one to guide sustainability development.

The Impact of Human Behavior

People's understanding of feedback systems is often very limited. Their mental models of how feedback systems work and their instinctual emphasis on simple causality lead to very inefficient use of energy. Let me illustrate this by sticking to the air conditioner example, but the phenomenon I am describing has a much larger impact.

People get impatient. When entering a very hot room, instead of setting the thermostat to the desired endpoint temperature, they are apt to “speed up the cooling” by setting the unit to a very cold setting. Why? Because many believe this causes the desired temperature to be reached more quickly. For most systems, this is a false conclusion. The fundamental difficulty is the combination of a false model of how the air conditioner works and the long delay between a temperature setting and the result. But that combination alone does not cause the inefficiency problem. Basically, we have two systems interacting with one another: a person's beliefs and the air conditioner's actions. Pitting two different systems against one another commonly causes inefficiency, waste, and failure.

An Australian colleague of mine, Terry Love, told me that people can understand one feedback loop, such as in the basic thermostat, but are incapable of understanding the impact of two or more feedback loops in tandem. Even a simple thermostat has at least four feedback loops. One is built into the thermostat and its sensor. The other three are not visible, so they are easy to overlook: two reside in the brains of the people using the thermostat and one is that slow, invisible coupling to the atmosphere.

Consider the two feedback loops in the person's mind. Someone trying to adjust room temperature monitors the thermostat's performance by the comfort level experienced in the room. If the temperature change experienced is slower than expected, the likely result is that the person will make a further adjustment of the thermostat's temperature setting. The second internal feedback loop is self-monitoring of the person's perception of comfort. The differences between these two loops are subtle: one compares the rate at which actual room temperature changes; the other compares how rapidly the person's perceived warmth—comfort—changes. Warmth and comfort are not solely determined by temperature. So even when the room temperature is at the desired temperature level, people might be comfortable for a while but then decide they are too hot or too cold. As a result, they continually adjust the temperature setting on the thermostat. Partners often dispute the room temperature, especially as individuals’ metabolisms differ, so one might feel hot, while the other feels cold. Temperature is a physical variable, measured by scientists. Comfort is a psychological variable, determined by many factors: the physical variables such as air temperature, humidity, the amount of radiant heat coming from the walls, and the clothes being worn, as well as the personal variables, such as the normal metabolism rate, whether the person has recently eaten or is hungry, and, of course, how much activity the person has been doing.

As soon as people are part of any system, it immediately becomes more complex. This relationship frustrates physical scientists, who carefully design everything without considering the multiple and complex systems of human behavior. They want to know why people behave so irrationally or irresponsibly. When I advise companies designing systems for people, I explain that the people are acting intelligently given their lack of understanding of the technology. It is the designers' responsibility to make the system's operations intelligible. Companies tend to hide the state of their systems on the assumption that this only confuses people. In fact, the absence of any coherent information is what is confusing, leading people to inappropriate behavior. Yes, simply presenting measured values would be confusing, but communication designers know how to present simple graphical depictions of a system's operation that match human understanding and behavior. For example, automobile manufacturers display the complex distribution of power in their hybrid automobiles by showing the operation of the system and the flow of power among the internal-combustion engine, electric generator, electric motor, regenerative braking system, and of course the drive shaft to move the tires. Good communication design can make complex systems understandable through meaningful displays, which are not to be confused with the measurements that only engineers and expert fans of automobile technology love.

The crises that we face today are caused to a large extent by not considering our actions as part of a complex, global system. Systems are indeed difficult to understand. As the air conditioner example showed, the air conditioner itself is only a small part of the problem. I didn't mention the energy consumption and waste in the system that mines the basic materials, manufactures the parts, transports the different parts to the assembly plant, which in turn uses energy and produces its share of "externalities." Then the units have to be transported to the distributers and sellers and then to the homes. The story I told does cover usage and the hidden feedback loops, but I ignored the disposal of an air-conditioning system, not only a problem by itself but a major source of atmospheric pollution when the disposal and dismantling of the parts releases the refrigerant into the air.

But even this complex system is simple compared to other systems where people and societies play major roles: the systems we call "sociotechnical."

20
Working with Complex Sociotechnical Systems

The science of complex sociotechnical systems itself is complex. Scientists who study these systems are scattered across the world, representing numerous scientific disciplines. Some of their work is very technical, using mathematical models and computer simulations (especially "agent-based" simulations).

Some of these people are research scientists, trying to develop a deeper understanding of such systems. Some are practitioners, whose goal is to apply existing knowledge in real systems that help control the infrastructure of life. Practitioners have different goals than scientists, even though they use the same kinds of models. The job of a practitioner is to build, maintain, and improve things. This seldom requires a full system analysis or full understanding of all interactions and nonlinearities and time constants: the models and understanding need be only "good enough." Practitioners seek practical benefits. Science aims at complete, precise understanding, whereas practice needs only an approximate level of understanding. For science, extreme accuracy and precision are required. Scientists seek truth. Yet both scientists and practitioners are needed. Not only do they often work together, but the same person can be a researcher in one set of activities and a practitioner in another. In my own work, I switch back and forth between these two modes of thought and behavior.

In an examination of a complex sociotechnical system, the first critical job is to know which parts matter and which don't for any particular issue. In most complex sociotechnical systems, it is possible to decide which parts are relevant to the problem being examined. This allows the development of easy-to-understand explanatory models that, although imprecise from a scientific point of view, are "good enough" to improve or maintain the system. The process of simplifying a complex system into a “good enough” model is riddled with complexity, though. Different technologists and decision makers will wish for different simplifications, in which case it may be necessary to build several different models, each appropriate to a specific context.

Complex Systems: Feedback, Feedforward, Recursive

In chapter 19 (in the “Causality” section), I pointed out that it is easy for people to understand simple linear cause and effect but not when the effect is long delayed and definitely not when there are multiple feedback and feedforward loops and nonlinearities. The complexity stems from the combination of feedback and feedforward loops. Some loops take decades to complete, some only seconds. Some increase the effect being examined, some diminish it. Feedforwards are expectations, so a system can start preparing for an event before it happens, and just like feedback loops, some are negative, some are positive.

The analysis of systems with multiple feedback loops is part of the domain of cybernetics, a field started by the MIT mathematician Norbert Wiener with the publication of *Cybernetics, or Control and Communication in the Animal and the Machine* in 1948. The book covers an amazing set of topics, from the way feedback systems work to their application in technology, to the understanding of animal and social behavior, their role in learning, in languages, and in society, as well as to the ethics of these systems because once their principles are known, their use is no longer under the control of their original founders.

Complex cybernetic systems are recursive: that is, each system not only contains systems within it but also might even contain a copy of itself inside itself. (A simple example of recursion is given in the chapter 20 note in the back of this book.) Some of the systems monitor the performance of other systems, so that systems monitor themselves. There are loops inside of loops, circles inside of circles. Most biological systems have this property.

The human body has multiple systems that monitor how other systems are performing. These "watchers" are called "second-order cybernetic systems." First-order systems control our actions and understandings, while second-order systems control the controls. This is yet another example of recursion: a second-order cybernetic system contains a model of itself.

When you are in a conversation with others, have you ever stopped and corrected yourself—you said something wrong, perhaps the wrong word or something you shouldn't be talking about or whatever? Your ability to listen to what you are saying and then correct it is an example of you as a second-order cybernetic system, where your model of the world includes you and your model of the world: recursion. There is a second loop watching over the output of the first: a circle monitoring the circle.

Similar activities take place outside of the individual person. Electric power companies watch how much energy people use, and if it gets too high, they may ask people to reduce their use of high-energy devices. The energy company's monitoring of people's behavior is an example of an outer loop monitoring the inner loops. The company uses feedforward systems as well, recognizing that before major sports events people's behavior will change in predictable ways, so it makes sure its power systems are ready for the increased loads.

Feedforward is easiest to understand if we rename it "prediction." Here is a simple example of a prediction system inside your body. Stand on the floor with one leg lifted. How long can you keep upright? (You may move your arms to maintain balance.) Most people wobble a bit but can stay upright for 20 seconds or more. Now close your eyes while standing on one foot. Most people can now keep standing for only a few seconds. Why does shutting the eyes have such a dramatic impact?

The body's postural system is a wonderful network of integrated sensors and muscular systems involving the muscles, sensory input from the eyes, proprioception, the vestibular system of sensors located in the inner ears, and a complex set of feedback and feedforward signals. Cut out the visual information, and the system loses a major source of information about the body: it has trouble predicting how the body will wobble without visual information that can be used to feed-forward the needed control signals to the muscles.

Similarly, if the eyes are open but the vestibular system is impaired, people find it difficult to maintain posture, so not only do they fall down, but they feel nauseated and might even vomit. Children (and adults) often deliberately create this situation by spinning around in circles for a few minutes. When they stop, they are stationary, but the vestibular system has become over saturated. While the body was spinning, the world was stationary, and the vestibular system predicted the impact of the spin on the visual system, so that the spinners felt (properly) that they were spinning, and the world was stationary. But when they stop spinning, the chemical/mechanical system that comprises the sensor mechanism of the vestibular system keeps sending spinning signals. Because the body is now stationary, the apparent conflict in the perceptions of the prediction and reality lead the brain to interpret the information to mean that the earth is spinning. When confronted with a spinning world, people lose their balance and fall down.

Notice that the vestibular system affects not only balance but also visual perception. Whenever there is a mismatch between the information about the world provided by the ocular system (i.e., the eyes) and the vestibular system, problems arise, and the body often reacts with nausea and motion sickness. This is the cause of seasickness, which some people also experience while traveling by train or automobile. Roughly half of astronauts get space sickness during the first few days of space flight until they finally adapt. (NASA prefers not to use the word *sickness*, instead naming it *space adaptation syndrome*, but language is a loosely coupled system, so mere bureaucrats can't control language, and the term *space sickness* is apt to survive.)

The system that controls posture is complex, but it is only one of the many systems in the body that regulates the different parameters required to keep us alive, healthy, and safe. If the body's system is this complex, what must be the complexity of the world's ecosystem, with its living organisms, weather patterns, geographical and geological structures, oceans, winds, and so on?

Natural Sciences, Engineering, and the Social Sciences

Unfortunately, the very complexity of the earth's ecological systems impedes its human population's ability to respond to many of our current dangers. The problems are especially difficult with complex systems where the components not only interact but also adapt and modify their behavior according to the circumstances and the behavior of the other components. Such are the systems of all living creatures and the natural world.

Julio Ottino, dean of the School of Engineering at Northwestern University and cofounder of its Institute on Complex Systems, argues that much of what modern engineering does today is to work on complex systems. To me, one of the most interesting observations that he and his coauthor, Luís Amaral, have made concerns the differences between the problems and issues studied by engineers and those studied by social scientists. The natural sciences study simple systems, they say, leaving complicated systems to the engineers. But the really complex systems are those studied by social scientists, ecologists, and physicians, and these systems, Ottino and Amaral point out, seldom have discoverable quantitative laws but rather require qualitative analyses and heuristics.

“Engineering,” they point out, is about “optimum design and consistency of operation; the central metaphor [in engineering] is a clock.” Societal systems are not like clocks. They change continuously, adapting, self-organizing, and seldom stationary. And as we saw with the example of the home thermostat, the systems are continually examining their own behavior, predicting and responding, always changing. When there are nonlinearities and extensive time lags, the systems’ behavior can easily get out of synchrony with the world, making matters worse rather than better.

The social and behavioral sciences are far more complex than the natural sciences. Show some sympathy for the economists and engineers who use the natural sciences as their guide to the behavior of people. After all, they don't know any better. But don't be sympathetic to their insistence on simple, fixed models of behavior. Sure, those simple models enable the construction of mathematical or computer models, but we are back to the expression “garbage in, garbage out.” If the models are based on erroneous assumptions about people, why should we trust the results? The natural sciences are not good models for human and societal behavior.

Many social scientists have also succumbed to the neatness of models of the physical world, so they, too, oversimplify their assumptions, which allows them to use linear analyses and to ignore all those feedback and feedforward loops. Yes, they use psychological variables but with models of human behavior that are oversimplistic and therefore wrong.

Let me end this chapter on a more encouraging note. We do not have to give up in despair. There are many activities that can lead to major improvements in life and the entire ecosystem. These actions can help get us out of the world's current predicaments. It is already too late to stop the damage that will be caused by climate change, but it is *not* too late to *minimize* the damage. One example is to rethink the structure of buildings.

Rethinking the Modern Way of Constructing Homes and Buildings

Consider the temperature of the planet, not just of the individual home. Instead of trying to throw more and more complex technology at the issue, let us step back and understand alternative solutions.

Let me use a simple example: Why do we heat our rooms so much in winter when we could simply put on more clothes? Similarly, why do we cool rooms so much in the summer when we could wear cooler clothing? Both of these actions waste energy and lead to increased damage to the environment. Why do we do this? We have artificially acclimated our bodies to temperatures that require extensive heating and cooling systems.

Suppose that in our homes, offices, and schools we wear more and heavier clothes in winter, and lighter and more comfortable clothes in summer, just as we do when we go outside. Photographs and paintings of people in the 1800s indicate that both men and women bundled up in winter: men would wear four or more layers of clothes—an undershirt, a shirt, a vest, and a jacket—and in the evening they might add yet another layer as well as, of course, the ubiquitous tie and hat. Women also had multiple layers. Both men and women bundled up when they went to bed, including a nightcap.

The design of buildings themselves also makes a difference. Society has become so used to heating and air-conditioning systems that the buildings are no longer designed to mitigate against the sun or the cold. (This practice is now changing. Awareness of climate change and the harm to the ecosystem by excess energy usage is leading to changes in building design.) Awnings, balcony and roof overhangs, and even the planting of deciduous trees on the sunny sides of buildings can provide shade in summer and sunshine in winter (the sunny side being south in the Northern Hemisphere and north in the Southern Hemisphere).

In earlier times, before the development of modern heating, ventilation, and air-conditioning systems, Indigenous communities used clever methods to control the temperature of their dwellings. These methods worked remarkably well despite the lack of electricity or automated devices. One was to construct thick, heavy walls out of mud that both insulated the inside from outside temperatures and acted to maintain constant temperatures: the heavy walls did not change temperature rapidly, so during hot spells they cooled off during the evening and maintained relatively cool temperatures in the daytime, and in winter they did the opposite: heating up in the daytime and maintaining that temperature at night.

Homes also used shade in summer and sun in winter to regulate temperature, and substantial portions were built underground. Pit homes could have their ground floor as much as 1.5 meters (around 5 feet) below ground level, which greatly increased the homes' thermal efficiency: cooler in summer, warmer in winter, but also stabilizing the temperature all year. Finally, roofs could even include sod and plants, insulating the home from temperature changes.

I sent Maryam Fazeli, a graduate student in mechanical engineering at Shiraz University in Iran, a draft copy of part III of this book during an email correspondence about architectural design that takes account of natural cooling and heating. She said that “looking at Fig. 19.1 . . . and the horrible impact of the air conditioners on the planet, I couldn't stop thinking of the magical feeling of a cool pleasant breeze produced by a windcatcher . . . I experienced once when walking inside the traditional building of Dowlat Abad Garden, in a hot arid climate of Iran with summer temperatures around 40°C (104°F). It was like going to heaven in the middle of hell! The building is totally passive and does not have the operational costs that air conditioners impose on the environment.” She sent me several references that described ancient cooling systems, some of which I have used in the discussion below. She wondered “whether there are economic advantages to adopting windcatchers for buildings over conventional HVAC” and found that the *Wikipedia* article on windcatchers had the answer: “Generally, the cost of construction for a windcatcher-ventilated building is less than that of a similar building with conventional heating, ventilation, and air conditioning (HVAC) systems. The maintenance costs are also lower. (See chapter notes for references and more.)

Vinod Gupta from the School of Planning and Architecture in New Delhi has written many detailed articles about the construction of older buildings. Here is his description of one of the differences between modern and Indigenous architecture: “When architects talk of passive cooling, it is as if the maintenance of certain specified temperatures in a building is an end in itself. On the other hand, the Indigenous builder could not care less if the building was cool or warm so long as people could be comfortable within or without the building. And in this, the builder's task was simplified by the willingness of the building users to put up with minor inconveniences.”

In today's world, we have become so used to artificial heating and cooling systems that we build homes and buildings that are inefficient and do not try to optimize the capture of the sun's heat in winter or the cooling effect of winds or shade in summer. We instead build marvelous, attractive structures that require advanced technology to regulate temperatures, which requires more energy to heat and to cool than is necessary had the architects followed some of these Indigenous practices.

Modernity strikes again. Sealed buildings, lots of glass. Powerful heating and air-conditioning systems. The energy supply is infinite, isn't it?

The philosophy of building for the sake of appearances and not for the comfort of people, let alone for the benefit of the planet and its ecosystems, has spoiled us all. So many of us have gotten used to modern heating and cooling systems that we dress more heavily than makes sense in high heat, using complex technology to keep us comfortable rather than simply changing the kinds of clothing we wear. If people were used to higher temperatures in summer, cooler temperatures in winter, if they dressed appropriately for these temperatures, and if buildings were designed to use natural processes such as thick walls, proper ventilation, cooling towers, heat pumps, shade in the summer, and sun in the winter, interiors could stay at a comfortable temperature with a minimal reliance on energy-consuming systems. If only. Consider the title of another one of Vinod Gupta's articles: “You Don't Need Airconditioners to Cool Your Home; Traditional Architecture Does It Well.” Gupta also makes it clear that just changing the architecture to be more efficient is not enough. In addition to his examples of ecologically sensitive design, he also assumes that we will dress appropriately (varying the layers and types of clothing to match the weather) and acclimate our bodies to accept higher indoor temperatures in summer, cooler temperatures in winter.

I asked the Mexican architect Jorge Gracia, recipient of numerous international awards for his work and founder of the Escuela Libre de Arquitectura (Free school of architecture) in Tijuana, Mexico, to comment on the draft text for this section. In his email response, Gracia said that the path set by modernism has not been good for architecture. “Look for example,” he said, “at a traditional Mayan house. It has a multifunctional living space. The building is sustainable, resistant, and built with organic materials. The walls are built of dirt (Thermal efficient).” He also pointed out that architecture should be responsive to the community in which it is built: “Architecture shouldn't be global; architecture needs to be local to respond to specific microclimates and social conditions. I do believe in technology, but it needs to be rooted in the process of nature.” However, he also pointed out that architects are in a bind. Building regulations can hamper their desires to build appropriately: “Having the rules set for energy efficiency and quality control put today's architects in a position where they must design buildings that are artificially controlled by technology, and that is the path we shouldn't have gotten in the first place. Traditional techniques have been replaced unnecessarily.”

Are people paying attention? Yes. Architects are changing their designs. Cities around the world are revising building standards. New levels of energy-efficient certifications are available, such as Leadership in Energy and Environmental Design (LEED), which sets international standards for design, construction, and maintenance. (See the notes for chapter 20 for more references.)

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It Is Not Too Late

The activities of people throughout millennia have changed the balance of forces that sustain life on earth. Because of the interconnections in this complex system, both social and technical, it is extremely difficult to visualize and understand the complexity of the earth's interacting systems. They have feedback and feedforward loops, both negative in limiting activity and positive in enhancing activity—loops that could have immediate impact and loops whose impact might take months or years to be felt. The individual parts of the system cannot be studied in isolation of their interactions with the rest of the system; otherwise, our understanding will be incomplete.

Today's technologies have increased the importance of understanding the systems. The importance of information and the way that it is transmitted forward and backward, looping around, gives rise to the digital, virtual worlds that have made the communication and interaction patterns across the world so ubiquitous and so powerful. Unfortunately, this power is being used for both good and evil.

The treatment of the earth as a source of infinite resources and a sink for infinite disposal of waste, coupled with misguided economic systems, labor relations, and governmental policies, have led to an unstable condition, where runaway waste is leading to a disastrous climate crisis. The conflicting cultures and value systems of the world's nations make remedial action difficult and preventive action almost impossible.

The leaders of the world engage in and talk about these issues. That is good. But even the best, well-motivated leaders also face pressures and contradictions. Despite the insistence that the harmful activities must immediately stop and be replaced with beneficial ones, and despite all the wonderful words and resolutions passed at many United Nations meetings, the nations of the world have been slow to take action. Even once the action starts, though, decades will be needed to develop those beneficial activities, decades to be able to make them available across the world. If all harmful activities are stopped before the beneficial ones are ready, there will be grave harm to many existing societies. However, if they are not stopped, they will most definitely cause harm to almost everyone. There are no easy answers to this dilemma.

Alexander von Humboldt warned the world in 1800 and again in 1831 that the way people were treating the world—abusing the forests and lands, the oceans and animal life, and the Indigenous cultures—would lead us to major disasters. Warning us two centuries ago! Humboldt is infrequently remembered today, but in the 1800s he was known throughout Europe, North and South America, and elsewhere and of great influence on other thinkers such as Simón Bolivar, Charles Darwin, Henry David Thoreau, and John Muir. He was an explorer, biologist, and author of multiple, detailed books that combined science, meticulous analyses and data, and a poetic writing style. But even so, his predictions were not taken seriously. Finally, people are paying attention, taking the warning seriously. Is it too late?

The bad news is that it is too late to stop some of the major damage that has occurred—and will continue to occur for some time. However, it is not too late to stop and then reverse the trend. What needs to be done? Increase the speed of the transition to a sustainable, robust, resilient, and regenerative world, lifestyle, and way of being. All the required ideas and methods are already known, being discussed, and even being put into practice. But these efforts are too small and as a result too slow. They must be amplified to broaden their outreach, to ensure that there is truly a worldwide, coordinated, collaborative effort. We can do it.

IV
Humanity Centered
Address All Aspects of the World Relevant to Life

22
Moving from Humans to Humanity

Why the name “humanity-centered design”? How is it different from “human-centered design”? Don't the terms *human* and *humanity* mean very similar things?

The meaning of phrases cannot be inferred simply by the words in them: it is necessary to view the context—history matters. The term *human centered* was developed in the late 1980s, and at that time the focus was primarily on the individual people for whom the design was intended. This concept had many virtues, and it is the dominant approach today. But now, four decades later, it is time to change the emphasis. Design started out as a tool for industry, helping develop mass-produced products that were attractive, functional, and cost efficient. Naming the methods “human centered” refocused the attention on the people instead of on the efficiency of manufacturing. The introduction of computer chips inside more and more products increased the complexity of the devices and decreased the usability as programmers added features and control structures that were idiosyncratic and confusing to users: hence the need for and the arrival of “human-centered design.” This type of design emphasizes ease of use and people's ability to understand the product. “Human centered” has been an important aspect of design, and as long as products are mass-produced, it will remain a dominant design approach. This approach is so popular that it is usually simply called by its three-letter acronym: HCD.

So why the need to change? Today, designers are called upon to do far more than tailor products for the mass-consumption market. The phrase “human centered” fails to emphasize the larger concerns and the need for increased sensitivity to biases and prejudices against certain societal groups. The phrase “humanity centered” emphasizes designs that take into account the sociotechnical system in which people reside. Design must be sensitive to the impact on the environment created by the manufacture, use, and disposal of physical products. For all products, both physical and nonphysical, design must address the impact on fairness, equity, prejudice, and bias. The phrase “humanity centered” emphasizes the rights of all of humanity and addresses the entire ecosystem, including all living creatures and the earth's environment. When did the phrase “humanity-centered design” first get used? I was able to trace an early use to articles in 2005 and 2006, which state that in humanity-centered design, “designs are judged on the basis of how they have created or will create coherent improvements in the collective human condition,” which is close to my interpretation. Today we would add an explicit reference to the ecosystem. Is that the first usage? I don't know.

Note that designers must still follow the design principles of human-centered design, but now within the broader scope of the entire globe: all living things; the quality of the land, water, and air; the loss of species; the changes in climate. Human beings—people—are an integral part of the system called “Earth,” where changes in one component can impact every component. I consider human-centered design to be a subset of humanity-centered design. Here is how I view the two terms (the definitions are my own).

The Four Principles of Human-Centered Design

There are four basic principles of human-centered design.

1. Solve the core, root issues, not just the problem as presented (which is often the symptom, not the cause).
2. Focus on the people.
3. Take a systems point of view, realizing that most complications result from the interdependencies of the multiple parts.
4. Continually test and refine the proposed designs to ensure they truly meet the concerns of the people for whom they are

intended.

These principles are important, but they ignore the problems of sustainability, inequity, and bias. In addition, the emphasis is usually on the immediate issues, not on the long-term impact. In other words, they describe the way we did things in the past, not the way the issues described in this book must dealt with and not the way we need to do things in the future.

Humanity-centered design accepts the basic framework of these four principles but expands them so that they are far more explicit about coverage of all living things (not just people), the ecosystem, and about looking at the long-term future impact. To move from human-centered to humanity-centered design requires adding one new principle (principle 5) and slightly modifying 2, 3, and 4 to widen the range of issues being addressed.

The Five Principles of Humanity-Centered Design

Transforming Human-Centered into Humanity-Centered Design: Five Basic Principles

1. Solve the core, root issues, not just the problem as presented (which is often the symptom, not the cause).
2. Focus on the entire ecosystem of people, all living things, and the physical environment.
3. Take a long-term, systems point of view, realizing that most complications result from the interdependencies of the multiple parts and that many of the most damaging impacts on society and the ecosystem reveal themselves only years or even decades later.
4. Continually test and refine the proposed designs to ensure they truly meet the concerns of the people and ecosystem for whom they are intended.

5. Design with the community and as much as possible support designs by the community. Professional designers should serve as enablers, facilitators, and resources, aiding community members to meet their concerns.

Our Traditional Design Methods Are a Form of Colonization

For centuries, the nations of Europe colonized the world, traveling to what they called “unexplored lands,” many of which they claimed to “discover,” and as the first “discoverers” they planted their countries’ flags and claimed these territories. The fact that this unexplored land was already populated by people didn't stop the explorers from proclaiming that these lands were now owned by whatever country had sponsored their exploration.

Over time, the colonizing countries installed new armies and replaced the ways in which the Indigenous people governed, settled disputes, and lived. Even colonizers who saw themselves as well meaning (albeit based on their imperial view of things) did not serve the occupied countries well. In India, for 190 years (1757–1947) the British took over the government, legal systems, and political practices. The native Indians were treated as second-class citizens in their own country, becoming laborers and servants. Even those that were well educated (sometimes even in English universities) remained lower-level civil servants and clerks, always reporting to British overlords. The British Empire colonized more territories than any other nation: 90 in all. Of those 90 colonies, nearly all have now left the British Empire, including the United States, which then became an empire in its own right. Between the sixteenth century and the early twentieth century, the United States expelled Indigenous groups from their native lands, relegating them to small reservations, as it expanded westward in an act of what the colonizers considered their “manifest destiny.” In addition to the 50 states that the United States now calls its own, it also still maintains five colonies, now called “territories”: Guam, Puerto Rico, American Samoa, the US Virgin Islands, and the Northern Mariana Islands.

Foreign aid is sometimes characterized as a form of colonization when it imposes the culture of the aiding country upon those being aided. When countries that lack food, water, sanitation, health care, or education—to name the most popular ailments—request assistance from established, wealthy nations, huge aid programs are formed in response, some of it run by the countries of the Global North, some by private foundations (nongovernment organizations, or NGOs), and some through the United Nations. These institutions send in expert teams to help and lend money, which many countries are unable to repay.

Foreign aid is now a big business, practiced by most major nations, the United Nations, and many of the world's largest foundations. The standard aid package starts with the experts who arrive in large numbers, spend some time in the country studying the issues, and then produce lengthy “White Papers,” summarizing the ailments and recommending “cures.” Very few of the people from the countries being aided take part in this process.

The result of these preliminary activities is usually the recommendation of a massive plan that will cost billions of dollars (in US currency) and take many years. One of the difficulties with these programs is that well-intentioned foreign experts are, well, foreigners. Their expert knowledge is relevant, deep, and principled, so the discussion of the ailments is usually appropriate. But outside experts seldom have a deep understanding of the local people, their skills, resources, and what they consider to be important. As a result, the experts propose—and supervise—the construction of systems that in principle address the appropriate issues, but that cannot be maintained or repaired by the local inhabitants, and often take over large parts of cities and towns, causing forced relocations of thousands of people. These issues are well documented in William Easterly's book *The Tyranny of Experts: Economists, Dictators, and the Forgotten Rights of the Poor*.

Whether foreign aid can be considered a success or failure is subject to much debate. Charles Kenny's book *Getting Better: Why Global Development Is Succeeding: And How We Can Improve the*

World Even More presents “evidence of widespread improvements in health, education, peace, liberty—and even happiness,” to quote the publisher's praise for the book. More importantly, Easterly, one of the severest critics of foreign aid says that Kenny “shows convincingly that most trends in human well-being worldwide, and region by region, are happily, dramatically positive.”

In an earlier book, *Reinventing Foreign Aid*, Easterly suggests that the issue revolves around whether the actual aid is provided by “planners” or “searchers.” Planners, he states, set out a big goal and then develop a big plan, into which they throw an endless supply of resources. “A planner,” he says, “thinks he already knows the answers; he thinks of poverty as a technical engineering problem that his answers will solve.” Searchers, in contrast, are humble about what they know. They are “on the lookout for favorable opportunities to solve problems—any problem no matter how big or small, whose solution will benefit themselves or others. Searchers must learn enough about each little problem to solve it, which means they must get feedback from the people affected by problem.” I like these characterizations and do suspect that planners think big, lofty thoughts, which although sensible do not always match the resources and cultures of the target location and, moreover, ignore the fact that large, massive projects are difficult to manage and take such a long time that the requirements change several times in the interval and never really meet the local people's needs.

In this division between planners and searchers, I identify myself as a searcher. This is precisely the strategy that I recommend and that many designers are already taking all over the world: designing *with* and *by* people, not *for* them. This is the focus of the next chapter, “Democratizing Design and Development.” In that chapter, I discuss how design solutions must come from the local communities, from the people who are the intended recipients of the designs. This approach means that the underlying issues are being addressed and recognizes that all-important human problems are part of complex sociotechnical systems, acknowledging that many of the issues are subtle and have resulted from historical practice, culture, and a lack of understanding of the larger picture. As Easterly says, we searchers are firm believers in working with continual and repeated observing, thinking, and testing. Most people don't like it when an outside group shows up where they live and tells them what to do, what new tools to use, and how to live and behave. The answers have to come from the people themselves. This is where humanity-centered design comes in.

23
Democratizing Design and Development

Step One: Community-Driven Design

The first step toward humanity-centered design lies in principle 5: design with the community and as much as possible support designs by the community—practices often simplified as "Design *with* and *by*, not *for*."

The approach of designing with the people rather than for them is well established in today's design field, even if not yet a major component of the foreign aid groups. I emailed a query to the 3,000 members of the PhD design discussion list (phd-design), asking when these forms of collaborative design started. I received a large number of thoughtful, delightful answers. The consensus was that it was difficult to know the exact starting date or who was at the very beginning, so the answer is simply "a very long time ago." Over time, the design community developed many different ways in which to work with the people who are to receive the aid. Some of my correspondents placed an emphasis on designing *with* people instead of *for* them. Others emphasized the importance of having the design done *by* the people.

These approaches have been given numerous names: "cooperative design," "codesign," "participatory design," "citizen-led, community-driven, or citizen design," "design for social innovation," "convivial design," "regenerative design," and others. They differ in the details but are the same in spirit. And it isn't just the design community that does this work. Numerous disciplines do: social workers, public-health professionals, civic leaders, foundations, and nongovernment agencies. Elizabeth Sanders and Pieter Jan Stappers offer an excellent review and treatment of collaborative design in their book *Convivial Toolbox: Generative Research for the Front End of Design*, which is devoted to detailed illustrations of how to work with communities, including case histories.

There are many difficulties in working with communities. Community members themselves do not often agree upon the goals and potential solutions. There are often internal power struggles within the community, with different groups aiming for control over the activities. Quite often political and government officials become involved, not always for the benefit of the local communities. Those who work on community issues have to develop expert social and political skills to navigate the terrain. The danger is that the funding is apt to be dispersed in ways that distort the goal of the aid. All these difficulties reinforce my long-stated belief that important projects are always difficult. If they weren't difficult, the issues probably would have already been solved. Designers and other professionals who wish to help, support, and empower local communities have challenging but important jobs.

Step Two: Enable Everyone to Design

Everyone is a designer. What do I mean by that? People—all of us—are continually readjusting our activities and our belongings to accommodate changes in needs. When we decide where to sit at a public performance, we are designing: taking into account a variety of goals and the constraints of the situation. Everyone is a designer in the same way that everyone is a tennis player or cook. Professional designers have advanced skills, training, and knowledge to apply to complex problems and issues. It is not possible to become a professional designer by taking a short course on design thinking or reading a magazine article, just as it is not possible to become a professional tennis player or cook by a short course or article. Yes, do take those courses and read those articles because even though they will not suffice to make you an expert, you will gain a deeper understanding of and appreciation for the skills of professionals.

Eric von Hippel, professor of technological innovation at the MIT Sloan School of Management, suggests that we should reverse the traditional way of designing. Instead of professional engineers and designers designing products for others, why not find out what people are already doing: in other words, let the creative people all over the world design for us, the professionals. In the mid-1980s, he recommended that companies wishing to improve their products should go see what their customers were doing with them. Von Hippel pointed out that many creative people had already modified the products either to overcome deficiencies or to enable them to be used in new ways. In either case, these “lead users,” as von Hippel named them, provided valuable new ideas. He has since expanded these suggestions, pointing out that they democratize innovation and in his 2005 book by that name, demonstrates the ways that creative people can develop their own new products and services. Even more powerful, says von Hippel, is when the ideas of the creative nondesign community are freely shared with the world, an idea he pursued in more depth in his 2017 book *Free Innovation.* In recent years he has extended these ideas to innovation in medicine by patients as well as in households.

Ezio Manzini from the Politecnico di Milano in Italy is one of the leaders in social innovation projects run by communities. His many books and the social innovation network he founded (Design for Social Innovation and Sustainability) have been applied to many communities around the world. The title of his book published in 2015 makes his importance to this section clear: *Design, When Everybody Designs: An Introduction to Design for Social Innovation*. He followed up in 2019 with *Politics of the Everyday*, a powerful introduction to dealing with today's complex world and a contribution to collaborative design. It is, as he says in the preface, a book about the culture of design and how in collaboration with others it can help lead design culture. On the last pages of his book, he asks, “What is the role of design experts in the new design coalitions[,] . . . in building a collective design intelligence, one that cultivates diversity” and is able “to take us out of the environmental, social, and cultural catastrophe we are falling into?”

“Yes, wonderful!” I said to myself. “What is the role?”

Alas, Manzini's answer, stated in the last seven words of his book, is “another book is required to answer them.” That was some time ago, and he has not yet answered the question. He did write another book in 2022, *Livable Proximity*, an excellent treatment of how we can restructure existing cities to optimize the social dimensions of interaction. This addresses some of the issues, but is incomplete (deliberately) in covering the range. OK, it is up to me: that is what this book is about, most especially parts IV and VI (“Humanity Centered” and “Action,” respectively).

As Manzini makes clear, the bottom line is societal issues: the people living in particular societies understand the problems they face, and many have probably already thought of solutions, but they probably need assistance in implementing them. So why not build on the creativity and ingenuity of the people in these communities? Democratize design so that everyone has a voice, everyone can play a role.

The goal is to empower everyone globally by providing them with the resources to help themselves. These resources would include modular education—available just when needed—tool kits, modules of appropriate technology, items that people in different cultures can use and modify to better fit their needs. All modifications would also go back into the general open-source repository of tools and knowledge modules, thus continually expanding the range and utility.

What about all the people in the world who are not professionals, who lack the resources to find and hire professionals, or who simply have such unique problems that professionals are not interested in them? After all, there are roughly 8 billion people in the world but nowhere near enough professionals to help them. Designers can help by being mentors and facilitators, by developing tools and procedures that everyone has access to, that everyone can learn to use.

Tools that enable this work need to be accessible. For many people, the word *tool* brings forth images of physical devices such as hammers, saws, and shovels. Many dictionaries define a tool as something handheld, something used to accomplish a task. That is far too narrow a definition. I follow the definition from the *Cambridge Advanced Learner's Dictionary & Thesaurus*: a tool can be “anything that helps you to do something you want to do." So tools can be procedures, templates, organizational structures, and even credit cards. Here, I use the word to refer to the well-honed, carefully researched structures and procedures that designers and others can use to discover underlying meanings, needs, and desires.

Creating the tools that enable people to solve their own problems was the focus of Ivan Illich's work in the early 1970s, well stated in his book *Tools for Conviviality*, where he said that people "need above all the freedom to make things among which they can live, to give shape to them according to their own tastes, and to put them to use in caring for and about others." He explained that he used the word *convivial* "after many doubts, and against the advice of friends . . . as a technical term to designate a modern society of responsibly limited tools." His book was too polemical for the times, arguing against the dangers of the Industrial Revolution and, for example, the way that schooling had become a method of indoctrinating children to become good consumers and willing slaves of technology. As a result, many of the valuable messages in his book were ignored just when they should have been heeded.

Illich's work has been brought up to date (without the polemical rhetoric) by Elizabeth Sanders and Pieter Jan Stappers in *Convivial Toolbox*, a book I have already recommended for its detailed discussion of co-creation. They provide a variety of convivial tools that enable people to design for themselves and include case histories of how these tools can be applied.

Note that in many ways designing for oneself is simpler than designing for others. The traditional human-centered approach requires study of the people to be served, the better to understand the environment and their needs. But if people are designing for themselves, they don't need to do all that—after all, they have experienced their problems, so they do not need to do elaborate studies, although simple ones might be in order. Similarly, design goes through iterative testing and redesign cycles. With people designing for themselves, this cyclical approach becomes automatic. After all, if they design something to help solve their own needs, and it doesn't work as well as desired, they are strongly motivated to modify it.

Tools can be designed for and used by individuals, by designers, and even for communities or groups of communities. But not all societal needs can be solved by communities working in isolation. Many require much larger groups to work with them. Some may require considerable funding and expertise. Technical knowledge is required, even for apparently simple things such as providing electric distribution, a communication infrastructure, clean, continuous water, and workable sewers that don't contaminate existing clean water because they are poorly designed. Many issues are part of large sociotechnical systems that require active participation of a large number of people and disciplines to address. Yes, the projects need to come from the people, but the technical requirements are often greater than can be mastered by the community alone. The trick is to be able to use technical experts without allowing them to dominate and thus derail the community spirit and collaboration.

How Standards Amplify the Power of Tools

Convivial tools are even more powerful if instead of each tool being built in isolation of the others, it is built to follow some standard guidelines and principles, so that the knowledge of how to work one tool is readily transferred to others, thus amplifying the power of individual tools to empower everyone. Consider clothing: standardization and ease of assembly of clothing evolved over tens of thousands of years. It took expert skills to make animal skins flexible and appropriate for wearing, to take animal wool and plant fibers and make them into durable, large pieces of material that could be fabricated into clothing either by pressing the fibers together (making felt) or by weaving the wool. Transforming these materials into wearable items took skill. Developing useful textiles and fasteners took centuries. Clothes used to be made specifically for each individual either in the home or by going to specialists (tailors and dressmakers). In the early 1800s, the development of standard clothing sizes began. These size standards are imprecise, but for most people they are good enough for most purposes.

Clothing standards are still imperfect and, moreover, tend to be used for the manufacturing of clothes by the clothing industry, not by individuals. The easy manufacture of clothes has not stopped people from using home tools to make their own clothing. There is a large network of "do-it-yourself" tools for individuals, along with multiple videos, articles, and books to teach people how to make their own clothes; patterns for different kinds of clothing; tutorials; discussions about hand tools, such as the differences between home and industrial sewing machines; and so on. This is an excellent example of community-led do-it-yourself projects, where the tools, the way to use them, patterns to guide the fabrication, and materials for self-learning are combined to guide beginners through all the necessary steps. Experts make themselves available to offer guidance and to share their expertise, tips, and suggestions with everyone, both new learners and other experts.

This is a simple example of democratization of design. Given an appropriate set of basic components, the average person can assemble whatever finished product is desired. A simpler example is how the items purchased from a variety of stores are used to create individualized furniture arrangements and kitchens. Food markets provide the ingredients for meals. Hardware stores and lumber yards provide a different level of components, designed for people with a bit more expertise. Experts at the manufacturing and supply companies produce the basic components in a form that those of us who are equipping our homes can use them effectively. From the individuals' viewpoint, they all are creating (designing) their homes to fit their needs.

The term *democratization* as used here means to "make (something) accessible to everyone." The goal is to enable everyone to specify their own needs and then, either individually or collaboratively, to supply them. The phrase "democratizing design" refers to the extension of self-developed, designed, and constructed objects and methods to domains not yet covered. Consider health-care tools or citizen control over housing, city planning, public transportation, education, and communication. Why must everything be done by experts?

That most of us assemble store-bought components into the clothing we wear, food for meals, and entertainment systems in our homes sounds mundane, but making it possible for nonexperts to do this has required many years of development. To make it possible for people to self-assemble solutions to their needs requires standardized components and a platform of principles, policies, and standards to ensure that the different components work well with one another. This has led to the development of metatools: tools that make it easy to learn and to use other tools—for example, instructional material, video tutorials, and discussion groups—and thus help everyday people build their own solutions.

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People Designing for Themselves

The Do-It-Yourself Movement

The do-it-yourself (DIY) movement is a wonderful example of self-forming democratization. The history of DIY can be traced to the earliest days of human technology, thousands of years ago, from tinkering and innovation by everyday people to the cleverness in such things as New Zealand's proud heritage of creativity using commonly available wire for making fences (the most common size was called “number 8”) to repair household, home, farming, and industrial products and also to develop creative new devices and solutions to many of the issues they faced in the late nineteenth century. The creative use of this wire led to the rise of the term “number 8 wire” to refer to New Zealand creativity. (The equivalent tool today is probably duct tape.) The creativity displayed by people who can repair and create novel devices out of common, everyday materials can be found in cultures all across the world. These skills are what the French call “bricolage,” with those that exhibit them called a “bricoleur” or “bricoleuse.”

In the United States, the word *hacker* originally referred to people who were skilled enough to be able to “hack together” whatever it was that was needed. In the 1960s, the term was attributed to many of the early computer programmers, especially MIT students, and was used to denote programmers who could quickly create clever programs to accomplish their needs. (Confession: I was one of those MIT students. Today I am a proud member of the Hacker's Society, which still retains the goal of making clever, useful things for meritorious purposes.) Alas, journalists distorted the term by using it to refer solely to people who break into computer systems to cause malicious harm. No, the latter are not hackers; they are troublemakers, criminals, or, sometimes, just teenagers who don't know any better. (And sometimes they are government employees of the many spy agencies in the world, stealing the secrets of industries and governments all across the world and causing havoc.)

The traditional example of DIY describes people who gather together raw materials and construct whatever they need at the moment: in a word, *hackers*. Today, as my examples of the tools available for clothes making in chapter 23 illustrated, it is not necessary to start with the raw materials. For example, in devising a home or office, one could go to furniture stores and select the individual items and then populate a home or office. But what if the items in the stores are not precisely what is needed or perhaps are too expensive? Not everyone can construct their desires from the raw materials. Clever members of the DIY community came up with ways to repurpose existing furniture. Need a desk? Buy a wooden door, prop it up on some filing cabinets, and there you are. Bookcase? Some bricks and some long boards. Store writing instruments in old jars that used to hold jam or preserves.

IKEA, a Swedish company that sells a wide variety of furniture all over the world, minimizes the costs of shipping by packaging their disassembled units in space-saving flat packages, designed to occupy the least possible volume. Why minimum volume? Because the costs of transporting goods by ship or truck is determined by volume. Flat packing saves money for the company and enables it to keep its prices low but requires purchasers to assemble the units by themselves (or to hire someone). IKEA turned its flat packaging into a mark of distinction, helped by its well-known, carefully tailored pictorial instructions.

Soon the hackers and DIY communities discovered the power of IKEA's modularity, which enabled individual pieces to be configured into a wide variety of structures, not just the finished unit that IKEA thought it was selling. For example, it is possible to turn IKEA kitchen cabinets into a raised (lofted) bed with room for storage.

In 2006, Jules Yap (a pseudonym) looked into the creative ways that people had combined parts from different IKEA packages. She thought more people ought to know about these ideas and build the items them- selves, so she started the website IKEAhackers.net to show off their work. Through the powers of the internet and social media, the word spread, and soon people all over the world were purchasing IKEA's furniture just for the parts and creating their own objects, then sharing the methods with the world. In 2014, IKEA discovered what was happening and took legal action against Yap. When the people who used IKEAhackers discovered the legal threat, they raised such a popular uproar that the company reconsidered its legal action and reversed course. Instead of penalizing Yap, company executives invited her to Sweden and had (according to Jules Yap) a series of constructive and positive conversations. Now Yap operates the fan-run IKEAhackers.net website, which includes a statement from IKEA that says that it doesn't have anything to do with the website and is neutral about the ideas and usage. From my point of view, this story clearly illustrates the power of individual creativity to repurpose commercial objects in ways that better suit one's needs.

These self-generated examples of democratized design have expanded to almost every domain of human experience, including art and entertainment, education, health care, public health, and modern (confusing) technology.

Health, Medicine, and Public Health

DIY websites for medical problems may seem difficult and perhaps dangerous to rely on. Nonetheless, many groups have banded together to create their own communities of self-help, complete with informative websites where medical experts serve as consultants and advisers. The collection of knowledgeable and credible amateurs, describing how they have learned to live with their ailments, mitigated the symptoms, or even cured their own problems has proven to be helpful if done in a careful, methodical, and responsible manner. Whether one is concerned about diabetes or obesity, cancer or autism, there are many self-help sites available. For example, *The Rise of the Bio-citizen* report from the Wilson Center's Citizen Health Innovators Project points out that citizens, patients, and families have become a leading force in health research. It states that “patients often have in-depth experiential knowledge of their conditions along with a vested interest to make sure that a treatment or device will be effective, safe, and beneficial.”

Medicine and health can be complex areas to work in because they are so carefully regulated and fraught with unintended consequences. After all, faulty medical devices or treatments can seriously harm or kill people. Some people argue that medicine is a good example of a field not yet ready for democratization. However, that has not stopped numerous individuals from treating themselves, often better than the medical profession was able to. Slowly, the medical profession is learning that these self-guided experiments and innovations by nonmedically trained people can teach valuable lessons everyone can adopt.

New tools allow people to do their own experiments. Societies and groups have been formed to assist and to communicate their findings, as in, for example, the Quantified Self movement. Inspired by the rise of self-experimentation, Vineet Pandey in the Design Lab at the University of California, San Diego, devised several tools to allow people to do controlled experiments on themselves, centered mainly around their own hypotheses about how their eating behaviors have impacted their health. The project was called “Gut Instinct” and was supervised by Rob Knight, the director of UC San Diego's Center for Microbiome Innovation and a faculty member in the Medical School.

High costs, regulations, and advanced technical knowledge required for certain health projects limit the kind of things that average people can do themselves. Nonetheless, there is much that can be done that does not require government approval. For example, in 2017 I visited Pedro Oliveira in Lisbon, Portugal, to learn more about his Patient Innovations group, which has encouraged numerous patients to devise solutions to their ailments. The group's website shows the innovations conceived and built by patients or their families. Even simple innovations can provide a meaningful uplift to people suffering from serious ailments.

One strength of these health-care DIY projects is that people noticed needs not being addressed by medical, public-health, or social welfare organizations, so they took the matter into their own hands and tackled those needs. A similar strategy has been used by the group in Bengaluru, India, calling themselves the “Ugly Indians.” I spent a day with them in 2019 and was impressed with the way the all-volunteer group had built up a strong community network to reduce the spread of litter on the streets and to provide other solutions to local issues. But they have been able to address only the symptoms, not the underlying causes of litter and other issues. This is a common difficulty for volunteer organizations because they lack the resources required to attack the more substantive, underlying causal factors. As a result, the symptoms continually recur. Even if citizen groups recognize that the problems they are addressing are part of a larger, complex system, they lack the resources to address the system as a whole.

Democratizing Design Will Not Succeed by Itself

Democratizing design cannot succeed by itself. The people who need assistance also often need access to information, education, and often some expert advisers or coaches who can help them get over hurdles, such as a string of failures or the lack of specific expertise. The use of the negative term *failure* is enough to discourage people, but one role of the coach is to celebrate “failures,” to emphasize that they are learning opportunities, and to make sure that everyone benefits from a review of the project (sometimes called a “postmortem”) to see what learning has emerged. I tell people that scientists never say they failed at some attempt. They say, “That didn't work,” and then they try a different approach. It also helps to gather together in a group people who have similar problems, so they can share ideas and results and give each other mutual support. When people discover that others have similar problems, they can be more effective. The most powerful aspect of the many citizen organizations that exist around the world is the sharing of knowledge and emotional support.

Perhaps the most powerful effect of democratization is to dissolve the separation between experts and everyone else, especially between academics and nonacademics. Experts and academics are, after all, simply people, and everyone realizes that each group can help the other. It is necessary to break down the feeling that many people have that there is a distinction between "us," the poor, the uneducated, and "them," the wealthy, the highly educated. No, we all are in this together, and quite often the everyday knowledge and experience held by "us" are highly educational to "them."

Ideas and innovation can come from anywhere, anyone. Many "everyday" people can be amazingly creative and innovative if given the right opportunities, environment, and support. Citizens are strongly motivated to solve personal, local, and even global challenges: all they need is assistance, guidance, and support.

Multiple methods will be needed, multiple techniques devised for each of the many different kinds of problems encountered over geographical regions, cultures, and environmental conditions. Even within the same set of problems, the answers will vary from group to group. Public-health methods that work in countries that have access to electricity, clean water, sewerage, good communication, and efficient transportation systems will be very different from those required in areas that lack many (or even all) of these amenities. Medical treatments and instruments devised in countries that have advanced technology and infrastructures often fail in places where there is no access to power, clean water, and refrigeration. This doesn't mean that the answers cannot be found—it does mean that tools and methods have to be designed differently for different contexts.

It is an enormous challenge to introduce effective design principles to everyone in the world. Fortunately, the work has already started. Many groups are working in many domains to establish tools for design, for making, for education, and for dissemination. Every society already has a large number of creative individuals designing things for themselves and others. Many already have organized themselves into self-help communities. As a result, if one does an internet search for "makers" or "do it yourself," each search yields an enormous number of valuable results.

The advent of relatively inexpensive machine tools, sensors, and computational tools has allowed everyday tinkerers to produce impressive results. Systems such as inexpensive computer processors, sensors, and actuators (motors) as well as inexpensive 3D printers and laser cutters, platforms for sharing ideas, websites, and innovating groups have proliferated. There are too many to list. Many of the issues that need to be addressed are large and complex: the large scope of these problems requires a unified effort across the world. One reassuring sign is the existence of numerous efforts already in place. Their impact would be greater if they could join in partnerships and collaborations with one another, precisely as described in Sustainable Development Goal 17 of the UN Partnership for the Goals.

This is a huge, exciting challenge that requires resources from multiple disciplines, organizations, and groups across the world. Even if some problems remain unsolved, a coordinated effort would leave behind a better-educated populace who have available ever-improving tool kits, technological aids, and the knowledge of how to apply their new capabilities. The effort's legacy will be the empowerment of all.

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DesignX: An Approach to Large, Complex Systems

Although I am a champion of community-led efforts and the DIY movement, many of the world's problems are far too large and complex to be done by small, volunteer organizations. In the 2010s at an advisory committee meeting of the College of Design and Innovation at Tongji University in Shanghai, five of us (four advisers and the dean) discussed the need to recognize that many important societal issues are in actuality complex sociotechnical systems. We asked, How can design address them? The issues are not new: many people and disciplines have grappled with them for some time. But how can designers play a role?

We realized that few designers are equipped to tackle the kinds of problems that combine a variety of stakeholders, multiple technical and environmental factors, economic considerations, political sensitivities, and cultures, laws, and regulations. Other groups have worked in these areas. The Systemic Design Network and its series of conferences titled “Systems Thinking and Design” as well as the Transition Design program at the School of Design of Carnegie Mellon University, among others, address many of these concerns.

Our group of advisers named our work “DesignX” and decided to organize a small workshop on the topic. We published the resulting summary of our findings in the design journal *She Ji*.

Small, Incremental Steps: Incrementalism or "Muddling Through"

During the DesignX workshop, John Flach, a psychologist and author of an excellent introductory text on the power of feedback, suggested that we consider the "world-famous paper" "The Science of 'Muddling Through'" by the political scientist Charles Lindblom. Of course, none of us had ever heard of this paper, but we found a copy and were immediately convinced that it indeed described an important approach.

When I returned to UC San Diego, I asked all my friends if they had ever heard of the paper. "No," they all said. I asked the dean of engineering. "No," he said. I think I even asked the chancellor (in his former life a roboticist and dean of engineering at Carnegie Mellon University). Nope. I asked the dean of the Business School. Nope. I asked the dean of social sciences: nope. I gave up, but one day when I was meeting with the dean of the School of Global Strategy and Policy, I asked him. "Yes, of course," he said. "Everybody knows that paper."

Ah yes, the specialization within academia. Anyway, it was the key to our understanding how to approach the problems of implementing large complex projects. For me to explain what "muddling through" means, I must back up and discuss the underlying conditions. So please be patient: I will get there.

Implementation: The Difficulty in Addressing Issues within Complex Sociotechnical Systems

The DesignX workshop did the easy part: analyzing the issues. The hard part is doing something. Small teams of designers have been working across the globe on societal problems, usually at a relatively small scale using many collaborative-design methods. How can designers work with larger issues?

Describing these issues as lying within large complex sociotechnical systems also points to the major difficulty: the issues are many, large, and complex. Our usual design methods work well with projects that are relatively small. When these same methods are employed with complex problems, perhaps helping to develop a health-care system in a remote village in an impoverished nation or to resolve sanitation or education issues or to devise new forms of growing food, cooking, or transacting commerce, the implementation of the recommendations is difficult, drawn out, subject to repeated revisions, and in many cases impossible. The design process never ends.

Complex Sociotechnical Systems

When dealing with the large systems needed by societies, we are of necessity dealing with complex, interconnected systems. Suppose that a small village of a few thousand homes has asked for a modern sanitation system. The current scheme is no longer working because the village has grown over the years. Wells, hand-carried buckets of water, and local disposal of sewerage from outhouses and chamber pots are not working and are leading to discomfort and disease.

Modern sanitation systems, to use a deceptively simple example, involve a huge number of interrelated variables that are connected together in a web of complex feedback loops. Water has to be cleansed, made free of toxins and infectious germs. Water naturally travels downhill, so unless the community is at a lower level than the water source, pumps will be required. If the community is small, a well and hand-powered pumps will suffice to bring water to the surface, and then to get it to homes they will require either hand delivery or a pumping station. A larger community will almost definitely need powered pumps.

If water is delivered to homes through pipes, it is important to keep germs from entering the pipes. The simplest way is to always keep the water under pressure, so that any leakage goes from the inside out. But continuous pressure requires a constant, continuous supply of water, which is not practical with hand pumps.

If the community wishes to employ a sewerage system, there are similar issues. Sewerage cannot be allowed to contaminate the water supply. Septic tanks and latrines must be downhill from water supplies. If the sewerage is carried in pipes, it is even more important to keep the water pipes always under pressure, so that if sewerage leaks out of its pipes (and there is always leakage), it cannot enter the water pipes.

There are many different kinds of wastewater, some of which can be reused with little treatment, while others require more complex treatment. Ideally, the different kinds of sewerage would be separated at the source into different output pipes. Thus, the waste from toilets (black water) should not be allowed to mix with waste from showers, laundry, and sinks (gray water), but in many places the two types of wastewater are intermixed. Gray water is relatively easily cleansed and reused through biological purification and filtration and then reused in the home or business. Blackwater is much more difficult to treat, but after treatment, it can be reused as fertilizer, a source of energy, and even nutrients. Most communities need technical specialists to advise them.

The most difficult part of dealing with large, complex problems concerns the complexities of implementation. Most people think that the technical issues and the accompanying technology are the most difficult, but they are actually the simplest. The implementation of recommendations involves human beliefs and behavior, so most difficulties arise from four components:

1. System design that does not take into account human psychology
2. The human tendency to want simple answers, decomposable systems, and straightforward linear causality
3. Collaborative interaction with multiple disciplines and perspectives
4. Mutually incompatible requirements

Here's a highly simplified (but realistic) example that involves a new sanitation system for a medium-size village. After many months of collaborative work among designers, sanitation engineers, civil engineers, and dedicated local citizens, the results are presented at a public session. Now imagine that everyone nods their heads in excitement over water pumped directly to their homes and indoor toilets with efficient disposal of waste. Hurrah! It seems that the collaborative work has been a success.

Alas, collaborative work with communities is never that easy. One problem is that different community members want different solutions. Worse, even if an agreement is reached at one meeting, during the next meeting many members of the previous meeting will be absent, and new people with yet different ideas have taken their place.

Even if agreement is reached on what is to be done, that early agreement is often the easiest part. The community is apt to expect the solution to be implemented immediately. Now watch them rebel when they begin to realize that the plans that they excitedly helped create require the village to be torn apart to install pipes for water, different pipes for sewerage, and perhaps wires for electricity. Some homes might have to be moved, trenches will have to be dug in the streets and in the yards of homes, and each house will need considerable renovation to allow installation of pipes, sinks, and toilets. To keep the costs down, every home will get exactly the same facilities, even though their needs—and their cultures—might require differences. And consider how they, still in shock from this news, will react when they are told that the disruption may last many years.

All four categories of roadblocks come into play in my simple example. All of them create huge political difficulties that are apt to fracture the social structure of the village.

All four involve complex human and social elements and can be exacerbated by little understanding of fundamental human capabilities and limitations in the design and analyses of a system. Any solution requires collaboration and agreement of multiple social entities and political actors and almost always demands compromises. Even where progress is made, it may require so many compromises that the eventual implementation tends to be delayed or cancelled or, if completed, is unsatisfying to all.

Muddling Through, Satisficing, and Approximation

How can designers deal with the complexity of implementation with so many social, economic, and political issues? One method is to divide and conquer. Here is where Lindblom's proposal for “muddling through” applies: Do the project in small, incremental steps. Avoid trying to implement a large, complex system in one step. When each step is small, it has to satisfy a smaller group of constituents, and the disruption created by its implementation is also more limited in both time and space than the disruption caused by large projects. Smallness allows for flexibility. Later I'll show how modularity allows for different implementations for different groups (or homes, as in the example that I use).

Lindblom called this method of doing things in small steps “incrementalism.” His paper was published in the major journal of his field, *Public Administration Review*, with the title “The Science of ‘Muddling Through.’” It was the most frequently cited paper in that journal for quite a long time.

Small steps do not ignite the passions as much as large ones, so they can often be approved. Moreover, success in small steps simplifies the approval process for future steps, whereas failure one of the small steps does not lead to failure of the entire effort. The operations don't have to be perfect: they simply need to be approximations to the desired end result, to be “good enough,” or, in Herbert Simon's terms, they should “satisfice” rather than optimize. In 2015, Jonathan Bendor wrote a useful review of the application of the incremental approach to large systems, “Incrementalism: Dead yet Flourishing.”

This approach requires a different design philosophy than might be used when considering the project as a whole. Now the design must be modular, with multiple small, relatively independent parts and linkages that are designed for flexibility. The end result may not to be as comprehensive as the one idealistic total proposal, but at least some change and improvement will occur.

Remember the discussion on foreign aid in chapter 22? There I described the nomenclature of “planners”—who love large projects—and “searchers”—who like small ones. Searchers were defined to be “on the lookout for favorable opportunities to solve problems—any problem no matter how big or small, whose solution will benefit themselves or others. Searchers must learn enough about each little problem to solve it, which means they must get feedback from the people affected by the problem.” That is an excellent description of what Lindblom meant by “muddling through.”

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Where Incrementalism (Muddling Through) Fails

Slow approaches, such as those championed by the proponents of incrementalism (muddling through), conflict both with the way organizations allocate funds as well as with their lack of patience for results. I discuss these two issues separately because they reflect two different pressures: funding and time.

Funding Pressures

It often is easier to get large sums of money than small sums. Why? Different people are involved in the large decisions. People who give out large amounts of money are higher up in the chain of command. They are used to making decisions about expensive projects. They will ask questions, but usually about the financial projections. When small amounts of money are requested, the people awarding the money are lower in the organizational hierarchy of decision makers. They are more likely to read proposals with care than are the people higher up in the organization and to question numerous small, inconsequential things in great detail.

Reporting on the progress and finances of projects to the funders is always a great burden, but the burden is similar whether a project is large or small. As a result, people whose livelihood depends on grants, whether from the government, companies, or foundations, always prefer one large grant over numerous small ones, even when the total sum received is the same.

With large proposals, oftentimes the larger they are, the more audacious the claims, the easier it is to get the money. Why?

Because all that is needed is for one very high-level person to fall in love with the concept. If this is the case, couldn't the community accept the single large grant of money but still do incremental development? In theory yes, but quite often in practice no. Why not? Ah, here is where time pressures arise.

Time Pressures

Hofstadter's Law: It always takes longer than you expect, even when you take into account Hofstadter's Law.
—Doug R. Hofstadter, *Gödel, Escher, Bach: An Eternal Golden Braid*

Norman's Law: The day the product team is announced, it is behind schedule and over its budget.
—Don Norman, *The Design of Everyday Things*, rev. and exp. edition

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People who fund a project often have great enthusiasm for it, especially for big, audacious projects that require a lot of money. They must be enthusiastic, else they would not have funded it. And yes, they have been briefed numerous times on the complexity, on how long it will take to see meaningful results, but still the funders get anxious. What has been happening? How come we haven't seen any results yet? Sometimes they will report having visited the project and seeing only a few small efforts being made.

Incrementalism is a superior strategy. It is small, flexible, and more likely to succeed. Quality takes time. New ideas take a long time to mature enough to be evaluated and then even longer to become usable. But in this modern world of continual financial and economic pressures, immediate results are the key to success. Incrementalism is a superior strategy, but it often fails to meet the political requirements of funding large projects.

The Conflicting Pressures: A Big Project versus Numerous Small Ones

When a serious societal issue reaches the attention of political powers, there is a desire for a full-scale attack on the problem. Whether it is hunger, homelessness, health, education, or transportation, once the problem is sufficiently large that it is brought to the attention of decision makers, the pressure is on them to provide an immediate answer. This is why large foundations are called in or, worse, major consulting firms. These people in turn call in the experts. The result is a huge, expensive report, recommending a huge, expensive project.

The chance for funding occurs only once. The politicians need to be able to tell the citizens of the area that the problem will be solved. If the incremental approach requires continual new grants, even if small, not only is that a lot of work for the requester and the evaluator, but the stream of money could end at any point. The temptation is to go for the big money, which means a big project, even though it is slated to fail. Is it possible to get all the money but then to proceed incrementally with small projects? Maybe, but it can be difficult politically. Politics plays a major role here: politics cannot be avoided when dealing with major societal issues. But let me try anyway: I suggest a merger of the one large project with a multitude of small, incremental “searcher” steps.

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Incremental Modular Design

Although I strongly recommend keeping projects small and incremental, maintaining flexibility to change as project requirements change, sometimes this approach is simply not possible. Funders and sponsoring agencies often prefer one large homogeneous project. How can this conflict be resolved?

Software-development projects have this same predicament, and as a result developers have created ways of working on large complex projects in small incremental steps. Societal projects have different variables than software ones, especially the complex mix of stakeholders ranging from community members, aid groups, and local foundations to government and large bureaucratic agencies. Nevertheless, for community projects I suggest an approach that borrows from the world of software as well as from other strategies: I call the resulting mix of methods "incremental modular design." At its heart is the software-development method called "Agile," which uses multiple small modules that are assembled to fulfill the requirements of the large, complex project being worked on.

Here is how incremental modular design unfolds. Everyone—the sponsoring organization, the communities, and all the stakeholders—agrees on the goals of the large project. They also agree that to maintain flexibility and to ensure that the project is indeed going in the correct direction, it must be divided into many small modules, each module being the smallest size that can actually produce a useful result. Software designers call this result a "minimum viable product," or MVP.

Because each module is small, it can be implemented relatively quickly, enabling its performance and utility to be assessed. Any concerns or inappropriate behavior can be rapidly modified. Because each module actually works to deliver some value (the MVP philosophy), it is relatively easy for the system managers and funding agencies to see progress. By being small, it avoids the large bureaucracies and the major political arguments. And because it is modular, it enables considerable flexibility.

Each module is constructed by a team effort in a process called a “sprint.” One of the important components of modular design and its implementation is a strict set of standards about the nature of the module's inputs and outputs. An important part of the implementation process is that each module is tested as soon as it is completed to ensure quality standards are met. The team develops the tests before the module construction begins, and the sprint is not considered over until the module passes the tests.

One powerful principle from object-oriented programming languages is used here. The performance requirements for each module are restricted to the inputs required and the outputs delivered. What goes on inside the module is of little concern as long as the input and output specifications are met. If the focus is only on the standardization of inputs and outputs, each module can be modified, rebuilt, or even replaced by a new module without the need for permission or revision of plans. From the system's point of view, as long as the inputs and outputs meet the specifications, the change in internal operation is irrelevant.

As I noted earlier, in most Agile assemblages each module is developed through a process called a “sprint.” In software, sprints are done quickly, in days or a few weeks. For the kind of large, community projects we are talking about, even small modules can take much longer than modules created for computer programming, in part because they often require collaborative efforts from different groups and may possibly rely on human activities inside the module. It is easier to develop procedures for the behavior of semiconductor circuits and software modules than for human behavior, so I suggest allowing modules to spend months in development. The Agile process also contains numerous guidelines about team organization and meeting structures, which I skip over here. These guidelines will require modification to fit the precise needs of the project.

Flexibility is important for two reasons. First, it is required to determine whether the system design is truly appropriate for the need. The advantage of Agile is that even if a ten-year project is the goal, it is not necessary to wait until the end to determine if it will meet the actual requirements. Instead, as each of the MVP modules is completed, it can be assessed. As the number of modules increases, the assessment can be more precise, which allows for continual reassessment of the overall project plan. Second, as goals and requirements change, the modular approach allows for rapid and smooth adjustment. And, of course, as the number of modules increases, the operations can move from "minimum viability" to "desired viability."

Traditional project management, often called a "waterfall process" because it tends to proceed in a linear fashion, has many virtues, especially in ensuring that the project does not lose its focus on the final goals and keeps to its schedule and budget. Modern waterfall techniques divide the project into different stages, where each stage is assessed by management to see how well it is progressing. This type of management can be combined with the incremental modular design approach so that each stage of the work is done using Agile techniques via small modules. This combination allows goals and requirements to be under continual adjustment, something that has traditionally been difficult with large projects, where even slight adjustments can cause massive delays and spending. At the same time, it maintains a top-down focus on meeting the final requirements (even as they are modified), something that can get lost in a traditional Agile project development.

Summarizing the Properties of Incremental Modular Design

Incremental modular design has four requirements:

- The project is divided into relatively small modules.
- The input and output requirements are clearly stated.
- Each module produces a useful result, certified by formal testing to ensure that it meets the input and output

requirements.

- The internal workings are not critical as long as the requirements are met.

The design yields several advantages:

- Each module can be replaced without harm to the system as long as the replacement satisfies the input and output requirements.
- Each module can vary internally to be appropriate to local cultures, thus providing equity and fairness to all of the community.
- The modularity that results from this process provides great flexibility to the project, allowing for goals and other requirements to be changed as necessary throughout the project.

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When Large, Multidisciplinary Projects Are Necessary

Even if everyone agrees that large, multidisciplinary projects with their accompanying large budgets, long timescales, and bureaucratic complexities are not ideal, sometimes for all the reasons stated in chapter 26 (and probably others as well) the funders may reject the modular approach and insist on a more traditional management structure for the large, expensive project under consideration. What then?

The design profession has very little experience with these large projects. However, as I was writing this book, Peter Jones and Kristel van Ael sent me their new book *Design Journeys Through Complex Systems: Practice Tools for Systemic Design*. (Peter Jones was a participant in the DesignX conference in Shanghai and has long been involved with the Systemic Design Network and its series of conferences.) It is an excellent book, and its proposals need to be studied, taught, and applied. In this book, the two authors present a process and recommendations for taming the otherwise overwhelming complexity of methods and meetings into a coherent, sense-making process that sets the standard for these new skills and methods. The process isn't easy. The major problems lie in the politics and organizational pressures. As a result, the authors divide the work into four different locations where co-creation—that is, creative solutions from all the stakeholders—can be developed, shared, and implemented. They call the four locations Lab, Studio, Arena, and Agora. They have also developed a seven-stage process that works its way through five different styles of workshops. Jones and van Ael have done a masterful job of proposing a process to manage the issues encountered in managing big projects. I strongly recommend their book.

Does this sound complex? Yes, it is. Why the complexity? Because social issues involve societal systems, governance, and a huge variety of people and their opinions, beliefs, activities, power, and influence. The difficulties posed by the large number of groups of stakeholders, each group with its own perspectives and requirements, leads to very complex decision-making. Moreover, during the extensive time frame for these projects, the people and the organizations change, altering the requirements and resources. Managing and accommodating the conflicting requirements of these large groups becomes the most difficult part of the project. This process is complex because the problems being faced are complex. As I state in *Living with Complexity*, "Our tools must match the complexity of the world."

So which process is better, the large, complex organizational structure described by Jones and van Ael or the large number of small, incremental projects detailed in chapter 27's description of the incremental modular design process? The answer? It all depends. That is, different requirements, funders, and political constituencies will require different approaches. Sometimes one will be best, sometimes the other. And sometimes it will require a combination of the two. In some complex situations, especially where the mix of stakeholders cannot agree among themselves what the project is about, or when in the middle of the project the cast of characters changes (perhaps a new mayor is elected), neither approach will work. The two approaches follow similar overarching principles, so it may be possible to merge the two. More importantly neither has sufficient experience attacking societal problems to be able to fully evaluate them yet. As each approach is tried in different applications, each will most likely have to be modified to be appropriate for the particular requirements of the project. Over time, each of these approaches will evolve and new methods will arise.

The Role of the Designer in Large, Multidisciplinary Projects

Today's designers have many truly critical skills: design thinking, the awareness to heed people's and societal groups' needs, and a unique mix of creativity and practicality. These traits allow novel, creative ideas to emerge, designs that can be built at an acceptable cost using existing technologies, are highly functional, robust, and reliable, and can be maintained and repaired by the communities where they are applied.

How can design play a role in complex sociotechnical systems such as those identified in the UN Sustainable Development Goals? The techniques of humanity-centered design are extremely well tailored for these large, sociotechnical systems. They provide a way of thinking that can be applied to almost any project, from the traditional design activities of creating products, services, and experiences to developing company strategy and organization, creating product families and platforms, modifying the company to work within the circular economy by reconsidering materials and supply chains, and so on—all very different activities. Design processes can be applied to governments, education, and recreation. There are no limits. One important theme is the emphasis on products, services, procedures, and concepts intended to enhance humanity.

Why is the field of design relevant when so many other disciplines are already at work on these issues? Because design is unique in the breadth of its coverage, especially with its interest in optimizing for the benefit of the people, in collaborative, community-based design, in systems thinking, and in proceeding through prototypes, tests, trials, and modifications based on early results. Designers, of course, cannot do these tasks alone: they must join forces with those who work in many other disciplines and teams, while providing a powerful framework for addressing the issues.

Designers as Orchestral Conductors

In an email to me on May 28, 2021, Gianandrea Giacoma and Sigurdur Thorsteinsson expressed a great metaphor for design: “Design is not the best musical instrument for every situation, but we believe it is the best candidate for the role of orchestral conductor.” Giacoma is an Italian psychologist working in design with Thorsteinsson, an Icelandic designer, in Milan, Italy. They are part of a fascinating research group run jointly between MIT in Cambridge, Massachusetts, and the Politecnico di Milano in Italy. They told me that current designers have the appropriate skills for many tasks, but extremely complex problems that involve multiple disciplines from science, technology, health care, and the humanities require new types of designers with different training.

In these very large projects, they suggested, in addition to performing the traditional role of helping the community and the other disciplines construct the design of the system, design can act like the conductor of a symphony orchestra. Modern orchestras consist of around 100 players, some playing the exact same music, but others with different musical parts, even if they play the same instrument as those near them. The conductor's real work occurs behind the scenes, during rehearsals. Musical scores show what notes are to be played but not the manner of playing, the phrasing, tempo, and especially the emotional feelings to express. Although different musicians may have their own interpretations, it is the job of the conductor to determine just *how* the piece will be played. The conductor's job during rehearsals is to ensure that all the members of the orchestra understand the overall theme and during the performance keep in harmony. This is made more difficult not only because players prefer their own interpretations but also because the players cannot hear all the others, and even the ones they do hear are usually just the ones adjacent to them. The conductor has to ensure that the sound that reaches the audience is balanced.

During a performance, the conductor's role is to help with the synchrony and sound balance of the instruments and to put on a performance for the enjoyment of the audience. Yes, I mean that there are two performances going on: that of the orchestra and that of the conductor. The conductor's gestures, body action, and grimaces are partially meant for the orchestra but often more for the enjoyment of the audience, and in today's era of televised performances the conductor's performance is an essential part of the entertainment. My colleagues in the Music Department at my university have told me, half seriously, half jokingly, that if conductors have done the rehearsals properly, there is little need for them to be there during the performance.

How is this metaphor relevant to design? Any large design project, especially those dealing with complex societal issues, requires synchronizing and balancing the opinions and talents of the many disciplines involved in the project and the needs and wants of the people for whom the project is being done. There can be hundreds of people involved (thousands in large projects), and it is essential that the different parts be done in the necessary sequence so that they can be brought together smoothly at the end. The design team quite often is the only participant that has a unique overall view of the project. Designers do not have the technical skills of all the other disciplines, but because of this overall understanding they are in a good position to orchestrate the organization, structure, and activities of the many different experts in the different disciplines necessary for the project.

To take a simple example, difficulties may arise in the implementation process of the designed work, requiring some changes. They might include a cost issue, the lack of availability or delivery delay of some critical component, or even a new set of requirements or constraints. Many disciplines might be required to restructure the project. The key knowledge that the designers possess is a deep understanding of the reason for the project and the needs of the community that it must serve. This higher level of understanding can ensure that whatever new procedure is selected, it is still compatible with the project's goals. This may sound like an obvious requirement, but it is not uncommon for a midproject modification to interfere with a project's purpose. Giacoma and Thorsteinsson are absolutely correct: designers should be conductors—facilitators, supervisors, and mentors.

“Why design?” you may ask. “Aren't there other fields that lay claim to this expertise—for example, product management?”

“Yes,” I would answer, “there are other fields,” and now I would use one of the arguments that Roberto Verganti and I proposed for design in the *Harvard Business Review*: “Yes and, but.”

The “Yes and” part of the answer is this: **Yes**, there are many fields that are good at managing large projects, **and** design should partner with them. Why do we need design? For all the reasons that I just described regarding the role of the design team as a conductor. Design has one overarching focus that is quite unique and essential when dealing with projects that address societal issues: the needs and goals of both the project and the people, the target population with whom the project is being done.

That's the “yes” part of the answer, and now comes the important “but.” **But** most designers cannot fulfill the role of conductors. In most current educational programs for design, designers are seldom taught how to work collaboratively and effectively in multidisciplinary teams where most of the participants are not designers. Even when designers receive some training with people outside of design, those people are from business or engineering, the “usual” disciplines involved in such projects. As a result, designers don't learn the managerial skills required to work with many different subject-matter experts, especially when these experts may give conflicting advice and recommend activities that would negatively impact the schedule or budget. That's what conductors are for: to restore harmony and progress without losing the participants’ trust.

Accountants and politicians are essential to the success of any large project, but project leaders often view their questions and requirements as disruptive. The various disciplines are often insensitive as to the costs of their activities, however, so accountants are necessary to ensure that the project does not exceed the amount of money allocated. Politicians have to continually act as communicators between their constituents and the project, often demanding things that were originally not part of the project, perhaps requesting a change in goals because of their constituents’ persistent objectives. Neither party is an enemy: they simply have different jobs to do. The project leader has to learn to work with these groups in part by maintaining friendship, in part by helping the project workers understand the need for keeping within budgets and responding to the voices of the community. Is this task easy? No. Is it necessary? Yes.

Forming Partnerships with Project Management

The field of project management (and the subfield called "complex project management") is responsible for ensuring that projects get carried out properly. This is an enormous responsibility for the kinds of large, complex projects I am talking about. Project managers must work with a wide range of people in multidisciplinary settings. They are especially skilled at ensuring that all the details of a large project are carried out with high quality, on schedule, and on budget. These are skills that most designers do not have, and the best way to ensure that a project makes use of these skills is for designers to partner with project management.

The requirements for designers and project managers are synergistic. The designers' goal is to ensure that the project actually meets the needs of the people for whom it is designed—the target population and all the many stakeholders—while the project managers focus on proper completion. The two disciplines of design and project management are therefore tasked with different aspects of the same goal. They need to learn to become a collaborative team. To my knowledge, such collaboration is not yet taught to students in either field.

Project management, for example, is responsible for the planning, budgets, timetables, and schedules for the many different operations required to complete the project. This includes managing the supply chain of parts and supplies, which in turn, includes the hiring, training, and management of the workers. With large projects, it is essential to have qualified project managers.

The design team specifies the structure and details of the project implementation. The project manager's job is to ensure that the project actually gets done. But what happens when there are difficulties, impasses, when the design turns out not to work or when the budget suddenly changes? In large projects that have huge budgets (measured in billions of euros or dollars) and take considerable time to complete (years or even decades), goals will change, and budgets will change. Even the politicians involved will change, with the new ones not necessarily sympathizing with the project or perhaps becoming too sympathetic and urging all sorts of radical "enhancements."

When changes are required, it is best to let those who designed the system in the first place make the changes in collaboration with the community for whom the project is intended. These groups understand all the numerous constraints and trade-offs involved. All too often, the changes are done by the working crew, and the original designers are never consulted. For small modifications, the latter approach is often satisfactory, but not for major ones.

In addition, many managers and administrative stakeholders are very concerned with optimization of time and money. That is all very well, but what about the people for whom the project was intended to serve? It will fall upon the designers to champion the end users of the project and ensure that their needs are always considered. If changes must be made, the end-user population should be involved in deciding how to carry out the changes in a manner that is satisfactory to them.

Many proposals for large-scale, societal projects optimize the recommendations for efficiency, productivity, cost, or reliability, not recognizing that these measures often take a toll on the people involved in the system. They also end up producing the opposite effects: reducing efficiency and productivity in the medium to long run while increasing cost. Many disciplines are good at problem solving, but few ask whether they are solving the correct problems. Well, that's what designers are trained to do: when given a problem, they ask, "Why is this the problem?" When given a proposed solution, they ask, "Why?" Designers can be difficult to work with because they are always questioning. But questioning can lead to better ideas.

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Dealing with Scale

The word *scale* has many different definitions. In the world of products, scale frequently has to do with the size of the item or the size of the market that is being addressed. The word *scale* is used in the world of business to represent the relative size of the business and the company's workforce. “To scale up” (often simplified as “to scale”) is to increase the size by a significant amount. Scaling up can be difficult. A company that produces only a few hundred items has to undergo massive change to be able to produce thousands, let alone millions.

Scale is one of the critical beginnings of the mechanization of society. Many of the ailments of modern society can be traced to the problem of supplying food, transport, clothing, shelter, education, and all the necessities and nonessentials of life to a large population. Scale probably started as people began to live in larger communities and introduced the first early stages of specialization in making clothes, cooking, hunting, fishing, farming, making tools, and so on. Each activity had to scale up from supporting only a small group of people to supporting large numbers. Agriculture is perhaps the earliest example: crops and animals were domesticated, and land was deliberately tilled, planted, and harvested to serve an ever-increasing number of people. The increased scale of agriculture destroyed the forests as trees were cut down to clear fields for planting. The quality of the soil was also harmed as the constant tilling and growing of the same plants season after season destroyed the soil's nutrients. Cities amplify the problems of scale, for they concentrate many people into a small area, leading to social problems and health issues as well as an increased need for food, clean water, waste disposal, and so on.

Scaling up from small to large is a daunting task. Some companies have grown larger and larger to the point that they exceed the net worth of many nations. Some manufacturing and distribution companies employ more than one million people. Today, large international companies supply products to the world. Some companies have hundreds of millions of customers, and large nations such as China and India have to provide services to billions of their citizens.

Scale has its problems. Management is difficult, for no single person can understand all that is happening in large organizations. There is a rule of thumb in management that no person should have more than roughly ten direct "reports" (workers who report to a manager), which means that as the size of a company grows, the number of managers grows logarithmically with the number of employees, with every tenfold increase in employees requiring a new level of management. Every thousand workers require one hundred managers, and those one hundred in turn require ten managers, who in turn have one manager: four levels of employees and managers for a thousand workers.

The limited number of people that any one person can manage automatically leads to a hierarchical structure of control, and as the number of employees grows, the traditional way of maintaining control is to develop official policies—more and more of them to the point that so many policies are needed to control the development, distribution, and overseeing of the policies that some of the policies will be about how and when to develop policies. This huge bureaucratic structure impedes the development of novel ideas.

Many companies have tried to develop other forms of management to avoid the stifling effect of large bureaucracies, but alternative structures where employees have less supervision and more freedom (in other words, loose coupling) are rare. The two oldest examples of hierarchical organizations are armies and religions. It is noteworthy that small, nimble startup companies can sometimes outperform large, slow-moving companies and even succeed in overtaking the large companies. Then the small, nimble company itself starts becoming large and ponderous as it scales up. Similarly, large, well-organized, and well-controlled armies can sometimes be defeated by much smaller forces of small, self-managing groups of warriors through guerilla warfare.

Scale implicates the world of monoculture, for it is easier and less expensive to make clothing, appliances, transportation, gardening, and woodworking tools the same across the entire world. As scale increases, the need for efficiency increases, which tends to dehumanize the workers.

I'm telling you here, though, that scaling up is the way of the past. Now consider a future in which we instead *scale down*.

Scaling Down: The Rise of the Small

Can the ailments of scale be avoided? Probably not, because with 8 billion people on the planet, a number growing rapidly, the requirements of feeding, housing, educating, and caring for everyone requires massive organizations. Can new technologies help, especially communication technologies that allow for flexible ways of communicating with and keeping track of multiple people? Jeremy Myerson, founding director of the Helen Hamlyn Centre for Design at the Royal College of Art in London (now retired), yet another participant in the DesignX seminars, has proposed that the way to scale up is to scale down. Scaling down, says Myerson, requires a new mindset. Among other things, it means “a participatory mindset, which means creating with people rather than for them.” He argues that designs should be for individuals, not abstractions. Scaling up, he points out, usually requires designers to avoid friction to maximize positive acceptance and tolerance. If the aim is to design for a small number of individuals, then the design can emphasize positive aspects, building on what individual people and communities offer. Scaling down removes many constraints on design, allowing for more freedom and creativity.

I argue that we can combine incremental modular design with Myerson's proposal to stop designing for the masses and instead scale down by designing for the small, for individuals. How? New technologies now allow a large number of small, independent firms to serve people's needs. They would be local, so the customers could go directly to the company. New additive manufacturing methods (for example, 3D printing) are inexpensive enough for individuals to own them. These accessible manufacturing tools can be coupled with plans that are easily available on the internet, some free, some at low cost, and that can be fed directly to the printers, which will then construct the needed item. Optimally, the creators of the plans would be available to answer questions or to customize the plans for specific needs.

There are other exciting new developments. Small farms using hydroponics, or so-called vertical farms, allow the plants to be hung on vertical supports so that food can be grown in a relatively small space. New telecommunication methods, especially the development of video conferencing, minimize the need to travel. People can live wherever they wish, eat food grown locally, and buy from small manufacturing shops available in the community or even in homes. People can do many of the tasks required by their jobs from their homes, avoiding lengthy, polluting commutes to and from work.

Small-scale, customized manufacturing moves away from the mass manufacturing of physical products with its need for huge, expensive manufacturing tools and efficient assembly lines. Instead, we could have small, local developers and designers. Imagine small machine tools and more use of additive manufacturing such as 3D printers, laser cutters, and many new developments for producing physical objects quickly in small numbers. This approach maximizes flexibility, allowing designs to be made for individuals. Similar ideas work for numerous activities. Think about just-in-time learning of a variety of subjects: short, simple lessons available whenever they are needed.

Mass-manufacturing will still exist for many products because it is by far the most efficient way of making them at high quality and low cost. But as new technologies and means of making are developed, the need for mass-produced objects will surely diminish. Mass production yields inexpensive items, but only if they are produced in huge quantities because this type of production requires huge investments. The machines and specialized dies and tools are expensive. The supply chain is large and complex, with high transportation charges, which is bad news for the planet. Local manufacturing avoids such wasteful activities.

New technologies might very well allow us to overcome the problems caused by the old technologies of mass production and the assembly line. We might be able to make things in small lots, crafted individually for the needs and desires of the audience. The designs of these small batches would be good matches for people instead of today's homogenization of the world, with everyone, everywhere wearing the same clothes, eating the same foods, learning similar things from their schools. We need to scale up the creation of goods to cover the world, but do so by scaling up the number of different efforts while scaling down the size of each effort. Myerson suggests five principles to follow:

1. Cultivate a Participatory Mindset—Not an Expert One
2. Make the Process Design-Infused—Not Design-Led
3. Design for People—Not Personas
4. Aim for Engagement—Not Abstraction
5. Build on Assets—Don't Just Minimize Deficits

Today's world has large, global companies that serve millions or even billions of people. Let's reverse this trend: have small, local companies, each one serving a small number of satisfied customers. In other words, scale up the number of small, scaled-down businesses and activities: lots of small, personalized businesses, perhaps collaborating with each other in the purchase of raw materials and sharing of ideas. A particularly attractive feature of small, scaled-down businesses is that it puts power back into the hands of local individuals within their own communities.

Technologists tend to think that all problems in the world can be solved with yet more technology. Challenging that belief allows everything else to change for the better because more technology can also mean more headaches, more requirements for maintenance, and more repairs. Because the world contains so many different cultures, each with its own belief systems, let's scale down the societies of the world instead of continuing to follow the destructive precepts of modernity. Why not create a pluriverse of societies, where the differences between them are celebrated, and each society accepts and tolerates the others. Designers need to allow multiple designs, or “designs for the pluriverse,” as Arturo Escobar puts it in the title of his 2018 book. Note that Escobar uses the word *designs*—the plural form of design. There must be different designs for different groups. Mass production assumes one design for all: we must change that.

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Design: Necessary but Not Sufficient

The major issues facing the world today can be solved only by a concerted, multidisciplinary, multifaceted mobilization of all peoples: a pluriverse of approaches, of styles, of methods. No single discipline can solve these problems; all of the disciplines are necessary.

One designer told me that he was asked to work on a health problem. After much investigation, he concluded that the real issue was simple: Americans' lack of good health care and public health facilities and practices. The United States needs to fix its health-care system. "Yes," I responded. "Are you going to address health care?" He said no. After all, he said, "I pointed out that the solution is political, not design." So he stopped working on the problem. I disagree with both parts of his conclusion. First, the solution is not simple, and, second, the problem *is* a design problem.

Health care is an amazingly difficult societal problem. This is true in all countries of the world, even those with advanced social programs for payment and support. I agree that the system in the United States is badly deficient, but this problem can't be easily resolved. There are numerous complexities, each involving a different dimension of interests and concerns. Moreover, payment for services is not the only issue. The United States has excellent physicians and hospitals. It spends more money per person on health care than other comparable nations but is nevertheless rated low in health outcomes.

This is absolutely a design problem. Just because a problem is difficult and involves politics and economics doesn't mean designers should give up on it. Designers need multiple collaborators to tackle health care, but that is no different than tackling any other complex problem. Complex sociotechnical systems always require multiple collaborators. Designers will have to stretch their knowledge and understanding, learn new topics and new approaches. Yes, the medical system issues must be fixed by political action. Yes, health care is also an economic issue because costs are driven up by the high costs of pharmaceuticals and health insurance, much higher than in nations that have better health care. Work on these issues obviously requires a different sort of skill than that of the traditional designer, but all societal issues require a mix of skills that are different from those traditionally used in design—skills such as economics, social sciences, technology, business, politics, and new methods within design.

Many of today's designers believe that their only responsibility is to do “the design,” which means the plan. They often have little say in selecting what problem they will work on or on the goals and constraints of the problem. And then when they finish their plan, the design, they pass it on to those who are charged with implementation. Is it any wonder that designers complain that their ideas were badly implemented or that they were not given the proper goals or constraints?

Today, in the twenty-first century, designers must face these new challenges. These complex issues will have new kinds of goals and constraints that will require the involvement of people in many disciplines. Does this make the process unwieldy? Yes, but it is design's function to transform an unwieldy, contentious process into a smooth-running, collaborative one. After all, everyone involved should have the same goal—successful health care in this example—even though they may have different philosophies and approaches. By focusing on the common goal, by agreeing to work together, the different players can come to agreement. Will everyone be satisfied? Of course not. Quite often the goals and constraints are contradictory, so that it is impossible to satisfy them all. Compromise and shared interests can lead to a system that all agree is better than what exists, and because of the dynamic nature of complex systems, the system can evolve and slowly improve over time.

Most of the serious issues of the world are within complex systems that combine economic, legal, educational, and political dimensions. But who else should deal with these issues if not designers? During the COVID-19 outbreak, many people refused to wear masks or get vaccinated. Are those issues for designers? I say yes. Preparing for future pandemics is another task for designers.

Consider climate change. Technologists are ready to respond with their billion- or trillion-dollar projects, and although they may succeed in countering the major issues of climate change, all that effort will go toward fighting the symptoms, not the causes. Fighting the symptoms of climate change with technology is difficult, but getting at the root causes is far more complicated, for it involves how the wealthy live and work at the expense of others in all the nations of the world. To stop this excess requires changing our entire structure of government, of the way we live, of the economics of life, and of the politics that lies beneath them and supports the disastrous policies that got us here. Everything has to change.

To drop out because the problem starts to involve politics or economics, supply-chain management, or labor relations, none of which is part of the traditional design training, is to guarantee failure. How do we mobilize the proper groups to work together for the betterment of all? Now that's a worthy design problem.

What Design Has to Offer

What is it that the design profession can add? Part of the answer is that design's major weakness is also its greatest strength. The weakness? Design has little content expertise outside of the design process itself. Its greatest strength? Designers have no prior knowledge or preconceived notions to get in the way or to suggest solutions to issues far too quickly. When called in to help with a problem, designers have to learn entire new fields. Designers will have to find the relevant subject-matter experts and form partnerships. But because they know so little about these areas of expertise, they have to ask many questions, usually of the type that are generally called “stupid.” Why stupid? Because to the experts they seem so obvious that no intelligent person would ever ask them.

Stupid questions are often the most powerful questions of all, however, because they often get at the fundamental assumptions of a field—so fundamental that nobody in the field has ever thought to question them. Here, the Japanese system of the Five Whys is useful. When told that a question is stupid, the response has to be to ask why: "Why is it stupid?" This often leads to the answer: "Because everyone knows the answer." That's interesting, so ask again: "Why is that?" When forced to explain why an assumption is made and to give reasons and examples, experts often stumble. The magic happens when eventually someone answers the question "Why do you do it that way?" with "We have always done it that way." This is the key spot because the next question should then be "Is there some other way we might do it?" Once that point is reached ("we have always done it that way"), a wonderful opportunity has arisen to challenge the assumptions and the history of behavior. Here is where the design method shines: it rethinks the issues and, surprisingly often, is able to point out a different way of doing things, a way so different that the original problem simply disappears.

In chapter 19 (in the "Causality" section), I point out the limitations of the Five Whys process: it assumes a linear system of causality, where there is one major causal path. With complex systems, there are multiple factors. This means that the designers must never be satisfied with the discovery of a single underlying causal factor. Designers must be sensitive to the complexity of systems and track down the most important critical paths. Nonetheless, the continual probing of experts with "why" questions can be a powerful tool.

The second major virtue of designers is the ability to view the issue from the point of view of humanity, ensuring that all the system-wide implications are considered and that the result above all fits human needs, not just the traditional time, efficiency, productivity, and cost requirements. Most projects are done for some purpose or for some high-level goal. Our twenty-first-century designers do projects with and by people.

And finally, modern designers do not force solutions upon others. We let the solutions be driven by the community. Designers work with the community as mentors, advisers, and facilitators. Community members understand their own problems and have often devised simple effective approaches to deal with them, but they lack the resources to go beyond attacking the symptoms. That is where outside people (designers) can help.

That's what makes design unique: it is necessary, but not sufficient. It is necessary enough to have a major voice, but because design alone is not sufficient, designers must work in partnership with the other experts and critical stakeholders, especially the local governments and the funding agencies.

As Pieter Jan Stappers and I state in our DesignX paper,

> A difference between the design point of view and that of the traditional analyst can be seen in the language used to describe the same behavior. Traditional analyses often blame system failures upon human error, such as “lack of attention” or “failure to follow procedures.” The solution is admonishment or retraining. To the designer, however, these are not causes: they are symptoms of underlying difficulties. From the design perspective, the proper solution is to discover the underlying causes of the human behavior and redesign the system so as to eliminate them.

V
Human Behavior
The Major Challenge

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Why Change Is Difficult

Change: The Most Difficult Task of All

The world has multiple political views that range across the extremes, with little common ground and an increasing reluctance among people with different opinions to communicate or even listen, let alone collaborate: Why do we think change is possible?

Trying to change people's beliefs and actions is extremely difficult, and although it can be done, history has shown that it takes a very long time. For example, the first public campaign to give women in the United States the right to vote started in 1848, but it wasn't until 1920 that the US Constitution was amended to permit this: 72 years (probably more because the effort clearly started before the first national campaign). While the United States was a laggard (New Zealand gave women the right to vote in 1893, when it was still a British colony), some nations still do not permit it.

Why is change so difficult? People dislike changes, especially in the way they live. When people are born into an existing society and experience no other way of living, the long-held beliefs and behaviors of that society seem so natural and reasonable that they seldom question or even think it reasonable to change. Underlying this problem are the fundamental properties of human cognition. People naturally look for simple explanations; they have great difficulty understanding what they cannot perceive, and, as a result, they find it easy and often compelling to listen to and believe the voices of tyrants who spread false information to gain power, control, and benefits for themselves at the expense of society. People often use simplistic and linear logic and think only of immediate results, not long-term consequences. Some ways of thinking can be overcome through education, but others, such as the belief in linear, simple notions of causality, work so well in dealing with everyday events and behaviors that it is difficult to convince people that not every action has an obvious cause. In a complex system, with its multiple loops, nonlinearities, time delays, and dynamical structure in which the system is continually undergoing change, it simply is not possible to specify precisely what one event or variable led to a particular state.

People do tend to behave lawfully, but that is not the same as behaving logically or behaving for the greater good. The fields of cognitive sciences, psychology, and neuroscience are beginning to work out the mechanisms underlying our thoughts and behavior. But note that human behavior is controlled by numerous factors, so it will always be difficult to predict, dependent on each individual's unique history (that is, path dependent), and dependent on each person's interpretation of a situation. Moreover, human actions and decisions are the result of different mechanisms that operate simultaneously, some consciously and some subconsciously. These decision systems often reach opposing conclusions. Human emotional systems play a major role in controlling beliefs and actions, and they may be in opposition to the cognitive system. Emotions are deployed chemically throughout the body, changing the way the nervous system functions.

In my book *Emotional Design*, I point out that the emotional system is just as powerful and important as our systems for reasoning. Moreover, our reasoning has at least two components. First, there is a fast-reacting system based on reflexive responses to situations, which is also driven by emotions. It is mostly subconscious and uses the emotional systems that I call “visceral” and “behavioral.” Most people believe that they make decisions consciously, with reasoning. I call this conscious system “reflective” and, whereas the subconscious states work rapidly, often in fractions of a second, the conscious stage of reflection can take tens of seconds or even minutes. Moreover, even the slow, reflective stages of thought don't follow traditional logic: logic is an artificial way of reasoning invented by philosophers and mathematicians.

Danny Kahneman, whom we encountered earlier because of his Nobel Prize for the study of economic decision-making, uses a similar analysis in his book *Thinking, Fast and Slow*, but for his purposes, he collapses the two subconscious, emotional stages into one system that he labels “fast.” I needed to distinguish between two stages of subconscious responses. One, the visceral response, is driven by the perception of the state of the world: safe or dangerous. The other, the behavioral response, is involved anytime we perform an action upon the world and the result differs from expectations. Kahneman talks about cognitive decision-making, so he does not have to distinguish these different fast stages. Kahneman's “slow” system is basically the same as the one I call “reflective.” He and I do slightly different analyses because we are interested in different kinds of phenomena.

Emotional values play a very large role in our understanding of and reactions to the world. The human cognitive system seeks understanding, whereas the emotional system provides value judgments. Both are required for intelligent decision-making. However, when time is limited or considerable stress limits the traditional brain processes, the fast system takes over, governed in part by quick assessments by the visceral and behavioral-processing structure of the brain. Alas, responses made via the fast system are often regretted later.

Emotionally, the fast system takes charge whenever time is limited. This happens whenever the situation is perceived to be one of the three D's: difficult, dangerous, or distasteful. These situations lead to a feeling of stress and the necessity to act. Often they are associated with one of the negative emotions, such as fear, anxiety, or anger. When these situations occur, when decisions are made quickly without time spent on reflection, they can be useful, especially in responding to possible danger. Moreover, the automatic reactions are often appropriate for the situation because they represent quick responses to things we have encountered frequently and, in some cases, are predetermined through evolutionary processes. Sometimes, however, responses made in haste without time to consider the implications or alternatives can lead to unfortunate outcomes. Decisions made while angry or stressed, for example, seldom turn out well.

The slow system of reflection happens when we are in a good mood, when there is time to think through issues with care, caution, and concern. These situations are perceived to be friendly, relaxed, and free from danger. The resulting decisions are often thoughtful and constructive, but they too can run into difficulties because nobody can foresee all the possible implications of a decision, nor predict what the future might bring. When we are in a positive mood, we sometimes rely on an overly optimistic view of the world and false perceptions of safety or trust. When in a negative mood, we can respond too quickly, without consideration of the full implications of our actions. We live in an imperfect world, where we must make decisions and take actions with imperfect information, biased to act too quickly in some situations and too slowly in others, with insufficient caution and inappropriate trust in some situations and too much caution and trust in others. We live in a difficult, complex world.

The human mind is both powerful and limited. It has evolved over millennia to operate well within the world, giving human beings the ability to survive, to create great civilizations, and to accomplish many wonderful things. But human belief systems also make it difficult for us to comprehend complex systems with feedback loops that require years to have an impact, where the implications of our actions may not be visible for a long time. The evolutionary forces that have shaped our minds and behaviors work quite well under the assumptions of simple linear causality and an infinite earth. This has led to many benefits, but with an unfortunate, slow, and deleterious impact on the environment. Even though warnings of the dangers that will result from the impending climate change have been with us for decades, it has been difficult to mobilize the world to take these concerns seriously.

People Will Respond to a Disaster but Not to Prevent It

People respond well when the negative impact of a situation is visible. Once a disaster strikes, people are quick to mobilize to repair the damage and help those who are suffering but not to mobilize to prevent it in the first place. Well, climate change is here: When will the governments of the world respond? The current impact of climate change is starting to convince many who previously were skeptical, but, so far, short-term interests still override the ability to make the major changes that are necessary to prevent future disasters. It may be necessary for people to experience even greater calamities before they will respond. And even then it will take years to reverse the continuing and ever-increasing severity of the impact of climate change. The sooner we act, the better our chances of preventing the most serious disasters.

As for epidemics and pandemics, public-health officials have been warning the world about the possibility of pandemics for many decades. Moreover, it isn't as if pandemics are rare: *Wikipedia* lists 18 major pandemics in the world, each of which has resulted in more than one million deaths. In addition, infectious diseases have killed tens and hundreds of millions, including some that killed as much as half the population of Europe and 50 to 80 percent of the population of Mexico. With all that warning plus the history of severe episodes, why did we do so little when the COVID-19 pandemic struck and killed more than 6 million people (as of mid-2022)?

The inappropriate responses to the COVID-19 pandemic may seem to contradict the statement that people respond to a disaster, but in this case the lack of universal response came about because many of the major causes and actions recommended to prevent the disease from spreading, and even the power of vaccinations, were all relatively complex and invisible. First, the coronavirus that was responsible for the disease is invisible. Second, even after exposure, not everyone catches the disease, which made many suspicious of the claims of the medical establishment. And third, even if one is infected, symptoms show up many days later. It is difficult for people to take precautions against an invisible threat that does not impact everyone. These issues are compounded by the powerful, worldwide information networks. Radio, television, and the social media permitted people who disbelieved that the disease existed or in many cases people who simply lied to spread false information about the disease and about public-health officials’ attempts to reduce its impact through physical separation, mask wearing, and vaccination.

Two more factors entered into this complex scenario. First, the immediate scientific recommendations by public-health officials and infectious-disease physicians were for everyone to wear medical masks, maintain at least a 2-meter (6-foot) separation from other people, and close all nonessential businesses to prevent mass exposure. Not only were these recommendations inconvenient, but the closure of businesses also created huge economic hardships. Now people had to decide whether to suffer the immediate results of economic loss or to prevent the encroachment of an invisible disease that many prominent people were calling a hoax.

Second, there were huge racial and ethnic disparities. The US Centers for Disease Control and Prevention (CDC) reported that the COVID-19 pandemic exposed inequity within society: racial and ethnic minority groups, which the CDC defined to include “people of color with a wide variety of backgrounds and experiences,” had disproportionately more COVID-19 cases, hospitalizations, and deaths than nonminority groups. The CDC listed five major causes of the differences: discrimination; lack of health-care access and use; occupation; education, income, and wealth gaps; and poor housing. Economic factors also played a larger role because people with lower incomes and job security suffered disproportionately from the lockdowns and business closures. Finally, the CDC pointed out that historical and current discrimination contributed to minority groups’ mistrust of the health-care system, including distrust of “vaccines, vaccination providers, and institutions that make recommendations for the use of vaccines.” Note that the issues of invisibility and uneven spread played into this second factor as well. Who are people to trust? Who is speaking with the voice of authority, and given the history of discrimination by authorities, why should people trust them?

Voices of authority? It was not possible to distinguish actual authorities from fake ones. The repeated false statements by Donald Trump, who was at that point the president of the United States, and by a few other prominent people didn't help. It was easy for these figures to influence people's beliefs by telling them what they wanted to hear as opposed to the truth. People believe what they wish to believe.

Governments build roads and bridges, railroads and airports, police and fire departments, hospitals and public-health agencies. This creates a huge infrastructure of pipes and wires, some underground, some above ground, to provide water and sewerage, gas and electricity, telephone and internet. But once installed, getting funds and permission to keep them going can be extremely difficult. People are reluctant to support funds for maintenance in part because they do not like to pay for things but also in part because although the benefits of infrastructure are visible, the need for maintenance is not. Why spend money strengthening bridges and dams when that money could be used for so many other valuable needs that seem more pressing? After all, the infrastructure is working just fine: Why is that money needed now? Maintenance can certainly wait.

Yes, it is possible to delay infrastructure maintenance, but eventually the lack will lead to calamity: floods, fires, crashes, loss of drinking water, electricity stoppages, nondelivery of goods, and deaths due to preventable causes—preventable had the infrastructure been maintained.

Why are we like that? People are highly attuned to feedback about the consequences of their actions. Sometimes we simply perceive a result and then believe that whatever event preceded the action is the cause. When something we do leads to a delayed result, it is challenging for us to understand the causal relationship. Many natural systems in the world have feedback mechanisms. However, sometimes the feedback takes place hours, months, or even years after the causal events. The outcome from a lack of maintenance might not reveal itself for decades.

If all these issues are not bad enough, consider the situations where successful actions show no visible result. Here are some examples:

Example 1: Effective safety responses lead to no result. I learned SCUBA diving in a two-week, professional course taught at the Scripps Institute of Oceanography (I needed the training to do research on hearing underwater). Our instructor insisted that we always be safe and that when we thought we were in trouble, especially if we were exhausting ourselves trying to fight the problem while underwater, we should immediately release our weight belt and drop it. (Divers would wear 3–5 kilograms [6–10 pounds] of weights to have neutral buoyancy in the water, compensating for the buoyancy from our bodies and wetsuits.) The problem, he said, was that once we dropped the weights, we would easily get to the surface and to safety, but it would be so easy, we would wonder why we had to drop those expensive weights. As he explained, if you do something to avoid danger, and you do avoid the danger, then you wonder why you had to do it. Worse, your friends might make fun of you—"Ha, you dropped your weight belt!" So we all were taught to congratulate and praise the person who dropped the weights, and the instructor even promised to pay for the new set. That lesson potentially saved lives. We were also required to read accident reports, learning that sometimes divers who had caught fish would drown in just a few feet of water, battered by waves, exhausted, but refusing to let go of the bag of fish or their equipment and weight belts. I interpret this behavior to be a predictable consequence of the emotional value of the fish that had been caught, triggering the fast system of decision-making, while the exhaustion prevented the slow, more rational system from operating.

Example 2: Effective vaccination, use of masks, and social distancing lead to no result. During the COVID-19 pandemic, people were forced to quarantine themselves, to close many businesses, facilities, and schools, and to wear a face mask and keep several meters apart from others. When the emergency was declared to be over, instead of being relieved, many were angry. "My family suffered enormous consequences in income, schooling, business," they would say, "and for what purpose? Nothing happened!" Rationally, the fact that nothing happened is a positive sign that the precautions worked. Emotionally, all the suffering seemed to have had no impact and therefore was meaningless and unnecessary. It is difficult to accept the absence of something as evidence. To make matters worse, the virus kept reemerging, adapting itself to create new variations that restarted the cycle of increased infections, hospitalizations, and deaths. But the emotions that had followed the early announcement of victory over the virus persisted, and so with each cycle of new variations, people were less and less obedient to the ever-changing medical advice.

Example 3: Assessing blame after an accident. When accidents occur, it is common for people to seek a simple explanation, the most common being human error. This makes it easy to find someone to blame: “Aha, we found the culprit!” Invariably, the culprit is simply the last person in a complex chain of events; if any one of those events had not happened, the accident would not have occurred. In such a situation, why would only a single person or event be named as the unique cause of the accident? When it comes to blaming a person, the argument is even less convincing because of what is called “hindsight bias.” Once something has happened, it is easy to invent a story of a simple act that would have prevented it. However, before the event, understanding the situation sufficiently well to be able to do the one correct action is often impossible. During the event, not all the conditions are known. Often there are high stress and time pressure. Yet these variables aren't considered in the after-accident analysis. In many cases, the person being charged did little or nothing relevant to cause the accident but is guilty of being present at the wrong time. Most accidents are part of a complex system of interacting components and events (compounded by faulty design of equipment or procedures). The person who is blamed is often a victim of the system, despite being blamed as the cause.

The moral of the stories. The various dangers facing the earth today are the result of interactions within the complex sociotechnical systems of the planet. They are difficult to understand, even by the experts who have devoted their lives to the task. Given people's relatively impoverished ability to disentangle causal factors in complex situations and their simple beliefs about causality that have been quite successful in guiding human behavior in everyday occurrences, we can now understand why it is so difficult to convince people to take immediate, difficult action to stop the invisible, slow-acting, unintelligible global events that are leading the planet to disaster.

Why should people spend money or take steps to prevent a catastrophe when there are no visible signs of the catastrophe to the average person, just to the experts? Once a catastrophe happens, then there is lots of evidence, so people will respond, but they stop responding as soon as the emergency has been taken care of, which is often before things have returned to normal. Without an emergency, people lose interest. It is difficult to get them to finish the recovery and even more difficult to get them to prepare for the next calamity.

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People Will Mobilize for a Common Goal

In an article subtitled “Finding Meaning in the Climate Fight,” Greg Jackson points out that people will bond and work fervidly for a common goal. Small groups combine for athletic sports. In warfare, millions of people gather together to fight a common enemy. Sacrifices deemed impossible prior to the war become necessary. People literally die to help defeat the common enemy. Why, asks Jackson, can we not have such a bonding of people in the common cause of climate change and the disasters that are already occurring? Why is there so much resistance to the preventative measures required to stop pandemics? Why are we so willing to fight wars and kill one another? Why not wars against disease and against the changes in our climate, wars that would enable us to save lives?

Jackson's interpretation makes sense, but people have different ways of making sense of the world's current difficulties. Perhaps the pandemic was a plot, filled with lies, false information, and deceit. Maybe the pandemic was started by a foreign country or maybe the other political party. Vaccinations, some believe, not only poison people but are part of a plot by pharmaceutical companies. One large conspiracy group explains events quite simply: Satan-worshipping pedophiles control the US government, media, and banks. Is this a small group? No, according to *U.S News and World Report*, a respected magazine, 25 percent of American Republicans agree with those sentiments—around 44 million people.

Everyone makes sense of events in their own way, aided by the rise of modern communication channels that make it easy to spread false information across the world. When people are bombarded by conflicting stories, which should they believe? They follow their friends. They follow their instincts. Facts? They don't matter in this age of fabricated facts. Scientists? Look at how they disagree with one another and how they continually change their recommendations.

One common complaint during the pandemic was that after a year and a half of careful behavior, just when the world thought the COVID-19 crisis was over, a new variant of the virus appeared that infected even those who were already vaccinated, causing the requirements for isolation and masking to be reinstated. “Why,” they asked, “did we do all those activities that destroyed our social lives and caused many of us to lose jobs when in the end nothing was prevented, the pandemic was still with us? Doesn't that prove that we were doing things without any value?” Sometimes the causes of events are complex and invisible. And if activities that might prevent future such events are difficult and expensive and seem to have no perceivable impact for years or decades, why are we surprised when people fail to act?

Can Climate Change Become a Common Goal?

Whatever we do today to combat climate change will not show any result for decades. The scientific community says this is why we must act with urgency—the longer the delay, the worse the problem will become. But ordinary citizens say, “What's the rush?”

To overcome the problems of climate change, many things must change: how people go about their jobs; the educational system; economic measurements; reward structures; the ability of large companies—some wealthier than many nations—to profit by policies that cause misery and harm. The world's economy has been based on short-term profits, large rewards to the heads of companies for these short-term results, and financial markets that perpetuate the quest for immediate gain without consideration of the long-term impact to the companies, to the people, or to the world. Self-interest trumps long-term, responsible action.

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What Must Change?

Climate change is not the only ailment facing the world, and, like climate change, most of the ailments we face are the symptoms of underlying issues, not the causes. To solve these issues, we must attack the causes, and they are many and vast. I list some of them in this chapter, but as you read about them, you will soon think that they seem impossible to change. Things that seem impossible can be done, however: impossible often means just difficult. So let us do the impossible.

The Balance between Work and Family

Today, as a direct result of policies introduced at the dawn of the Industrial Revolution in the 1700s, we find our lives dictated by the workforce requirement that we all congregate together for a full day of work. This workweek is artificial. The number of days in a week has varied across historical time, with some societies not even having the concept of a week. The choice of seven days for a week is attributed to the fact that early astrologers believed there were seven celestial bodies: the Sun (Sunday), Moon (Monday), Mars (Tuesday), Mercury (Wednesday), Jupiter (Thursday), Venus (Friday), and Saturn (Saturday).

If the week is artificial, so too is the workweek. If the number of hours in the day is artificial, so are the number of working hours in the day as well as the number of hours that must be worked per week. Right now, our days are governed by the artificial patterns laid down at the start of the Industrial Revolution. In many Western countries, the work week is five days, with Saturdays and Sundays devoted to religious ceremonies and recreation. Other areas of the world have different conventions. The reduction of the work week from six to five days owes much to Henry Ford and the factory he built to construct the mass-produced Model T automobile at the turn of the twentieth century, but not because he was a benevolent leader. By giving his workers a two-day break (rather than the one-day break that was common then), the workers had more time to travel to the country or surrounding cities, which meant they had more reason to buy automobiles—in particular, Ford automobiles.

Work: Consider Changing Work from a Job to an Occupation

The nature of work and employment is artificial, based primarily on the artificially determined dictates of the Industrial Revolution, although work hours have been shortened since then, and now the normal workweek has been reduced to five days instead of earlier regimes of six or even seven days. Nonetheless, the traditional eight hours of work each day mean that people spend more time at work than with their families. And there are huge disparities in power and reward between the few at the very top and the massive numbers below them. As Kathi Weeks says in the introduction to her scholarly, academic tome on the nature of work, “What is perplexing is less the acceptance of the present reality that one must work to live than the willingness to live for work.”

Right now, the time designated for work is determined through a work-centric philosophy. Suppose this could be shifted to a life-centric one, where workers have more flexibility as to when they do the work and where. The critical variable ought to be whether the required tasks are done and done well. Even better, change the name of this activity from “job” to “occupation”: occupations rather than jobs, occupations that people enjoy doing, where they can take pride in the work they do, in having a voice in what is done.

Work patterns vary considerably across industries and locations in the world. Fixed working hours and days are common for many, but there are an increasing number of occupations where work can be sporadic, as in the “gig” economy, where there are no permanent jobs but instead pick-up employment, freelance work, and short-term contracts. This includes low-paying work such as taxi or rideshare industries. For many, the continual search for employment, often for low wages, can be stressful, whereas for the few who are highly sought after and well paid, the lack of fixed employment gives great flexibility. Some occupations are unpredictable, subject to external conditions such as weather and emergency calls (e.g., massive fires, storms, transportation accidents, and a host of other unexpected and unplanned events).

The demands of the factory are for the factories of old. New means exist today to allow greater freedom and flexibility. Yes, industry will always play important roles in lives, producing clothes, food, the media that inform and educate us, and the many small items and appliances that we use to make our lives more comfortable and stimulating. But much of the work and jobs within these industries could be switched to occupations, with a more collaborative, flexible manner of interacting.

The COVID-19 pandemic, which prevented people from gathering in the workplace, has demonstrated that it is quite possible for many jobs to be done by a distributed workforce, not only in terms of location—frequently in workers’ homes—but also in terms of time. Personal life and work life are now separated, with the daytime hours spent at work and evenings at home. Suppose this was made more flexible in a productive fashion, good for business, good for the people, good for wellness and happiness. Flexibility of time and place gives more autonomy and control to the workers. Different occupations, of course, would have to tailor the flexibility to the requirements of the business; for instance, assembly lines do require people to work together at the same time and place. But do we really need assembly lines? These are a century-old idea of increased efficiency and productivity, but at the cost of creating large numbers of meaningless, repetitive jobs with little flexibility. Workers do the tasks required of them because the economics of today's world requires these activities for them to earn sufficient money to support themselves and their families.

It is time to rethink how people live their lives, why so many people have meaningless jobs, why the work environment destroys families and places an artificial division between work life and family life. Employers might object that giving the workforce more flexibility in their jobs and scheduling might threaten productivity, but they often use a misleading measure of productivity, counting what gets accomplished per hour or per employee but ignoring lost hours due to accidents, errors, sick days, and employee turnover. All of these factors reduce real productivity measures. Specifically, employee turnover has a huge impact, often not measured. When workers leave, not only is the company now missing employees, but there are also extra costs and time delays in finding, hiring, and training the replacements. New employees may require considerable time to come up to speed in their job. These negative sides of high-productivity measures are ignored. Even worse, high employee turnover leads to a positive feedback cycle (i.e., an increase in the size of the problem): it lowers the morale of the remaining workers, which in turn leads to a higher rate of dissatisfaction and then even higher turnover. Treating workers fairly and giving them a say in their activities are good for their lives, good for the company, good for business, and good for the world.

Education: Move from the Rigidity of Today's Education

In addition to rethinking work, it is also time to rethink education. Rethink its purpose, when it should be experienced, and the pedagogical principles behind it. What topics are required for the citizens of the twenty-first century? Why is education concentrated during one's youth? Isn't it critically important throughout life? After all, in today's world new knowledge is continually required.

Once we recognize the value of lifelong learning, we might question the value of degrees. Why not switch to modular education? Notice the rise of nontraditional courses being delivered outside of the traditional educational system. These are nondegree courses, where the students decide what they wish to learn: advanced mathematics, computer programming, art, history, literature, foreign language. Or perhaps painting, woodworking, and other crafts. Amateur groups get together for pleasure to perform plays, music, or sports—not for money, not attempting to match professional levels of skills, but instead for pleasure and enjoyment (with the positive side effect of an enhanced appreciation of the abilities of professionals). The skills required to do these voluntary activities are taught outside of the formal educational process to students who have a need and desire to learn. Is this the future? We could use these same voluntary educational pursuits, undertaken by people who truly want to learn the material, as a model of the future of education.

I am a cofounder of a large, global initiative to rethink design education for the twenty-first century. Nondegree courses might very well be the keys to lifelong learning for the future. Let people take courses when they need and want them or perhaps simply when they become interested. This is not a new idea: it has been called “just-in-time learning” or JITL. I have traced it to 1996, but I suspect that the concept originated long before then. The courses might be short or long, depending on the material to be learned and the learner's background. They might come from universities, which would require universities to rethink their educational methods to make it possible for a student (of any age) to simply enroll for a single course. It is more likely to come from community schools, which in the United States are called community colleges, or from private institutions that offer certificates for the courses they provide. (It is unlikely to come from the large, well-established research universities of the world that are among the most conservative of all existing institutions, except perhaps for the long-established religions.) Today there are many courses being offered by a wide variety of institutions, some taught by volunteers, some by established professionals. The quality varies from wonderful to horrid. The one thing missing is a credentialing system to assure quality standards. Today it is hit or miss: some programs are excellent, others are not.

Will the teachers and parents miss the lectures that teachers are forced to give and students forced to listen to? Yes, they probably will. But give them time. Teachers will learn to enjoy the new structures and the fact that they can spend more time helping students learn. And imagine—educating students who want to learn! What an amazing situation that would be. In fact, let's stop talking about teaching. Let's emphasize learning instead. Don't call people teachers and students. Call those leading the activities “educators.” Call those learning “learners.” Forget age segregation. Educate those who are ready for the level of knowledge being presented, regardless of their age and background: if they have the proper prerequisites, let them in.

Parents will notice that their children are more excited to go to school. Does every student need to learn the quadratic equation? No, I say. (And I am not math-phobic: I have had six years of graduate courses in engineering mathematics.) Don't believe me? Ask yourself: How many times in your life since you left school have you had to solve quadratic equations or use other tools of algebra, trigonometry, and calculus? Mathematics is an important tool, but the choice of how and what is taught is often left to academic mathematicians who do not think about what math is actually needed to be a productive citizen. Schools still teach classical mathematics as developed in the Western countries of Europe instead of modern concepts from statistics, computer programming, and decision theory. Students learn how to solve obscure equations but not how to understand statistics and simple probability, how to read a balance sheet, or how to use the mathematics that really does impact their lives. I recommend the book *The Math Myth: And Other STEM Delusions* by Andrew Hacker, a professor of mathematics.

What about teamwork? Very few activities are done in isolation by a single person. Almost all real activities are done through teamwork, but the school systems insist on individual work. Most occupations require people to collaborate with one another, to work in teams, yet schools somehow assume that all people work by themselves. Students need to learn how to work together as teams, to help each other, even to copy from one another. Copy? Yes, because we should also teach how to give credit to the person they copy from.

The student's grade (if we still give these things) can be determined by whether that student puts the material together in a useful manner, perhaps with new interpretations. Copying is fine—the real question is whether the students know which students to copy from, which material to include. And the students whose work is copied should also get a bonus. After all, even in this book almost all the material comes from the work of others: my contribution comprises the novel combinations and interpretations I provide. And yes, as the many pages of references in the bibliography and the many people I list in the acknowledgments show, I always credit the sources of ideas. My behavior is common to all serious journalists and authors of nonfiction works. Why not help students master these skills?

Grades for Students

Economic variables are not the only example of flawed measurements. Here is a very different example: student grades. Teachers give students grades even though grades are highly subjective and often rather meaningless. The grades assigned by teachers are suspect, for they vary by all sorts of factors—how tired the teachers were when they assigned each particular grade, whether they were grading the work of the first student or the twentieth or the fortieth. Grades also vary depending on the time of day or whether the teacher has just eaten (or not yet eaten). And even then, what does the number mean? If a student has received a grade of 85, does that mean that this person knows 85 percent of the material or all of the material, but only 85 percent of it well?

Different countries grade students differently. Some, like the United States, use letter scores. Others give numerical scores, often as a percentage or simply a number within a specified range, such as 0 to 10 or 100. The boundary between one grade level and the next can make a huge difference to the life of the student, but the boundary is set for all sorts of spurious reasons, so a one-point difference can mean moving from one letter grade to another.

University courses are artificial divisions of a body of knowledge that fit into the arbitrary time given for the class. In the United States, university classes usually have three hours of classroom time per week, with the class lasting 15 weeks for schools that have two semesters per academic year or 10 weeks for schools that have three quarters per academic year (three quarters have the same number of hours of classroom work as two semesters). This is yet another artificial thing in our lives, dividing a huge amount of material into the fixed duration of a course split up into numerous small class sessions. It is all very arbitrary. Notice that the length of the course is fixed not by the amount of time it takes to teach or learn the material but to make the task of scheduling the classes and classroom easier and more uniform. Letting professors determine the length of the class based on how long it should take to learn the material would make much more sense: two weeks to learn some material, a full year for other. (This approach is usually rejected because of, among other objections, the scheduling difficulties.)

I have an alternative suggestion: couple learning modules with mastery learning. I recommend that we divide the material into modules, each module requiring anywhere from a few days to a week of study and learning experiences. A module should cover a single topic, and on completion one could say the student either understands it and can apply the understanding or not. Complex topics would get divided into numerous single-topic modules. I am a fan of project-based courses, where the learning is immediately applied to some situations that students find engaging. At the end of each module, students are given an examination on the material, for which they receive one of two grades: pass or nothing. No grade at all. Students then have a choice: they could decide that they are not interested in that module and go on to others they prefer, or they could go back and study the material once more, perhaps taking advantage of the tutorial services offered by the instructor and take the exam again (a slightly different exam each time). At the end of the course, students receive a list of modules they have passed. If the modules are small enough and the tests good enough, passing guarantees knowledge and understanding of the material. If a student didn't pass a module, so what? The subject of that module simply doesn't get listed among the topics the student knows. When students graduate, their record shows all the modules that they know. Good students will have more modules, including more advanced ones.

Courses as they are currently set up are also modules, but they are deficient because each course is generally too long, so the grade that is assigned does not clearly state which topics the student understands. Courses are forced to be of equal length, regardless of the amount or difficulty of the material, with the length usually selected to simplify the scheduling of classes and rooms. The result is an educational system that is designed for the convenience of the administrators and faculty rather than for the effectiveness of the education for the learning by students. Having smaller modules, each based on a relatively self-contained material, is a fairer way of gauging performance. Modules can be structured so that advanced modules require other modules as prerequisites. In many ways, modular education would look very much like today's education, except for the added flexibility of course length and the use of much more meaningful assessments of student understanding of the material covered.

What are grades good for? Not much. Once people are several years out of school, few care what grades they received: what matters is how well they are performing now. Grades are usually used as a screening criterion by those trying to decide whether to employ or admit people to graduate education. Grades primarily reveal how well students have performed on examinations, which has little to do with predicting success in life. At least with mastery modular learning, future employers or schools can look over the modules an individual has passed and decide if the list fits their needs.

How would schools issue degrees if there were no grades? Personally, I see no reason for degrees. If a school wants to issue degrees, it could require that any graduate have a required number of total modules, maybe different numbers of modules for different categories of material, with a certain number of modules required from humanities, sciences, engineering, the arts, computer skills, mathematics, and so on, depending on the student's major interests.

Grades—and degrees—would not be necessary because anyone wishing to determine a student's qualifications would look at the list of modules. The modules provide meaningful measures of the student's real knowledge.

Politics

There are many reasons why so many people dislike politics. They may dislike particular politicians or parties, dislike the laws that get enacted, and, most of all, dislike the fact that these political acts seem outside of their own control. Many think that politicians are corrupt or self-serving and prey to lobbyists. Sometimes these beliefs are correct. Some politicians do put their desire to stay in power and their desire to curry the favor of wealthy donors and lobbyists above the interests of the people they are supposed to be serving.

The concerns are real, but even where they are justified, the criticism should not be about politics in general but rather about particular politicians or institutions. Politics is a necessary and important part of how people manage to work together, despite differing opinions. Part of this dislike stems from ignorance. Few people understand how many serious politicians there are and the amount of work they spend behind the scenes, in committee work, with advisers, and often with a group of dedicated staff.

When I was vice president of advanced technology at Apple, I was responsible for working with the US Congress and the Federal Communications Commission to establish the network allocation that started the availability of free Wi-Fi networks and to determine the high-definition television (HDTV) standards for the United States. My staff ensured I was well briefed beforehand, and the Washington office told me about each of the people I would meet, what their interests were, and what their biases might be. I met privately with many of the key politicians and their staffs. In most large governments, the staff play a critically important role because politicians usually do not have the time or training to understand all the many issues before them: they rely on the opinions and wisdom of their staff. So I had to spend considerable time visiting congressional offices, explaining the issues, both technology and economics, as well as its benefits to the American people.

In the end, we got the Wi-Fi bands, and we ended up with HDTV standards that were not precisely what we wanted but nonetheless have worked well. I was lobbying for the good of the people of the United States, and, yes, Apple also stood to benefit from free wireless connectivity for its machines (Wi-Fi networks are now essential for many people and activities). I had to work with politicians and understand their needs. I learned to appreciate the way the system works when the legislators and commission members listen to evidence, when there are highly skilled staff members to support their analyses, and where compromises are a necessary and reasonable part of decision-making.

I define politics as the means by which disagreeing people can agree to work together in part to change opinions, in part to form compromises that allow progress to be made. A search of the internet, however, seems to indicate that my definition is in the minority and is labeled “political moralism.” I take that as a compliment.

Reward Structures

Almost all activities in life are governed by the rewards at the outcome. Most people seek the feeling of accomplishment from contributions that are recognized by colleagues and management. Rewards do not have to be monetary. Being given control over one's own activities is a reward. In fact, monetary awards, although appreciated and useful, are one-time events. Intrinsic awards that yield recognition and acknowledgment of the work's value are more powerful and longer lasting, especially as they can also lead to more rapid promotion and enhanced duties and titles. The reward could be personal satisfaction or just plain pleasure. Sometimes rewards are promotions in rank, whether in a company or in politics. Sometimes the reward is monetary. Almost every action has a reward structure associated with it, and those structures have considerable control over what activities are attempted and how they are accomplished.

A favorite saying among business executives is "*What gets measured gets managed*," although the saying has many variations. It has been attributed to the well-known business academic Peter Drucker. Some people trace it to Lord Kelvin, whose statement in 1889 about the power of measurement I discussed in chapter 8. There I pointed out how difficult it is to measure the things that truly matter, and, as a result, when something cannot be measured, it is either dismissed as unimportant or replaced by something else that can be measured, which is considered satisfactory as long as it has some relationship to the desired measurement.

The problem is that when measurement is insisted upon, it soon becomes the basis for the reward structure: people whose measurement numbers are high get rewarded with praise, monetary bonuses, and promotion. Workers soon learn to concentrate on getting a high score on the defined measurements, neglecting other things that might actually be more important for the business. Many reward systems tend to reward short-term activities, not what is good for the long term, in part because long-term results are difficult to get and, well, take a long time to get: managers want to reward the daily activities, even though the long-term results are more important. In addition, those who work hard behind the scenes to make things possible are often neglected. In a badly structured reward process, the perception often is that the wrong people get rewarded, and that may be true. Moreover, people focus their activities on what is most likely to be rewarded instead of on what is of most value. It is always easier to give rewards for short-term values rather than for long-term benefits.

University Reward Structures Are Wrong

Research universities reward professors by the measures that are easy to count: the number of publications in journals of high prestige, the number of books published, the number of paper presentations at conferences—all by a single author. Citations are counted—that is, the number of other papers (excluding those by the person being judged) that refence each paper. In the real world, work is done by teams. In the university, this is also true, but those who evaluate faculty for promotion worry that if a person publishes numerous papers with other people, they won't know how much of the work was done by the person under review. When I was a department chair, I was once told that I should tell a faculty member to stop publishing so many papers with others because it was weakening that person's chance for promotion. I consider this a criminal recommendation: the world's best work is done by teams. In modern science and engineering—that is, in the world of technology—it takes teams of people to do the work. Why do we punish professors when they work in teams?

The current reward structure produces other anomalies. The major research universities of the world require outside letters from experts to evaluate their own faculty: the more exotic the work, the more detailed and specialized, the more likely that it produces strong, positive recommendations from other specialists in the same field. This approach tends to turn faculty into specialists. Someone who is a generalist and publishes in different research areas and fields is not so well known by the specialists who write letters because much of the work is published and presented in journals and conferences that the specialists do not read or attend. Work that is important to the general public—and to people who work in industry—is often not well respected by academics. Specialism leads to silos.

In farming, the word *silo* refers to the tall, cylindrical building that a farmer uses to store grain, being careful never to mix different grains in the same silo. In academia, it is used as a metaphor to describes how one discipline is narrow and deep (like a silo) and so focused on the ever-increasing details of its one field of work that members can be completely unaware of what neighboring disciplines do, even though they all reside at the same university or even within the same department. Just as a farm might have multiple silos, each with different material in it that is carefully not allowed to mix with the materials in other silos, a department or a university has multiple silos in which discipline specialists are isolated and never mix with academics in other fields.

Silos in the ranks of professors lead to silos in the teaching of courses, so that even though the material in one course might be highly relevant to that of another course, especially in another discipline, neither professor knows about the relationship. And when the students graduate and begin their careers, they enter the silos of industry, where all engineers are put together in pods according to their specialty, designers in another, marketing in another, and so on. I blame the universities for the spread of silos across the world and the lack of teamwork in many industries.

The Reward Structure of Industry Executives Is Wrong

Unfortunately, the reward structures of industry often reinforce the lack of teamwork across departments and divisions. Yes, teamwork among members of the same department is encouraged, but managers and executives are rewarded by the profitability of their unit or division. When reward time arrives, the more money one unit gets, the less money others can get. This leads divisions to compete with one another, in some cases refusing to help each other.

Reward structures dominate our lives, whether they are monetary or verbal. It is no wonder that many people forget the true purpose of their work and lives and instead focus on the short-term benefits of the reward system. Many a company top executive, the CEO, has gotten huge bonuses and praise for cutting costs by firing a lot of workers, spinning off divisions, and raising prices to get higher profitability for the company. Then the CEO leaves the company, receiving a nice bonus, while the company falters with demoralized employees, deteriorating products, and increasing losses.

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The Dominance of Technology

Since the beginning of humankind, people have designed and constructed the technology for living, eating, protecting themselves with clothing and shelter, and carrying out their daily activities as well as the forms of governing, behavior, mathematics, currency, and laws that define and regulate their societies across the globe. The construction of artifacts (designs) made technology a fundamental part of life, so much so that it altered how people think and behave, hence the charge that design is in part responsible for the mess the world is in. And that mess is not just climate change: it also includes war, slavery, class domination, racism, many forms of discrimination, and colonization. The climate crisis is only one of the many results of the way the world has been designed, whether intentionally or unintentionally, consciously or not.

Treating People as Objects, as Servants of Machines

The history of treating people as objects, subservient to the needs and requirements of machines, is often ascribed to the rise of the Industrial Revolution and of factories in England in the 1700s. But way before then, thousands of years ago, the Egyptians, Aztecs, and Chinese built massive structures by putting hundreds or thousands of people to work: tombs, temples, cities, and, of course, the pyramids and the Great Wall of China. The construction workers were treated as expendable, replaceable machines. The same is true for soldiers in the millennia of wars, where they were treated like pawns in a game of chess: strategic tools to be sacrificed as needed in the path toward victory (or loss).

In the early 1900s, Frederick Winslow Taylor studied the efficiency of laborers, demonstrating that he was able to get more work out of them by refining their movements, minimizing superfluous travel of the hands, limbs, or body (sometimes by devising simple rigs that helped). He was proud of his ability to treat workers like machines, instructing them in “the one best way” to do every task and reconstituting the tasks to require a minimum of steps, a minimum of thought, and zero creativity—all in the name of increasing workforce “productivity.” Industry applauded his notion of productivity even though it treated the workers as if they were machines, incapable of thought, most efficient when they did the same job repeatedly. He destroyed their humanity to increase productivity. He called the workers “brutes,” and he insisted that he was making them happier by taking away the need to think.

When mass production in the factory arrived in the early 1900s, the disassembly line for slaughtering animals and turning them into packaged slices of meat and the assembly line for turning a bunch of small parts into automobiles furthered the advancement of people as machines. Of course, if we treat people like machines, we should not be surprised when they rebel, suffer medically, and lack motivation for or interest in the work they do.

Technology-centered approaches to the development of labor practices force people to behave according to the needs and requirements of the technology. Why can't we reverse that relationship, so that people are in charge, and technology and machines work according to *our* needs and requirements? When our lives are dictated by technological requirements, we are forced to behave like machines. When we act like machines, we do badly.

People are not machines. People don't behave the same way as machines, even the most intelligent of machines. Today the field that is responsible for the creation and design of intelligent machines is called artificial intelligence (AI). The word *artificial* applies both to the obviously constructed machine or computer program and to the concept of “intelligence.” The intelligence that results is very different than real, natural human intelligence—hence “artificial.” Machine intelligence is far superior to people's intelligence in doing some things, such as solving mathematical equations, but far worse in doing many other things, such as developing the equations that are to be solved or, for that matter, doing anything involving human values. The calculator built into my mobile phone can do arithmetic more quickly and accurately than I can. I don't find that threatening; I find it useful. What I want is to make sure that all of our artificial devices are the same: useful, not threatening.

Artificial intelligence has valuable powers and horrible weaknesses. AI-powered technology can provide valuable tools to aid our activities, but we must always remember that devices with AI are machines, not people, and that we are people, not machines. We are different from one another, and diversity can be valuable. A good example of the limitations of our artificially intelligent machines comes in the field of automobile automation. Many of the algorithms that drive these autos use neural-net-based algorithms. These powerful systems do better than most humans, including experts in some tasks, but they work by recognizing patterns not by deep understanding. Modern AI systems excel at tasks that are basically pattern recognition, such as finding medical anomalies (tumors and cancers) in medical images (X-rays and MRIs), performing them better than many well-trained radiologists. But lacking understanding, they can only point out these areas to the physicians. When used in automobiles, they have trouble understanding unusual situations, such as an emergency vehicle parked in the middle of a major highway. Trouble seeing and understanding things can also mean ignoring them (which has already led to accidents and deaths). On the other hand, some AI systems are overly cautious, hesitating to go through an intersection when it cannot determine whether a vehicle, pedestrian, or bicyclist is about to cross. This caution can lead to accidents when vehicles behind the automated ones, not expecting such slow, hesitant behavior, crash into the automated car.

The automobile manufacturers use the legal defense of explaining that they warn the drivers that they must always be alert, ready to take over when the automated system has difficulties. This strategy may work in a court of law, but it defies decades of studies by researchers (including me) in the field of human factors and psychology. Careful monitoring of a system for hours and even days when nothing goes wrong means that the monitoring cannot be continued: people cannot keep their attention focused on the possibility of some unexpected event when nothing has been happening for a while. I return to this point in the next section.

In everyday life, fads such as concepts, memes, ways of dressing, and genres of music go in and out of fashion. The same is true in academic discourse: science and technology are no exception. In the early days of the field of AI—which today is referred to as "good old-fashioned AI"—the emphasis was on symbolic systems and reasoning. The field had many successes, but they were always limited in scope: they did not scale well. Then, in the 1980s early work on neural networks was revived, leading to a short-lived fad for these systems, variously called "parallel-distributed networks" or "connectionist networks" and then finally "neural networks." This fad was short-lived because these early networks had only three levels and were quite small. As a result, they had many limitations, but some people kept working on them anyway, and today such networks have hundreds or thousands of layers and have both feedforward and feedback loops: they have become extremely sophisticated. They are now called "deep-learning networks." When I asked my former postdoctoral student Geoffrey Hinton, an early developer of the three-layer neural network and today's deep-learning systems, what theoretical breakthrough enabled the difference, he said, "Nothing: the breakthrough was that computers today are roughly 30 million times more powerful than they were then." (He was being modest: the early back-propagation algorithms for learning did not work well with the increased number of layers, and he helped develop new algorithms that do work.)

Deep learning is today's fad. The limitation, as I noted earlier, is that although neural networks are excellent at the important function of pattern recognition, which is why they are so good at recognizing faces, molecular structure, and other patterns, they cannot reason. In addition, they work by adjusting numerical weights on millions of internal nodes, which makes it very difficult—often impossible—to understand how they work. The mechanisms by which neural networks recognize things are not programmed into them: the networks learn the millions of numerical weights when they are presented with multiple situations. In most cases, they are trained to duplicate human activities by using people's past judgments in tasks such as issuing loans, determining bail, or setting sentences in court sessions, and they do a good job of duplicating human judgments, complete with human biases and prejudices. At first, it was thought that the use of machines would eliminate human biases, but that prediction soon proved to be faulty: the networks simply replicated human prejudices about race, gender, social class, educational level, and so on because the training sets were (unintentionally) biased and networks had learned based on those sets.

What is the solution? People are working on it in groups such as those exploring "explainable AI": new ways of generating learning sets, including in some cases not using real human data but rather synthetic situations developed by computer-simulation systems that are programmed to behave ethically. Ben Shneiderman, a professor of computer science, has long championed the design of human-centered systems that are ethical, fair, and humane. His book *Human-Centered AI* is an important contribution to the development of humanizing technologies. It offers design alternatives that prevent people from being treated as objects and as servants to machines, transforming human-centeredness to humanity-centeredness. My personal bias is that we must combine the good parts of old-fashioned symbolic reasoning with neural networks to allow the power of each to overcome the other's deficiencies.

The Technology-Centered Approach Redefines Human Virtues as Deficits

We are stuck in a world where technologists construct machines and tools—even automobiles and aircraft—to be completely controlled by other machines, usually AI machines, which do whatever task they are able to do but expect people to take over the things they cannot do. This forces people to continually attend to the behavior of these devices, with little to do but watch—that is, until the machine is no longer capable of doing the job, at which point the workers must quickly take over. Our automated automobiles of today have precisely this quality. The cars can almost drive themselves: it's the "almost" that makes them so dangerous. For most of the time, they drive so well all by themselves that people grow used to trusting them, which means their attention lags: curiosity, day-dreaming, and, the worst horror of all, thinking creep in, all usually positive things in their own right but distractions when the driver is supposed to be monitoring the automated car's performance. Even with simpler machines, people are constrained to be the servant. Just think of the clothes-washing machines and dishwashers. They automatically clean clothes and dishes, but people must load and unload them, add the cleaning materials, and take over when the machine gets into trouble (for example, the washing machine gets out of balance and makes banging noises). As a result, people must satisfy the needs of the machines. Almost all of our activities are constrained by this technological bias of putting all the effort on the design of the machine with little or no attention to the weird requirements that this bias poses for people. The philosophy seems natural, but it is wrong.

People are wonderful at creativity, at problem solving, and at qualitative acts such as drawing but bad at paying careful attention to something when nothing seems to require our attention, and so when something urgent occurs, we are, unsurprisingly, surprised. If the machine wants us to take over, why not give us some warning? Oh, because it doesn't know when it will fail. So machines require us to do the things we are bad at simply because that fits the machines' needs.

I propose that we design things by starting with an understanding of people's needs and abilities and create our devices to supplement and enhance them. When developing manufacturing methods, we must do the same: start with people's abilities and use machines to do the parts of the job that people cannot do or that are dull or dangerous. In this way, the machines that we use will be our servants rather than we being theirs.

The philosophy underlying the design of the factory and its assembly lines is fundamentally wrong, inhumane, and possibly immoral. People are especially bad at doing repetitious tasks, monitoring something for extended periods of time, and doing tasks that require high accuracy and precision for long periods. We err in spelling and simple arithmetic. Machines can do any of these things better than we can. But when the parts come in different sizes and shapes, or even if they are simply oriented on the assembly line in different positions, machines cannot cope (until very recently)—hence the need for people. In some manufacturing tasks, the machines cannot fit into a location where fasteners must be inserted or welding done, so people must contort themselves to fit the machines.

The current attitude forces us to do things on technology's terms, which means we are forced to do things we are bad at. And then when we turn out to be bad at doing the things we are bad at, we are blamed: human error, distraction, lack of attention, sloppiness—all negative terms thrown at us. Nonsense, I say.

The dominance of technology in the workplace and people's everyday lives means that the needs of the technology take priority over the needs of people. Cities have been modified or designed to accommodate automobiles, not pedestrians. Customers are treated as objects that can be seduced, manipulated, and addicted to products. When some of the most positive, creative aspects of people get in the way of the performance of machines, we relabel those positive attributes as negative. One of the things people are good at is noticing changes in the environment. We are naturally curious, and the nervous system has exquisitely evolved to be sensitive to change. But in a technology-centered world, curiosity as well as the dull and tedious nature of the tasks we are assigned to do lead us to think, look, or listen to something new and novel, so we can be distracted from those tasks, resulting in error and perhaps accident. We are blamed. "Human error" it is called: we allowed ourselves to be distracted.

I don't call this result "human error," I call it "design error." Requiring people to do things that are dreary and dull is to require people to do things they are genetically unsuited for. Complete, perfect automation of tasks that requires human judgment and thinking only in dealing with unexpected situations is extremely difficult. Automation of these tasks, such as driving a car, is only partially effective for many years until it becomes sufficiently good that a person is not required to oversee the operation. For some tasks, this condition may never be reached. The problem here is that although the automation gets better and better, it is still imperfect, and so it requires people to monitor it for hours on end. The better the automation, the worse the monitoring job is because as automation gets better, events that require human intervention are less frequent, hence the curiosity/distractibility curse.

The only good automation is perfect automation. Perfect automation does exist. Think of a home dishwashing machine: in its early years it had to be watched, but not today. This is true for most automation in the home. But automation of the activities done in business and commerce is more difficult, for the tasks are more complex, and the chances of unexpected difficulties higher. It is there that the phrase “human error” becomes commonly used.

Focus, Curiosity, Mind Wandering, and Mental Addiction (Flow)

We display the behavior of curiosity whenever some externally perceived thing triggers our attention, moving our conscious focus from whatever it was concentrating on before to the interrupting event as we try to understand what we are perceiving and what its implications and meanings might be. Mind wandering is precisely the same, except in this case the interruption comes from within, from some internal trigger that diverts our conscious attention to pursue it at length. Curiosity and mind wandering are good for creativity but bad for getting work done. To complete something requires focus, and focus means intense concentration with no detours, no triggers, no curiosity. The two are great partners: curiosity and focus. But this means that each of us must learn how to control which state we are in. Are curiosity and mind wandering human deficits? No, they are deficits only when we are forced to do some mind-numbing activity, so that the bored mind creates more interesting and important things to think about.

Mental addiction is the opposite of curiosity and mind wandering. It is when we are so deeply engrossed in an activity that we completely lose track of the world, including other things we are supposed to be doing at the moment. In some cases, this deep concentration is powerfully productive, whether in doing some creative work, reading an engrossing book, or working on a difficult but intriguing problem. It is the focus that gets things done but often to the detriment of other activities. The mind has been captured. Mihaly Csikszentmihalyi called this mental state “flow,” a concept that has been widely studied. But as is usual in our world, any concept can be used in either positive or negative ways, so “flow” has also been turned into “addiction,” where companies try their best to get the mind into an addictive state where people find it difficult to stop what they are doing: playing a computer game, gambling, or spending time on the tiny, tantalizing tidbits in social media.

Suppose you have been puzzling over a difficult issue but cannot find an answer, no matter how long and hard you have tried. You may then consciously give up and move on to other things, but the hard work and effort you already performed has now primed the subconscious mind to keep working on the same issue, sometimes for hours, weeks, or months. Psychologists call this period “incubation.” Then suddenly, without warning, the subconscious processes deliver a potential answer into the conscious mind. The subconscious is actually pretty intelligent: it won't alert you if the conscious mind is deeply focused on something. It waits until little is going on in your mind. When that happens, whatever you are doing at the moment gets replaced as you get diverted into rethinking the important issue you thought could not be solved. Like all behaviors, when done in the right amount at the right time, engaging in curiosity, paying attention to change, and letting your mind wander can be powerful tools. Of course, if curiosity or mind wandering takes our attention away from an important activity, it can be very annoying to others who are working with us and even dangerous if the task is critical. In these cases, the negative term *distraction* is warranted.

Technology is essential to our lives. After all, almost everything we have around us is artificial, and much of that artificiality is the technology that supports us, from providing us with clothing to writing and reading for us, cooking for us, sheltering us, and transporting us. The list is huge. New developments in technology promise to aid our lives in many beneficial ways. Some technologies are being developed to clean the air, earth, and oceans of the waste that contaminates them and leads to the destructive gases in the atmosphere, and some are being devised to offer ways of living to maintain life all over the planet without harming the ecosystem. We all must ensure that these technologies are humanity centered—that is, that they pay attention to the needs of all living things and the environment, pay attention to both unintentional and intentional biases and prejudices. I look forward to the future of technology.

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The Future of Technology

Science, engineering, technology: those who work in these fields never stop creating, dreaming, inventing. I know because I am one of them. I am always thinking of new ideas and new ways of doing things. When I was an engineer, I built things. When I was a psychologist and cognitive scientist, I did experiments, trying to uncover new findings, new interpretations, and new ways of thinking about the issues. When I was an industry executive and adviser to startup companies and then a designer, I thought of new methods, new tools, and new techniques. I'm still doing all of these things, except now most of my time is spent advising, developing new curricula for designers across the world, and writing about what I have learned—this book being an example.

Science and technology never stop, which means that our technology will always be changing. So whatever we say about technology today will be out of date tomorrow. (Except for doors, water faucets, and light switches: I predict that in 20-plus years we will still have trouble with all three.)

What new advances in science will eventually lead to new technologies? It takes a long time for principles discovered by science to find some use in a product, and then it takes many years for engineers and technologists to be able to produce a reliable, affordable, and useful product. Even after a new product is introduced, it can easily take another decade before it is accepted and absorbed into everyday life. These long lead times make it seem quite easy to predict what might come to market in the next decade or two: simply look into the research labs and see what they are working on. How do you do that? Read the journals, go to conferences, and peruse the patent applications.

A word of caution: most of the ideas in the research laboratories never become products. And even among those that do, most do not succeed in the marketplace. New companies, start-ups, mostly fail. This doesn't mean that their product was not good. Failure can occur for many reasons.

I can predict with great assurance that all of the technological breakthroughs of the future will have first appeared in the research-and-development laboratories 10 to 20 years earlier. I also predict that people who tell us just which ones will succeed will be wrong most of the time. “Predicting the future is easy,” Herb Simon once told me. “People do it all the time. The hard part is getting it right.”

I am a technologist and a designer, so even though I know the difficulties, I cannot stop thinking about the future, not by predicting but by forecasting. What is the difference? A forecaster explores possibilities, sometimes by presenting multiple reasonable scenarios, even though all are different, sometimes (as in the case of weather forecasts) discussing the likelihood of an event happening. The goal is not to be accurate (because that is not possible) but instead to be ready, to be prepared. Technological breakthroughs are infrequent and often unexpected. Moreover, the time between a breakthrough and its appearance in the world as something useful, reliable, and affordable is measured in decades. The activities in research laboratories around the world are important sources of forecasts. Many of these great breakthroughs will never become practical enough for use, but some will. By preparing for all of them, though, we will be ready no matter what happens.

Here are some general forecasts about the future of technology. The advances in micro-and nanoelectronics will make things smaller and smaller, thus requiring less and less power while increasing speed and accuracy. This decrease in size enables advanced processing by small portable devices that will in turn enable many other new technologies to be affordable and useful. New sensors will enable the measurement of more and more physical and biological variables, including deductions from observations of human and animal behavior. Communication technologies will become so small and ubiquitous that, when coupled with small, powerful computational devices and a wide variety of sensors, almost everything of importance in the world will be monitored and observed. Many of these new devices will be mobile, even airborne, flying by the use of propellers as in today's drones, or, for tiny objects, using a variety of biological mechanisms, especially flapping wings. The world of invisible and ubiquitous computing, discussed and predicted in the 1990s, will be with us even more than it is today. Advances in batteries and motors will enable many things to be motorized: baby carriages, shopping carts, as well as walkers and rollators for those who have difficulty with gait or balance. Actually, some of these devices may become unnecessary, as exoskeletons become smaller, more powerful, and intelligent. The modern automobile already has hundreds of motors, most of them electric. We already have motorized surfboards and skateboards, and if we can make these things, think of what we can do with almost anything that has moving parts.

Exoskeleton? What is that? It is an external structure—often called an external skeleton, hence “exoskeleton”—worn by a person to compensate for injury to the body to enable walking or the use of limbs, or as a device to allow people to carry heavy objects or exert far more strength than is possible by the unaided human. This is yet another technology that has been worked on for over a hundred years (a US patent was issued in 1890 for an “apparatus for facilitating walking”), and is the dream of many science fiction novels and movies. But if the problems of the excessive weight of the skeleton and of energy supplies can be overcome, exoskeletons will one day become common, everyday devices. The most common energy source is batteries, but the power demands are so high that the battery weight and restricted lifetime are severe impediments. Today, there are commercial units used in hospitals for rehabilitation and in some manufacturing facilities to allow people to manipulate heavy loads, and there are units in various testing stages for the military.

What about biological advances? They will be numerous: sensors for all sorts of ailments; genomic sequencing done inexpensively in a few hours; new biological markers and sensors; new smart systems for diagnosing, monitoring, relieving pain, and improving muscular control—all of which will not be cures but will nonetheless minimize the negative impacts of the underlying disease. Many of these devices will be used in the home or worn on or in the body without the necessity for supervision by medical assistants.

New understanding of biological and neurological processing will enable many advances. The use of biologically inspired products will increase. Advances in medicine will lead to better predictions and treatment of medical conditions, with home analysis and monitoring and individualized medicine appropriate to the needs of each patient.

There will be new sources of energy and increased efficiency in all energy-using devices. And there will be new ways of producing the technologies the world needs to continue to be sheltered and to have productive occupations, education, and lives—all without harm to the world's ecosystems.

The concept of money may very well change. Consider cryptocurrency—a new and rather confusing idea. It is a digital form of money, but with a difference. Money itself is confusing, and very few people understand the concept. Before you insist that it is simple, try to answer why people assume that pieces of paper have any value. They have value mainly because of trust in the government, but what is that trust based on? So although cryptocurrency is confusing, it combines confusion about the nature of money with an entirely new way of generating, transferring, and spending money. Today, at least in the Global North, people have moved away from physical objects as money. I have traveled from the United States to Europe for week-long trips and never had to use paper or coin-based money. Much of the money in the world is digital, not physical, whether in the form of credit cards or wire transfers, and today can be directly and digitally transferred from person to person or from person to store by using a smart device such as a mobile phone rather than by taking it out of your wallet and handing it over: “mobile money” it is often called. Numerous low-income countries still use cash for most purposes, but the use of mobile money is increasing rapidly. Because the cellphone revolution has covered the world with simple, inexpensive phones, even in the poorest of communities, more and more people use the phone for such information as agricultural prices (thereby getting rid of exorbitant fees by the brokers) and digital transfer of funds. The International Monetary Fund reports that mobile-money accounts are widespread in many low- and middle-income countries. The Brookings Institution reports that digitizing cash delivery can address the UN Sustainable Development Goal of ending poverty. The future of money, it is becoming increasingly clear, is digital.

Today, all that digital currency still flows through the banking system, which includes governmental agencies. Does cryptocurrency change this process in a satisfactory, trustworthy way? Some people believe that cryptocurrency is still in its infancy, so what you hear about it today is not what it has the potential to do. Money is simply paper or digital figures in a database. Money works because people have trust in the government that has issued it, which means that the money of different countries has different values of trust. Many people who are deeply involved in cryptocurrency believe that it represents a fundamentally new perspective and view of currency.

Many of the advances in technology, biotech, sensors, and motors will be driven by artificial intelligence. How threatening is AI? The main threat is unintelligent use of technology—any technology. It's like a gorilla. Gorillas in the wild are peaceful. They are vegetarians, so they spend much of the day chewing on plants, leaves, seeds, and fruits. If you approach them quietly, intelligently, your interaction with them will be an enjoyable experience. AI need not be a threat, but it has to be approached, designed, and implemented intelligently, with full understanding of the people with whom it will interact and with the aim to enhance their activities, not to replace them.

Unfortunately, technologists who design and release AI—and most new technologies—are proud of their expert technical skills but overlook ethical considerations such as enabling equity, eradicating bias, and ensuring that people are in control. They seldom will think of how to design the new technologies so that people are comfortable with the way they work and perform. All too often, these societal and social issues lie outside the development team's technical expertise. This limitation is understandable, for these technologies require certain kinds of specialization, and human- and humanity-centered design principles are, like ethics and equity, quite a different specialty. This is why every team of technologists should have social scientists among them, people who do understand these principles and who will assist the technologists in addressing them. Quite often in the rush to deliver the new, amazing wonders of the world to the waiting public, technologists and designers push the social issues aside, giving AI the bad reputation it has today.

Automation will change how work is done, whether by AI, robots, or other technologies. This might be virtuous if it allows a change in the notion of work. As I discussed in chapter 33, what if we could move from the notion of work—an activity that many people consider a chore and a burden, necessary for the income provided but not something they look forward to every day—to the notion of an occupation. Let's replace jobs with occupations and professions. Give some meaning to income-earning activities, which will allow workers to have pride in their accomplishments.

It is also important to pay attention to the letter A of AI. The A stands for "artificial." Artificial intelligence and human intelligence are quite different, and that diversity can either be threatening or, if used properly, a source of powerful enhancement of people's abilities.

Some of you may remember when arithmetic and algebraic calculators became inexpensive and commonplace. The big fuss in schools was whether students would be allowed to use them. After all, the argument went, if students could always use a calculator, they would lose their arithmetic skills. Well, that fuss is over—mostly. Children still learn arithmetic, but when it comes to doing it, they (and most adults) turn to calculators. I use my computer's calculator frequently. Sure, I know how to add and subtract, multiply and divide, but I make errors. So why not use the computer?

In solving algebraic equations or even the integral or differential equations of calculus, why not use a calculator? When I was taught in college, we used to look up the answer to integrals in handbooks that had been compiled by hand, so they had errors in them, and even they didn't give all of the answers because it is impossible for a handbook to have every conceivable differential or integral equation. So it took a lot of manipulation to get an equation into a format that matched one of the equations in the handbook. Today we just type the equation into one of the many computer programs that can quickly solve the equation.

Worse, I had to learn logarithms and use big, clumsy tables of values to do complex operations. What a relief today that we no longer have to do things that way. I still have a collection of slide rules—mechanical devices used to find arithmetic and trigonometric solutions by sliding a wood rod back and forth beneath a movable glass cursor, which provided answers because the scales on the slide rule were laid out logarithmically. Note that the precision was limited to three or four digits—with no decimals: those had to be calculated mentally. I once took a course at MIT on the design of rockets in which a homework problem might take a full week to complete, not because it was difficult to understand but because we had to do so many calculations. We invariably got the wrong answers, not because we didn't understand what we were doing but because there were so many calculations that we were bound to make errors in some, and then each error would propagate through the problem. Today, the very same problem can be solved in 30 minutes using a calculator or, better yet, a computer program. Today, the use of computer tools and AI allows students to solve far more complex problems than my class could do and, moreover, lets the students concentrate on the science behind the issues, not just on the mechanics of grinding out numerical answers.

Is it bad to use a computer to solve our equations for us? No! This tool allows engineers to concentrate on the real problems, to try multiple alternatives, to rethink assumptions—to be engineers instead of wasting their time and energy on hours and days of dull, tedious, nonproductive mathematical equation crunching. Solving the equations? That involves mechanical operations best left to machines.

However, just because the computer can solve the problem doesn't mean that engineers are out of work. On the contrary, engineers are necessary to develop the equations in the first place and then to evaluate the answer to determine if it actually meets their needs. If not, they must go back to their equations and reexamine the assumptions they made in developing them and then try again. When we had to solve the equations manually, it might take hours for each interaction: today it can be done in minutes. This speed leads to improved solutions, including solutions to problems that couldn't have been solved before. Replacing the mechanical tedium of solving equations lets engineers focus on their real skills: formulating the equations and interpreting the results. The advanced tools changed the jobs in a good way.

A person plus an intelligent machine can have enhanced abilities, just as the power of engineers is enhanced when they use computers and calculators. I see the same for many activities. John Markoff, the *New York Times* technology columnist, wrote a wonderful book on this topic, *Machines of Loving Grace*, reviewing the history of machines and technology and suggesting that the real benefits will come when instead of doing AI, we do IA—not artificial intelligence but intelligence amplification. Yes, augmenting and amplifying human capabilities, becoming collaborators and assistants. The National Academies of Sciences, Engineering, and Medicine did a study of the use of AI and concluded that “there will be an increased need for AI systems to function effectively as teammates with humans.” Moreover, they continued, “when considering an AI system as a part of a team, rather than simply a tool capable of limited actions, the need for a framework for improving the design of AI systems to enhance the overall success of human-AI teams becomes apparent.”

We live in an artificial world where we can no longer survive without the artificial artifacts that govern our lives. But today, technology often has priority, forcing people to behave according to its artificial and arbitrary requirements. It is time to change this perspective, time to put people first, to allow and encourage people to do the activities they want to do and have technology do whatever the tasks they don't. What tasks? They are known as the three D's: dull, dangerous, or dirty tasks.

VI
Action
Learn, Reflect, Decide, Act

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What Can Be Done?

The issues facing the planet are overwhelming. The average person can contribute by changing personal activities, but the task is so large, the magnitude of change required so immense, that many of the tasks must be accomplished through large foundations, countries, and companies. The huge costs associated with the needed changes produce major political conflicts. Moreover, many powerful individuals and institutions with enormous influence feel that change will hurt their interests. All of these factors make many governments hesitate to move forward. Here is where citizen involvement is absolutely essential, for votes are what keep politicians in power, and sales are essential for businesses, so massive public opinion can have a powerful impact.

The problems facing the world affect all of us, so our actions must also be worldwide. There are multiple ways of categorizing all the various ways of having influence, but the distinction between top-down and bottom-up approaches is essential: both will be needed.

The title of this book is *Design for a Better World*. What is the role of design in all of this? In this concluding part, I review the topics of the book with a view to understanding the role that individuals and organized groups can have, the role that multiple disciplines can play, and, because I do think that design has something special to offer, what the designers of the world might be able to contribute. I start in this chapter by examining top-down versus bottom-up involvement, the former usually done by governments and international organizations such as the United Nations, the World Bank, and the many multinational organizations and meetings or by large private foundations. Bottom-up activities are undertaken by individuals or small groups, some of which might be NGOs or small foundations, sometimes even officially dedicated units within a small city or town government. Of course, not all activities fit neatly into these two major categories. After that discussion, I examine the major sorts of things that can be done by both top-down and bottom-up involvement.

In chapter 37, I address different ways of working. And, finally, chapter 38 provides a review of the major points of the book.

Top-Down Activity

Top-down activities are those that come from the highest level of government (the top). The goal here is to enable governments, civic leaders, and powerful people and organizations in all walks of life—foundations, industry, academia—to take a stance and to take positive actions and curtail negative ones. These actions start at the very top of an organization and then move downward through its bureaucracy to enact and enforce the actions. This movement requires institutions to establish directives and governments to institute policies and laws. These activities will move downward through the bureaucracies until they reach every citizen. The United Nations' meetings on climate change and initiative targeting the seventeen Sustainable Development Goals, discussed in chapter 11, are perfect examples of top-down activity.

Top-down structures usually create large, expensive, multiyear projects that are often unwieldy, hence my strong recommendation about ways to avoid such projects in chapter 25 through 27. If it is not possible to avoid these large projects, chapter 28 discusses how they might be handled.

In some cases, top-down action is essential. This is especially the case when the action desired by citizens is too expensive for industry, either in actual expenditures or because the companies fear a loss of sales or profit. Similarly, important trade groups, labor unions, or other large groups with persuasive lobbying and voting power may object because of the perceived negative impact on their members. No company wishes to be the first to undertake an expensive operation, thus causing its costs to rise, leading to higher prices.

The attempts to switch from fossil fuels to ecologically sound energy sources illustrate many of the difficulties. Despite the pleas from many organizations and the United Nations and even the passage of international statements, little will actually get done without government action. Government action will force industry and all other users to comply. This will also have the benefit of spreading costs equally among all. Even so, look at the relative inaction of the United States and the flip-flop that happened when one president, Donald Trump, denied that there was any problem at all and tried to destroy all the actions that had been taken to protect against climate change, and then the next president, Joseph Biden, tried to reinstate all the actions but was unable to pass many of his proposals through a bitterly divided and cantankerous Senate. International crises in the normal distribution of oil and natural gas stopped normal flow of these fuels into countries in western Europe, forcing them to reopen coal-powered plants and stop the conversion to ecologically friendly, non-fossil fuels. The United States has now had to encourage more development of oil supplies through new wells because the war in Ukraine has led to a slowdown of the flow of oil and gas from Russia to countries in northern Europe. The reversal from discouraging and stopping the use of natural gas and petroleum for fuel was because the governments felt they had no choice: it was either that or have their constituents go without electricity.

The slow pace of government intervention is quite common. Consider the history of seat belts in automobiles. Seat belts were in use in the 1930s in a few isolated situations, but it wasn't until 1966 that they were required to be installed in all new vehicles, despite strong evidence that they saved lives and minimized the severity of injuries. It took 31 years, until 1997, before all 50 states required them to be used, and even then New Hampshire required seat belts to be worn only by minor children, not by adults. So something as simple as implementing a seat-belt law took more than half a century.

What does that suggest for the needed changes described in this book? If seat belts are an example, it might take 20 years for any meaningful governmental activity to take place and perhaps another few decades before any specific actions are under way. Remember, we are not talking about something as relatively simple as seats belts in automobiles. We are talking about major changes in what and how we measure things and then in the way that those measurements are used. We are talking about major changes in the mining, designing, manufacturing, and use of products. We are talking about major changes in the material being used, from material that must be mined across the world to more biological materials that are grown, processed, and then returned to earth. These changes are not easy to make. Yes, I can easily imagine huge success by the year 2100. By then, many of us will not be around, some countries will no longer exist, and many parts of the world will be devastated by sea-level rise, by flooding in the interiors of countries, by droughts, windstorms, fires, and major storms—tornados, hurricanes, typhoons.

Bottom-Up Activities

Bottom-up activities originate with individual citizens who band together, mobilizing popular sentiment and putting upward pressure on their nation's bureaucracies to act. Numerous groups are organizing themselves for action to influence governmental action. Greta Thunberg may be the most successful example of a single individual activating millions of people across the world, from students to leaders and prime ministers of major countries to high officials of the United Nations and the pope. However, numerous other students and citizens are banding together. This kind of mass mobilization is important. Some groups are successful—some have convinced their governments to take small steps, and some have been able to impact local communities—but some have failed.

Individuals can start movements. Clint Borgen, who had served as a volunteer in refugee camps during the Kosovo war and then interned at the United Nations, used money he made by working on fishing boats in Alaska to start a national campaign that today has two US congresspersons on its board (nicely balanced: one a

Republican, the other a Democrat). He calls it the “Borgen Project.” It started in Tacoma, Washington, in the northwestern corner of the United States. It works primarily on food security, newborn-infant survival, clean water and sanitation, and food-aid reform by meeting with members of Congress and mobilizing people across the globe to achieve these goals.

Other Levels of Involvement

The world is not neatly organized into a hierarchical structure that fits the top-down/bottom-up categorization. Many companies are international in scope with budgets and global impacts larger than those of many countries. Many NGOs and foundations ranging from quite small to quite large are nonhierarchical in structure. One NGO, Global Justice, ranked the top-100 global economic entities and found that 69 were companies, and only 31 were countries. (Beware, though: you know how I feel about these global rankings.) These are the big, top-down organizations. What about everyday people, though? What can we do?

37
What Can We Do?

The problems of today's world cannot be solved simply by raising awareness or by stating what needs to be done. The solutions require action. What positive actions can we take? Here are some actions that could be done and, even better, some actions already being taken.

Band Together: Combine Existing Projects

One gratifying observation is that a large number of individuals, groups, organizations, and governments are taking action. There are so many articles, books, and approaches that I cannot keep up with all of them. Better yet, many are forming coalitions, joining forces to amplify their individual efforts, helping compel critical decision makers to respond: politicians, corporate leaders, and other influencers.

One major difficulty is that many of the changes that must be made to safeguard the planet, reduce inequities, and improve the environment will at first have a negative impact on some people's lives. Many changes that are essential and beneficial for the world in the long run will create local harm in the immediate future. Consider all those people whose jobs depend on coal mining or on the mining, drilling, processing, and distribution of fossil fuels. Or, if you will, consider the impact on companies that must now pay for and eliminate their externalities, the waste products that result from their manufacturing method. Will the remedies for the planet cause these companies to expend considerable amount of time, labor, and money? Yes, of course. These expenses will add costs to the companies' products, which will cause them to lose competitiveness, which in turn will mean a loss of sales and income required to keep them in operation. Humanity must confront all of these legitimate issues.

The negative impacts on the lives of people and the viability of businesses must be addressed, else the path forward will be too slow. Here is where the creativity of people—professional designers, yes, but also everyone—can contribute. A special goal for the creative people and entrepreneurs reading this is to create new occupations to replace the jobs that are lost, occupations that take advantage of workers' existing competencies and interests. Many proposals for retraining people are completely unrealistic in the kinds of skills they expect displaced workers to be able to learn. It would be better to find ways of using their existing talents. People who lose their jobs need ways to find activities that use their abilities, excite them, and provide them with a sense of accomplishment, control over their lives, as well as sufficient funds to care for themselves and their families.

Industries will have to change their design and manufacturing processes, which in turn is likely to require changes in their products. The cost of these changes to industry can be handled in two ways. First, follow the lead of Amory Lovins, cofounder of the Rocky Mountain Institute, in demonstrating how much of the waste created by industry actually represents lost income, so that through the restructuring of factories and their procedures, the waste material can be both reduced and reused to lower costs or bring in extra income. Create national collaborative bodies to hold constructive sessions with critical decision makers to develop policies that are favorable to all. Meetings among competing companies have to be handled with delicacy to avoid, among other things, violating antitrust regulations. This process can be started by the already existing network of societies and associations that do bring together similar industries and occupations and that allow constructive meetings to come up with ways to advance all parties' initiatives. Such meetings must be open to all companies but include no discussion of specific business, tactics, prices, or proprietary information.

Even where extra costs cannot be avoided, companies might agree to national legislation that requires all companies to meet the same standards. This policy was followed by the automobile industry to enable manufacturers to add safety features such as crumple zones in cars, seat belts, air bags, various improvements in braking, skid controls, and so on. If policies impact all companies the same way, all companies share the same economic burden.

These policies are not necessarily easy to introduce. As I already pointed out in chapter 36 in the section on top-down activity, the only way seat belts got installed on all American vehicles was through government regulations that forced car manufacturers to install them, but it took a long time. The struggles we must make now will be even more complex. It is essential to start now.

Warn Everyone

The first essential steps in solving a problem are recognition and acceptance of the problem. Here is where communication is the key, starting with the scientific community, then moving on to the scientists with the most public visibility, to politicians, and then to citizens not in these groups. Concern over the human impact on climate was voiced in 1957 when Roger Revelle and Hans Suess sounded the alarm about climate change and the general degradation of the earth's environment. They were not minor scientists. Roger Revelle was director of the Scripps Institute of Oceanography and science adviser to the secretary of the interior in the Kennedy administration. Hans Suess helped develop radiocarbon dating. Revelle went on to drive the establishment of the University of California at San Diego. Suess became one of the first four faculty members there. (I joined them in 1966 and am a faculty member of Revelle College.) The Revelle and Suess paper was considered extremely important by the scientific community but did not have any impact on the government. Then the politicians came on board, perhaps the best-known being Al Gore, vice president of the United States from 1993 to 2001, who has long championed action against climate change, with powerful lectures, books, and films. Finally, everyday citizens began to weigh in, some of whom have reached worldwide prominence. Roughly 65 years after the Revelle and Suess paper, in the late 2010s and 2020s Greta Thunberg, the young Swedish student whom I mentioned in chapter 36, mobilized millions of people across the world.

Call this the "Warn Everyone" approach. Warning and spreading the word about current and impending problems are important, but as the timeline for global warming indicates, warnings are only the first of many steps necessary for action to begin. Unfortunately, in the case of climate change, although the warnings have been sounded for more than six decades, very little action has been accomplished.

For three decades, the United Nations has been sounding the alarm in the annual Conference of Parties (COP) for the UN Climate Change Conference: COP1 was in 1995, COP26 in November 2021. Twenty-six years of expensive meetings of world powers, producing various agreements and accords that were severely weakened just to obtain unanimous agreement. And after each meeting? The parties failed to fulfill the agreements, even though they had been severely weakened.

COP26 was held in Glasgow, Scotland, and produced numerous nice-sounding but heavily watered-down statements about intentions. Yes, several treaties have been written, but look at how the United States has responded to them. Bill Clinton signed the Kyoto Protocol in 1998. His successor, George W. Bush, renounced the treaty. At the 2015 COP, Barack Obama embraced the Paris Agreement, but his successor, Donald Trump, not only abandoned the treaty but repealed more than 100 of the regulations that Obama had put into place to eliminate our dependence on fossil fuels. In their place, Trump expanded drilling and mining. In 2021, President Biden rejoined the conference, promising that the United States will be a leader, but back in Washington, DC, Congress has stalled and refused to back the ambitious plans Biden proposed for this purpose. The US Supreme Court has ruled that most of the regulations imposed by the current president to enhance efforts to reduce pollution and global warning are illegal because they were made by presidential proclamations and not by legislative law, but the legislature is deadlocked and will not act. Some sort of bill will probably pass through the legislative process, but it will be dramatically reduced in the amount of money and number of activities it will put into place to make the needed changes. This oscillation of opinions in the United States demonstrates the extreme difficulties that major projects face. Presidents and world leaders may agree to act, but those leaders do not control the policies of a nation—the elected representatives to the various parliaments and congresses do, and leaders can be replaced at the next election.

What important results have there been from the attention that has been paid to climate change? In 2021, fossil fuels were named as a major culprit of climate change for the first time, even though Al Gore had called for an end to the use of fossil fuels years earlier, in 2008. Why did it take 13 years for the COP meeting to name fossil fuels as a major threat to climate? Because the coal-burning nations of the world objected. Indeed, even at COP26 the original language requiring that coal burning be phased out was changed. After two weeks of long sessions lasting late into the night, at India's insistence the wording was changed from the proposed “phased out” to the weaker “phasing down.”

The dilemma faced by developing nations is indeed real: India—and all nations that burn coal—do not have much choice. They need more and more electricity for the needs of their growing populations who today lack reliable electric power. Despite the harm that the coal burning does to inhabitants of the large cities with the overpowering, dangerous smog it produces alongside the long-term impact of global warming, these countries have no choice but to use this fossil fuel. It takes considerable time and resources to build and deploy cleaner sources of energy. As we reduce the use of fossil fuels for transportation and heating of factories, the cost of electrical generation and transmission increases. New generation plants have to be built, and new transmission lines and transformers will be required to handle the increased demand for electric power.

Moreover, the nonpolluting energy sources for generating electricity (wind, sun, water) are far from the location of today's electric power plants and from the places where the electricity is needed. They require more transmission lines, which will not only require fossil-fuel energies to manufacture but raise many objections from those who live near the proposed power lines. These people may object to the loss of land when it is taken to be used for the towers, to the aesthetic loss when the natural landscape is covered with tall structures and wires, and the fear of radiation from the power lines (the fear does not have to be based on evidence; the perception is enough to stop or delay a project). All these factors create severe political and economic challenges in the attempt to cut down on the use of coal, oil, and natural gas, the major emitters of CO_2 and methane. The actual costs might not be as large as many fear. The Rocky Mountain Institute has prepared the optimistic report *Profitably Decarbonizing Heavy Transport and Industrial Heat* to argue the transformation is not difficult and can be quite profitable.

What makes countries respond? Major disasters that cause death and destruction and that require expensive reconstruction can motivate action, although obviously too late for those who were killed, injured, or made homeless and for the many nations that are barely above sea level. Warnings are necessary but insufficient. I pointed out in chapter 31 that people will help to recover from a disaster but not to prevent the disaster.

The Technological Approach

Problems in complex systems must be attacked by a wide variety of approaches: social, political, economic, and technical, among others. Each is essential, but each alone is insufficient. Bill Gates, cofounder of both Microsoft and the Gates Foundation is a brilliant technologist. In 2020, he published his approach to climate change: *How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need*. Gates presents an extensive list of technological challenges that must be overcome and for each challenge provides a thorough review of the issues and the various technological suggestions that have been made to address them. These ideas, if they could be implemented, would produce the type of powerful attack on climate change that is required to deal with the symptoms. Gates's recommendations are appropriate and possible but only if there is sufficient political and economic support.

In similar fashion, John Doerr, a well-known venture capitalist in California's Silicon Valley, proposes in his book *Speed & Scale: An Action Plan for Solving Our Climate Crisis Now* that we use his method of setting objectives and identifying key results (OKRs) to show if the objectives have been satisfied.

Both books are excellent, well informed, and populated with considerable evidence about the issues and the proposed solutions. The two, moreover, are quite consistent in their identification of the issues and the possible solutions. Gates is heavier on the technology, and Doerr heavier on the procedures for setting meaningful goals and measuring the success or failure of the effort. He developed the OKR method for use by startup companies, and the most successful use happened when he convinced a tiny two-person startup company that it would work. That company today is known as Google: the cofounders Larry Page and Sergey Brin attribute much of their success to the OKR method, which the company is still using today.

Both these and other, similar technology-centered suggestions, however, fail to address the behavioral and societal issues that I describe in this book. Both have attacked the symptoms of climate change with the tool they are experts at, technology, but they do not address the underlying discrimination, prejudice, deep divisions between political parties, and other social barriers that prevent any meaningful reform from being carried out. These issues can be found in countries all across the world—in Africa, the Americas, Asia, Europe, and Oceania. The major problem is human behavior. Yes, we need to solve the technological issues, but unless we address people's behavior, the problems will persist.

Both books discuss the role of laws, regulations, and government spending practices for addressing climate change and devote considerable attention to the kinds of policies that are required. But neither book dives into the core issues that have led to the practices doing so much harm today and have prevented us from adopting appropriate policies (for example, the short-term harm that these policies would do to people, business, and livelihoods), and neither book addresses how these issues can be overcome.

Today there are two powerful forces in conflict with one another. On the one hand, we have the strong voices of the many workers who will lose their jobs and therefore their incomes and homes and of the many small business owners and companies that will suffer as we address the perils of climate change. On the other hand, we have the voices of the scientific community who predict even worse hardships to everyone if we do not act quickly. Follow the first set of choices to maintain the status quo, and any response we make will be too little, too late. Follow the second set to act now, and the ecosystem will be stabilized, but at the cost of many who will be harmed economically.

What is the appropriate approach to the major issues facing us? There is no simple answer. They require a concerted effort on all sides for a multipronged approach to get at the causal issues: technological, economic, political, legal, organizational, commercial, industrial, and societal. The issues are huge and complex, involving not only sociotechnical and environmental systems but also the human behavior at the center of everything.

There is a third book that, to me, offers more hope. So, in addition to the books that focus on technology, the environmentalist Paul Hawken's book has the optimistic title *Regeneration: Ending the Climate Crisis in One Generation*. Moreover, although his book is quite different from mine, he has similar views. For example, I point out that this is a complex problem, involving many disciplines, but the most important and difficult component is human behavior. Here is how Hawken addresses a similar view in his book: “The climate crisis is not a science problem. It is a human problem. The ultimate power to change the world does not reside in technologies. It relies on reverence, respect, and compassion—for ourselves, all people, all life. This is regeneration.”

I agree with Hawken. Human behavior is a major part of the problem, and therefore key to the solution. This is why I suggest that the field of design can play a major role in that solution. Why? Let me restate the case. Design does not have expertise in all of the important technical issues that are involved, which is why it can be a neutral party in making use of the world's best experts. Designers have other attributes. First, design is an applied profession, not noted for writing well-cited papers in technical journals but rather for building and creating solutions to issues. The design approach is based on a deep understanding of human behavior, following the dictates of humanity-centered design: design *with* the people, not *for* them. The world must act quickly while also assisting those who are harmed by the actions taken to solve the environmental issues we confront. This assistance will add to the expenses but is an essential investment to help mitigate the impact of the disasters that are already happening: both issues must be addressed simultaneously. The disasters that are now taking place ironically have one positive benefit: they make clear to the world that action is required, thus making it easier to reach agreements.

Dealing with Complex Sociotechnical Systems

Complex sociotechnical systems are often called “wicked problems,” a term created in 1973 by Horst Rittel, a design professor, and Melvin Webber, a professor of city planning, both at the University of California, Berkeley. The complex societal issues we are trying to deal with now are “wicked problems.” These problems are ill defined —but that does not make them any less real. Wicked problems do not have completely satisfactory answers or solutions, but nonetheless it is possible to make things better.

The reason it is so difficult to address the issues raised in this book is that they all are wicked problems. They are economic, political, legal, organizational, business, and societal issues. However, we do not have to solve all aspects of all of them: we simply need to start with what is called “low-hanging fruit.” That is, for each issue, we find the causes that are relatively easy to address but that in being addressed also lead to significant, meaningful improvement. Each thing we address successfully in this way can make it easier to do the next thing. Eliminating the problems during the first round often lowers the difficulty of the second round—the other fruit start to sag, getting easier to reach. And note that this description is precisely the way that the "searchers" from chapter 22 and the "incremental modular design" process of chapter 27 work: step by step, always producing a "minimum viable outcome" so that even the relatively small projects quickly lead to a beneficial result.

How should designers attack these issues? My suggestions are presented in part IV on humanity-centered design, especially chapters 25–30. I suggest we follow the behavior of William Easterly's "searchers" and Charles Lindblom's concept of "muddling through" (introduced in chapter 25). The two concepts are similar, involving incremental, small attacks on the issues, enabling continual flexibility guided by the feedback from the early results, working slowly and incrementally, always aiming for the betterment of humanity, but making change cautiously. Small changes are easier to make than large ones. It is easier to mobilize people who agree with the small changes than to get everyone to agree and act on massive changes. Moreover, small changes impact less people: yes, less people benefit, but, more importantly, less people are apt to feel damaged, ignored, or otherwise harmed. And finally, by working slowly with thousands or even millions of small groups, the work is flexible. It can change as situations change. It can change as the impact of previous changes are assessed, building on the ones that worked, learning from the ones that failed.

I also highly recommend a few sources for more in-depth reading, which I have listed in the notes for this chapter.

Many different groups are already pursuing similar aims, which is a very positive sign. The mobilization of the many who are required to attack these issues requires a large group of mobilizers, people who already have established projects. Goal number seventeen of the UN Sustainable Development Goals is to "strengthen global partnerships." Yes, the United Nations is absolutely correct: global partnerships are essential. The more people with similar goals and methods, the better. The large number of demonstration projects and organizations now involved is wonderful, but they mostly operate independently of one another. Most are unaware of the many other organizations doing similar work. As a result, the power of the many is lost. Instead, we have only the power of individuals, which is always limited.

To make the existing efforts visible and thereby more effective and powerful, we all must collaborate. Success requires flexibility. The problem is massive, but massive projects will fail. Massive projects lead to a tightly coupled system, which is precisely the wrong way to deal with complex behavioral issues, where conditions are always in a state of flux. Making incre-mental progress, searching, and muddling are the ways to go. The incremental modular design approach that I suggest in chapter 27 is one suggestion for a proper procedure to follow. It requires loosely coupled systems that permit initiative, flexibility, and freedom. It requires the sharing of methods, of successes, and of failures. It requires a communication medium for enlisting others. And it requires making the entire effort visible to the influencers and decision makers of the world to manage peaceful and constructive ways to combat the challenges we all face.

38
The Major Points of This Book

Artificiality

Much of the world we live in has been designed, not by today's design profession but by people over the history of civilization. The objects we wear, live in, and rely on and the ideas, beliefs, laws, and forms of government and customs that govern our lives are all artificial. This is both good and bad. It is good because if the critical issues facing us revolve around these artificial ways of being, of living, and of governing, their very artificiality means they can be changed, redesigned. It is bad because all these artificial designs have been in existence for so long that for most people they no longer seem artificial: they seem natural, so natural that the thought of change is, first, inconceivable and, second, if they accept the possibility, frightening. Path dependency, the impact of decisions made at some earlier time, affects the way decisions are made today.

Different societies, groups, and religions have developed with different underlying beliefs, all of which seem perfectly natural to those in the group (who often wonder how others can hold such discrepant views). The conflicts between short-term and long-term values show up strongly as the world considers how to adapt to the requirements of developing a sustainable world, where changes in behavior might bring immediate suffering, and the long-term benefits seem far away and insubstantial. And finally, many people are motivated by personal gain and seem reluctant to give up their comfortable lives simply to save the world for future generations.

Meaningful

Today's world is dominated by the remaining traces of modernism, where technology, science, and economics rule. The result is an unwise emphasis on measuring everything, even things that cannot be measured, which turns almost all aspects of life into abstract, meaningless numbers. The result is that decisions and actions are made through abstractions, calculations, algorithms, and other mechanisms that are devoid of meaning and unintelligible to nonexperts. Worse, phenomena are studied in artificial situations, in the cold, isolated world of the laboratory, where the conditions can be completely controlled, or, in the case of economics, either through abstract "logical" thinking devoid of observation of the real phenomena or by analyzing the abstract numerical statistics that economists are so enamored of. As a result, when the findings of these analyses are applied to the real world of multiple, uncontrollable, messy variables, they fail.

The dogma from the physical sciences that everything must be measured on a numerical scale, even things that cannot be measured in this way, must be modified. The physical and economic sciences are of clear importance to the world of human beings, yet their studies seldom focus on people. The focus should shift to an emphasis on people's needs, resources, and abilities—to an emphasis on the critical needs of humanity for all people across the globe. This is especially important now when massive pollution affects the air we breathe, the water we drink, and the land we use to sustain our needs for food. In principle we need more emphasis on the social sciences, except that these disciplines have also been infected with the need to study things in isolation, in carefully controlled, artificial environments, with an emphasis on measurement, even of things that are not measurable.

Many of the most important aspects of life cannot be readily measured, but this does not minimize their importance. The whole point of humanity-centered design is to focus on humanity's needs: if these needs cannot be measured by traditional metrics, we must develop nontraditional measures. The social science community already has ways to do this: we must make use of them. Psychologists have devised different measurement scales that allow the study of issues that cannot be specified by the numerical methods of the physical sciences. Qualitative measures, stories, and narratives can greatly expand our knowledge and understanding, but they tend to be discounted by the scientific community that is dominated by those trained or working in the physical sciences. Yes, it is true that measurement is essential for assessing the impact of our work, but the measurements must be of things that matter to the lives of people as opposed to the general focus on costs, productivity, efficiency, and schedules.

Qualitative methods can describe and make prominent the important issues, showing where things are in good shape, where some attention must be given to problem areas, as well as where the situation is dire and requires considerable attention. The qualitative statements of where one's attention should be focused do not require numerical values. Items can be ranked in importance without them. Obviously, the ordering must be done rigorously, and some of it might require precise measurements, but qualitative presentations avoid the difficulties of attempting to summarize complex, multidimensional factors into a single numerical measure that hides the underlying issues, each of which must be addressed separately. These summary numerical measures are extremely misleading. Finally, the use of monetary measures for things that are not fundamentally about money (such as the state of the nation or the value of human lives) is hurting the ability to place emphasis on the factors that really matter. Many of these apparently reasonable numerical methods of measuring an organization, a community, a state, a nation, or a system, such as cost–benefit analysis, result in reinforcing inequity. These measures are long overdue for change.

Among the many methods of summarizing the issues that need immediate attention in complex systems, the discussion on information dashboards in chapter 11 demonstrates how graphical depictions can provide meaningful representations of the activities of companies, cities, states, and countries. These depictions make it easy for viewers to quickly assess many of the critical dimensions of a problem and to decide where attention and effort are required.

Instead of using a single numerical value to represent a complex state of affairs, we can show the critical components. Of course, any summary of the top few issues impacting the world leaves much to be specified. Nonetheless, the high-level summary may be sufficient for the highest level of decision makers. Once the important areas are identified, each can then be broken down into multiple components, displayed on their own, more specialized dashboard, which in turn can lead to the perusal of more detailed specialized displays. The important point is to avoid the joint dangers of oversimplification and, by showing too little information and over-specifying, burying decision makers in reams of data, numerical tables, diagrams, and graphs.

Dashboards are important ways of summarizing complex problems so that decision makers know where to focus their attention. Once attention is focused on one sector, then it will have to be represented by its own, specialized dashboard, which in turn might lead to multiple other dashboards, each dealing with different components of the issues. Yes, the complexity of the world requires a complex representation, but the complexity can be managed by dividing the analyses into different levels, each with a meaningful, understandable representation. Thus, the highest levels of analyses will be the ones discussed in chapter 11. As problems are handed off to specialized teams, each will have their own specialized ways of viewing the issues. If we were to put all the different analyses together, it would indeed be cumbersome, complicated, and unmanageable, but this is probably never necessary. We make meaningful sense by breaking complexity down into meaningful segments.

Narratives and stories are essential additions to the usual emphasis on graphs and tables of numbers, helping translate the abstraction of quantitative and qualitative measures into the described impact of a problem on people's lives and experiences. Stories and narratives do not replace the need for measurements, but they supplement them and make them understandable.

Finally, the domination by technology and STEM in assessing and dealing with problems has to be undone. Yes, STEM is important, but so too are all areas of knowledge. Technology has too much control over our lives. We have to change the entire fundamentals of technology to ensure that machines and technology are the servants of people, not the other way around.

These changes impact primarily governments and industry through changes to the measures used by financial analysts, investors, and government monitors, including international alliances. These changes are on the one hand relatively straightforward but on the other hand a huge policy challenge, one that must be met to win over those who are reluctant to change their methods. The cost will be substantial, but it will be a one-time expense because once the new ways of measuring actions are in place, they will be no more difficult or expensive than current methods.

Changes in the way that governments and industry operate will have major impacts on everyone, but people's lives will be affected primarily for the better. They will have to do little once the changes are implemented. Moreover, the new methods of measurement by companies and governments will be considerably easier to understand than today's highly abstract procedures.

Sustainability

Waste must go away. Here is where the design profession can play a major role by switching to materials that are sustainable and by designing for repair, regeneration, and reuse. Waste destroys the environment, leads to climate change, and kills innumerable species of animals and plants, making life unbearable for millions of people and creatures on earth. For the best approach to this problem, I recommend the principles of a circular economy.

Today people have been trained to recycle, although it is the rare group that does it properly. How could they? To recycle properly, one has to be a materials scientist as well as knowledgeable about the actual capabilities of the local recycling facilities. But with the circular economy in play, recycling will become reusing and returning. There may still be some small amount of pure recycling, but it will be much simpler to do, much easier to understand.

Homes will change their sources of energy, but this will not have much impact on how people live their lives. The major source of energy for homes will probably become electricity as coal, oil, natural gas, and wood-fired heating systems and fireplaces go away. The amount of electric power required to run heat pumps is much less than that for more traditional heating and cooling methods. That electricity can come from a variety of sources, some on the property (the most logical being solar panels for homes and perhaps wind turbines for those in remote areas), others at a distance, where geothermal, hydroelectric, solar, and wind technology can be used. Nuclear power still has a role to play if the waste-disposal problem can be solved. Power by nuclear fusion might play a role years in the future, as has been predicted every year for the past 60 years. Research is ongoing.

A better way to reduce the dependency on energy sources is to reduce the need for energy. Buildings can be designed to use natural ventilation and sunlight to be cooler in summer and warmer in winter. The general rise in temperatures across the world might also lead to a change in the clothing we wear: lighter and cooler throughout the year. These relatively simple actions can yield enormous reductions in the use of energy.

Increasing the efforts to make every citizen better at recycling waste is addressing only the symptoms, not the causes. The problems of waste orig-inate in the methods of industry: in how raw materials are accessed, how things are manufactured, how purchasers cannot efficiently and easily repair, upgrade, or reuse products, and how the design and manufacturing methods make it difficult to reuse basic materials. Once industry implements circular-economy methods, waste problems will be greatly reduced, replaced by much easier abilities to repair, reuse, and return. If designers attack the underlying cause of a problem, the problem becomes manageable and, in the best of cases, disappears.

Humanity Centered

Designers, governments, and industry must broaden the notion of design from human centered to humanity centered. Consistent with respecting the rights of all humanity, they must change the design process to be supportive rather than prescriptive. In other words, when involved in projects to solve a community's specific problem, it is essential to gain the trust and cooperation of the people being served. This is done by resisting the temptation of experts to tell people what to do or what is good for them. Instead, let them live their own life. Designers and other experts must act as facilitators and mentors. Designers and the many NGOs and foundations seeking to help distressed communities must avoid imposing their ideas on the communities. Let those who live in the community have a say in the changes. This can be difficult because even in relatively small communities there will not be full agreement. However, if the design team works with the community organizers and recognizes the humanity of the people they intend to help, their designs could start to change the world on a local level.

The Need for Change

There are many barriers to change, the most difficult of all being human behavior and the human resistance to change, sometimes for legitimate reasons, sometimes owing to a lack of understanding or reluctance to give up the current way of living, but also sometimes because of individual selfishness, fear of losing wealth or power. Here is where great social skills are required. The need for meaningful and sustainable actions must be satisfied primarily by groups of people, the organizations of the world, not by individual citizens here or there. This gives us hope because many major, international organizations have already indicated their willingness to modify their behavior and governments have begun to change policies to encourage or in some cases force the needed changes.

The more major organizations and governments change, the more pressure there will be on the stragglers.

Why am I optimistic even after my remarks about the slow progress by the United Nations on climate change, especially the minimal progress made at its international meetings? Because I believe there has been some progress, but the very structure of the meetings hides it. Any statement must be approved by 100 percent of the parties participating, all 194 of them. This is a high bar, so it is amazing that there is any agreement. These votes suffer from the same problem as the economic models discussed in part II: the attempt to have a single number (in this case, a single vote) represent the complexities of all the underlying issues. Why should these complex bylaws be voted on as a whole? Why not make the bylaws themselves modular? And why not have a normal vote, in which individual nations' responses are tabulated? Then for each of the modular parts of the bylaws, everyone would know how each nation voted.

If the bylaws were to stipulate some defined threshold for passing a module (a simple majority of those voting or perhaps 60 percent of the eligible countries or of those voting), some agreements would be passed and would look much stronger because of the number of important countries that agree to a strong statement. Nations that do not agree would be visible through their vote (or abstention), which might cause their citizens to pressure them to vote positively. The United Nations is well known for the weakness of having to accommodate every nation and every point of view. As a result, so many compromises are necessary that statements are often reduced to nice-sounding platitudes with no actual substance. Today is the time for action. It is time to overcome these weaknesses.

What is preventing us, the people of the world, from staving off the disasters that are bound to befall us? The scientific evidence is clear. The early signs of harm are already present—signifiers of the growing difficulties we will face. The problem is the human tendency to postpone action until after a calamity. These are the issues that define this book.

In 1972, Donella Meadows and the Club of Rome released a pioneering work, *The Limits to Growth; a Report for the Club of*

Rome's Project on the Predicament of Mankind, which argued that the activities of people were starting to exceed the world's capacity to sustain life. The book caused a sensation that died down after numerous savage attacks by the academic community, which focused on the details while ignoring the book's important message. Today, a half century later, the authors' predictions prove to be correct—in spirit, even if the numerical estimates are wrong. In 2022, the Club of Rome published a follow-up book, *Limits and Beyond: 50 Years On from* The Limits to Growth*, What Did We Learn and What's Next?*, to discuss the impact (and lack of impact) of the original publication, despite the veracity of its predictions. This book points to one of the great weaknesses of the original study: the absence of any discussion of human behavior. Both the very first chapter and the concluding chapter, written by the book's editors, Ugo Bardi from the University of Florence and Carlos Alvarez Pereira from the Club of Rome, respectively, point out that the major missing component in the earlier study was a realistic understanding and treatment of human behavior.

Human behavior is the most critical aspect of today's difficulties. Yes, there are numerous physical, technological, and economic issues that must be solved. But underlying all of these is human behavior and its impact on what can be done. Human behavior is what drives political behavior, and without appropriate political responses to the ongoing crises, we will fail. We, the people of earth, have work to do: we must make our voices heard. We must mobilize to insist on change, so that all political and industrial leaders understand and act on the necessity of changing their policies.

Acknowledgments

My books never get written without significant help from many sources and people. This book is no exception. I started it during the racial and gender disturbances in the 2010s symbolized by the hashtags #BlackLivesMatter, #MeToo, and #WhatAboutUs (combining race and gender). The more I read, listened, and learned, the more I realized that the entire world was part of the problem. During a video call, the uncategorizable designer Leyla Acaroglu urged me to expand my point of view: write about “designing for the well-being of humans, for each other, and for the planet,” she said. She explained that my longtime emphasis on human-centered design was too narrow, too closely aligned with the notion that design is a tool of industry. I needed to expand my point of view to cover humanity. As the title of this book indicates, I followed her advice. (Why is she “uncategorizable”? She calls herself a provocateur. I recommend her website, https://www.leylaacaroglu.com/, so you can form your own categorization. She is from Australia, New York, London, and elsewhere.)

While writing the book, I met weekly with a small, five-person discussion group: Arturo Escobar, an anthropologist, best known for his influential book *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*; Terry Winograd, a longtime colleague and friend and a professor of computer science at Stanford; Fernando Flores, a leader in management with a PhD in philosophy whose book with Winograd, *Understanding Computers and Cognition: A New Foundation for Design*, introduced the powerful notion of ontological design; and B. Rousse, a philosopher who is director of research at Pluralistic Networks, the consulting firm founded by Fernando. We call the group Beyond Computers and Cognition (BCC) because we started

with the Winograd and Flores book and then moved far beyond it. The BCC group motto is "to mobilize for a new way of dwelling on the earth." One goal of this book is to urge readers to join us in this goal, perhaps through the activities discussed in part VI.

The students and faculty of the Design Lab at the University of California, San Diego, have deeply enrichened and informed my views. They helped me understand design's contribution to the numerous injustices of the world, not by intent, but through its origins as a profession in service to industry and its acceptance of this practice. These interactions, coupled with the weekly discussions in the BCC group, led directly to part I of this book.

And so the book began. Here is an incomplete list of people I owe thanks to, incomplete because of my faulty record keeping and memory, for which I apologize:

The work of Eric von Hippel has been an extremely important source of inspiration. I knew of his early work on lead users from the mid-1980s, but we didn't meet until 2008 when he and his wife Jessie visited me at my home in La Jolla, California. We immediately bonded. We continued to interact for 10 years, including visits to his home in Cambridge, Massachusetts, next door to his office at the Sloan School of Management at MIT. Unfortunately, the COVID pandemic put an end to travel. His work (and enthusiasm) has been inspiring and is one of the key motivations for my work on humanity-centered design and all of part IV of this book.

Simon Widmer, Anna Queralt, Emily Marsh, Mark Buckley, and Joe Iles of the Ellen MacArthur Foundation for the long discussions during my three-day visit to London in October 2021 and their comments on my draft writings on sustainability.

The initial members of the Sustainability Working Group of the Future of Design Education Initiative: Jeremy Faludi, Leyla Acaroglu, Ana Rapela, Deborah Sumter, Cindy Cooper, and Paul Gardien.

Erica Robles of New York University and Scott Ferguson of the University of South Florida, who made me realize the importance of my own work on affordances for much of the book. One result is the section on how "artificiality creates societal antiaffordances" in chapter 4.

One of my neighbors, Tom English, an environmentalist who directed a large program on energy and the environment for the Environmental Protection Agency, had numerous long discussions with me about the issues. Ken Friedman, a longtime friend and colleague who is editor in chief of *She Ji: The Journal of Design, Economics, and Innovation* and edits a design series for the MIT Press, has given considerable advice. (He was a part of the DesignX team mentioned in chapter 25.)

David Conover of the graphic design firm Studio Conover and I spent numerous weeks prototyping and iterating the figures for the book, putting our design credentials to work.

Sandy Dijkstra, Elise Capron, and Andrea Cavallaro of the Sandra Dijkstra Literary Agency provided support, especially with the early, very rough drafts, and then discussions about publishers that led me to my top choice, MIT Press.

I used an early draft as a text for an advanced seminar at the University of California, San Diego, in the winter academic quarter of 2021, analyzing the content, finding new sources of information, and developing note boards that amplified the ideas. The final reports on the book were so engaging we decided to create a website to accompany the book, which 13 of those students and I worked on in the spring academic quarter of 2022. Thanks to Kimberly Alonzo, Sravya Balasa, Andrew Caballero, *William Duan, Cyrus Gonzalez, Julie Han, *Donna Kim, *Juna Kim, Mylinh Lac, Evan Lam, *Jiaming (Jessy) Li, Meishi Li, *Rajvir Logani, *Corey Lopez, *Huimeng Lu, *Amanda Mark, *Steven Molotnikov, Tracy Nguyen, *Omar Ortega, Nia Page, *Deyshna Pai, Karam Singh, *Adam Syed, Michelle Tenin, *Boyang Wang, and *Tiffany Zhong (asterisks indicate those who took part in the second project to design the architecture and material for the website, primarily based on the final reports that the winter quarter class had prepared).

The website is at DBW.jnd.org. Thanks to Matt Goddard and his team at UX Consultancy in England for bringing the Figma prototype to life.

At MIT Press, acquisitions editor Noah Springer made numerous useful suggestions and guided me through a major revision of the entire book, starting with the title. I thank the four anonymous reviewers who gave feedback to the MIT Press on that early draft: their criticisms were appropriate, and I followed their advice. Virginia Crossman and Anne M. Barva, of the editorial staff, at first frightened me by the density of the red corrections they scattered over every paragraph of the manuscript, but as I realized how well the corrections tightened the writing and improved the flow, while still keeping my informal writing voice, I became delighted with the results (and thankful for their efforts).

Eric Norman provided continual feedback and suggestions, derived from his extensive experience in the world of products and finance in the high-tech world of Silicon Valley. The most important person of all is Julie Norman, who has advised me on all the many revisions of this book (as she has for all my books).

Notes

These notes not only include some comments about each chapter but also, more importantly, provide sufficient information to find a reference for an author or topic mentioned in the text. Where the description in the text is sufficiently clear, I usually do not offer a reference in the notes.

Chapter 1

“We design the world, and it, in turn designs us.” This mutual impact of design upon the world and that of design upon us is a basic tenet of the discipline called “Ontological Design,” introduced to the design community in the 1968 book by Terry Winograd and Fernando Flores, *Understanding Computers and Cognition.* The idea, however, is older, with the basic idea being stated by Winston Churchill in 1943 and by Marshall McLuhan in 1967 (Quote Investigator, “We Shape Our Tools”).

Why Does Our Artificial World Seem So Natural?

The quotes by Nehru come from Alex Von Tunzelmann, *Indian Summer*, p. 21, location 199 of 9840.

Gregory Bateson introduced the term schismogenesis in “199. Culture Contact and Schismogenesis.”

If Design Got Us into Today's Mess, Can Design Get Us Out?

The arbitrary nature of society, civilization, classes, and governments receives a very thorough treatment in David Graeber and David Wengrow's *The Dawn of Everything New*.

Chapter 2

Herb Simon, *The Sciences of the Artificial*.

Chapter 3

The quotations from Victor Papanek are from *Design for the Real World*.

Path Dependence: The Past Does Matter

The English East India Company statement comes from the entry "East India Company" in *Encyclopaedia Britannica Online*.

The Role of Technology and Modernity

The motto of the Chicago World's Fair of 1933 is from its pamphlet *Official Guide-Book of the 1933 Fair*, p. 11. The description of the impact of modernity is from the entry "Modernity" in *Encyclopedia Britannica Online*.

The Curse of Modernity

The quotations "There are professions more harmful than design" and "by creating whole species of permanent garbage" are from Victor Papanek's book *Design for the Real World*, first paragraph of the preface to the first edition (1971), page ix in the second edition (1984).

Make It New

The statement that artists "must break with the formal and contextual standards of their contemporaries" is quoted in E. Bledsoe's essay "Make It New." The works of the Confucian scholar Chu His and his impact on Pound are from Michael North, "The Making of 'Make It New.'" The quotation from Adam Smith, "People of the same trade seldom meet together," is from his well-known book *An Inquiry into the Nature and Causes of the Wealth of Nations*, book 1, chapter 10 (p. 152 of the 1976 edition).

Chapter 4

The concept and term *affordance* were invented by the perceptual psychologist J. J. Gibson in 1966, best described in the chapter on affordances in his 1979 book *The Ecological Approach to Visual Perception*.

The notion of affordances has been generalized far beyond Gibson's descriptions. I introduced the term to design in 1988 in my book the *Psychology of Everyday Things* (later retitled *The Design of Everyday Things*). Later, Bill Gaver published "Technology Affordances," and Erin Bradner's introduced the term *social affordances* in "Social Affordances of Computer-Mediated Communication Technology." Victor Kaptelinin's detailed entry "Affordances and Design" in the Interaction Design Foundation's *Encyclopedia of Human-Computer Interaction* gives a thorough analysis of affordances and the many nuances and related concepts that are of relevance to the design profession.

Chapter 5

The reduction in air pollution caused by travel restrictions due to COVID are discussed by Muhammad Usman and his colleagues in “Climate Change during the COVID-19 Outbreak.”

Where Capitalism Has Gone Wrong

Planned obsolescence has been the subject of numerous publications. Victor Papanek called it “a sort of Gresham's Law of Design” in his book *Design for the Real World* (“important values of real things have been driven out by phony false things, a sort of Gresham's Law of Design,” p. 17), and the index of the book points to numerous references to obsolescence. Gresham's law of economics is often simplified as “bad money drives out good.” Papanek clearly interpreted the design version of the law to be “bad design drives out good.” Giles Slade's treatment of obsolescence in his book *Made to Break* is more nuanced and scholarly (and makes for delightfully easy and informative reading). The definition of planned obsolescence as “the deliberate shortening of a lifespan of a product . . .” is from Slade's book, p. 5. The book edited by Justin McGuirk for the London Design Museum's exhibit *Waste Age* in some sense is entirely about planned obsolescence but covers it specifically in the chapters by Kate Soper (pp. 35–36, 60–61), Lee Vinsel (p. 78), and Julia Watson in conversation with McGuirk, who discuss a beneficial aspect of planned obsolescence. If things were built from natural, biological materials, they would ultimately self-decompose (that is, rot) in a useful biodegradable fashion. McGuirk suggests that the design industry needs “to get more comfortable with decomposition and with the aesthetics of imperfection” (p. 181).

Chapter 6

What Is Missing

I used several books to learn more about many of the social ailments of our times. In addition to these readings, I talked with many of the authors and met with numerous groups from around the world who are concerned with these issues. Many more books and articles are now available: these are the ones that most influenced me.

Sasha Costanza-Chock's powerful and important book *Design Justice: Community-Led Practices to Build the Worlds We Need* has played a major role in educating me about the issues that have been badly neglected in traditional design education (including neglected in my book *Design of Everyday Things*). I have used Kat Holmes's book *Mismatch: How Inclusion Shapes Design* in my research and consulting and have recommended it to many. Holmes shows how badly we ignore differently abled people, much to our own detriment. Do we think about designing things that can be used for people with only one hand? Usually not, yet many people frequently have only one hand available for use: a parent holding an infant with one hand while trying to use a computer or trying to open a car door and put packages inside with the other.

Safiya Umoja Noble's work *Algorithms of Oppression: How Search Engines Reinforce Racism* demonstrates the failings of our technology. It joins other books such as Shoshana Zuboff's *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*, which shows how companies deliberately exploit us. Noble discusses the harm done by lack of attention to the biases of our systems and the training sets used for search algorithms and neural networks; Zuboff shows how insidious this lack of attention can be when the offenses are deliberate.

I have learned from Ibram X. Kendi's books on racism and antiracism (e.g., *How to Be an Antiracist*) and Robin J. DiAngelo's *White Fragility: Why It's so Hard for White People to Talk about Racism*. The collected volume *The Black Experience in Design:*

Identity, Expression & Reflection edited by Anne Berry and others captures the impact of racism on the lives and works of Black designers. Chapter 1 of Kevin Bethune's book *Reimagining Design* tells some of the difficulties he faced as a young Black designer, and the other chapters tell how he made his way through corporate America so that today he is a respected major leader in design. Arturo Escobar's book *Designs for the Pluriverse* changed my views about culture and the role of the Indigenous peoples of the world, but I have also benefited immensely from his many other books and writings and especially through his frequent conversations with me.

This is only a brief survey of the many important books on the topic of what is missing in the field of design. Perhaps of greater importance are the many community groups aimed at overcoming and eradicating these prejudices and injustices. I urge readers to explore both the books and the communities.

Chapter 7

How Scientists Explain the Crisis of Climate Change

The predictions of the sea-level rise come from scientists at the NASA Earth Observatory Project, *Anticipating Future Sea Levels*, with a caution that it is “hard to see far into the coastal future.”

Chapter 8

The quotation from Lord Kelvin is from *Popular Lectures and Addresses*, listed under his given name, William Thomson, in the bibliography. An excellent description of the difficulty (and futility) of measuring the productivity of tasks such as programming is brilliantly described in Isaac Lyman's *Stack Overflow* blog, which is an excellent source of information for and by programmers. The witty quotation "*not everything that can be counted counts, and not everything that counts can be counted*" is most frequently attributed to Albert Einstein, but it probably comes from William Bruce Cameron's text *Informal Sociology*. (See the Quote Investigator entry "Not Everything That Counts" for more about this and other quotations on counting things). This is yet another example of Stigler's law, which states that things get named after the most famous person in the vicinity. Stigler points out that the law that carries his name was actually first stated by Robert Merton (see *Wikipedia*, "Stigler's Law of Eponymy").

The Scientist's Creed That Measurement Is Necessary, Even of the Unmeasurable

The quotation from Ruth Carlitz is from "Pandemic Data Collection," an interview with her in the *American Scientist* (p. 331), and is reproduced here by permission of the *American Scientist*.

Chapter 9

The original criticism of the dominance of studies of people who are WEIRD (Western/White, educated, industrialized, rich, and democratic) in psychology was by Joseph Henrich, Steven J. Heine, and Ara Norenzayan in their 2010 article “The WEIRDest People of All” in the journal *Behavioral and Brain Sciences*. A study trying to get at the origins of WEIRD appeared in 2018, with a massive analysis of all the contributions to WEIRD thinking: see Schulz et al., “The Origins of Weird Psychology.”

The Measures Used in Economics

Adam Smith's book is thought of as the first rigorous approach to economics, marking the start of the discipline. it is generally referred to as *The Wealth of Nations*, but its full title is *An Inquiry into the Nature and Causes of the Wealth of Nations*.

Chapter 10

Justin Fox's referenced *Harvard Business Review* article is "The Economics of Well-Being." The quote from David Pilling is given in Peter Coclanis's article in the *New Statesman*, "Why We Urgently Need a Real Alternative to GDP as an Economic Measure," reprinted with permission of the *New Statesman*, as well as on page 2 of Pilling's book *The Growth Delusion: Wealth, Poverty, and the Well-Being of Nations*. The British students of economics who complained about the courses they were getting and wrote the book *The Econocracy* to give voice to their complaints are Joe Earle, Cahal Moran, and Zach Ward-Perkins.

The Quest to Replace the GDP

The World Economic Forum's measures of growth are given in its article "Five Measures of Growth That Are Better Than GDP." The Gini index is nicely described in the article "The Gini Index" by Lisa Smith in the economics online forum *Investopedia*.

GPI and HDI

The description of the HDI comes from the United Nations article "Human Development Index (HDI)."

Chapter 11

The Seventeen Sustainable Development Goals of the United Nations

The seventeen goals are listed in the United Nations Development Programme's webpage “United Nations: Sustainable Development Goals.” The page is also a rich source of information about each goal and the progress that has been made.

Measuring Happiness and Well-Being

The University of Pennsylvania's Positive Psychology Center's website is at https://ppc.sas.upenn.edu/. The technique for measuring the well-being of nations is described in A. Adler and M. E. P. Seligman's article “Using Wellbeing for Public Policy.” Multi-Dimensional Scaling (MDS) is a well-established technique today. There are even tools that enable one to do an analysis in an Excel spreadsheet (see https://help.xlstat.com/6702-multidimensional-scaling-mds-excel-tutorial). The number of papers and books on the field is too large to provide even a short list of titles here.

A Meaningful Approach to Climate Change

The two papers that inspired figures 11.2a and 11.2b are “A Safe Operating Space for Humanity” by Johan Rockström and his colleagues in *Nature* and “Planetary Boundaries” by Will Steffen and his colleagues in *Science*.

The Doughnut Model: An Effective Dashboard

The website of the Doughnut Action Lab (https://doughnuteconomics.org/) contains a rich collection of tools, theory, videos, and case studies of groups and cities that are applying the doughnut economics model, shown in a wide variety of graphical renditions. Much of the material on the website is freely available, licensed under Creative Commons Attribution-ShareAlike 4.0 International License (https://creativecommons.org/licenses/by-sa/4.0/). Kate Raworth's book *Doughnut Economics* provides a succinct and powerful attack on much of contemporary economics (highly relevant to my arguments in part II of this book) as well as an introduction to the doughnut representation. Raworth points out that “the Doughnut is, itself, a global dashboard of living metrics” (p. 206).

Chapter 12

A typical paper from Ward Edwards's studies of gambling in Las Vegas is “Choices among Bets” by D. G. Fryback and W. Keith Edwards. To learn how findings from psychologists were used, see Natasha Schüll's book *Addiction by Design*: *Machine Gambling in Las Vegas*.

The Power of Stories

Shiller's quote from *Narrative Economics* is on p. 22.

I wrote about my experiences in the hospitals (Story 1 and Story 2) in Norman, “A Fetish for Numbers.”

Measurements Give Facts, Stories Provide Meaning

Hillier, Kelly, and Klinger studied narrative style in writing technical scientific reports and found that those written in a narrative style were cited more frequently. Moreover, the more important the journal (judged by its impact ratings), the more frequently narrative style was important. The quotation is from the introduction to their article “Narrative Style Influences Citation Frequency in Climate Change Science.”

The quotation “complexity is a fact of the world, whereas simplicity is in the mind” is from my book *Living with Complexity*, p. 53 (and is its theme).

Chapter 13

The catalog *Waste Age: What Can Design Do?* for the London Design Museum's exhibit is an interesting and valuable contribution to understanding the history of waste, its current status, and opportunities for the future with eight excellent chapters by topic authorities. It is edited by the museum's chief curator, Justin McGuirk.

Chapter 14

The Throwaway Economy
The three techniques of designing for obsolescence comes from multiple sources, but I recommend the book *Made to Break* by Giles Slade.

Chapter 15

Science, Technology, Engineering and Mathematics Are Not the Answer

The quotation “In New York in 1900, the population of 100,000 horses produced 2.5 million pounds of horse manure per day” is given in Edwin Burrows and Mike Wallace's book *Gotham: A History of New York City to 1898*, but I was unable to find the actual source of the quotation. Karl Popper's paradox of tolerance is given in the *Wikipedia* entry “The Paradox of Tolerance.”

Chapter 16

The Take-Make-Waste Philosophy of Manufacturing and Usage

The website for the Center for the Advancement of Steady State Economics is https://steadystate.org/ and the quotations come from their mission statement. On their "Meet CASSE" page, they state that they have "become the leading organization promoting the transition from unsustainable growth to a steady state economy."

The Circular Economy of Repair, Reuse, Regenerate

The Ellen MacArthur Foundation is located on the Isle of Wight in the United Kingdom and focuses on proselytizing for the circular economy and circular design. Its website contains much useful information: https://ellenmacarthurfoundation.org/. I spent three days with some of its staff at the opening of the *Waste Age* exhibit at the London Design Museum in October 2021: they are thanked by name in the book's acknowledgments.

It is difficult to trace the origins of the circular economy because so many people developed the underlying principles, probably starting in the 1960s. *Cradle to Cradle: Remaking the Way We Make Things* by William McDonough and Michael Braungart played an important role in publicizing the critical issues (the phrase "cradle to cradle" was invented earlier, probably by the Swiss architect Walter Stahel, who was instrumental in developing many of the key ideas of a circular economy). The Ellen MacArthur Foundation and some of the key people in its circular design section (named in the acknowledgments chapter of this book) have been a valuable source for me. Ken Webster's book, *The Circular Economy: A Wealth of Flows,* was written when Webster was head of innovation for the foundation (he was one of its founding members) and provides an excellent description of the fundamental ideas underlying the concept.

The Circular Economy Requires Circular Design

The comment that “Nature minimizes waste” is expanded upon in the *Vikaspedia* entry “Nature Wastes Nothing.”

Chapter 17

My two papers on recycling in the technology magazine *Fast Company* are “I'm an Expert on Complex Design Systems. Even I Can't Figure Out Recycling” and “Waste Is an Enormous Problem, but Recycling Is the Wrong Solution.”

The Legislative Route to Minimize Waste

The *New York Times* quote is from Soumya Karlamangla, “What to Know About California's Landmark Plastics Law.”

Why Businesses Might Balk at Implementing the Circular Economy

The quotation from Canon comes from their environmental brochure for the United States, *Canon Laser Consumable Return Program*.

Chapter 18

Loosely and Tightly Coupled Systems

The distinction between loosely and tightly coupled systems was first introduced by the sociologist Charles Perrow in his book *Normal Accidents: Living with High-Risk Technologies*. This book has been extremely influential in my own work on human error, aviation safety, and accidents. (Perrow is also well known for his book on complex organizations.)

Chapter 19

Causality

Conditioned taste aversion has received huge interest from the research community in several disciplines: psychology, biology, and medicine. The review article by Jian-You Lin, Joe Arthurs, and Steve Reilly, “Conditioned Taste Aversions” is a good place to start.

My favorite article on people's tendency to quickly blame others after a major incident by pointing to an obvious clue that seemed to have been missed is the classic “Hindsight ≠ Foresight” by Baruch Fischhoff. Yes, after some event has happened, it is easy to look back and see what appears to be the triggering event. But what if we could put those same people into a time machine and arrive at the scene before the incident? Would they predict properly? Nope.

How an Air Conditioner Heats the Entire Planet When It Cools Our Rooms

The quotations and many of the numerical estimates of the use of electrical power by air conditioners comes from the report *The Future of Cooling* by the International Energy Agency (quotes come from the foreword and pp. 29–30). The study of temperature rise in Phoenix due to air conditioning is Salamanca et al., “Anthropogenic Heating of the Urban Environment.”

For more about air conditioning, see Rushad Nanavatty's article for the World Economic Forum, “Air conditioning Is Threatening Our Ability to Tackle Climate Change” and Sneha Sachar and her colleagues’ report *Solving the Global Cooling Challenge: How to Counter the Climate Threat from Room Air Conditioners*.

Chapter 20

Complex Systems: Feedback, Feedforward, Recursive

The simplest example of recursion I can think of is the mathematical function called “factorial.” (Factorial *n*) means to multiply all the numbers up to and including *n*, so Factorial 4 = 4 * 3 * 2 * 1 (where * means “multiply”). So Factorial (*n*) = *n* * Factorial (*n* –1).

According to the *Wikipedia* entry “Space Adaptation Syndrome,” “Space adaptation syndrome (SAS) or space sickness is a condition experienced by as many as half of all space travelers during their adaptation to weightlessness once in orbit.”

Natural Sciences, Engineering, and the Social Science

The quotations come from Luís Amaral and Julio Ottino, “Complex Networks,” p. 159.

Rethinking the Modern Way of Constructing Homes and Buildings

All quotations from the emails with Maryam Fazeli are used with her permission (the relevant correspondence continued from March to July, 2022). She recommended the *Scientific American* article by Mehdi Bahadori on “passive cooling systems in Irani architecture,” which led me to the work of the Indian architect, Vinod Ghupta, “Indigenous Architecture and Natural Cooling” (p. 48), whose words I quote. And while you are at it, be sure to read his paper with the marvelous title “You Don't Need Airconditioners to Cool Your Home; Traditional Architecture Does It Well. Even Rich Families Living in the Heart of the Desert Cool Their Homes Naturally.” The paper is old (1984), but the fundamental principles still apply.

Fazeli also pointed to a recent World Economic Forum discussion about the use of wind towers (go to www.weforum.org/ and search for “wind tower”—include the quotation marks). The *Wikipedia* article “Windcatcher” (another name for wind tower) is a good review of the topic. I also recommend a search on the internet for the string “‘wind catcher' OR 'wind tower'" (again, use quote marks around the terms and include "OR") as quite productive. For example, it led me to this paper published by A. R. Dehghani-sanij, M. Soltani, and K. Raahemifar: "A New Design of Wind Tower for Passive Ventilation."

The more I searched, the more I found. Architects all over the world are responding to the climate crisis. In addition to the internet searches for windcatchers and towers described above, search for "passive cooling buildings" and note the wide variety of suggestions.

Jorge Gracia's comments are quotations from his email sent to me on July 1, 2022, used with his permission.

Chapter 22

The two papers that show the early usage of “humanity-centered design” are “Recognizing Individual Needs and Desires in the Case of Designing an Inventory of Humanity-Centered, Sustainability-Directed Concepts for Time and Travel” by Christy Reed, Hui-Wen Wang, and Eli Blevis and “Regarding Software as a Material of Design” by Eli Blevis, Youn-Kyung Lim, and Erik Stolterman. The quotation comes from the latter paper, p. 11.

The Four Principles of Human-Centered Design

There are numerous dialects of human-centered design, so the principles presented here are my own, taken in part from my book *The Design of Everyday Things* and much of my writing since then as well, of course, in part from my practical experience as a member of or a consultant to industry. I outlined the principles in “The Four Fundamental Principles of Human-Centered Design.”

Our Traditional Design Methods Are a Form of Colonization

Easterly's assessment of Kenny's 2011 book is printed in Charles Kenny, *Getting Better* (loc. 107 of 4250, Kindle). The quotes about searchers and planners come from Easterly's introduction to his collected volume *Reinventing Foreign Aid*, p. 13.

Chapter 23

Step Two: Enable Everyone to Design

The work of Eric von Hippel includes his very influential paper "Lead Users," in which he pointed out the great value in basing the design of products on the experiences of people who really use them and who make their own changes and modifications—people he called "lead" users. His work has since extended way beyond that initial concept, most importantly for me in his books *Democratizing Innovation* and *Free Innovation*, and today into other areas, including health care and households (see de Jong and von Hippel, "Household Innovation"). I cite the work on patient innovation in the notes for chapter 24.

The question of "what is the role of design experts?" is from the last page of Manzini's *Politics of the Everyday* (p. 127). His next book, which does not address the question but, nonetheless, offers a partial answer, is *Livable Proximity: Ideas for the City That Cares*. Here is what I said about it for the cover blurb: "Manzini redefines proximity for the digital age, revealing a delightful, creative, sociable way of living. . . . A powerful, important way to live sustainably in the 21st century."

The quotation from Ivan Illich's book, *Tools for Conviviality*, is from page 7 of the 2011 Kindle edition.

Chapter 24

Health, Medicine, and Public Health

The Rise of the Bio-citizen is by Eleonore Pauwels and Sarah W. Denton and the quotation comes from the end of the executive summary (page 4 of my PDF, but listed as page 8 in the table of contents).

Vineet Pandey's work can be found on his website: https://vineetp13.github.io/; I list two papers he wrote with others in the bibliography, one titled “Gut Instinct” and the other “Docent.”

Eric von Hippel introduced me to Pedro Oliveira, and I visited him in Lisbon in 2017. His Patient Innovations group in Lisbon has its website at https://patient-innovation.com/ See his recent paper with von Hippel: Demonaco et al., “When Patients Become Innovators.”

The Ugly Indian group has been written about in the newspapers in India, but perhaps the best source is the *Wikipedia* entry “The Ugly Indian.”

Chapter 25

The Systemic Design Network can be found at http://systemic-design.net/. “Transition Design” is described by the Carnegie Mellon University Transition Design effort at the design department's website: https://design.cmu.edu/tags/transition-design.

The DesignX paper I mention in the text is Don Norman and Pieter Jan Stappers, “DesignX: Complex Sociotechnical Systems.”

Small, Incremental Steps: Incrementalism, or “Muddling Through”

Various papers on muddling through include John Flach, “Complexity: Learning to Muddle Through”; Charles Lindblom, “The Science of ‘Muddling Through’” and “Still Muddling, Not Yet Through”; and Jonathan Bendor's review paper on the status of “muddling through,” “Incrementalism: Dead, yet Flourishing.”

Chapter 28

The book by Peter Jones and Kristel van Ael, *Design Journeys through Complex Systems* is the only publication I know about that treats the implementation stages of large design projects from the designer's point of view. Large projects are very well treated in the standard literature on project management and its subdiscipline complex project management, but although the books and the skills they discuss are essential, they don't make project managers part of design teams—whether it is a team of designers, architects, urban planners, and engineers or a team of designers with professionals from other disciplines.

Chapter 29

Scaling Down: The Rise of the Small
Jeremy Myerson's paper “Scaling Down” was published in *She Ji* in 2016. The quotation and the list of five principles are on pp. 291 and 292, respectively.

Scaling Up by Scaling Down
I give a slightly changed version of the title of Escobar's book *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*.

Chapter 30

What Design Has to Offer

The quotation at the end of the chapter comes from Norman and Stappers, “DesignX,” p. 91.

Chapter 31

Change: The Most Difficult Task of All

The title of Danny Kahneman's book sums up his main concept: *Thinking, Fast and Slow*.

People Will Respond to a Disaster but Not to Prevent It

On the CDC and inequity in the responses to COVID-19, see US Centers for Disease Control and Prevention, "Health Equity Considerations & Racial & Ethnic Minority Groups," and Saunders and Evans, "COVID-19, Tuberculosis, and Poverty."

On SCUBA deaths, see the *Wikipedia* entry "Scuba Diving Fatalities," especially the section titled "Failure to Ditch Weights." (The entire long article will be of interest to readers who are SCUBA divers.)

Chapter 32

Greg Jackson's article is “Prayer for a Just War.”

Regarding “one large conspiracy group,” I don't wish to name the group because it does not deserve more publicity. The frightening part is *U.S. News and World Report*'s finding that roughly 44 million people agree with that assessment. See Susan Milligan's article “A Quarter of Republicans Believe Central Views of QAnon Conspiracy Movement.”

Chapter 33

The Balance between Work and Family

If you were bothered by the statement that the names of weekdays are derived from the names of celestial bodies, yet the actual names seemed discrepant, it is because different languages and cultures associate different gods with the celestial bodies. The romance languages tend to use the Latin names assigned by the Romans, whereas English uses the Roman names for Sunday, Monday, and Saturday but Germanic and Norse names for the others.

In 1922, the Ford Motor company became the first very large company to change the six-day work week to five days. It wasn't because Ford wished to be nice. He had determined that reducing the work week by one day did not make much difference in productivity but provided his workers leisure time that they could use to purchase products. As Green, the president of the American Federation of Labor, said, “Unless the masses of people have leisure they cannot buy and use the products of science and industry. Mass production without mass consumption becomes a Frankenstein.” This history comes from Frank De Vyver's article “The Five-Day Week,” pp. 224 and 227.

Work: Consider Changing Work from a Job to an Occupation

The quote from Kathi Weeks is from *The Problem with Work*, p. 2.

Grades for Students

The *Wikipedia* entry on mastery learning is quite good. I wrote a paper on project-based learning with Paul Kirschner: “Teaching and Learning Design.” I also highly recommend the book by Jeroen van Merriënboer and Kirschner, *Ten Steps to Complex Learning*.

Chapter 34

Treating People as Objects, as Servants of Machines

Taylor's classic book about his methods is *The Principles of Scientific Management*. Also see Robert Kanigel's critical history of Taylor's work and impact, *The One Best Way: Frederick Winslow Taylor and the Enigma of Efficiency*.

My personal bias is to combine the good parts of GOFAI ("good old-fashioned AI"), which worked through symbolic reasoning, with the good parts of neural networks. I first said this in the very early days of neural networks, when the first systems developed at the University of California, San Diego, were called "parallel distributed processing" (soon replaced by "connectionist networks" and then by today's name, "neural nets"). I was asked to write the last chapter of a two-volume introduction to this new form of computation, *Parallel Distributed Processing*, and to reflect on how they worked and what they might yet become: My chapter is "Reflections on Cognition and Parallel Distributed Processing."

Focus, Curiosity, Mind Wandering, and Mental Addiction (Flow)

Mind wandering is treated by Benjamin Baird and his colleagues in the journal article "Inspired by Distraction: Mind Wandering Facilitates Creative Incubation." Adam Alter discusses addiction in his book *Irresistible: The Rise of Addictive Technology and the Business of Keeping Us Hooked*.

Mihaly Csikszentmihalyi discusses his "flow" concept in numerous books and articles. Start with the two books *Flow: The Psychology of Optimal Experience*, which gives the original presentation of the concept, and *Finding Flow: The Psychology of Engagement with Everyday Life*.

Chapter 35

The reports from the International Monetary Fund and the Brookings Institution are Bazarbash et al., “Mobile Money in the COVID-19 Pandemic,” and Chowdhury et al., “Accelerating Digital Cash Transfers to the World's Poorest.”

The world of ubiquitous computing and that of the invisible computer was first discussed in 1991 by Mark Weiser, then at the Xerox Palo Alto Research Center (PARC). The concept was expanded upon by Weiser and John Seely Brown, also at PARC, in 1997 (and renamed “calm computing”) and by me in my book *The Invisible Computer* (1998). For more on calm tech, see Seely Brown's review at http://www.johnseelybrown.com/calmtech.pdf.

The patent mentioned for facilitated walking was filed by Nicholas Yagn as “Apparatus for Facilitating Walking, Running, and Jumping.”

The National Academies of Sciences, Engineering, and Medicine report about the need to develop AI systems that collaborate with people as equals is *Human–AI Teaming: State-of-the-Art and Research Needs* (p. 85).

Chapter 36

Bottom-Up Activities
The Borgen project can be found at https://borgenproject.org.

Chapter 37

Band Together: Combine Existing Projects
Amory Lovins's report on profit from waste *Profitably Decarbonizing Heavy Transport and Industrial Heat*. The Rocky Mountain Institute's website has a huge library of relevant and important studies (https://rmi.org).

Warn Everyone

The paper by Roger Revelle and Hans Suess is “Carbon Dioxide Exchange between Atmosphere and Ocean and the Question of an Increase of Atmospheric CO_2 during the Past Decades.” Alas, the title amplifies the concerns I expressed about meaningfulness in part II. It prevents people outside the field of climate from reading it. The paper may have been an important warning, but people often decide whether to read a paper just by its title, and that tile is aimed at climate scientists, not at politicians and decision makers or other nonspecialists.

Al Gore's book *Our Choice: A Plan to Solve the Climate Crisis* is written in an easy-to-read style, and the electronic edition is dynamic, allowing the reader to get multiple insights.

The Technological Approach
For the quotation, see Paul Hawken, *Regeneration: Ending the Climate Crisis in One Generation*, p. 9.

Dealing with Complex Sociotechnical System
The original “wicked problems” paper is Rittel and Webber, “Dilemmas in a General Theory of Planning.”

Jeffrey Chan and Wei-Ning Xiang provide an excellent review of the issues of “wicked problems” in their guest editorial “Fifty Years after the Wicked-Problems Conception” in the journal *Socio-Ecological Practice Research*. I highly recommend this editorial because it does a nice job of reviewing the major issues involved in the attack on wicked problems. They are planning a special issue of the journal devoted to wicked thinking on its fiftieth anniversary in 2023. As I write, it is still 2022, so I can't use the articles in that issue, but many readers will be able to.

Jon Kolko has made his handbook *Wicked Problems: Problems Worth Solving* freely available on the internet (the URL is in the bibliography entry). The handbook describes multiple approaches and introduces the concept of the “social entrepreneur.” One of the classic design papers on wicked problems comes from Richard Buchanan, “Wicked Problems in Design Thinking.”

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